From Problems to Working Solutions

Applied Mathematics, Computing and AI for Resource-Constrained Environments

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An open textbook in mathematics and scientific problem-solving

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Preface

A useful solution is often less impressive than its description. It may be a better stock-ordering rule, a dependable timetable, a safer way to store field observations, or a calculation that tells an engineer not to trust an apparent pattern. None of these achievements requires the most fashionable algorithm. Each requires an understanding of the problem, adequate evidence and a method that can survive contact with the circumstances in which people will use it.

This book begins before the algorithm. It asks what someone is trying to accomplish, what information they possess, and what would count as a genuine improvement. Mathematics then has a clear job: to make assumptions visible, expose consequences, compare choices and identify uncertainty. Computing makes those arguments repeatable. Statistical and machine-learning methods enter when they supply information that simpler methods cannot provide sufficiently well.

Low- and middle-income countries provide many of the motivating examples because unreliable power, limited measurements, expensive connectivity and stretched institutions can make poor design decisions especially costly. The underlying principles are general. An expensive hospital system in a wealthy country and a remote health centre with one computer face the same basic question: does the proposed change address the cause of the problem, and is there evidence that it works?

The book has three parts. Chapters 1–3 teach how to identify a useful problem, construct a meaningful model and judge the evidence. Chapters 4–8 develop prediction, optimisation, machine learning, resilient computing and a sustained health-service case. Chapters 9–12 ask whether interventions improve outcomes, how organisations maintain them, what transfers to a new context, and how a reader can deliver a complete solution dossier. The examples are chosen so a motivated reader can check the central arithmetic on modest equipment. Their value lies in the reasoning they make possible, not the size of the datasets.

Evidence convention. A mathematical result derived from stated assumptions is labelled as such. A calculation on invented data is an illustration. A computation on publicly documented observations is a retrospective empirical analysis. A prototype evaluated with an organisation is a different form of evidence again. None should be described as a demonstrated operational benefit without a suitable field evaluation.

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Part I

Understanding Problems and Evidence

Chapter 1

Problems Worth Solving

A problem has not been understood merely because someone can describe what is going wrong. We need to know which decision can change the situation, whose interests matter, and what evidence would justify that decision.

1.1 The queue outside the clinic

A patient arrives at a rural health centre shortly after eight in the morning. Several people are already waiting. One worker is registering patients, a clinical officer is seeing those who have completed registration, and the laboratory is working through yesterday's machine fault. The electricity has returned after an overnight interruption, but one computer is still restarting. By midday, the waiting room is full. At a meeting the following week, someone proposes an artificial-intelligence system to predict tomorrow's attendance. If the facility knew how many patients would arrive, the argument goes, staff could prepare for them.

Perhaps that prediction would help. Yet the patients might still wait just as long. An excellent forecast of arrivals does not repair a laboratory instrument, increase the number of people seeing patients, or shorten a registration procedure. The statement "patients are waiting too long" describes an undesirable outcome. It does not identify the reason, and it certainly does not identify the remedy.

We shall use this fictional facility throughout the chapter. No waiting-time measurements were collected from a real health centre for this illustration. The point of the example is to examine the reasoning that would be required before making a recommendation. In an actual facility, staff, patients and administrators would need to confirm the description, the costs of delay and the options they can realistically implement.

There are at least three plausible explanations for a long queue. Arrivals may be concentrated into a short period. Registration may be too slow. Or patients may reach a later stage that cannot process them quickly enough. These explanations are not interchangeable. An extra registration desk would be of little value if every patient subsequently joined a much longer laboratory queue. A better arrival forecast could be useful for scheduling, but only if someone had the authority and resources to change the schedule.

This is the first discipline of applied problem solving: resist choosing a method before establishing

what it is supposed to change.

1.2 Need, symptom, decision and objective

In ordinary conversation, all of the following may be called "the problem": the need for timely care, the symptom of long queues, the decision about staffing, the wish to reduce delay, and the shortage of available workers. Their differences matter in mathematics.

A *need* describes what people have reason to value: receiving necessary care within an acceptable time. A *symptom* is what we observe: a long queue. A *decision* is a choice available to someone: open a second service desk on busy mornings, move part of registration to another time, or modify appointment arrangements. An *objective* states how choices will be judged: perhaps reduce waiting while keeping staffing expenditure within budget. A *constraint* rules out some choices: the facility may have only one qualified person available to perform a particular procedure.

These distinctions prevent an important kind of mistake. Suppose a manager says, "Our objective is to introduce an electronic registration system." Introducing a system is an action, not the public-service outcome that motivates it. The relevant objective might instead be to reduce registration time without compromising record accuracy, privacy or staff workload. A digital system is one candidate intervention. It may turn out to be unnecessary.

Table 1.1: Different statements about the same visible difficulty

Kind of statement	Example at the fictional health centre
Need	Patients should receive appropriate care without avoidable delay.
Symptom	Waiting time is long on some mornings.
Possible mechanism	Registration capacity is low when many patients arrive together.
Decision	Change the service schedule or allocate an additional registration worker.
Objective	Reduce the total time patients spend before completing the required stages.
Constraint	Staff time, available rooms, cash, equipment and privacy requirements.
Evidence needed	Arrival and service timestamps at each stage; staff schedules; workflow observations.

The table contains a hypothesis about the mechanism, not a finding. The first visit to the facility should challenge it. If queueing is caused mainly by delayed laboratory results, focusing on registration could merely move patients from one queue to another.

1.3 From an everyday choice to a mathematical decision

Let a represent a decision. The set $\mathcal{A}$ contains the decisions that are actually available. This is the *feasible decision set*. It can be small: for example $\mathcal{A} = \{a_0, a_1, a_2\}$, corresponding to the current arrangement, a changed staff timetable and an additional service desk. Or it can be large, containing every admissible combination of staffing levels and shift times.

The outcome of a decision depends on circumstances we do not completely control. Denote these circumstances by ξ: the number and timing of arrivals, interruptions to electricity, unusually

difficult cases, staff absences and other disturbances. Write the loss arising from decision a under circumstances ξ as

$$L(a, \xi). \tag{1.1}$$

The word *loss* need not refer only to money. It might measure waiting hours, missed appointments, stock-outs or some carefully justified combination of such outcomes. If the uncertainty can be represented by a credible probability distribution, a possible decision rule is

$$a^{*} \in \arg\min_{a \in \mathcal{A}} \mathbb{E}\big[L(a, \xi)\big]. \tag{1.2}$$

Equation (1.2) is not a claim that every social problem can be reduced to one number. It simply makes a proposed criterion explicit. A real decision may require several objectives and constraints that should never be traded away merely because they are difficult to price.

For example, a staffing decision may minimise a weighted combination of time and expense,

$$J(a) = c_{\mathrm{w}}\, \mathbb{E}[W(a)] + C_{staff}(a), \tag{1.3}$$

where $W(a)$ is waiting time, c_{w} translates delay into a monetary or utility measure, and C_{staff} is staffing cost. The units must be compatible. If W is measured in hours per patient, then c_{w} should express cost per patient-hour and the objective should consistently represent the same population or period. Writing down (1.3) exposes a question that casual enthusiasm for a new system may conceal: how much does a reduction in waiting matter, and to whom?

Instead of assigning a single value to every consequence, the decision-maker could use a constrained formulation,

$$\min_{a \in \mathcal{A}} \mathbb{E}[W(a)] \quad \text{subject to} \quad C_{staff}(a) \le B, \tag{1.4}$$

where B is the available staffing budget. One may also require a maximum probability of unacceptable delay. A solution that looks optimal after ignoring patient safety or minimum staffing regulations is not feasible, regardless of its numerical score.

1.4 Who decides, and who bears the consequences?

An engineer may be able to calculate an efficient shift timetable without being authorised to change staff schedules. A hospital administrator may control staffing but not the electricity supply. Patients may experience the cost of delay but have no role in defining the performance measure. Before modelling, identify who makes the decision, who must implement it and who experiences its consequences.

It is useful to distinguish *decision authority* from *technical capability*. A prediction service may have perfect technical performance and no practical effect if staff cannot respond to its output. Conversely, a modest rule for switching duties between staff members may be valuable when it fits their existing authority and working relationships.

Three questions help expose this distinction. What action can the user take after receiving a prediction? How soon can that action be taken? What happens if the prediction is wrong?

If there is no useful answer to the first question, there is little reason to build the prediction system yet.

This is also where equity enters the calculation. An average wait of twenty minutes can conceal an hour-long wait for people who arrive from remote villages or require additional assistance. A service objective might therefore include an upper quantile of delay, or a requirement that vulnerable groups not be systematically worse off. Neither rule can be evaluated without the corresponding observations and careful thought about privacy.

1.5 A small queueing model that we can check

We now make the health-centre problem deliberately narrower. Imagine a single service desk through which patients arrive at an average rate of λ patients per hour. Each service takes on average $1/\mu$ hours, so the desk can serve at mean rate μ patients per hour. Assume arrivals are independent and occur according to a Poisson process, that service times are independent exponential random variables, and that patients form one queue served first-come-first-served. These assumptions define the familiar $M/M/1$ model (Little, 1961; Law, 2015). They are convenient and may be wrong for a real clinic.

The *traffic intensity* is $\rho = \lambda/\mu$. If $\rho \geq 1$, the idealised infinite-capacity system does not have a finite steady-state mean waiting time. If $0 \leq \rho < 1$, the stationary number of patients in the system has probabilities

$$\Pr\{N = n\} = (1 - \rho)\rho^n, \qquad n = 0, 1, 2, \ldots. \tag{1.5}$$

The expected number in the system follows by differentiating a geometric series:

$$\mathbb{E}[N] = (1 - \rho) \sum_{n=0}^{\infty} n\rho^n = \frac{\rho}{1 - \rho} = \frac{\lambda}{\mu - \lambda}. \tag{1.6}$$

Under the applicable stable, long-run conditions, Little's law relates the average number in a system to the arrival rate and average time spent there: $L = \lambda W$ (Little, 1961). Combining it with (1.6) gives

$$W = \frac{1}{\mu - \lambda}, \qquad W_q = W - \frac{1}{\mu} = \frac{\lambda}{\mu(\mu - \lambda)}. \tag{1.7}$$

W includes service; W_q counts waiting *before* service. Confusing the two would lead to a misleading claim about what the facility can improve.

Consider invented rates of six arrivals per hour and eight completed services per hour. Equation (1.7) gives a mean time in the system of thirty minutes even though the average service itself takes only $60/8 = 7.5$ minutes. The remaining 22.5 minutes is average queueing delay. The result illustrates congestion: as the arrival rate approaches service capacity, mean waiting time rises rapidly. A clinic may therefore struggle during a short morning rush even when its daily total number of patients seems manageable.

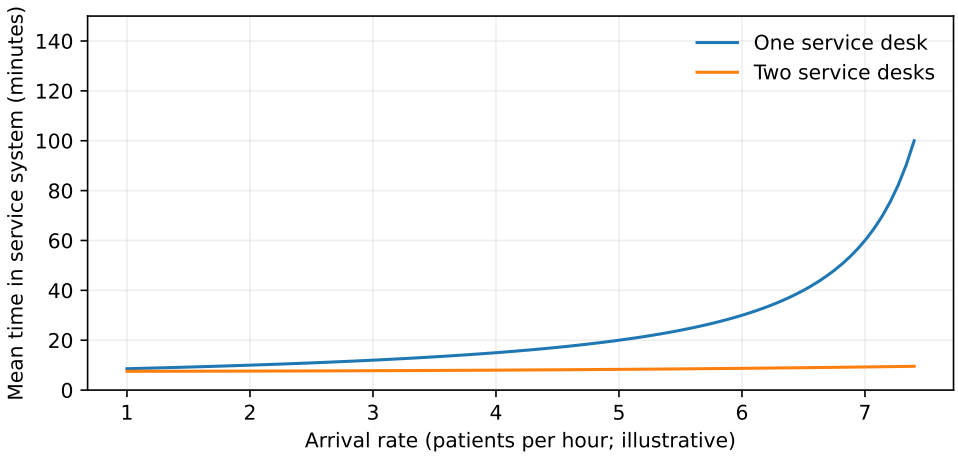


Figure 1.1: Analytical queueing times for a fictional service desk as the assumed arrival rate changes. The curves are computed from an idealised $M/M/c$ model, not fitted to any health-facility observations.

1.6 Comparing interventions without claiming field impact

What if the facility could process registrations a little faster? Or assign a second person to the same task? We can compare these choices under the assumed model. The two-server case uses a common queue with two identical exponential-rate servers. With c servers, define $a = \lambda/\mu$ and $\rho = a/c$. When $\rho < 1$, the Erlang-C probability that an arrival must wait is

$$P_0 = \left[\sum_{k=0}^{c-1} \frac{a^k}{k!} + \frac{a^c}{c!(1-\rho)}\right]^{-1}, \tag{1.8}$$

$$P_{\text{wait}} = \frac{a^c}{c!(1-\rho)} P_0, \qquad W_q = \frac{P_{\text{wait}}}{c\mu - \lambda}, \qquad W = W_q + \frac{1}{\mu}. \tag{1.9}$$

The calculations below are generated by the companion program. The assumptions are exactly those stated; none of the numbers is a measured waiting time.

Table 1.2: Three illustrative queueing arrangements with six patient arrivals per hour

Assumed arrangement	W_q (min)	W (min)	ρ per server
Existing single desk	22.5	30.0	0.75
Faster desk	13.3	20.0	0.67
Two service desks	1.2	8.7	0.38

The hypothetical two-desk system has a much shorter calculated wait. That does not show it is the best real-world decision. A second worker has a cost. The two desks may share equipment that becomes a bottleneck. The assumption of identical service times may be false. Patients may move from registration straight into a queue at the laboratory. A proper evaluation would trace the complete journey and the intended outcomes.

What does the result say about arrival prediction? Under fixed λ, μ and c, an arrival forecast

alone does not alter the queueing equations. The forecast could matter if it were used to vary staffing, appointments or some other controllable resource. The value would then come through the *action taken in response to information*. This distinction will recur throughout the book.

1.7 The value of a measurement depends on the decision

Suppose staffing can be changed tomorrow morning, but the manager must choose tonight. Let the morning be quiet with probability $1 - p$ or busy with probability p. Two actions are available: keep one worker, a_1, or provide two, a_2. If the losses in the two states are known, the expected loss without further information is

$$R_0 = \min_{a \in \{a_1, a_2\}} \big[(1-p)L(a, \text{quiet}) + pL(a, \text{busy})\big]. \tag{1.10}$$

If someone reveals the true state before the decision, the risk becomes

$$R_{\text{perfect}} = (1-p) \min_a L(a, \text{quiet}) + p \min_a L(a, \text{busy}). \tag{1.11}$$

Thus the *expected value of perfect information* is

$$\text{EVPI} = R_0 - R_{\text{perfect}} \geq 0. \tag{1.12}$$

The inequality follows because the decision-maker who sees the state can always choose the same action as before. This is a mathematical bound, not evidence that a real forecasting system will attain the bound. Forecasts are imperfect, and acquiring and using them may cost money.

To see the idea in numbers, use an entirely invented loss table measured in common notional units: With $p = 0.4$, the expected loss of one worker is $0.6(2) + 0.4(14) = 6.8$; of two workers it

Table 1.3: Illustrative loss units for staffing under two possible demand states

Action	Quiet morning	Busy morning
One worker	2	14
Two workers	6	7

is $0.6(6) + 0.4(7) = 6.4$. Without information, two workers are preferred. Perfect information gives $0.6(2) + 0.4(7) = 4.0$, so EVPI $= 2.4$ loss units. A forecast costing more than that amount to obtain and act upon cannot be justified by this objective even if it were perfect. More realistically, we should calculate the reduction in expected loss under the forecast's actual error rates and then subtract implementation costs.

Notice how differently the same forecast might matter at two facilities. If one can alter staffing and the other cannot, the forecast has different operational value even if its statistical accuracy is identical. Information has value in relation to a decision, not in isolation.

1.8 Tracing a service before writing software

A queueing model is useful only if the modelled queue corresponds to the real process. An initial field visit should therefore map the sequence of tasks. A simple process could look like this:



Figure 1.2: An illustrative sequence of service stages. Real facilities may have different pathways, return visits and parallel services.

For each stage, we need to distinguish arrival time, time service begins and time service ends. A single timestamp saying that a patient visited a clinic cannot establish waiting at individual stages. We would also need to distinguish observed service demand from patients who left before being seen. Otherwise the recorded data may describe the behaviour of the queue rather than the unmet need.

The WHO Service Availability and Readiness Assessment provides an established framework for documenting facility availability and readiness (World Health Organization, 2015). It is not a substitute for time-stamped patient journeys or a measured queue at the specific facility. Nor should a facility survey be silently interpreted as evidence of the effectiveness of an appointment system. Different questions require different data.

1.9 When should the modeller stop and investigate?

There are situations in which the best early decision is to postpone technical development. If nobody can agree which outcome matters, testing software would be premature. If the staff cannot change the relevant procedure, the proposed decision tool may have no use. If the observations confuse patients waiting for a test with patients waiting for a consultation, the data do not yet describe the system accurately.

A short investigation can be more valuable than a complex algorithm. Ask the person experiencing the problem to describe the last specific failure. Observe what happened just before it. Compare records with the actual process. Seek an example in which the proposed explanation would be false. These steps are not a rejection of mathematics. They help ensure that the mathematical quantities will correspond to something real.

1.10 Diagnosing the cause rather than naming the symptom

Suppose a clinic manager tells us that the registration queue is intolerable every Monday. We have already heard the complaint. What would it take to understand it? One temptation is to request a spreadsheet of arrival counts and immediately fit a forecasting model. Another is to install more registration computers. Both responses presuppose a cause. A careful investigation begins with a map of the service and with the people who work in it.

Walk through a patient's journey, keeping ordinary activities separate from waiting: arrival, registration, clinical consultation, laboratory test if ordered, collection of results, treatment and departure. Follow some patients who do not need a test. Ask where records are entered twice and which steps depend on electricity. The exercise is observational, and it must be conducted with appropriate permission and privacy protection in a real facility. A staff member may report that the computers are slow. What appears on the screen as slow computing may be a delay in locating a patient file, or it may be the time it takes to negotiate an unreliable network. These are different mechanisms, requiring different interventions.

We can make the argument precise with a process graph. Let each node denote a service activity and each edge a possible transition. A patient who follows path p experiences

$$T_p = \sum_{j \in p} S_j + \sum_{j \in p} W_j + \sum_{j \in p} D_j, \tag{1.13}$$

where S_j is the time receiving service, W_j is waiting time, and D_j is a transfer or administrative delay. Each quantity is patient-specific even when we omit the index. Improving one S_j may hardly change T_p if the associated W_j is small, or if another queue absorbs the released workload. This is the practical reason to measure several stages instead of collecting only the patient's arrival and departure time.

1.10.1 Five competing explanations of the same queue

Consider these explanations. First, registration capacity may be too low, so work accumulates while arrivals exceed service. Second, the clinical consultation may be the constraint, and the registration queue may reflect staff deliberately slowing entry to prevent unsafe congestion later. Third, arrivals may be strongly concentrated between 8:00 and 9:00, while workers have enough capacity over the day. Fourth, unusually complex cases may increase service time, even when arrivals are stable. Fifth, a power interruption may stop a task that depends on a single electronic record system. None of these explanations can be distinguished by daily attendance totals alone.

A useful diagnosis therefore describes *what observations would differ if competing explanations were true.* If we believe registration is the bottleneck, then during busy intervals a registration queue should grow while its worker is continuously occupied. If we suspect the laboratory, patients who require tests should account for much of the excess time. If interruptions are the cause, delays should coincide with documented periods of unavailability and the same service step should recover when power returns. These are hypotheses, not conclusions. We should record both confirming and contradicting observations.

The table is not an instruction to gather every possible variable. The question is which measurements would change a real decision. A two-day structured observation may establish that a bottleneck lies downstream, allowing the team to avoid an expensive attendance-prediction project. Alternatively, it may establish that arrivals are the main uncertainty, justifying more sustained measurement. A small discriminating study can be worth more than a large undirected dataset.

Table 1.4: Distinguishing plausible causes of delay in the illustrative clinic

Proposed cause	Observation that would support it	Observation that would weaken it
Registration bottleneck	Queue grows while registration staff are busy	Registration queue is short while delays persist elsewhere
Arrival concentration	Large within-day peaks with quiet intervals later	Arrivals are nearly level throughout the day
Laboratory bottleneck	Long waits concentrated among patients receiving tests	Similar delays among patients with no tests
Unreliable electricity	Delay episodes match interruption or system-restart logs	Delays remain unchanged during uninterrupted service
Repeated paperwork	Multiple data-entry steps and long handling times	Service records are already entered once, rapidly

1.10.2 The difference between a process measure and an outcome

One count in a register can serve several purposes, and those purposes should not be confused. The number of patients registered per hour is a measure of *throughput*. The time required to register one patient is a *service-time measure*. The time from entering the queue to receiving service is a *waiting-time measure*. The number of patients who left without care is a distinct outcome, with consequences that an average waiting time cannot capture. A change that raises throughput but also increases record errors has not been unambiguously successful.

The distinction is familiar in health-services research: resources and arrangements, activities, and outcomes require different kinds of evidence. Donabedian (1988) offered a foundational framework for assessing quality through structure, process and outcome. We use that distinction here to guard against a common error, not to suggest that our fictional queue model assesses clinical quality.

1.11 Queues are mathematical systems with human consequences

The elementary queueing calculation in this chapter is a starting point, not a model to deploy without checking. The standard $M/M/1$ formula assumes a Poisson arrival process, independent exponentially distributed service times, one server, an effectively unlimited waiting room, first-come-first-served service and steady operation with $\lambda < \mu$. The formula becomes attractive because it is easy to calculate. That ease must never be mistaken for a reason to assume that clinics satisfy all its conditions. For foundations see Little (1961), Gross et al. (2008) and Law (2015).

Write λ for mean arrivals per hour and μ for mean services per hour. Traffic intensity is $\rho = \lambda/\mu$. Under the $M/M/1$ assumptions, the stationary probability of n people in the system is

$$\Pr(N = n) = (1 - \rho)\rho^n, \qquad n = 0, 1, 2, \ldots, \tag{1.14}$$

which is a geometric distribution. Its mean is $L = \rho/(1 - \rho)$. Little's relation $L = \lambda W$ then

yields

$$W = \frac{1}{\mu - \lambda}, \qquad W_q = \frac{\rho}{\mu - \lambda}, \qquad L_q = \lambda W_q = \frac{\rho^2}{1 - \rho}. \tag{1.15}$$

Here W includes service, whereas W_q is the time waiting before service. The distinction matters: someone might report that a patient has a thirty-minute "waiting time" even though ten minutes were spent in consultation.

1.11.1 Why small changes near full capacity matter so much

Imagine $\mu = 12$ registrations per hour. With $\lambda = 6$, the model gives $\rho = 0.5$ and $W = 10$ minutes. With $\lambda = 10$, the utilisation rises to $5/6$ and $W = 30$ minutes. With $\lambda = 11$, it predicts 60 minutes. The arrival rate has not doubled between the last two cases. Yet the predicted time in the system has doubled, because there is only one registration per hour of spare capacity at the margin.

The derivative makes the mechanism visible:

$$\frac{\partial W}{\partial \lambda} = \frac{1}{(\mu - \lambda)^2}, \qquad \frac{\partial W}{\partial \mu} = -\frac{1}{(\mu - \lambda)^2}. \tag{1.16}$$

Sensitivity grows rapidly as the difference $\mu - \lambda$ shrinks. A planner should therefore not use the average daily number of arrivals to infer that a clinic can tolerate short bursts without delay. Nor should they treat high utilisation as a universal sign of efficiency: near-saturation may be extremely expensive in patient time.

The formulas also show an important limitation. If $\lambda \geq \mu$, no stationary finite expected queue length exists under these assumptions. It is incorrect to insert $\lambda > \mu$ into the formula and interpret a negative waiting time. The calculation has left its region of validity. A competent model is one whose failure conditions are stated before the results are used.

1.11.2 A counterexample to using average arrival rates

Suppose twenty-four people arrive over four hours and a worker processes eight per hour. The average arrival rate is six per hour, below service capacity. The day appears manageable. Now suppose all twenty-four patients arrive in the first hour and almost none arrive later. The mean rate across four hours remains six, but a queue forms during the first hour. Its size, length and duration depend on within-day timing. A steady-state formula using the average cannot recover this transient pattern.

To investigate the peak properly we might use a discrete-event simulation, a time-varying queueing model, or an actual time-stamped queue trace. All involve assumptions about service order, priorities, breaks and disruptions. The better choice depends on the decision and the resources. For a one-day scheduling change, a carefully recorded trace may answer the question without any advanced model.

1.12 Alternatives, baselines and the burden of proof

An intervention should be compared with an existing practice and at least one plausible simpler alternative. Otherwise, we are measuring whether a prototype works at all, not whether it improves the decision. Imagine three possible changes to the fictional clinic: a new arrival-forecasting model, a revised staff start time and a temporary second registration desk during the morning peak.

The forecasting option has more than one component. We must estimate arrivals, communicate the prediction, decide which changes the forecast will trigger, and put those changes into effect. If staffing cannot be changed, forecast accuracy has little operational value. A revised start time may need no software, only an agreed schedule. The second desk may reduce waiting but increase costs and possibly take staff away from clinical duties. A serious comparison includes those consequences rather than presenting one technology as the "proposed method" against no alternative.

1.12.1 From prediction error to decision loss

Suppose a model estimates tomorrow's peak as $\widehat{d}$. A planner chooses the number of morning staff a using that estimate. Define actual arrivals as d and a cost function

$$C(a, d) = c_s a + c_o(d - ka)_+ + c_i(ka - d)_+, \tag{1.17}$$

where k is the number of arrivals one staff member can handle in the relevant interval; c_s is staffing cost per person; c_o penalises unmet or delayed demand, and c_i penalises unused capacity. The positive-part notation $z_+ = \max(z, 0)$ is important. It ensures that we charge for shortage only when demand exceeds capacity and for idle capacity only when capacity exceeds demand.

Consider $k = 10$, $c_s = 20$, $c_o = 5$ and $c_i = 0.5$ in illustrative cost units. If $d = 24$, then staffing two people costs $40 + 5(4) = 60$, while staffing three costs $60 + 0.5(6) = 63$. Under these particular weights, the lower-capacity choice is preferred by three cost units. If delayed care had a larger penalty, the ranking could reverse. The mathematics does not decide the ethically acceptable cost of delay; that has to be justified by the organisation and the people affected.

Now imagine two forecasts, $\widehat{d}_A = 23$ and $\widehat{d}_B = 27$, when the realised demand is 24. Model A has smaller absolute error. But if a decision rule allocates two staff for estimates below 25 and three otherwise, the forecasts lead to different actions. Which is better depends on the decision costs and on the actual operating context. Comparing only mean absolute prediction error omits that question. The broader study of predictive scoring and probabilistic forecasts is developed by Gneiting and Raftery (2007); our purpose here is to show why scoring a model and evaluating the decision it informs are distinct tasks.

1.12.2 What should count as the existing practice?

A baseline is not necessarily "do nothing." Staff may already know which mornings are busy and adjust informally. A notebook may contain a useful ordering rule. A person may call ahead to check whether a medicine is in stock. Replacing these practices with a complicated system and comparing only with a naive constant forecast can exaggerate the apparent gain.

Document the present practice as faithfully as possible. Who makes the choice? What information do they see? How much time does the choice take? What happens during a network outage? The existing system may be weak in ways that are not visible in a historical dataset, but it may also be surprisingly adaptive. A fair comparison is based on what people actually do, not on a caricature designed to make a new method look attractive.

1.13 A decision does not have to be made at once

Some problems reward a staged response. A hospital might first observe workflow, then test a revised timetable for a week, and only afterwards decide whether a forecasting system is needed. This is not indecision. It is a way of buying evidence before committing resources.

Let θ denote an unknown condition, such as whether registration or laboratory capacity is the dominant source of delay. With no further information, the best attainable expected loss is

$$R_0 = \min_a \mathbb{E}_\theta[L(a, \theta)]. \tag{1.18}$$

If we could learn θ perfectly before acting, the expected loss would be

$$R_{\mathrm{PI}} = \mathbb{E}_\theta\Big[\min_a L(a, \theta)\Big]. \tag{1.19}$$

The *expected value of perfect information* is $\mathrm{EVPI} = R_0 - R_{\mathrm{PI}} \geq 0$. It is nonnegative because having information does not force us to use it. For practical decision analysis see Raiffa (1968).

1.13.1 A complete two-state example

Assume that the clinic is in one of two states. In state R, registration is the main bottleneck; in state L, the laboratory is. To keep the arithmetic transparent suppose our prior probabilities are $\Pr(R) = 0.6$ and $\Pr(L) = 0.4$. There are two affordable actions: a_R, to improve registration, and a_L, to improve laboratory workflow. Illustrative losses are

Action	State R	State L
action a_R	2	12
action a_L	9	3

With present information, the expected losses are $0.6(2)+0.4(12) = 6$ for a_R and $0.6(9)+0.4(3) = 6.6$ for a_L. Thus $R_0 = 6$. With perfect information we would choose a_R in state R and a_L in state L, yielding $R_{\mathrm{PI}} = 0.6(2) + 0.4(3) = 2.4$. Hence $\mathrm{EVPI} = 3.6$ loss units per decision period.

This does *not* mean a diagnostic study will save exactly 3.6. Perfect information is an upper bound on the value of an actual, fallible study under these assumptions. If measuring the bottleneck would cost five loss units for the same period, it would not be justified by this one-period calculation. If the decision would govern a year of operations, its value may be much greater. The units and time horizon have to be made comparable before interpreting the answer.

1.13.2 An imperfect test and Bayes' rule

Suppose a short observation exercise reports "registration bottleneck" with sensitivity 0.8 and specificity 0.75 relative to the hypothetical true state. If the report is positive, the posterior probability of state R is

$$\Pr(R \mid +) = \frac{0.8 \times 0.6}{0.8 \times 0.6 + 0.25 \times 0.4} = \frac{0.48}{0.58} \approx 0.828. \tag{1.20}$$

If the report is negative, $\Pr(R \mid -) = 0.2(0.6)/[0.2(0.6) + 0.75(0.4)] \approx 0.286$. The test changes the evidence available to the decision-maker, but not with certainty. In each test-outcome state we should recompute the expected losses and choose an action. This is more defensible than saying simply that "we gathered more data, therefore the decision improved." The improvement depends on how the measurements alter choices and whether the change in losses exceeds the cost of measurement.

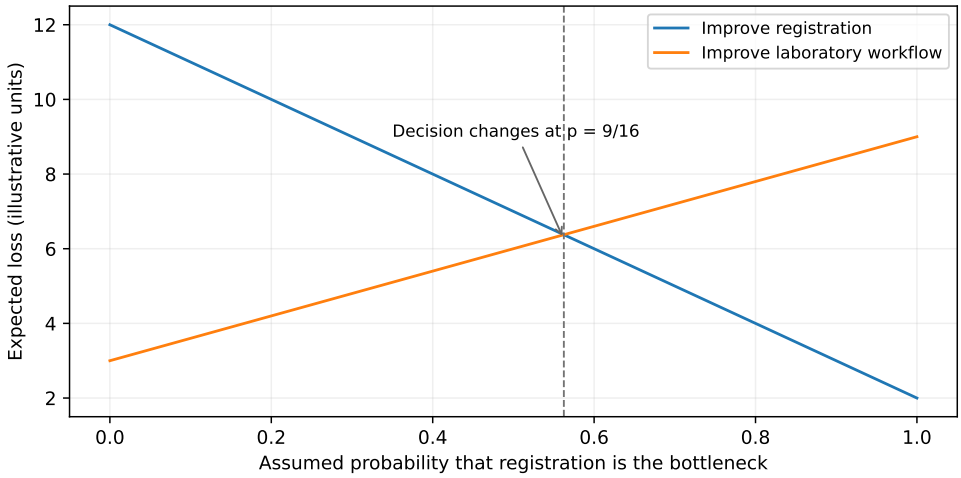


Figure 1.3: Illustrative decision losses against the assumed probability that registration is the main bottleneck. The threshold depends on the stated loss table; it is not an empirical probability estimate.

1.14 Fairness, risk and the averages we hide behind

Suppose a registration change cuts mean waiting from forty to thirty minutes. That sounds unequivocally beneficial. But what if the new queue allocates appointments mostly through a phone service available only to people with reliable internet? Some patients may wait ten minutes while people who arrive without a booking wait two hours. A single average conceals who benefits and who bears the delay.

For a waiting-time distribution, report a median, a high quantile and, where appropriate, comparisons across clinically and operationally justified groups. One possible requirement is

$$\Pr(W(a) > w_{\max} \mid G = g) \leq \alpha_g \quad \text{for each relevant group } g. \tag{1.21}$$

The thresholds $w_{\max}$ and α_g are not purely mathematical discoveries. They involve service standards, patient safety and practical judgement. Where observations are scarce, subgroup

estimates may be unstable; the right response is to report that uncertainty rather than invent confidence.

Risk is also not the same as average performance. Two interventions may have equal expected waiting time, but one may occasionally fail catastrophically. A risk-sensitive objective could include a quantile of waiting, a worst-case scenario, or conditional value at risk. If Z denotes loss, the conditional value at risk at level α can be represented, under ordinary integrability assumptions, as

$$\mathrm{CVaR}_\alpha(Z) = \min_{\eta \in \mathbb{R}} \left\{ \eta + \frac{1}{1-\alpha} \mathbb{E}[(Z-\eta)_+] \right\}. \tag{1.22}$$

It measures loss in an adverse tail rather than only at the mean. We do not need to calculate this for a small teaching clinic to learn its lesson: when failure has serious consequences, “good on average” is an incomplete claim.

1.15 How to talk to people without turning interviews into numbers too soon

A useful problem definition begins in conversation. The nurse knows which tasks cannot be postponed. The records officer knows which reports have to be submitted and what happens during an outage. Patients can describe aspects of delay that time stamps miss. The district accountant may know which expenditures can be approved this quarter. These perspectives can contradict one another without any person being dishonest. Their roles expose different parts of the system.

A good interview asks for events, decisions and exceptions, not merely opinions about a proposed app. “Tell me about the last day when the queue became unmanageable” is often more informative than “Would an AI attendance predictor help you?” The latter asks a respondent to evaluate a technology whose assumptions may be unfamiliar. Ask what they did when the delay began; what information they lacked; what would have been possible if they had known the arrival count earlier; and what prevented a different response.

Document disagreement instead of smoothing it away. If patients describe a long wait while the register records only a short service interval, that disagreement is not a nuisance to be averaged. It is evidence that the register measures something different from the experience at issue. Such discrepancies become central when we discuss measurement in Chapter 3.

1.16 From a proposal to a testable claim

Before requesting funds or writing a program, formulate a claim that could be wrong. A vague claim such as “digital registration will modernise the health centre” cannot be refuted in a useful way. A stronger claim names the intervention, the target, the comparison, the time horizon and the conditions: “On mornings when arrivals exceed the observed registration capacity, moving one qualified staff member to registration for the first ninety minutes will reduce the median registration wait relative to the current schedule without increasing laboratory delays or record errors.” This statement contains several things the evaluator must measure.

The counterfactual matters. We do not observe the same morning under two staffing schedules. A before–after comparison can be distorted by seasonality, patient mix, holiday patterns, changes in equipment and staff absences. A small randomised timetable experiment might be possible if safe and authorised; a carefully structured matched comparison may be needed otherwise. Hernán and Robins (2020) explain the importance of specifying the causal question and its identifying assumptions. Here we are not conducting such a study, but we should recognise what kind of evidence would be needed before claiming that an intervention caused an improvement.

A field study also needs stopping and escalation rules. If waiting worsens, patients are harmed, or the system becomes inaccessible to a subgroup, staff must be able to return to the previous practice. The fact that an algorithm continues to produce predictions is not a reason to continue a harmful operational experiment.

1.17 Extended laboratory: deciding whether to collect data

This laboratory is deliberately small. It should be possible on an ordinary laptop, without an internet connection or specialised machine-learning software. Define two alternative actions and two possible states using the loss table above. A short program computes their expected losses, the value of perfect information, and the posterior probability after an imperfect diagnostic signal. The student should change the prior probability and the cost of the test, then identify when the best action changes.

```
import numpy as np

states = np.array([0.6, 0.4])
loss = np.array([[2.0, 12.0], [9.0, 3.0]])
expected = loss @ states
best_without_test = expected.min()
perfect_information = (states * loss.min(axis=0)).sum()
evpi = best_without_test - perfect_information
print(expected, best_without_test, evpi)

# Probability that the test says registration is the bottleneck.
signal_plus = np.array([0.8, 0.25])
prob_plus = (states * signal_plus).sum()
posterior_plus = states * signal_plus / prob_plus
print(prob_plus, posterior_plus)
```

The first printout should give expected losses $[6, 6.6]$, current best loss 6 and EVPI 3.6. The probability of a positive report is 0.58, and the posterior probability of registration being the bottleneck is approximately 0.828. These are checks on the calculation, not observations from a real facility. A student who obtains different numbers should examine the meaning of each row and column before modifying the code.

To complete the laboratory, derive the expected value of the imperfect test by weighting the best posterior expected loss under each possible report. Then subtract the study cost. Compare your result with EVPI. It must not exceed EVPI under the same decision model. If it does, the cost or conditional-probability calculation needs to be reconsidered.

Further questions. Why would an observed reduction in average queue length not necessarily mean improved patient access? What if the arrival process is not Poisson? What if the staff needed for the second desk must leave laboratory work? What observation would make the proposed forecast irrelevant? Which consequences are difficult to express honestly in a monetary cost function? Write your answers in ordinary language before constructing another mathematical model.

1.18 When the bottleneck moves, the first improvement may disappoint

A registration station, consultation room and laboratory form a sequence of processes rather than three independent queues. Increasing capacity at registration may send patients faster to consultation, raising its congestion. A project that measures only the first queue can report success while patients experience no meaningful reduction in total visit time. To study this, write the patient's total time as the sum of time spent at all required stages, not the time at whichever stage is easiest to measure.

Consider two service stages in series. Suppose patients arrive at an average rate of eight per hour and both stages initially serve ten per hour under simplified stationary $M/M/1$ assumptions. Each stage has expected time in system $1/(10-8) = 0.5$ hour. If every patient visits both stages, the expected total is one hour, assuming the modelling conditions needed for each stage are met. Now increase the first stage's capacity to sixteen per hour. Its expected time becomes $1/(16-8) = 0.125$ hour, but the second remains at 0.5 hour, so total expected time is 0.625 hour. The first stage's time fell by 75 percent, while total time fell by only 37.5 percent. Those percentages are not interchangeable.

The calculation is schematic. In an actual tandem queue, the arrival process reaching each stage may depend on the upstream service process, and a simple $M/M/1$ formula for both stations requires additional assumptions. We use it to teach a distinction, not to describe a specific clinic. The difference remains important under more realistic models: improving a stage changes the flows and workloads faced by the rest of the system.

1.18.1 Flow conservation through a service process

Let $A(t)$ be cumulative arrivals by time t, $D(t)$ cumulative completed departures, and $N(t)$ the number still in the system. With an initially empty system and no abandonment or loss of records,

$$N(t) = A(t) - D(t). \tag{1.23}$$

If some patients abandon their visit, include an abandonment count $B(t)$, giving $N(t) = A(t) - D(t) - B(t)$. These are accounting identities. A log of arrivals and departures that violates them has a missing, duplicated or inconsistent record somewhere. Before estimating a sophisticated queueing model, check that the basic count balances.

When customers pass through several stages, a similar identity applies at each: incoming transitions minus outgoing transitions equal the change in those waiting or receiving service. A process diagram and a small transition table are often sufficient to detect impossible paths, such

as a recorded laboratory completion for a patient whose laboratory visit was never registered. The point is not to police clerical mistakes; it is to understand what happened and prevent mathematical calculations from inheriting unexplained inconsistencies.

1.18.2 The useful limits of Little's relation

Little's law connects mean number in the system, throughput and mean time in system under appropriate stable long-run conditions. It is more general than the $M/M/1$ formula, which needs stronger assumptions about arrivals and service. One may be able to estimate the mean number present and average throughput from actual records and use Little's relation as an internal consistency check even when service times are not exponential. The analyst must still define the system boundary, what counts as an arrival and completion, and whether the observation window is sufficiently representative of steady operation (Little, 1961).

For example, if stable long-run throughput is eight patients per hour and mean number present is sixteen, Little's relation implies mean time in system of two hours. But if the clinic closes at noon and carries a backlog into the following day, a one-morning arithmetic average may not satisfy the steady assumptions. That is not a reason to discard the relation; it is a reason to identify the correct period and system boundary.

1.19 How much measurement is enough to begin?

An applied team can become trapped in a different mistake: collecting data indefinitely because further evidence would always be helpful. The stopping rule should depend on the decision's sensitivity and the cost of another observation. A small study may already establish that a popular intervention cannot work under the current staffing constraint. It may show that two alternatives have almost identical expected outcomes over every plausible input range. Or it may reveal that a crucial assumption changes the preferred option, making targeted measurement essential.

Let $\mathcal{I}$ represent current information and Z a possible new observation. A rational one-step comparison is between the expected loss of the best action using $\mathcal{I}$ and the expected loss after observing Z and choosing an action conditionally, including the observation cost. Symbolically,

$$\mathrm{EVSI}(Z) = \min_a \mathbb{E}[L(a,\xi) \mid \mathcal{I}] - \mathbb{E}_Z\Big[\min_a \mathbb{E}[L(a,\xi) \mid \mathcal{I}, Z]\Big]. \tag{1.24}$$

The value is nonnegative if the new information can be ignored and it has no decision-side cost. Its *net* value subtracts collection, delay and implementation costs. The full calculation may require a probability model whose parameters are themselves uncertain. In small projects, scenario comparisons and clearly bounded estimates may be more defensible than spurious precision.

A practical stopping conversation therefore asks three things. Is there a feasible choice that is already preferable under every credible scenario? Would one more observation plausibly change the choice? And what happens if we delay action while waiting for information? For urgent decisions, the cost of waiting can dominate; for irreversible infrastructure purchases, the cost of acting on inadequate evidence may be larger. The answer should come from the decision context, not a fixed rule that all projects must gather a certain number of data points.

1.20 Case study: choosing between a timetable and a forecasting system

Let us now conduct a complete small design exercise with the fictional clinic. The aim is not to win an accuracy competition. It is to decide what should be done next and to show which missing evidence prevents a stronger claim. Suppose registration is performed by one worker for four hours each morning. Twenty-four patients arrive on an average morning, but on some mornings arrivals are concentrated during the first hour. The manager can move a qualified colleague to registration for ninety minutes, revise appointment times, or commission a computer-based arrival forecast. We shall treat every number in this exercise as a teaching assumption.

The first task is to separate the choices from their enabling conditions. Moving a colleague requires that the other service can tolerate the temporary absence. Revising appointment times requires a meaningful ability to assign times and communicate them to patients. Forecasting requires a history of arrivals at useful time resolution, a procedure for checking predictions, and someone able to act when the forecast signals a busy day. It also requires an alternative if the computer or network is unavailable. The three projects are not equally costly simply because they all appear as options on a slide.

1.20.1 What should an arrival record contain?

A day-level total does not tell us when congestion occurred. If the proposed decision changes staffing in ninety-minute blocks, collect arrival times at roughly the same or finer resolution. For each service visit record a de-identified visit code, arrival timestamp, registration start and end times, and where ethically permitted timestamps for subsequent stages. An interruption log should include when the service stopped, why it stopped and when it resumed. A staff timetable provides the actual service capacity available at each interval. Avoid recording names, clinical diagnoses or identifiers not needed for the operational question.

Before collecting these timestamps, define a clock standard. Two computers whose clocks differ by eight minutes can create a negative waiting time when records are joined. A timestamp without a timezone can become ambiguous when exported. Manual paper records may round times to the nearest five minutes. These details affect conclusions about waiting as surely as a missing coefficient affects a regression.

1.20.2 Three levels of analysis

At the first level, describe the observed work. For each hour report arrivals, service starts, completions, number of available workers and recorded interruptions. Plot the observed queue length through the day when the data support reconstruction. Do not begin with a machine-learning model. At the second level, fit a simple service-capacity approximation and compare its predictions with the trace; record where it fails. At the third level, test counterfactual staff schedules in a transparent queue simulation, varying the estimated service times and the assumed behaviour of patients. This third level depends on assumptions that were not directly observed: how the worker would perform after being reassigned, whether patients would arrive

at the same times, and whether upstream or downstream delays would change.

The levels should not be collapsed into a single line saying that "the simulation shows a twenty-percent improvement." The descriptive trace is a record. The queue model is an approximation. The counterfactual simulation is a conditional prediction under a proposed policy. None is proof that a timetable change will improve care when implemented. The proof requires an appropriately designed operational evaluation.

1.20.3 A transparent decision table

Suppose the decision group assigns each option four values: estimated direct cost, expected change in registration wait, potential delay transferred to other services, and ability to keep operating without the internet. We do not have credible numeric estimates for all four. The proper table would leave those cells visibly unfilled rather than inventing an overall score. A preliminary record might be:

Option	What it requires	Decisive uncertainty
Temporary second worker	Qualified spare staff and appropriate supervision	Does diverting staff worsen later services?
Altered appointments	Booking process and communication with patients	Will arrival behaviour follow the schedule?
Arrival forecast	Timestamp history, tested model, and action rule	Can staffing or appointments change in response?

Which option should be piloted first? The table alone does not determine it. If a qualified worker is available and laboratory capacity is unaffected, a short staffing pilot may be low-cost and informative. If no spare worker exists but many patients already receive appointment times, schedule revision may be more plausible. A forecast may be justified if advance information consistently changes a feasible staffing decision. This kind of conditional conclusion is more valuable than an unsupported announcement that AI is the answer.

1.20.4 The quiet cost of a complicated solution

The purchase price of a device or software licence is not the total cost of a system. The clinic may need data-entry time, recurring connectivity, training, maintenance, backups, a person to resolve failed uploads and replacement equipment. Changes in reporting requirements may make the code obsolete. The useful life of a system can be shorter than the useful life of the mathematical insight on which it is based.

A lifecycle estimate can be written as

$$C_{\text{life}} = C_0 + \sum_{t=1}^{T} \frac{C_{\text{operate},t} + C_{\text{maintain},t} + C_{\text{train},t}}{(1+r)^t} + C_{\text{exit}}, \tag{1.25}$$

where r is a discount rate and C_{exit} represents any responsible migration or shutdown cost. The equation is not difficult; collecting defensible inputs is. A prototype that appears cheap because

it relies on unpaid manual effort or a student who will soon graduate may not be maintainable. Conversely, a low-tech timetable adjustment can be inexpensive while demanding significant organisational cooperation. These costs should be recognised, not assumed away.

1.21 A careful experiment is a question with a comparison

After a design has passed initial feasibility checks, the next task is to decide whether the intervention helps in practice. A comparison must be specified before the result is seen. The simplest comparison is current practice versus a revised timetable, measured on similar operating days. Unfortunately, days are never perfectly similar: patient volume, case severity and staffing can change. A randomised allocation of safe alternatives across appropriate days or service periods may sometimes be feasible, but only after consultation and ethical approval where required. Where randomisation is impractical, a structured comparison can document what was matched and what remains uncontrolled.

A measurable primary outcome is preferable to an undefined promise to "improve efficiency." For example, the primary outcome might be median registration waiting time, while secondary outcomes include the high-percentile wait, patients leaving before registration, record errors and delays in other services. If twenty indicators are inspected and only the most favourable is published, the evaluation has changed its question after seeing the data. Predefining the principal result disciplines interpretation.

1.21.1 Why the unit of comparison matters

If whole mornings are assigned to different timetables, patients within the same morning share staff, equipment, weather and interruptions. They are not independently assigned units. A standard error calculated as though every patient received an independently randomised staffing intervention can be far too small. The number of independently comparable mornings may matter more than the number of patient visits. The same principle applies to schools, wards, clinics and districts. Collecting more rows inside one unit does not create new units of assignment.

Suppose a pilot compares five intervention mornings with five control mornings. This may provide useful feasibility evidence, but ten morning-level units do not justify highly precise claims about year-round effects. The honest report should show the outcomes for individual mornings, context and any interruptions, while identifying the uncertainty and seasonality that remain. An experiment is not good because it produces a small p-value; it is good because its design allows an interpretable comparison.

1.21.2 A decision rule for continuation

Before a pilot, agree on what would cause the project to continue, stop or be revised. One rule might require a meaningful decrease in waiting, no unacceptable transfer of delays, no rise in record errors and an implementation cost within the available budget. Results could be mixed. A timetable may improve registration waiting but overburden the clinical team. A forecasting

model may improve predictions but leave the schedule unchanged. The evaluation must be allowed to conclude that the proposed solution should not be deployed.

1.22 Problem formulation as a craft that can be taught

What, finally, would a student learn from the clinic? They should be able to walk into an unfamiliar institution without immediately suggesting a model, ask which decision is made and by whom, distinguish observations from assumptions, propose competing mechanisms and design the smallest investigation that would discriminate among them. They should know how to express a measurable loss without pretending all consequences have the same unit. They should anticipate what could go wrong when an intervention is implemented.

One way to practise is to rewrite a vague request in three versions. The request "Build AI to improve patient flow" becomes first a concrete observation: "Some patients spend more than two hours at this facility before seeing a clinician." It becomes next a causal hypothesis that can be challenged: "The longest delays appear on mornings when registration arrivals exceed available service capacity." Finally it becomes an operational question: "Would moving one qualified worker to registration during the first ninety minutes reduce the relevant waiting measures without worsening downstream care?" Each rewrite reduces vagueness and increases the amount of evidence that can genuinely refute it.

A second practice exercise begins with the opposite problem. Present a beautifully written objective such as "Minimise the expected value of system inefficiency using artificial intelligence." Ask the reader to identify what is missing: an observable outcome, a feasible decision set, the source of uncertainty, a unit of measurement, a baseline, implementation authority, and consequences of failure. The equation might be elegant, but the problem remains undefined.

For a final assessment, give students three short institutional scenarios: a cooperative with repeated failed deliveries, a school whose examination records are frequently late, and a clinic where medicine issues vary sharply. Ask each student to produce a decision diagram, two competing explanations, a feasible action, a missing measurement and a credible baseline. They may use mathematics or computing in their solutions, but they should be marked on whether they have identified the right question, not on whether they mention the most advanced technique.

1.23 Worked investigations for independent study

The following investigations are longer than ordinary end-of-section exercises. They show how to follow a decision from a verbal complaint through mathematical assumptions to a conclusion that can be defended. Each is solved with the information given; none should be read as an empirical account of a real health service.

1.23.1 Investigation 1: what is a diagnostic observation worth?

Return to the two-state decision problem. The prior probabilities are $\Pr(R) = 0.6$ and $\Pr(L) = 0.4$. Registration improvement gives losses $(2, 12)$ in the two states, while laboratory

improvement gives $(9, 3)$. The current minimum expected loss is six. A short diagnostic test detects state R with probability 0.8 and reports R incorrectly in state L with probability 0.25.

We computed the probability of a positive report as 0.58. The joint probability of being in state R and receiving a positive report is $0.6(0.8) = 0.48$, whereas that of state L and a positive report is $0.4(0.25) = 0.10$. Conditional on a positive report, the expected loss of improving registration is

$$\frac{0.48(2) + 0.10(12)}{0.58} = \frac{2.16}{0.58} \approx 3.724.$$

For laboratory improvement it is $[0.48(9) + 0.10(3)]/0.58 \approx 7.966$, so registration improvement remains preferable. If the report is negative, the joint probabilities of state R and L are $0.6(0.2) = 0.12$ and $0.4(0.75) = 0.30$. Registration improvement would then have conditional expected loss $[0.12(2)+0.30(12)]/0.42 \approx 9.143$. Laboratory improvement gives $[0.12(9)+0.30(3)]/0.42 \approx 4.714$ and is preferred.

The expected loss when the test result is used optimally is obtained most simply from the joint, rather than conditional, probabilities:

$$R_{\text{test}} = [0.48(2) + 0.10(12)] + [0.12(9) + 0.30(3)] = 2.16 + 1.98 = 4.14.$$

The expected value of sample information is therefore $6 - 4.14 = 1.86$ loss units for the same decision period. This is less than EVPI 3.6, as it must be. If the diagnostic costs two loss units, the net gain is negative for a single period under the assumptions. If the findings govern many future periods, the test may be worthwhile, but only after discounting and checking whether the underlying state remains stable. Thus a test can be informative yet not economical for a particular decision horizon.

A useful extension is to change the prior probability of state R. Derive the point at which action a_R becomes preferable before a test. Solve

$$2p + 12(1 - p) \leq 9p + 3(1 - p),$$

which reduces to $9 \leq 16p$, or $p \geq 9/16$. At $p = 0.6$ registration improvement is chosen, but if credible new information lowers p below 0.5625, the recommendation changes. The numerical threshold explains which measurement might matter. A tiny improvement in estimating a parameter far from any decision threshold may be less useful than a modest observation that resolves this particular uncertainty.

1.23.2 Investigation 2: why a queue can look calm until it does not

Assume an $M/M/1$ registration station with $\mu = 15$ patients per hour. Calculate expected time in system for arrival rates $\lambda = 6, 10, 12, 14$. The formula $W = 1/(\mu - \lambda)$ gives 1/9, 1/5, 1/3 and 1 hour respectively: roughly 6.67, 12, 20 and 60 minutes. The service rate did not change. The largest increase in waiting arises as available spare capacity approaches zero.

Compute also the queueing component $W_q = \lambda/[\mu(\mu-\lambda)]$. At $\lambda = 12$, this is $12/(15\times3) = 0.2667$ hours, or 16 minutes; mean service time is 1/15 hour, or four minutes. The total is twenty minutes. Reporting all twenty as waiting *before* registration would misdescribe the model. The calculation is simple, but it requires precise meanings for service, waiting and time in system.

What policy conclusion follows? The formula suggests that operating near full capacity is risky when arrivals are variable. It does not establish that adding staff is always the best response. Appointment timing, variation in service durations and bottlenecks at later stages may matter. Nor does an expected waiting time describe the high tail, patients who leave the queue, or those requiring priority care. The decision to add staff still requires data on these considerations and an evaluation of costs and consequences.

1.23.3 Investigation 3: two averages, two ethical conclusions

Imagine an intervention with ten patients in each of two groups. Group A has mean wait ten minutes; Group B has mean wait ninety minutes. The overall mean is fifty minutes. After a scheduling change, A waits five minutes on average while B waits ninety-five, preserving the overall mean at fifty minutes. A report containing only the average would claim no change. Yet the change benefits one group and harms another. Neither the overall mean nor its stability tells us whether the change is acceptable.

Now suppose the intervention instead reduces Group A to five minutes and Group B to seventy-five. The overall mean becomes forty minutes. This is a real improvement in the aggregate of the stated sample, but it does not establish that every individual benefits or that clinical outcomes improve. If the groups differ in urgency, travel distance, vulnerability or ability to use a digital booking service, those factors deserve explicit attention. Students should propose a meaningful distributional outcome and describe how it would be measured, rather than inventing a universal fairness score.

1.23.4 Investigation 4: when should the project end?

A clinic has completed a ten-morning pilot of an electronic arrival dashboard. The dashboard predicts the number of arrivals well, but staffing did not change on any morning because the facility had no spare qualified workers. The system has a recurring subscription charge and depends on a reliable network. Is the pilot a failure? The prediction component may have met its technical objective. The operational objective of reducing waits has not been demonstrated. Those statements can both be true.

A sensible final memo would separate technical verification (the code produced the specified forecasts), predictive validation (the forecasts matched held-out observations to a stated degree), decision evaluation (the dashboard did or did not change actions) and operational impact (patient outcomes changed, or remain unmeasured). The team might retain the dataset for planning, change the decision question or stop deployment. Continuing solely because considerable effort has already been spent is not a mathematical reason to proceed.

1.24 Laboratory 1: reproduce the decision calculations

The companion script `code/reproduce_chapters.py` generates Table 1.2 and Figure 1.1. It uses only standard mathematical formulas for the stated queue. Students should begin with the one-server formula and verify thirty minutes for $\lambda = 6$, $\mu = 8$, before inspecting the two-server calculation. They should then change the arrival rate while keeping the service rate fixed.

A good laboratory submission will not stop after drawing the curve. It will describe which assumptions are required, where the curve becomes implausible for the intended application, and which measurement would most improve the analysis. If the waiting time is extremely sensitive to a small change in arrival rate, estimating λ inaccurately could reverse the practical conclusion. Such sensitivity is a reason to collect better evidence, not a reason to hide uncertainty (Saltelli et al., 2008).

1.25 A Problem Definition Record

Before starting a full solution, write a two-page record that includes the intended user, decision authority, need, observed symptom, competing explanations, proposed choices, success measures, constraints, costs of failure, available evidence and what remains unknown. This should be revised after investigating the process. It is not a proposal form to fill in mechanically; its purpose is to expose assumptions and disagreements early.

For the fictional clinic, the record might state that the proposed decision is a morning staffing schedule; the desired outcome is shorter time from arrival to completed clinical service; the constraint is staff availability; and the unresolved questions concern the location of the main bottleneck and the distribution of service times. It would explicitly say that the $M/M/c$ calculation is an illustration rather than a validated staffing recommendation.

Exercises

1. A clinic reports delays, and a supplier proposes a chatbot for appointments. Identify the symptom, two competing mechanisms, one measurable outcome and two observations needed before selecting a solution.
2. Derive $\mathbb{E}[N] = \rho/(1-\rho)$ from equation (1.5). State the conditions under which the derivation is useful.
3. For a stable $M/M/1$ queue with $\mu = 10$ services per hour, calculate W at $\lambda = 5$, 8 and 9 arrivals per hour. Explain why the waiting times do not increase linearly.
4. Using Table 1.3, repeat the decision calculation for $p = 0.1$ and $p = 0.8$. Explain why the preferred staffing rule changes.
5. Construct a case where improved prediction accuracy does not change any feasible decision. State what new action would be needed to make the prediction valuable.
6. **Project exercise.** Interview a willing adult about an everyday service problem without recording private or sensitive details. Write a Problem Definition Record containing at least two alternative explanations and a plan to distinguish them.

What this chapter establishes. A numerical model is useful after the decision, outcome and constraints have been made explicit. The next chapter begins where this one ends: having identified a genuine decision, how should we represent the physical or organisational system whose behaviour matters?

Chapter 2

Turning Reality into a Mathematical Model

A model does not have to contain everything that exists. It must contain what is necessary to answer a particular question, in units and relationships that can be checked.

2.1 A water tank serves several neighbourhoods

Consider a town that distributes water from a storage tank. Some days enough water arrives from the source; on others it does not. Demand varies. Leaks can occur between the source and the customers. The town must decide whether to increase storage, repair a pipe, ration deliveries or change its pumping schedule. It may have only a few measurements: the volume entering storage, the water-level indicator and a daily estimate of the amount delivered.

An engineer could begin with a sophisticated network model. The engineer could also begin with the simplest meaningful equation: water cannot appear from nowhere. If S_t is stored water at the beginning of day t, I_t is water entering the tank, D_t is water delivered and ℓ_t is water lost, then a basic balance is

$$S_{t+1} = S_t + I_t - D_t - \ell_t. \tag{2.1}$$

The symbols turn a verbal description into a claim that can be tested. Yet the equation is incomplete unless we say what each term means, which time interval it covers and whether the tank can run empty or overflow.

A model is therefore not just an equation. It is an equation together with definitions, assumptions, boundary conditions, a domain of validity and a purpose. This chapter develops those ingredients through the water system, then shows how the same principles lead to networks, differential equations, probability models and constrained optimisation.

2.2 Choose the boundary before counting what crosses it

Imagine drawing a line around the tank. Water that crosses the line into the tank contributes to I_t. Water that crosses out contributes to D_t or ℓ_t. This simple act of choosing a *system*

boundary is essential. If the outlet meter sits downstream from an unmeasured leak, what the meter reads may differ from the amount leaving the tank. If pumps return treated water to storage, counting the return as fresh supply can double-count part of the flow.

To apply equation (2.1), the quantities must refer to the same system and period. The storage S_t is a volume, measured here in cubic metres (m^3). Daily inflow, delivery and loss are volumes over one day, not flow rates. If the inflow is recorded as a rate in litres per second, it must be integrated over the day before it can be added to storage. If q_t denotes the numerical value of a flow rate measured in litres per second and that rate is constant for one day, the volume entering over that day is:

$$I_t = \left(q_t \,\frac{\mathrm{L}}{\mathrm{s}}\right)\left(86\,400\,\frac{\mathrm{s}}{\mathrm{day}}\right)\left(\frac{1\,\mathrm{m}^3}{1000\,\mathrm{L}}\right)(1\,\mathrm{day}) = 86.4\,q_t\,\mathrm{m}^3, \tag{2.2}$$

where q_t is the numerical rate in litres per second. With a time-varying rate $q_t(\tau)$, where τ is measured in seconds from the start of day t, the volume is instead $I_t = (1/1000)\int_0^{86\,400} q_t(\tau)\,\mathrm{d}\tau$ cubic metres. The daily quantities I_t, D_t and ℓ_t in the balance are all volumes in m^3, not rates. Units are not decoration: their agreement is one of the easiest ways to detect a wrong model before running it.

Table 2.1: A minimal data dictionary for a daily storage balance

Symbol	Unit	Meaning and measurement caveat
S_t	m^3	Water in the tank at the start of day t; depends on an accurately calibrated level indicator.
I_t	m^3	Total volume entering the chosen tank boundary during day t.
D_t	m^3	Actual volume delivered from the tank, not necessarily requested demand.
ℓ_t	m^3	Volume lost inside the boundary; may include physical leaks and unmeasured releases.
K	m^3	Maximum physical storage capacity.

A good data dictionary also records time zone, collection interval, sensor resolution, missing-value codes and whether values are estimated or directly observed. Without such details, matching datasets can introduce errors even when each dataset is internally correct.

2.3 A feasible storage model

Equation (2.1) allows negative stored water if the requested delivery exceeds what is available. Negative volume is mathematically possible as a number but physically meaningless. The correction is not to set negative predictions to zero after the analysis and move on. We must distinguish *demand requested* from *water actually served*.

Let Q_t denote demand during day t, and let K denote tank capacity. For a simple ordering of

events in which inflow arrives, a prescribed loss occurs and delivery follows, define

$$A_t = \max\{0, S_t + I_t - \ell_t\}, \tag{2.3}$$
$$D_t = \min\{Q_t, A_t\}, \tag{2.4}$$
$$U_t = Q_t - D_t, \tag{2.5}$$
$$E_t = \max\{0, A_t - D_t - K\}, \tag{2.6}$$
$$S_{t+1} = \min\{K, A_t - D_t\}. \tag{2.7}$$

Here A_t is water available before delivery, U_t is unmet demand and E_t is overflow or spill. The timing is an assumption; another system may overflow before some daily demand is supplied. A finer time step may be needed to model that process correctly. The important point is to state the sequence of events instead of hiding it inside a programming convention.

An equivalent conservation check for this ordering is

$$S_{t+1} = A_t - D_t - E_t, \tag{2.8}$$

with an additional check that $A_t = S_t + I_t - \ell_t$ whenever $S_t + I_t \geq \ell_t$. If the assumed loss exceeds the available water, the simple clipping in (2.3) needs to be interpreted explicitly: the system cannot physically lose more water than it has. A practical implementation would cap realised loss and report the shortfall or revisit the loss model.

Proposition 2.1. *Suppose $K \geq 0$, $0 \leq S_0 \leq K$, and $Q_t \geq 0$. Under equations* (2.3)*, storage stays within its physical bounds: $0 \leq S_t \leq K$ for every subsequent day.*

Proof. Assume $0 \leq S_t \leq K$. By construction $A_t \geq 0$, and $D_t = \min\{Q_t, A_t\}$ satisfies $0 \leq D_t \leq A_t$. Hence $A_t - D_t \geq 0$. Its minimum with K is between zero and K. The statement follows by induction from S_0. □

Notice what has and has not been proved. We have proved a property of a proposed mathematical update rule. We have not proved that the real tank will operate safely. The rule may omit pressure constraints, contamination, minimum operating reserves or the actual sequence of pumping decisions. Verification of equations and validation against the physical system answer different questions.

2.4 A reproducible twenty-day illustration

Consider a fictional tank with initial storage $55\,\mathrm{m}^3$, capacity $75\,\mathrm{m}^3$ and a prescribed loss of $2\,\mathrm{m}^3$ each day. We invent twenty days of inflows, then compare constant demands of 16 and $22\,\mathrm{m}^3$ per day. The companion program computes equations (2.3) without importing any observed utility data.

The larger-demand scenario experiences shortages in this artificial sequence. This does not show that households consume too much or that rationing is desirable. It shows that a decision about demand, storage or supply is connected to an accounting identity. If the recorded inflows are uncertain, a sensible analysis should repeat the calculation for plausible alternatives and find where the preferred decision changes.

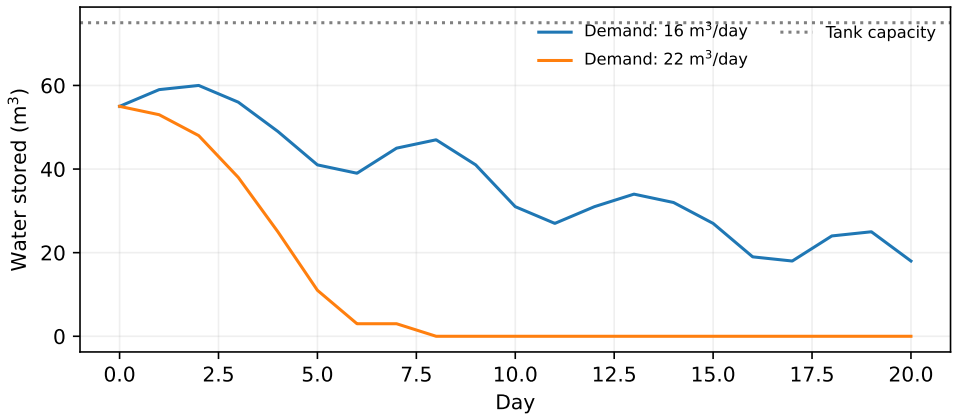


Figure 2.1: Simulated storage under two constant demand assumptions. The inflows and losses are invented for teaching. The plot makes no claim about an actual Ugandan water system.

Table 2.2: Calculated outcomes for the invented twenty-day tank scenario

Demand (m³/day)	Final storage (m³)	Total unmet (m³)	Total spill (m³)
16	18.0	0.0	0.0
22	0.0	102.0	0.0

The example also illustrates a danger of using aggregate measurements. Suppose a daily balance suggests that $3\,\mathrm{m}^3$ is disappearing between inlet and outlet measurements. The equation can show an unexplained difference, but it cannot say which pipe is leaking. To make that distinction, we need additional information about the network.

2.5 When a network is needed

A water distribution system, road map, electricity network and delivery route can all be represented by a graph. Let V be the nodes and E the edges. Choose an arbitrary orientation for each edge and let q_e represent flow in that orientation. The node-edge incidence matrix B contains -1 at the starting node of an edge, $+1$ at the receiving node and zero elsewhere. A signed net injection vector s must satisfy

$$Bq = s. \tag{2.9}$$

The sign convention matters. Under the convention just stated, a positive component of s represents net flow *into* a node from the directed edges. If instead injections from reservoirs are defined positive, the right-hand side changes sign. Both conventions are valid; silently switching between them is not.

For a network with m edges and n nodes, B is an $n \times m$ matrix. Every column sums to zero. Consequently,

$$\mathbf{1}^{\mathsf{T}} s = \mathbf{1}^{\mathsf{T}} B q = 0. \tag{2.10}$$

Total net flow must balance in the modeled closed network. If measured injections do not sum to zero, the difference may indicate a leak, measurement error, unrecorded consumption or simply a boundary that omitted part of the system. The equation identifies an inconsistency, not its cause.

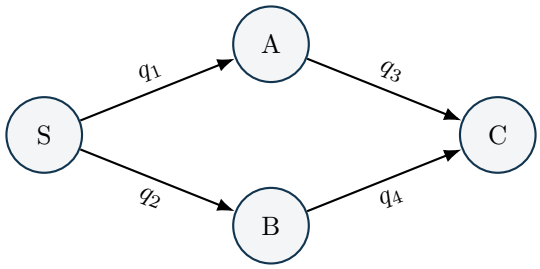


Figure 2.2: A small invented network. Its equations express conservation but do not determine flows without enough observations and physical relations.

For the diagram, conservation alone allows different choices of q_1 and q_2 that sum to the same total. Without pressures, pipe characteristics, flow meters or other constraints, the internal distribution is not uniquely determined. This is an example of *identifiability*: two different internal states can generate the same observations. A model cannot infer which state is real merely by using a more complicated algorithm.

2.6 State variables, parameters and decisions

Water storage changes from day to day, so S_t is a *state variable*. Tank capacity K is a *parameter* of the model, usually taken as fixed during a particular study. Pumping duration or the opening of a valve might be a *decision variable*. Rainfall or unplanned outages might be uncertain external inputs. The categories can change with the purpose. If we are designing a new tank, capacity becomes a decision rather than a fixed parameter.

A general discrete-time system can be written as

$$x_{t+1} = F(x_t, a_t, \xi_t; \theta), \qquad y_t = H(x_t; \theta) + \varepsilon_t, \tag{2.11}$$

where x_t is the state, a_t the action, ξ_t an external disturbance, θ fixed parameters, y_t observations and ε_t measurement noise. This notation separates how the system is believed to evolve from how it is observed. It will be useful in the next chapter because the measured quantity y_t is not necessarily the underlying state x_t.

A continuous-time balance may be more appropriate when processes change rapidly. If $q_{\text{in}}(t)$, $q_{\text{out}}(t)$ and $q_{\text{loss}}(t)$ are flow rates, then

$$\frac{dS}{dt} = q_{\text{in}}(t) - q_{\text{out}}(t) - q_{\text{loss}}(t). \tag{2.12}$$

Integrating over a day gives a balance in volumes. A discrete-time model is therefore not necessarily a different physical theory; it can be an integrated account over an observation interval. But if the rate and the decisions vary within the day, replacing every rate with a single daily average may hide important short-lived shortages.

2.7 Probability when the future is not known

Demand tomorrow is not generally known. One approach is to assume a fixed demand such as $20\,\mathrm{m}^3$. That is a scenario, not a probability model. A probabilistic representation instead assigns a distribution to uncertain demand Q_t. For illustration, suppose

$$Q_t = \begin{cases} 16, & \text{with probability } 0.7, \\ 22, & \text{with probability } 0.3. \end{cases} \tag{2.13}$$

Then $\mathbb{E}[Q_t] = 17.8\,\mathrm{m}^3$. Planning only for mean demand would ignore the possibility of high-demand days. If shortages carry a nonlinear cost, the expected cost cannot generally be obtained by substituting the expected demand into the loss function:

$$\mathbb{E}[L(a, Q_t)] \neq L(a, \mathbb{E}[Q_t]) \tag{2.14}$$

in general. This is why uncertain inputs, chance constraints and robust decisions require care beyond a single average.

The example probabilities are invented. With observed demand data, the modeller would have to examine seasonality, population served, demand suppressed by previous shortages and possible changes after new water connections. Recorded deliveries are not automatically equal to underlying demand. This distinction mirrors the clinic example in Chapter 1, where patients who leave a queue before service may vanish from service records without their need vanishing.

2.8 From understanding the system to choosing an action

Suppose a municipality wants to choose daily pumping p_t under a limit $0 \leq p_t \leq P_{\max}$ and an operating budget. If pumping contributes to inflow and carries an energy cost $c_t p_t$, a simple finite-horizon objective could be

$$\min_{p_0,\ldots,p_{T-1}} \sum_{t=0}^{T-1} \left[c_t p_t + c_u U_t \right], \tag{2.15}$$

subject to a balance equation, capacity limits, pumping limits and any service requirements. Here c_u is the modelled cost of unmet demand. This is no longer merely a prediction problem. It is an optimisation problem: among feasible actions, find one that performs well according to the stated criterion.

The optimisation is only as trustworthy as its constraints. If electricity interruptions disable the pump for several hours, an optimisation that assumes uninterrupted pumping can recommend an impossible schedule. If the pump cannot restart without a technician, that operational limitation belongs in the feasible set. Good modelling keeps physical and institutional realities visible rather than treating them as inconvenient details.

There are different levels of complexity. A simple fixed pumping rule may be sufficient. A linear programme may help when costs and constraints are linear. A nonlinear model may be needed for pressure-dependent flows. The additional complexity is justified only when it represents a mechanism that affects the decision and can be supported by data.

2.9 Assumptions are claims to be challenged

Every model includes simplifying assumptions. The responsible question is not "Are there assumptions?" but "Which assumption could change the answer?" An assumption audit should identify the consequence of each choice and a practical way of testing its importance.

Table 2.3: An assumption audit for the fictional water model

Assumption	Possible failure	A useful check
Daily inflow is known	Inlet meter drifts or lacks power	Compare independent readings and meter calibration
Loss is fixed at $2\,\mathrm{m}^3$	Leakage varies with pressure or pipe faults	Test a range of losses and seek pressure/flow data
All demand is known	Unserved customers disappear from delivery records	Distinguish requests and deliveries; document shortage days
Tank capacity is usable in full	Minimum reserve or pump intake prevents complete emptying	Add minimum operating volume to constraints
One-day time step is adequate	Shortage or overflow occurs during the day	Recompute at shorter intervals if observations permit

Saltelli et al. (2008) develop sensitivity analysis as a systematic way to investigate how model outputs depend on uncertain inputs. We will return to these methods in later chapters. For now, even a small scenario table can be revealing: vary initial storage, loss and demand, and observe whether the recommendation changes. If it does, the conclusion should be communicated conditionally.

2.10 Verification, validation and the limits of the model

Verification asks whether equations and code implement the intended mathematical description. For our storage model, a basic test checks that storage never becomes negative and never exceeds capacity; another tests conservation. These are valuable even before any real data are obtained.

Validation asks whether the description represents the intended physical system sufficiently well for its use. That requires evidence about actual storage, inflows, demand, deliveries and operational constraints. An error-free program can model the wrong system. An accurate prediction on one operating regime may fail during an unusual dry spell. The intended use determines which evidence matters most.

An important consequence is that successful reproduction of Figure 2.1 does not validate a water-management system. It verifies a particular illustrative computation. A later field study would need suitable measurements, a baseline, error reporting and operational review.

2.11 A model begins with a question, not an equation

A district water officer may want to know whether a storage tank can keep a health facility supplied during an interruption. A plumber may want to locate a leak. An engineer may want to size a pump. A finance officer may want to compare the cost of adding storage with the cost of improving delivery schedules. All four people are interested in the same water system, but the same mathematical model will not answer all four questions.

The tank balance introduced earlier is adequate for an accounting question: how much water entered, left or remained in storage during each period? It is not by itself adequate for locating a leak. A network model is needed to represent how water moves through pipes. A hydraulic model may be needed to predict pressure. A cost model is needed before deciding whether a pump or storage upgrade is affordable. The lesson is not that increasingly complicated models are always better. It is that a model must contain the variables and mechanisms required by its intended question.

This can be written with a simple observation map. Let z represent the physical state and h the measurement process. The observed data are $y = h(z)+\varepsilon$, where ε represents measurement error. A proposed quantity of interest is $q(z)$. If two different states z_1 and z_2 satisfy $h(z_1) = h(z_2)$ but $q(z_1) \neq q(z_2)$, then exact noiseless observations of $h(z)$ cannot distinguish the answers to the question. The ambiguity comes before any computation. No optimisation algorithm can infer an unmeasured distinction without introducing new assumptions or data.

Proposition 2.2 (A basic test for distinguishability)**.** *Let $h : \mathcal{Z} \to \mathcal{Y}$ be a deterministic observation map and let $q : \mathcal{Z} \to \mathcal{Q}$ be the desired answer. There is a function g on the observed image $h(\mathcal{Z})$ such that $q(z) = g(h(z))$ for every $z \in \mathcal{Z}$ if and only if q is constant on each set $\{z : h(z) = y\}$.*

Proof. If $q = g \circ h$ and $h(z_1) = h(z_2)$, then $q(z_1) = g(h(z_1)) = g(h(z_2)) = q(z_2)$. Conversely, assume q is constant on every fibre of h. For any observed value $y \in h(\mathcal{Z})$, choose any z with $h(z) = y$ and set $g(y) = q(z)$. Constancy ensures that the definition does not depend on which z was chosen. Then $g(h(z)) = q(z)$ for all z. □

The result is elementary, but its practical force is considerable. We may not be able to reconstruct every hidden pipe flow. Yet we might still recover a district-wide quantity such as total supply if that quantity is the same across every flow pattern consistent with the available meters. A decision often needs less information than a complete reconstruction of the system. The important question is which differences between states would change the decision.

2.11.1 A leak that an aggregate meter cannot locate

Consider two pipelines leaving a tank, with flows f_1 and f_2. The tank meter measures $y = f_1+f_2$. A reading of 100 litres per minute is consistent with $(f_1, f_2) = (60, 40)$ and with $(80, 20)$. If our question is total outflow, $q(f_1, f_2) = f_1 + f_2$, it is identifiable from the meter. If the question is how much water flows through pipe 1, $q = f_1$, it is not. A second independent branch meter can resolve the ambiguity, subject to its accuracy. This is why specifying the decision before ordering sensors can save money.

2.12 Units, time scales and dimensional reasoning

A plausible mathematical expression is not necessarily a meaningful one. Units reveal many errors before code is written. If S_t is measured in cubic metres and I_t in cubic metres per hour, then $S_{t+1} = S_t + I_t - D_t$ is wrong unless I_t and D_t already represent volumes over the time step. With flow rates q_t^{in} and q_t^{out}, a consistent balance is

$$S_{t+1} = S_t + (q_t^{\text{in}} - q_t^{\text{out}})\Delta t - \ell_t, \tag{2.16}$$

where Δt is measured in hours and ℓ_t is a loss *volume* over the interval. Every additive term now has units of cubic metres.

2.12.1 A realistic distinction between a volume and a rate

Imagine a tank containing 14 m^3. A pump supplies 2.5 m^3/h and customers withdraw 1.8 m^3/h. If the net loss is 0.1 m^3/h, then in a two-hour interval the expected balance is $14 + (2.5 - 1.8 - 0.1) \times 2 = 15.2$ m^3. Adding the flow rates without multiplying by the two-hour interval would incorrectly give 14.6 m^3. The error would be silent in a program that stores all quantities as floating-point numbers.

Changing the time step also changes the assumptions. A daily balance may be good enough for long-term supply planning but too coarse to describe pressure during a brief morning peak. A ten-minute model can resolve peaks, but requires more detailed inflow and demand data. Making Δt small is not automatically an improvement if the input rates were measured only once per day. Fine numerical resolution cannot create fine physical information.

The same question appears in statistics. Daily rainfall, monthly household spending and yearly business counts are different temporal measurements. Rescaling may be possible, but it must reflect what the variable represents. A monthly total can be divided into daily *averages*; it cannot be turned into observed rainfall on individual days without further evidence.

2.13 From a discrete balance to a differential equation

The discrete tank equation may be understood as an accounting identity over successive intervals. If inflows and outflows vary continuously with time, a continuous formulation is

$$\frac{dS}{dt} = q_{\text{in}}(t) - q_{\text{out}}(t) - q_{\text{loss}}(t). \tag{2.17}$$

Integrating from t_0 to t_1 gives

$$S(t_1) - S(t_0) = \int_{t_0}^{t_1} \big[q_{\text{in}}(t) - q_{\text{out}}(t) - q_{\text{loss}}(t)\big]\,dt. \tag{2.18}$$

The integral expresses conservation. It does not presume that rates stay constant inside the interval. The earlier discrete formula is exact when the terms I_t, D_t, ℓ_t are measured interval volumes; it becomes a numerical approximation when we insert instantaneous rates and multiply them by Δt.

2.13.1 A leakage model with a closed-form answer

Suppose water enters at a constant rate u, customers withdraw at constant rate d, and leakage is proportional to stored volume, $q_{\text{loss}} = kS$ with $k > 0$ in units of inverse time. Then

$$\frac{dS}{dt} = u - d - kS. \tag{2.19}$$

Let $a = u - d$. The integrating factor method yields

$$S(t) = \frac{a}{k} + \left(S_0 - \frac{a}{k}\right) e^{-kt}. \tag{2.20}$$

Substituting into the differential equation verifies the result: $S'(t) = -k(S_0 - a/k)e^{-kt} = a - kS(t)$. With constant positive net inflow, the model approaches the equilibrium a/k. If $a < 0$ it predicts an eventual negative storage level, which is physically impossible. The solution is mathematically correct for the unconstrained ODE, but the operating model ceases to be valid before the tank reaches negative volume. A tank controller must stop withdrawal, alter flows or introduce a nonnegative-state condition. This is an example of mathematical correctness not guaranteeing physical feasibility.

2.13.2 Euler's method and an error we can measure

For illustration, use a forward step

$$S_{n+1} = S_n + h(a - kS_n). \tag{2.21}$$

Let e_n be the difference between this numerical solution and the exact solution of (2.20) at the same time. For the scalar linear problem, errors in an existing perturbation propagate as $e_{n+1} \approx (1 - hk)e_n$, in addition to local discretisation error. Thus $|1 - hk| < 1$ is the usual linear stability condition for perturbations, or $0 < hk < 2$. But satisfying it does not automatically preserve nonnegative storage, especially with constraints and withdrawals. A stable computation may still violate a physical requirement.

A smaller step should improve the discretisation error for sufficiently smooth inputs, but it does not repair wrong pump rates or leakage parameters. We will return to verification in more detail later; for now, one should compare a numerical result with the exact solution above before trusting the same code on a system for which no exact answer is available.

2.14 Networks: conservation is local before it is global

A storage tank is part of a distribution network. Denote nodes by $1, \dots, n$ and directed links by $1, \dots, m$. Choose an orientation for each link, even if the real flow may later reverse. Let $B \in \mathbb{R}^{n \times m}$ be the node–edge incidence matrix, with -1 at the link's starting node, $+1$ at its receiving node, and zero elsewhere. If $f \in \mathbb{R}^m$ contains oriented flows, then

$$Bf = s, \tag{2.22}$$

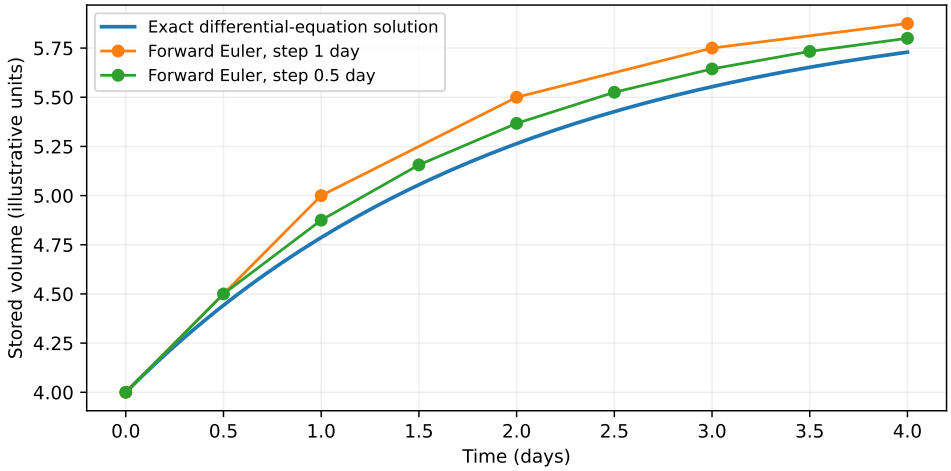


Figure 2.3: An exact linear storage equation compared with forward Euler calculations at two time steps. The numerical illustration checks the algorithm, not a measured water system.

where s describes net arrival of flow at nodes under the same sign convention. The columns of B sum to zero. Therefore $\mathbf{1}^\top B = 0$ and any feasible s must satisfy $\mathbf{1}^\top s = 0$ on a closed network with no unrepresented storage changes. This simple necessary condition often catches errors in a spreadsheet of node balances.

2.14.1 Why conservation alone does not determine every pipe flow

Consider a network shaped like a triangle, with links $1 \to 2$, $2 \to 3$ and $3 \to 1$. Its incidence matrix is

$$B = \begin{pmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}. \tag{2.23}$$

The vector $(1, 1, 1)^\top$ belongs to the nullspace of B, because a circulation around the loop has no net effect on node balances. Therefore, if f satisfies $Bf = s$, then so does $f + c(1, 1, 1)^\top$ for any c. Node balances alone do not distinguish the flow around the loop. Hydraulic pressure relations, additional observations or engineering constraints must supply the missing information.

This is a useful example for students accustomed to solving linear equations. The issue is not an inability to invert a matrix due to poor software. It is that the physical equations included so far do not uniquely determine the quantities requested. The rank of B is at most $n - 1$, and each independent cycle typically introduces a flow degree of freedom before constitutive relations are imposed.

2.14.2 Location models and why the geometry matters

A pipe network represents more than lengths. Different links may have different diameters, elevations, operational restrictions and failure risks. A simple road network may use edge weights representing driving time, while a water network might use pressure losses or allowed

capacities. A shortest path computed using distance may not be fastest, safest or even passable. The graph is meaningful only after its edge values are defined and checked.

For a transport example, let $G = (V, E)$ be a directed graph with travel-time weights $c_e \geq 0$. A route from node s to t is a sequence of connected edges and has cost $\sum_{e \in p} c_e$. The shortest-path problem selects p to minimise this sum. If rainy-season travel times are uncertain, the true cost becomes $c_e(\xi)$; the shortest path under dry conditions need not remain optimal. One can compare expected costs, conservative scenarios or the regret of choosing a route that later becomes unusable. We should not call a map-derived route an observed journey time unless the edge costs actually came from travel observations.

2.15 Constraints are part of the system, not decoration

Many mathematical models produce attractive but impossible recommendations because constraints were added after the objective had been selected. Suppose a district operates a pump with rated capacity $u_{\max}$ and a tank of capacity K. If u_t is the volume pumped in period t, a feasible decision must satisfy

$$0 \leq S_t \leq K, \qquad 0 \leq u_t \leq u_{\max}, \tag{2.24}$$

$$S_{t+1} = S_t + u_t - D_t - \ell_t. \tag{2.25}$$

Depending on units, pump capacity may be specified per period rather than per hour. If the pump operates only when electricity is available, introduce an availability variable $z_t \in \{0, 1\}$ and require $0 \leq u_t \leq z_t u_{\max}$. This is a mathematical way to express the operating fact that a working pump cannot supply water during a power outage unless it has independent power.

A shortage variable $r_t \geq 0$ may represent unmet customer demand. If d_t is desired delivery, then actual delivery is $D_t = d_t - r_t$ with $0 \leq r_t \leq d_t$. The system's objective might be to minimise $\sum_t c_r r_t + c_e u_t$ subject to the balance and resource limits. Here c_r penalises shortages and c_e represents a pumping cost per unit. Without a shortage variable, an infeasible model may simply have no solution on low-supply days. With it, the model can show what is impossible and how much demand cannot be met. We have to state whether unmet demand is permitted at all for a critical service.

2.15.1 The difference between infeasible and expensive

Suppose a tank initially holds 5 m^3. During an interruption, the pump is unavailable, and the facility demands 8 m^3 without incoming deliveries. Even with no leakage, a requirement to meet all demand is infeasible. Raising the financial penalty for unmet demand cannot create three cubic metres of water. A solver returning "infeasible" may be giving the most valuable possible warning. In an emergency setting, the next step is to consider a new action—reserve supply, delivery by truck, demand restriction or backup pumping—rather than hiding the deficit inside the objective function.

Mathematical optimisation distinguishes a hard constraint from a preference. Patient safety limits, maximum tank capacity and legal operating limits belong in the feasible set. Cost and convenience can then be compared among feasible choices. Their relative weights must be

documented, because the optimiser will pursue the exact mathematical criterion we give it, not the values we intended but did not write down.

2.16 Linear optimisation, duality and the value of scarce capacity

Consider a simplified supply problem with two neighbourhoods. Let x_1, x_2 denote delivered cubic metres. Benefits per cubic metre are $v_1 = 5$ and $v_2 = 3$ in illustrative units. A total of ten cubic metres can be delivered, and neighbourhood 1 can receive at most six. We solve

$$\begin{aligned} \max \quad & 5x_1 + 3x_2, \\ \text{subject to} \quad & x_1 + x_2 \le 10, \\ & 0 \le x_1 \le 6, \qquad x_2 \ge 0. \end{aligned}$$

Because the first neighbourhood yields larger marginal benefit, allocate six to it and four to the second: $(x_1, x_2) = (6, 4)$, objective 42. If total supply increases slightly from ten while the six-unit cap remains, an additional unit goes to neighbourhood 2 and adds three benefit units. Locally, the *shadow value* of total supply is therefore three per extra cubic metre. This interpretation depends on the active constraints and the range of the change; it is not a permanent economic price of water.

We can see the same value through a dual programme. For the primal above, introduce nonnegative dual variables u for total supply and v for the neighbourhood 1 cap. One dual form is

$$\begin{aligned} \min \quad & 10u + 6v, \\ \text{subject to} \quad & u + v \ge 5, \\ & u \ge 3, \quad u, v \ge 0. \end{aligned}$$

The solution $u = 3, v = 2$ has dual objective $30 + 12 = 42$, equal to the primal optimum. This equality confirms optimality by strong duality for the feasible, bounded linear programme. The mathematics reveals a valuable question to take back to the water officer: is an extra unit of water worth the cost of obtaining it under the present operating constraints? The answer may change when demand, service priorities or capacity limits change. For systematic treatments of linear optimisation see Hillier and Lieberman (2021).

2.17 Uncertainty should change how we formulate a decision

An average-demand model can understate shortages when demand fluctuates. Let next period's demand D be uncertain with mean μ_D and standard deviation σ_D. Suppose the planner chooses supply a and wants to satisfy

$$\Pr(D \le a) \ge 0.95. \tag{2.26}$$

If, and only if, demand is credibly approximately normal with those parameters, one would use $a \ge \mu_D + 1.645\sigma_D$. If $\mu_D = 10$ and $\sigma_D = 2$ m^3, the illustrative required supply is about

13.29 m^3. The numerical quantile is not a universal safety factor. If demand is strongly skewed, seasonal or measured poorly, the normal assumption may misstate the tail.

An alternative is scenario planning. Construct a small set of clearly declared demand and outage scenarios $\xi_1, \ldots, \xi_J$, and assess how a candidate pump schedule performs in each. A robust formulation may minimise the maximum loss,

$$\min_{a \in \mathcal{A}} \max_{j=1,\ldots,J} L(a, \xi_j). \tag{2.27}$$

This protects against the worst specified scenario, not every event that nature could possibly produce. A scenario set invented for teaching cannot be called a measured distribution of outages. A practical study should obtain operational logs, expert review or sensitivity ranges justified by the system's history.

2.17.1 What a sensitivity experiment really asks

Suppose we do not know the leakage coefficient k. Rather than choosing one value and reporting a precise optimal pump schedule, vary k over a declared range. Calculate how storage, shortages and costs respond. If the preferred design changes sharply with a small variation in k, measuring leakage more accurately may be more valuable than making the optimiser more sophisticated. Sensitivity analysis examines how conclusions depend on uncertain inputs; it does not turn arbitrary input ranges into evidence. Saltelli et al. (2008) and Morris (1991) provide broader approaches to analysing sensitivity in computational models.

A simple dimensionless sensitivity for an outcome $Q(\theta)$ is

$$E_\theta = \frac{\theta}{Q(\theta)} \frac{\partial Q}{\partial \theta}, \tag{2.28}$$

when Q and θ are nonzero. It measures the relative change in Q per relative change in θ. In the leakage model's positive equilibrium $S^* = (u - d)/k$, this elasticity with respect to k is -1: a one-percent increase in the assumed leakage coefficient produces approximately a one-percent decrease in equilibrium storage, holding the other quantities fixed. An elasticity is local. It says nothing about whether the operating system can actually remain near equilibrium or whether the coefficient has been measured correctly.

2.18 Numerical verification is different from checking the model

A computer program may reproduce the equations perfectly while the equations omit a critical mechanism. Conversely, a well-chosen mathematical model can be implemented incorrectly. These are separate sources of failure. The terms *verification* and *validation* are used in scientific computing to distinguish checking whether we solved the equations correctly from checking whether those equations adequately represent the intended reality. Oberkampf and Roy (2010) provides a detailed treatment.

2.18.1 Four tests for the tank model

First, *dimensional consistency*: all terms added to storage must be volumes. Second, *conservation*: the difference in stored water must equal net inflow less delivered water and losses. Third, *bounds*: physical storage must remain within its admissible range, and shortage must never be negative. Fourth, *analytical reference*: in the special constant-rate leakage case, a numerical solver should approach equation (2.20) as the time step is refined.

A useful coding practice is to put tests next to the implementation. An example with a two-hour interval is:

```
def tank_step(stock, inflow_rate, outflow_rate, loss_rate, dt):
    assert dt > 0
    return stock + dt * (inflow_rate - outflow_rate - loss_rate)

next_stock = tank_step(14.0, 2.5, 1.8, 0.1, 2.0)
assert abs(next_stock - 15.2) < 1e-12
```

This test establishes that the function implements the stated arithmetic for one case. It does not show that the physical tank's inflow and loss rates are correct, that its capacity has been respected, or that it can operate safely during failure. These require separate evidence and additional tests.

2.18.2 A conservation ledger for a longer simulation

For any sequence of periods, compute a cumulative residual

$$R_T = S_T - S_0 - \sum_{t=0}^{T-1} (I_t - D_t - \ell_t). \tag{2.29}$$

If the discrete update was implemented consistently and uses the same values, R_T should be approximately zero up to floating-point rounding. A nonzero residual can reveal an indexing error, a hidden intervention, or inconsistent accounting. But an exact zero can coexist with seriously wrong input measurements. A balance check certifies internal consistency, not the truth of every variable.

2.19 A model-card exercise for engineers and decision-makers

Before choosing a model for a real project, write one page describing its purpose and another documenting its limits. State the decision, the mathematical outputs required, the state variables, each input's unit and origin, the geographical and temporal scales, and the operating constraints. Identify at least two plausible mechanisms that the model excludes. Name the tests that would detect a computational bug, and the observations that would challenge the model's assumptions.

For a water tank this record might say: the model predicts daily storage under measured deliveries and estimated demand; it does not predict pressure at individual taps. Its primary output is a shortage indicator; its most sensitive parameter is an uncertain leakage coefficient;

its validation requires actual tank-level and delivery observations. That statement is modest, but it allows a colleague to judge whether the model is appropriate and what would be required before deployment.

Extended exercises. Derive the equilibrium of the leakage model and explain when it is physically feasible. Show why the triangle network's incidence matrix is singular and identify its nullspace. Recalculate the two-neighbourhood optimisation if total supply falls to five cubic metres; explain which marginal value changes. Construct a demand scenario in which the expected-cost solution violates a critical minimum supply constraint. Explain how one would distinguish numerical discretisation error from a biased inflow meter. A strong answer should make the question, assumptions, mathematics and evidence visible, rather than supplying only a final number.

2.20 Estimating parameters is an inverse problem

Even a simple conservation equation requires numerical inputs that may not be known. A tank's capacity can perhaps be measured from its geometry, but its leakage coefficient may not be printed on any engineering label. Demand may vary by season. If we use measurements of storage to estimate leakage, we are no longer merely predicting the state from parameters. We are working backwards from observations to infer a parameter. That is an inverse problem, and it has its own conditions for a reliable answer.

In the continuous leakage model $S' = a - kS$, suppose a is known and storage is observed at several times. If the observations were exact and sufficiently informative, the solution

$$S(t) = \frac{a}{k} + \left(S_0 - \frac{a}{k}\right) e^{-kt}$$

would allow a fit for $k > 0$. In practice a may be only approximately known, water levels are noisy, and some operating intervals contain unrecorded withdrawals. We must therefore ask whether two different parameter settings can produce nearly the same observed trajectory. A highly precise numerical optimiser may return a parameter with many digits even when the data cannot distinguish those settings.

2.20.1 A direct identifiability calculation

Take the simpler case of no input or withdrawal, $S' = -kS$ with $S(0) = S_0 > 0$. Then $S(t) = S_0 e^{-kt}$. If S_0 is known and we observe exact storage at $t > 0$, we can recover

$$k = -\frac{1}{t} \log\left(\frac{S(t)}{S_0}\right). \tag{2.30}$$

But the measurement must satisfy $S(t) > 0$. If S_0 is unknown, a single storage observation is insufficient to distinguish S_0 from k: infinitely many pairs can give the same reading. Two exact observations $S(t_1), S(t_2)$ with $t_2 > t_1$ can identify k through their ratio, provided the assumptions truly apply:

$$k = -\frac{\log S(t_2) - \log S(t_1)}{t_2 - t_1}.$$

The equation explains why the measurement schedule matters. Two readings taken almost at the same time have a small denominator, so ordinary measurement error can produce a large parameter error. Waiting longer increases the observable decay, but also creates opportunities for unrecorded pumping, withdrawal or changing conditions. Experimental design is a balance between statistical information and the reliability of the assumptions.

2.20.2 How uncertainty in storage affects the fitted parameter

Differentiate (2.30) with respect to the later observed storage:

$$\frac{\partial k}{\partial S(t)} = -\frac{1}{tS(t)}.$$

Thus small measurement errors can strongly affect k when $S(t)$ is small or the observation interval t is short. A first-order approximate variance for independent errors in S_0 and $S(t)$ is

$$\operatorname{Var}(\widehat{k}) \approx \frac{1}{t^2}\left(\frac{\sigma_t^2}{S(t)^2} + \frac{\sigma_0^2}{S_0^2}\right).$$

If the errors are correlated, a covariance term must also be included. This expression is not a complete uncertainty analysis when the model structure is uncertain. It nevertheless shows why apparently harmless measurement choices can dominate the stability of the inverse calculation.

2.20.3 Why fitting several unknowns at once can fail

Suppose both a and k are unknown in $S' = a - kS$. A storage record that remains almost constant near its equilibrium may identify approximately the ratio a/k without identifying either parameter separately. For example, many pairs (a, k) with the same ratio yield the same equilibrium level. A sequence of measured transients, or independent measurements of inflow and withdrawal, may help separate them. If the tank was observed only after it had settled, a more complicated optimiser is unlikely to create the missing transient information.

The lesson for resource-constrained work is practical. If the current evidence identifies only a ratio that is sufficient for a planning decision, perhaps nothing more is needed. If the decision requires the separate leakage coefficient because a repair cost depends on it, targeted measurement is justified. Model identifiability should be assessed against the decision, not against an automatic requirement to estimate every parameter in an elaborate model.

2.21 What counts as a sufficient model for an engineering choice?

It is common to describe models as either simple or sophisticated. A more useful distinction is whether a model is adequate for a particular decision. A reservoir balance with one state may be adequate to decide whether existing storage can bridge a scheduled six-hour interruption. The same model is inadequate for determining pressure losses in a branched network. A high-dimensional hydraulic simulator may be essential for an engineering design and unnecessary for a weekly accounting report.

Adequacy must be tested on outputs that matter. If a new pump is selected because a model predicts a maximum daily shortage of one cubic metre, compare its predictions with observations on high-demand and outage days, not just quiet days. If a pipe size is selected because pressure must exceed a safety threshold, check error near that threshold. A small average prediction error over ordinary conditions may conceal unacceptable errors during the very conditions the intervention is meant to address.

For the modeler, this creates a disciplined sequence: specify the decision threshold, identify the necessary physical outputs, derive the simplest model that represents them, establish which measurements constrain its uncertain components, and compare its outputs with independent evidence. Only after these steps does increased numerical complexity have a clear purpose. A model can be appropriately simple because the decision is simple, or appropriately complex because the missing mechanisms are consequential. Mathematical elegance is not a substitute for that judgement.

2.22 Case study: a water system with several possible models

Consider a small district service supplying a health centre and two nearby settlements. The source feeds one tank; water leaves through separate branches. Some days the source is available, other days pumping is interrupted. The manager wants to know whether to purchase more storage, change pumping hours or improve distribution. No direct pipe-pressure records are available for this teaching case. Every numerical value below is assumed and should be treated as a scenario, not an observed Ugandan utility record.

At the simplest level, the system is a tank with total demand. A daily balance can answer whether enough volume is available across a sequence of days. But it cannot show whether one settlement receives little water while another receives most of the supply. For that we must split demand and delivery by branch. If the decision is to locate an undetected leak, a branch-flow model alone may be insufficient unless there are meters at informative positions. If the decision concerns minimum pressure at taps, a hydraulic relation linking flows, pressure and pipe attributes is necessary.

2.22.1 A model ladder

We will write four versions, each retaining only what is needed for a different decision. Version A uses total tank storage S_t and total inflow I_t, demand D_t and loss ℓ_t. Version B adds branch deliveries d_{1t} and d_{2t} and allocation rules. Version C treats the physical system as a network with nodes and links, with flow conservation. Version D adds uncertain demand, electricity availability and planned pumping. This progression does not imply that the fourth model should always be built. Version A may be all a manager needs to decide whether tomorrow's planned delivery volume exceeds available storage.

2.22.2 Version A: an exact accounting model

Let the initial storage be $S_0 = 12$ m^3. On day one, inflow is $I_0 = 8$ m^3, total delivery is $D_0 = 10$ m^3 and measured or assumed losses are $\ell_0 = 1$ m^3. The resulting storage is

$S_1 = 12 + 8 - 10 - 1 = 9$ m^3. If an outage stops pumping on day two, with the same delivery and loss, the unconstrained equation gives $S_2 = 9 + 0 - 10 - 1 = -2$ m^3. That is a warning that the desired schedule cannot be supplied from the tank. It is not a physical prediction that the tank will contain negative water.

A physically meaningful balance introduces actual delivery $D_1^{\text{served}} \le 10$ and unmet demand $U_1 = 10 - D_1^{\text{served}}$. With storage nine and one unit lost, at most eight units can be served if no inflow occurs and storage may reach zero. Thus $U_1 \ge 2$. If some water must be kept in reserve, unmet demand will be greater. The formulation reveals what policy needs to decide: how to allocate unavoidable shortages among uses with different consequences.

2.22.3 Version B: deliveries are not interchangeable

Suppose the health centre requires at least three cubic metres per day for critical services and two settlements request four and three respectively. Total desired delivery is ten cubic metres. With only eight available for distribution, a policy that allocates proportionally would give the health centre 2.4, below its required minimum. A policy with a hard minimum for critical care could first allocate three to the health centre, then distribute the remaining five according to a declared criterion. The mathematical model needs a constraint $d_H \ge 3$ and an explicit rule for the other demands; merely minimising total unmet cubic metres cannot distinguish which community was shorted.

One option is to minimise weighted shortage,

$$\min \quad w_H u_H + w_A u_A + w_B u_B, \tag{2.31}$$

subject to $d_j + u_j = r_j$, nonnegative deliveries and shortages, and $d_H + d_A + d_B \le 8$. Weights can express priorities but should not quietly replace a non-negotiable minimum. A care minimum should remain a constraint, while any differences among noncritical shortages may be debated with the affected communities.

2.22.4 Version C: the same total can conceal different flows

Suppose the tank has one outlet meter reporting eight cubic metres and two distribution branches with no separate measurement. The readings $d_A = 5, d_B = 3$ and $d_A = 2, d_B = 6$ are observationally identical at the outlet meter. A manager deciding where to repair a leaking branch cannot infer the answer from the total. Adding another day of total readings may help only if flows or demands vary in a way that distinguishes the branches under a justified model. A single reliable branch meter may be more informative than another month of aggregate tank readings. This is the distinguishability principle in action.

2.22.5 Version D: decisions before uncertainty is resolved

The manager may need to choose tomorrow's pumping hours today, before knowing whether grid electricity will be available. Let $z \in \{0, 1\}$ indicate whether pumping is available. An action a fixes a schedule in advance; actual inflow becomes $I(a, z)$. One could minimise expected shortage or a risk-sensitive loss over a documented distribution for z. If there is no defensible

outage probability, compare a small set of scenarios without presenting them as frequencies observed in the district. Robust planning may choose a conservative schedule, but it should show the extra cost of protection.

2.23 A complete reproducible tank experiment

A useful teaching programme must reveal how assumptions affect results. The following compact code simulates a tank under a schedule and records unmet demand. It implements a clearly stated operating policy: inflow enters at the start of each period, losses are then deducted up to the water available, and the remaining water is allocated to demand. The programme is not a hydraulic simulator; it ignores pressure, pipe capacity and within-period timing.

```
def simulate_tank(initial, capacity, inflows, demand, losses):
    assert len(inflows) == len(demand) == len(losses)
    stock = float(initial)
    records = []
    for t, (incoming, wanted, loss) in enumerate(
            zip(inflows, demand, losses)):
        assert incoming >= 0 and wanted >= 0 and loss >= 0
        available = min(capacity, stock + incoming)
        spill = max(0.0, stock + incoming - capacity)
        lost = min(loss, available)
        available -= lost
        served = min(wanted, available)
        unmet = wanted - served
        stock = available - served
        assert 0 <= stock <= capacity
        records.append((t, stock, served, unmet, spill, lost))
    return records

out = simulate_tank(
    initial=12, capacity=20,
    inflows=[8, 0, 6], demand=[10, 10, 8], losses=[1, 1, 1])
for row in out:
    print(row)
```

Work through the programme by hand before trusting it. At the end of period one, $12+8-1-10 = 9$ remains. Period two begins with nine, loses one, serves eight and leaves zero: unmet demand is two. Period three receives six, loses one, serves five and leaves zero: unmet demand is three. The program should reproduce those results. Note that the output depends on the assumed order of events. If a pump supplies water gradually throughout the day rather than at the beginning, the real service and its shortages may differ. A correct implementation of this policy does not prove that the policy represents actual pumping practice.

2.23.1 How to verify the calculation

For each period, mass conservation requires previous stock plus inflow to equal new stock plus served demand, losses and spill, provided all terms refer to the same sequence of events. A suitable property test computes the residual independently and checks that it is nearly zero.

Additional tests should cover an empty tank, full tank, zero demand, inflow far above capacity, and a loss request larger than the available water. These boundary cases find errors that an ordinary three-day example can miss.

2.23.2 How to validate the model

Validation is a different project. With permission, record tank-level observations, pumping activity, water deliveries, significant leakage incidents and interruptions across a useful period. Convert level to volume using a verified tank geometry; check that clocks align and that all major inflows and outflows are included. Compare predicted stock with held-out periods, ideally including at least one outage. The discrepancy might reveal wrong demand, wrong leakage, poor timing assumptions or measurement error. It does not automatically tell us which cause dominates.

2.23.3 What would justify a purchase?

A larger tank creates storage capacity, but not water. If the source is chronically insufficient, increasing tank capacity alone cannot close a persistent average deficit. If outages are short and production is plentiful when power returns, extra storage may help. If losses are the main issue, repairing leaks may dominate investment in storage. Compare at least a do-nothing baseline, a revised schedule and a capacity change under the same measured scenarios. Include costs, maintenance, local technical skills and failure risks. Only then does a numerical optimum become a potentially useful recommendation.

2.24 An invitation to examine the mathematics critically

There are several ways for this case to fail, and finding them is part of learning applied mathematics. The objective might penalise shortages but overlook the consequences of contaminated water. The tank capacity may be known but the source yield may have been guessed. The model may assume one daily time step even though interruptions occur for two-hour periods. The branch-flow variables may be mathematically feasible but hydraulically impossible because the pipes cannot deliver the specified rates. The financial model may omit replacement costs or staff time.

For each failure, ask whether we need new equations, new measurements, a different decision variable or a restriction on what we claim. A missing pressure equation is a modelling problem. An uncertain demand value is a data problem. A negative storage caused by an omitted nonnegative constraint is a formulation problem. An incorrect update index is a programming problem. A schedule that staff cannot implement is an operational problem. These failures are not interchangeable, and a good engineer should be able to name them before attempting a solution.

Project investigation. Write a short engineering memo for an imaginary district considering a backup water tank. State the purpose and the allowed decisions. Identify the simplest model

adequate for the first decision and one more detailed model that would be needed for a later question. Give a numerical reference case that checks conservation, a failure case that exposes infeasibility, and a list of real measurements required before procurement. Conclude with one decision you would not allow the model to make without additional evidence.

2.25 Worked investigations: from equations to trustworthy decisions

A useful mathematical education should train a reader to judge the result as well as to obtain it. Each investigation below has a numerical or symbolic answer and a second, interpretive answer. Stopping at the first is not enough for a working solution.

2.25.1 Investigation 1: a numerical method and a physical constraint

Consider $S'(t) = 3 - 0.5S(t)$ with $S(0) = 4$ in a stated volume unit, and measure time in days. The exact solution is $S(t) = 6 - 2e^{-0.5t}$. At day one the exact storage is $6 - 2e^{-0.5} \approx 4.78694$. A forward Euler step of one day gives $S_1 = 4 + 1(3 - 0.5 \times 4) = 5$. The absolute error is approximately 0.21306. With two half-day Euler steps, the first state is $4 + 0.5(1) = 4.5$ and the second is $4.5 + 0.5(3 - 0.5 \times 4.5) = 4.875$. The error is then approximately 0.08806.

The smaller time step improves the numerical approximation in this example. It does not mean we have measured the inflow or leakage coefficient accurately. If the true inflow is 2 rather than 3, reducing the time step further converges to the wrong physical trajectory with great numerical precision. If the physical tank capacity is 4.8, the continuous model above also eventually predicts an impossible level unless overflow or pump control is represented. These are three distinct issues: discretisation error, parameter uncertainty and a missing physical constraint.

2.25.2 Investigation 2: a network balance with infinitely many flows

Take the three-node cycle with incidence matrix

$$B = \begin{pmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}.$$

Let net nodal flow $s = (-4, 4, 0)^\top$. Solve $Bf = s$ using $f = (f_1, f_2, f_3)^\top$. The equations are $-f_1 + f_3 = -4$, $f_1 - f_2 = 4$, and $f_2 - f_3 = 0$. Put $f_3 = c$. Then $f_2 = c$ and $f_1 = 4 + c$, giving the entire family

$$f = (4 + c, c, c)^\top, \qquad c \in \mathbb{R}.$$

The family contains $f = (4, 0, 0)$ and $f = (5, 1, 1)$, which yield the same net node balances. If our question concerns how much water reaches node 2 in total, the flow conservation may answer it. If the decision concerns which pipe carries excess flow, we need additional information. Imposing an arbitrary minimum-norm solution does select a unique number, but it does so by introducing a mathematical preference, not by observing physical flow.

For a water network, pressure differences and pipe laws are possible additional equations. If some pipes permit reverse flows, the sign and physical meaning of negative oriented flows must be understood; choosing an orientation is a bookkeeping convention, not a valve. If there are bounds on pipe capacity, the admissible interval for c may shrink. The task of a modeller is to make those added physical assumptions visible rather than bury them in a solver's defaults.

2.25.3 Investigation 3: why an apparently reasonable allocation fails

A region has 10 units of water and two customer groups. The first can use at most six. Benefit is 5 per unit delivered to the first group and 3 to the second. We found the optimal allocation $(6, 4)$, with objective 42. Now a new minimum supply requirement is introduced: group 2 must receive at least five units. The former optimum violates this hard requirement. The best feasible allocation becomes $(5, 5)$, objective 40.

Notice what happened. The mathematical optimiser did not become less competent. The decision question changed because a policy requirement was stated explicitly. A system that continues to display $(6, 4)$ after the policy change is solving an outdated model. If the demand of group 2 has a legal or health requirement, that requirement belongs in the constraints, not in a footnote next to an old optimum.

Now reduce total available water to four units while insisting that group 2 receive at least five. The problem becomes infeasible. There is no numerical algorithm that can make four units satisfy a five-unit minimum. The system should report infeasibility and the missing unit, or offer an appropriately authorised emergency alternative. An optimisation tool that silently relaxes the minimum to print an answer has crossed an important boundary.

2.25.4 Investigation 4: what can a sensor actually tell us?

Consider the observation $y = z_1 + z_2$ for nonnegative quantities z_1, z_2. If the sensor reads $y = 8$, the pair $(8, 0)$, $(4, 4)$ or $(0, 8)$ could all be correct. The observation identifies the total but not the components. Suppose we add a second measurement $y_2 = z_1 - z_2$. The two observations form

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}.$$

The matrix is invertible, with determinant -2. Thus $z_1 = (y_1 + y_2)/2$ and $z_2 = (y_1 - y_2)/2$. If $y_1 = 8$ and $y_2 = 2$, we infer $(z_1, z_2) = (5, 3)$.

But if the second measurement is instead $y_2 = 2z_1 + 2z_2$, it contributes no independent distinction. Its matrix has determinant zero. Adding more observations of the same mixture improves precision under suitable noise assumptions, but cannot separate the underlying components. This is a reason to plan sensors around the decisions they must support. It is also a reason to resist high-resolution maps of quantities that were never independently observed.

2.25.5 Investigation 5: evidence required before recommending storage

Suppose the model predicts that installing a larger tank reduces shortages in simulated outage scenarios. What does that justify? It may justify additional engineering investigation. Before

procurement we still need measurements of daily and subdaily water demand, source replenishment, historical outages, tank integrity, pipe capacity, installation cost and maintenance needs. We should identify the service objective: minimum supply to a critical facility may matter more than the average volume delivered.

A transparent engineering report should therefore have at least three parts. The first states what was mathematically established under the model. The second lists operational measurements actually collected and describes how they were used to assess the model. The third specifies a decision rule, constraints, alternatives and what information remains missing. A result derived from invented input ranges is an informative scenario calculation; it is not a field-validated estimate of how many people will receive additional water.

2.26 Laboratory 2: build and test the balance

The companion program produces Figure 2.1 and Table 2.2. Students should first compute two days by hand and verify that the code uses the same ordering of inflow, loss, delivery and overflow. Then they should test the program with zero inflow, zero demand, high initial storage and extreme demand. The assertions in the script check the basic storage bounds.

A useful extension is to replace the fixed loss with $\ell_t = \alpha S_t$, where α is a small nonnegative daily fraction. Explain the physical interpretation, check dimensional consistency and determine whether the change affects the choice between increasing capacity and reducing leakage. A more demanding extension adds pumping as a bounded decision and compares a fixed schedule with a rule responding to storage measurements. Neither exercise requires neural networks.

Exercises

1. A tank contains $30\,\mathrm{m}^3$ of water. During a day it receives 12, loses 2 and has requested demand of $45\,\mathrm{m}^3$. Determine delivery, unmet demand and end storage under the ordering in (2.3).

2. Explain why $S_{t+1} = S_t + I_t - Q_t - \ell_t$ may be unsuitable if Q_t denotes requested rather than delivered water. Supply a counterexample.

3. Derive the relation between continuous flow rates and daily volume balance. State the units of each term.

4. For the network in Figure 2.2, exhibit two different nonnegative flow vectors that have the same flow into C. Explain what extra measurements could distinguish them.

5. Construct an optimisation objective for a water service that values reliable delivery but is restricted by electricity cost. Identify an assumption that could make the computed optimum infeasible.

6. **Programming exercise.** Modify the companion program to allow time-varying demand and test conservation numerically for every time step.

What this chapter establishes. A useful model relates variables and decisions through explicit assumptions and physical or organisational constraints. But the next difficulty is unavoidable: even a well-constructed model can be misled by what we have measured. We must now examine the production and quality of the data themselves.

Chapter 3

Data That Can Be Trusted

Before asking a computer to discover a pattern, ask how each number came to exist. Sampling, measurement, missing records and changing definitions can alter a conclusion more than the choice between two prediction algorithms.

3.1 A number is not the thing it describes

A household survey says that a proportion of households use electricity. A satellite image shows light emitted at night. A business register counts establishments. Each number may be useful, but none is a perfect substitute for the underlying process. A household can have a connection but experience frequent outages. An area can be brightly lit because streets were electrified even if the number of factories did not change. A business may operate without appearing in a current register. Numbers are produced by instruments, rules and institutions, and those processes leave traces in the data.

The distinction matters because a model can fit a measured number extremely well and still fail to describe the quantity of interest. If our question is whether a district needs additional health workers, a count of recorded consultations will be misleading if patients turn away because the waiting time is already intolerable. If our question is whether farmers need better rainfall information, a gridded weather product may describe a large area while missing conditions on a particular hillside.

We begin by separating three concepts. The *target quantity* is what we want to know: for example the proportion of households in a stated population that had access to electricity during a defined period. The *measurement* is what an instrument, questionnaire or administrative process reports. The *estimand* is the precise numerical feature of the target population that an analysis intends to estimate, such as a population-weighted proportion for a particular survey year.

These concepts sound abstract until they disagree. A researcher who wants a national household proportion but computes the ordinary average of an intentionally unbalanced survey sample has answered a different question. A researcher who treats an observed medicine issue as a patient's unmet demand has silently changed the meaning of the outcome. Good analysis begins by naming the quantity of interest before selecting a statistical technique.

3.2 What a measurement equation tells us

A simple model of observation is

$$y_i = x_i + b_i + \varepsilon_i, \tag{3.1}$$

where x_i is the underlying quantity of interest, b_i represents systematic error and ε_i represents random error. The equation is not a universal law of instruments. Some measurements involve thresholds, censoring or multiplicative errors instead. Its value lies in separating two ways a reported value can be wrong.

Suppose a temperature sensor reads consistently two degrees above a trustworthy reference. That is bias. Repeated readings can also vary around their average because of random noise. Taking more readings can reduce the uncertainty in an average when noise averages out, but it does not automatically remove the two-degree offset. If $b_i = b$ is constant and $\mathbb{E}[\varepsilon_i] = 0$, then for n readings of a stable quantity x,

$$\overline{y} = x + b + \overline{\varepsilon}, \tag{3.2}$$

$$\mathbb{E}[\overline{y}] = x + b, \tag{3.3}$$

$$\mathrm{Var}(\overline{y}) = \mathrm{Var}(\overline{\varepsilon}). \tag{3.4}$$

Under independent errors with common variance σ^2, the last variance is σ^2/n. Under positively correlated errors, the reduction may be smaller. Repetition is therefore not the same as calibration, and additional data points do not always compensate for poor measurement design.

An obvious difficulty remains. Without a reference or structural restriction, the observed value y_i cannot reveal exactly how much came from x_i and how much from bias b_i. For any arbitrary adjustment δ_i, the pair $x_i + \delta_i$ and $b_i - \delta_i$ produces the same y_i. This is a small example of non-identifiability. To distinguish the contributions, one needs new information, stronger assumptions or an appropriately designed measurement comparison.

3.3 Why the method of collecting data changes the estimate

Imagine a country with two hypothetical regions. Region A contains three quarters of households and has an electricity-access proportion of 0.30; Region B contains one quarter and has an access proportion of 0.80. The population proportion is

$$P = 0.75(0.30) + 0.25(0.80) = 0.425. \tag{3.5}$$

That is 42.5%, not the simple average $(0.30 + 0.80)/2 = 55\%$. The latter gives the two regions equal weight even though their household populations differ.

Suppose a survey intentionally selects the same number of households from each region. This could be sensible: researchers may want enough observations to report results for the smaller region. But if we ignore the unequal sampling fractions, the combined average will tend toward an equal-region estimate instead of the national household estimate. A solution is to use survey weights that reflect selection probabilities and the actual survey design.

For observations $z_i \in \{0, 1\}$ and nonnegative weights w_i, a basic weighted proportion is

$$\widehat{P}_w = \frac{\sum_{i \in s} w_i z_i}{\sum_{i \in s} w_i}. \tag{3.6}$$

For simple unequal-probability sampling, the initial inverse inclusion-probability weight is $w_i = 1/\pi_i$, where π_i is the probability that unit i enters the sample. Real surveys may adjust these weights for nonresponse and calibration; their documentation governs which weight belongs to which analysis. The sampling design is not optional metadata. It is part of how the observations should be interpreted (Cochran, 1977; Groves et al., 2009).

3.4 A synthetic survey that shows the problem

The companion program constructs a *fictional finite population* of 10,000 households: 7,500 in one region and 2,500 in another. It draws electricity-access indicators from the two hypothetical probabilities, then selects 250 households from each region. The simulation is reproducible because its pseudorandom generator uses a fixed seed. It is not a sample from Uganda and should never be used as evidence about Ugandan electricity access.

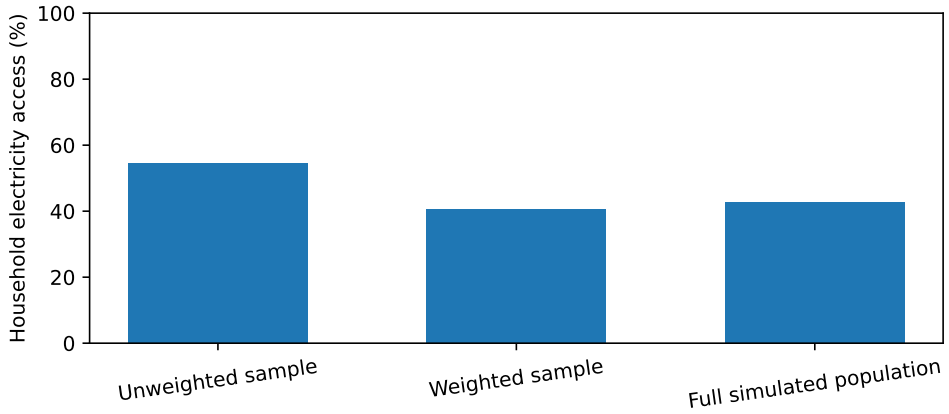


Figure 3.1: An intentionally disproportionate sample from an invented household population. Weighting changes the estimate toward the population composition; the differences here are not empirical estimates for Uganda.

Table 3.1: Electricity access in the reproducible synthetic population

Quantity	Access (%)
Unweighted sample mean	54.4
Population-weighted sample mean	40.6
Synthetic finite-population mean (known here)	42.8

In this particular simulated sample, the weighted proportion is closer to the finite-population value than the unweighted one. That is the expected direction in this design, not a universal guarantee for every random sample. If a rare subgroup is heavily weighted and has few observations, variance can be substantial. A sound analysis reports precision, not just the difference between two point estimates.

What does the code teach that a table of numbers alone cannot? It makes the selection

mechanism visible. The two regions are sampled at different fractions, so treating the combined observations as equally representative distorts the target. Changing the random seed will change the estimates but not the underlying population-definition problem.

3.5 Clustered observations and uncertainty

Not every record is an independent piece of evidence. Ten households interviewed within one village may share electricity infrastructure, weather, local prices or exposure to the same interviewer. Their answers can be more similar to one another than answers drawn from different villages. Treating every household as an independent draw can make confidence intervals too narrow.

A familiar approximation for a cluster sample with roughly equal cluster size m and intracluster correlation ρ_c is the design effect

$$\mathrm{DEFF} \approx 1 + (m - 1)\rho_c. \tag{3.7}$$

It illustrates how dependence inflates variance relative to independent observations. It is not a substitute for using the actual stratification, clustering and weights in a real survey. When survey software offers design-based standard errors, the analyst should specify the relevant primary sampling units, strata and weights from the technical documentation.

The same caution applies to spatial and temporal prediction. If observations from one monitor or one school appear in both training and test sets, model performance may be overly optimistic because the test data are not as independent as the intended future application. We will study validation designs more fully in later chapters. For now the lesson is that the unit supplying independent information is not always the row of a spreadsheet.

3.6 Missing does not mean zero

A dataset is rarely complete. Sensors fail, people decline to answer, administrative files omit facilities, and some quantities are never recorded. Replacing every blank with zero can create a plausible-looking dataset that answers the wrong question.

Consider a health centre recording daily medicine issues. A blank on a day of equipment failure means that the number issued is unknown. A recorded zero may mean that no medicine was issued. If the facility had no stock, a zero issue count may conceal patient demand. These possibilities have different implications for forecasting and stock planning. Before cleaning anything, the analyst must know what a missing value means in the originating system.

The framework associated with Rubin (1976) distinguishes missing completely at random (MCAR), missing at random (MAR) and missing not at random (MNAR). These terms describe assumptions about how the probability of missingness depends on observed and unobserved information. They do not label a dataset automatically, and they cannot generally be determined from observed records alone.

Let $R_i = 1$ if a value is observed and $R_i = 0$ if it is missing, X_i denote fully observed information,

and Y_i the quantity that may be missing. A simplified description is:

$$\begin{aligned}
\text{MCAR: } & \Pr(R_i = 1 \mid X_i, Y_i) = \Pr(R_i = 1), \\
\text{MAR: } & \Pr(R_i = 1 \mid X_i, Y_i) = \Pr(R_i = 1 \mid X_i), \\
\text{MNAR: } & \Pr(R_i = 1 \mid X_i, Y_i) \text{ can still depend on } Y_i \text{ after conditioning on } X_i.
\end{aligned}$$

These expressions should be interpreted within a clearly stated joint-data model. For instance, if remote facilities have more missing measurements and remoteness is recorded, the probability of missingness may depend on an observed characteristic. If outages are more likely on precisely the days when consumption is highest and that consumption is unobserved, the problem may involve nonignorable missingness.

3.7 An elementary sensitivity calculation for missing values

Suppose n_o of n measurements are observed. We wish to estimate the mean of an underlying quantity known to lie between L and U. Let S_o be the sum of observed values. Without assuming anything about the missing values beyond these bounds, the full-sample mean must lie within

$$\frac{S_o + (n - n_o)L}{n} \;\le\; \overline{y} \;\le\; \frac{S_o + (n - n_o)U}{n}. \tag{3.8}$$

This bound is simple but useful: it states what the data *guarantee* under the declared range assumption. It may be wide. A wide interval is sometimes the correct scientific result because the missing observations matter too much.

Take ten households, of which six report electricity access and two report no access; the remaining two responses are missing. Here $n = 10$, $n_o = 8$, $S_o = 6$, $L = 0$ and $U = 1$. The household proportion in the ten-record group could be between 0.6 and 0.8. Reporting $6/8 = 0.75$ assumes that the missing households can be ignored for this purpose. That might be reasonable under suitable assumptions, but it is not forced by the records. With two missing responses, the original data permit a range of outcomes.

A more advanced analysis might use multiple imputation or a joint model. Those methods make additional assumptions that should be inspected, not treated as a way of making missingness disappear. When the decision is sensitive to the missing values, collecting a small amount of better data may be more valuable than applying a sophisticated imputation method.

3.8 Measurement error, selection and the difference between prediction and explanation

Selection bias and measurement error can reinforce each other. Suppose internet-connected facilities submit reports regularly, while facilities with poor connectivity submit only when electricity is available. A dataset of completed submissions will not necessarily describe all facilities equally. A predictive model trained on these records may work especially well on the same well-connected facilities and poorly elsewhere.

Measurement error also changes which relationships a model can identify. Suppose night-time brightness L increases as either manufacturing activity M or electrification E increases. A conceptual model could be

$$L = \alpha M + \beta E + \varepsilon. \tag{3.9}$$

Without independent information on E or M, an observed increase in L cannot uniquely identify the change in manufacturing activity. This is not a weakness that a more powerful learner can automatically overcome; it is a lack of distinguishing information. A useful empirical study would test whether additional measurements such as business counts, electricity connections or production records reduce the ambiguity, while respecting the dates and spatial units of each source.

Equation (3.9) is an illustrative linear decomposition, not an empirically established relationship for Uganda. It is included to show why data integration may be necessary. An analyst should not claim firm openings or production growth from brightness changes unless independent outcome data support those claims.

3.9 Time and geography are part of a variable's meaning

A monthly rainfall total and a daily rainfall gauge measurement cannot be compared without aligning periods. Similarly, an administrative count organised by districts in one year cannot simply be joined to a later map after district boundaries have changed. A variable is not adequately documented by a column name such as `population` or `cases`. The record needs a geography, time, unit, definition and source.

There are at least three common failures of alignment. First, data from different periods are treated as contemporaneous. A business census measured in one year is used to validate satellite observations from another year without explaining the mismatch. Second, polygon boundaries differ across files and records are assigned to the wrong areas. Third, multiple measurements are aggregated at different levels without considering what is lost. A district mean may conceal large variation between villages, and village means do not automatically behave like individual household observations.

When two datasets cannot be aligned exactly, the researcher should choose a defensible common unit, disclose the loss of detail and conduct sensitivity checks where possible. Forced matching can produce a larger dataset but weaker evidence. The temptation to maximise row counts must not replace the obligation to understand what each row represents.

3.10 A real-data laboratory: Uganda's National Panel Survey

The World Bank's Living Standards Measurement Study–Integrated Surveys on Agriculture (LSMS-ISA) programme includes the Uganda National Panel Survey, with publicly listed downloadable rounds including 2018–19 and 2019–20 (World Bank, 2026). The programme documentation describes the survey as a panel rather than a census. It records household and agricultural information through a survey design, not through a complete enumeration of every household in Uganda.

The laboratory proposed here asks a limited, meaningful question: *How do selected household characteristics, agricultural activities and reported electricity access vary across the survey population, and how sensitive are descriptive estimates to weighting and missing records?* The chapter does not present numerical findings from that survey because the microdata have not been downloaded, harmonised or analysed as part of this manuscript draft.

The task begins with source selection. Students should select a clearly documented survey round, download the questionnaire, codebook, sampling documentation and public microdata under the applicable conditions. They should then construct a variable table that states the household identifier, relevant electricity-access variable, relevant agricultural variable, weight, stratum, cluster and permitted geographic classification. The exact variable names must be taken from the chosen round's codebook; guessing them from a different year is unacceptable.

3.10.1 Establish the unit of analysis

Is the target quantity a household proportion, a person-weighted proportion, the proportion of farms, or the proportion of sampled plots? They are not the same. If electricity access is reported at household level, combining a household-level access indicator with plot-level observations can duplicate some households unless the records are carefully structured. The analysis should state which unit each estimate represents.

For a household proportion, begin with the estimated weighted fraction using the appropriate household-level survey weight and the survey's documentation. Then calculate the naive unweighted mean as a comparison. Report the number of observed households and the proportion with missing access responses. Differences between weighted and unweighted estimates are findings about the sample and design, not proof that weights fix every possible bias.

3.10.2 Check what changes between rounds

Panel datasets can be valuable for studying change, but changes in questionnaire wording, sampling frames and definitions may prevent a simple like-for-like comparison. Repeated households can contribute information about within-household change, while new entrants and attrition raise additional questions. Do not treat two nationally weighted cross-sections as identical to a balanced household panel. Distinguish the estimand before choosing the calculation.

A careful exercise might compare electricity-access proportions in two survey rounds after confirming that the question, response coding and weights support that comparison. It should report survey uncertainty and attrition considerations. It must not present the resulting descriptive difference as a causal effect of an electrification policy without an appropriate identification strategy.

3.10.3 Prepare outputs that someone can audit

The laboratory should produce a data dictionary, record-count flow, missingness table, weighted and unweighted descriptive estimates, appropriately calculated uncertainty intervals and a short limitations note. If geographic identifiers are coarsened for confidentiality, map only

the geography that the release permits. Household point locations must not be inferred or reconstructed from restricted information.

A sensible workflow is to preserve the raw downloads in an unchanged input directory and create processed outputs through scripts. Every transformation should be traceable. This discipline is consistent with general reproducible-computation guidance (Sandve et al., 2013). A public tutorial should never redistribute restricted microdata merely because the author was able to access them.

3.11 When do we have enough data to decide?

There is no universal minimum number of rows that makes a dataset adequate. Adequacy depends on the intended decision and on whether the information distinguishes the alternatives. A thousand repeated measurements of the same biased sensor may provide less useful information than ten calibrated measurements from locations that reveal the missing mechanism.

Suppose a manager must decide whether to install another water-flow meter. The relevant question is not simply "Can we predict water use accurately?" It is whether the present observations can distinguish a leak from legitimate consumption well enough to place the new meter or target a repair. If several physical explanations fit the observed totals, the Data Readiness Assessment should identify the ambiguity and what additional observation would resolve it.

A useful assessment has six dimensions:

1. **Meaning.** Are the target, units, time period and population defined precisely?
2. **Coverage.** Which places, people and times are represented or systematically excluded?
3. **Measurement.** How were observations recorded, and what biases or detection limits are plausible?
4. **Alignment.** Can the datasets be combined at the required spatial and temporal scale?
5. **Uncertainty.** Are sampling error, missingness and possible model failures described?
6. **Decision relevance.** Could the measured information actually change a feasible action?

The final dimension connects this chapter back to Chapter 1. Data are not trustworthy merely because they are clean, and a model is not useful merely because it predicts an available label. The available measurements must support the question someone needs answered.

3.12 Trace a number backwards before using it

A spreadsheet reports "electricity access = 0.72" for a district. What, precisely, does that mean? It could mean that 72 percent of sampled households reported a grid connection, that 72 percent of households had access to some electricity source, or that an analyst used a model to estimate access for every small area in the district. These are not minor differences. A solar

engineer, a survey statistician and a public administrator could all use the phrase “electricity access” while referring to different measurements.

A useful data investigation runs backwards from the number to the observation that created it. Who defined access? Was the respondent asked about a physical connection, actual hours of supply, or the ability to use electricity when needed? Was the unit a household, building, service point or electricity meter? How were people selected? When was the question asked, and were answers revised later? If access was inferred from night-time brightness rather than measured independently, it cannot serve as an independent test of a satellite-based electrification estimate.

This trail of meaning is often called *data provenance*. It records where a value came from, which transformations it underwent, and which decisions were made while preparing it. A data dictionary is an initial part of that record; a reproducible processing script is another. The purpose is not bureaucratic completeness. It is to discover when two apparently identical columns mean different things.

3.12.1 An observation is a chain of transformations

Write an unobserved quantity of interest as X. A measurement device, questionnaire or administrative process produces raw record R, and a sequence of cleaning and aggregation steps produces the value Y used in analysis:

$$X \longrightarrow R \longrightarrow C(R) \longrightarrow A(C(R)) = Y. \tag{3.10}$$

The arrows are not all deterministic. People may misremember, instruments may drift, records may be missing, and staff may interpret questions differently. Cleaning can introduce additional decisions. Aggregation loses some distinctions by design. When we write a regression using Y, we inherit assumptions from every link in this chain, whether or not the regression code acknowledges them.

Imagine rainfall observations from a gauge that reports once per day. If the gauge is not emptied consistently, its reading may represent water accumulated over more than twenty-four hours. Replacing an unusually high measurement with the median may make a plot smoother while removing a real storm event. A data-quality procedure should first ask how the observation was produced and whether an anomaly is plausible. Statistical unusualness is not the same as error.

3.13 Measurement uncertainty is not only random noise

The elementary equation $Y = X + b + \varepsilon$ distinguishes bias b from random noise ε. Its usefulness depends on what information we have about those terms. Repeating the same measurement may reduce random variation in an average, but it does not remove a constant calibration bias. A survey administered in only one language may systematically miss some types of misunderstanding, even with a large sample. A satellite pixel may integrate light from roads, homes and industrial buildings and cannot label their contributions merely because its value is precise.

Consider independent repeated measurements $Y_j = X + b + \varepsilon_j$ with $\mathbb{E}[\varepsilon_j] = 0$ and $\mathrm{Var}(\varepsilon_j) = \sigma^2$.

Their mean satisfies

$$\mathbb{E}[\overline{Y}] = X + b, \qquad \operatorname{Var}(\overline{Y}) = \frac{\sigma^2}{n}. \tag{3.11}$$

As n grows, the variance shrinks, but the bias remains. It would be a mistake to claim that repeated measurement makes the estimate accurate without independently assessing the instrument's calibration. The distinction between precision, accuracy and traceability is central to the JCGM (2008) guidance on measurement uncertainty.

3.13.1 Independent errors and common errors

The σ^2/n formula relies on independent errors. If every reading shares a random calibration component U and separate random components ϵ_j, write $Y_j = X + U + \epsilon_j$ with $\operatorname{Var}(U) = \tau^2$ and $\operatorname{Var}(\epsilon_j) = \sigma^2$. Then

$$\operatorname{Var}(\overline{Y} - X) = \tau^2 + \frac{\sigma^2}{n}. \tag{3.12}$$

Collecting more readings cannot remove the common component. For example, hundreds of measurements from a thermometer calibrated incorrectly do not provide hundreds of independent checks of its calibration. This is a general warning for dense digital datasets: high frequency does not automatically mean high information.

3.13.2 Propagation of measurement uncertainty

Suppose we calculate a rate $r = 1000c/p$, where c is a recorded count and p a population estimate. If both have small errors, a first-order approximation gives

$$\operatorname{Var}(r) \approx \left(\frac{1000}{p}\right)^2 \operatorname{Var}(c) + \left(\frac{1000c}{p^2}\right)^2 \operatorname{Var}(p) - 2\frac{10^6 c}{p^3} \operatorname{Cov}(c, p). \tag{3.13}$$

One cannot merely ignore population uncertainty because the denominator appears in a spreadsheet cell that looks exact. The covariance term matters if both count and population depend on common sampling or modelling procedures. This calculation is a local approximation, not an all-purpose confidence interval; ratios with uncertain denominators can behave badly when the denominator is near zero.

3.14 Sampling: the population is not the spreadsheet

The target population is the collection about which we want to make a claim. The observed sample is the collection we happen to have. These coincide only in a true census of the stated population with complete and correct enumeration, which itself can be difficult to achieve. Many datasets that arrive as "national" contain records from a designed sample, not every household. A useful analysis makes the target, sampling unit, coverage period and selection probability explicit. Classical design-based methods are developed in Cochran (1977), Kish (1965), Lohr (2019) and Groves et al. (2009).

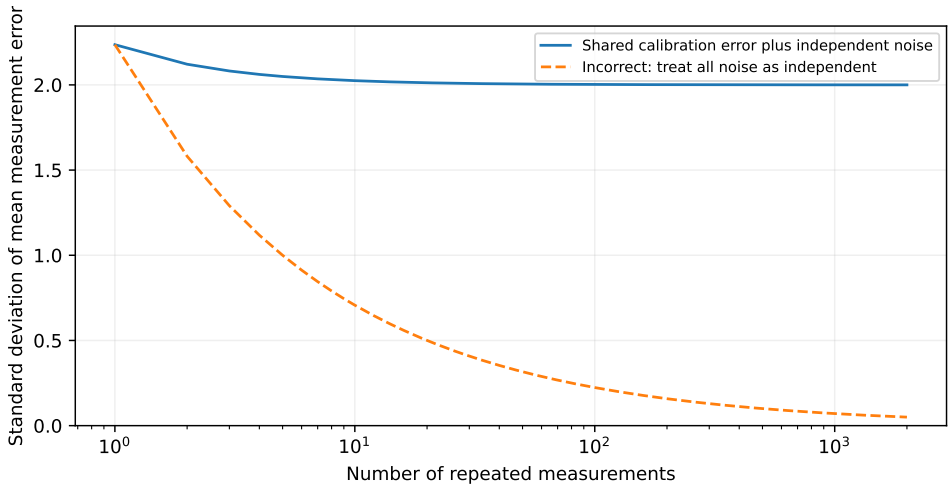


Figure 3.2: The fallacy of treating shared calibration error as independent random noise. Repeating readings reduces the independent component but not the common offset. The parameters are hypothetical.

3.14.1 Where sampling weights come from

Suppose a finite population contains N units and unit i is included with known probability $\pi_i > 0$. For a population total $T = \sum_{i=1}^{N} y_i$, the Horvitz–Thompson estimator is

$$\widehat{T}_{\mathrm{HT}} = \sum_{i \in s} \frac{y_i}{\pi_i}. \tag{3.14}$$

Under the stated probability design, $\mathbb{E}[I_i] = \pi_i$, where I_i is the indicator that the unit was selected. Therefore

$$\mathbb{E}[\widehat{T}_{\mathrm{HT}}] = \sum_{i=1}^{N} \frac{y_i}{\pi_i} \mathbb{E}[I_i] = \sum_{i=1}^{N} y_i = T. \tag{3.15}$$

The proof explains the weights: a sampled unit represents itself and others with similar inclusion probability. It also explains why weights are not optional decorations attached to a survey file. Ignoring disproportionate inclusion probabilities changes the target implicitly.

If N is known, $\widehat{T}_{\mathrm{HT}}/N$ estimates a population mean. If it is estimated too, a ratio estimator may be used:

$$\widehat{\mu}_R = \frac{\sum_{i \in s} w_i y_i}{\sum_{i \in s} w_i}. \tag{3.16}$$

This ratio is widely used but is not exactly design-unbiased in finite samples merely because the total estimator is unbiased. It requires appropriate design-based variance estimation. The choice between total, proportion, mean and ratio estimators should match the target quantity and the published survey design.

3.14.2 Stratification, clustering and multi-stage selection

A survey might first select districts, then enumeration areas, then households. Its records may contain strata, clusters and weights. Two households from the same enumeration area may

resemble each other because they share roads, electricity infrastructure and local conditions. Ten records from one area therefore need not provide the information of ten independently sampled areas.

The design effect approximation introduced earlier describes one route by which dependence enlarges uncertainty. For a wider account of multilevel structure and modelling choices, see Gelman and Hill (2007). For real multi-stage surveys it is better to use the actual sample design rather than insert an arbitrary intracluster correlation. Also distinguish the count of *records* from the count of independent primary sampling units. A table containing thousands of household rows may still contain relatively few independent clusters for a particular analysis. No confidence-interval algorithm can create independent sampled areas that were not observed.

3.14.3 Weights can affect both estimates and precision

A descriptive measure of how unequal weights reduce a sample's effective size is

$$n_{\text{eff},w} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}. \tag{3.17}$$

By Cauchy–Schwarz, $n_{\text{eff},w} \leq n$, with equality for equal positive weights. For example, with four sample weights $(1, 1, 1, 9)$, the effective-size expression is $12^2/(1 + 1 + 1 + 81) = 144/84 \approx 1.71$, much smaller than the four observed records. This is an instructive warning about concentration of weights. It is not the full survey effective sample size, because stratification, clustering, calibration and the particular estimator also matter.

3.15 Nonresponse changes the meaning of a comparison

A field team may interview most households in a district and miss the rest. The completed records are not automatically a random subset of the intended sample. People can be absent for reasons related to employment, migration, income or access to services. If response probabilities differ in ways that matter for an estimate, simply computing the average among respondents may be biased.

Let R_i be the response indicator and Y_i the target outcome. The observed mean estimates $\mathbb{E}[Y \mid R = 1]$. The intended population mean is $\mathbb{E}[Y]$. Their difference can be written

$$\mathbb{E}[Y \mid R = 1] - \mathbb{E}[Y] = \frac{\operatorname{Cov}(R, Y)}{\Pr(R = 1)} \tag{3.18}$$

when expectations are well defined. To see this, note that $\mathbb{E}[RY] = \Pr(R = 1)\mathbb{E}[Y \mid R = 1]$, while $\operatorname{Cov}(R, Y) = \mathbb{E}[RY] - \mathbb{E}[R]\mathbb{E}[Y]$. The formula makes the danger transparent: nonresponse creates selection bias when response and the outcome are associated.

This identity does not tell us the missing covariance from observed outcomes alone. It tells us what we need to reason about. Did connectivity affect response? Was nonresponse higher in remote areas? Can auxiliary variables from the sampling frame explain the differences? If we assume responses are missing at random conditional on recorded covariates, weighting or imputation may help. If missingness depends on the unobserved value even after conditioning,

sensitivity analysis and transparent uncertainty are needed. Rubin (1976) and Little and Rubin (2019) provide the statistical foundations.

3.15.1 A sharp worst-case bound for a proportion

When an outcome is binary, we can give an honest bound even without a model for nonresponse. If n_o of n sampled units respond, and r respondents report access, the proportion among all n intended records lies between r/n and $(r + n - n_o)/n$, if every missing outcome might be either zero or one. With $r = 60$, $n_o = 80$, $n = 100$, the range is $[0.60, 0.80]$. The observed-respondent proportion 0.75 sits inside it. A narrower interval requires assumptions or additional observations. For a properly weighted complex survey, the analogous exercise must respect the sampling weights and target estimand.

A good analyst might decide to call back a small targeted sample of nonrespondents rather than argue endlessly about competing imputation models. Whether such follow-up is permissible and worthwhile depends on ethics, costs and how strongly the uncertain observations affect the intended decision.

3.16 Imputation is an estimate, not a repair of the past

Missing observations can be filled with model-based values for some analyses. But an imputed number was not observed, and it should not be copied into a public administrative record as though it were. Imputation must be traceable. At minimum keep the original value, an indicator of missingness, the imputation method and the resulting analytic value separately.

Consider a linear relationship $Y = \alpha + \beta X + \epsilon$. If missing Y values are replaced by their conditional predictions $\widehat{\alpha} + \widehat{\beta} X$, the filled-in values omit the individual residual variability. A single deterministic imputation can make the distribution artificially concentrated and understate uncertainty if the completed data are analysed as though all values were observed. Multiple imputation introduces variation across plausible completed datasets under an explicit missing-data model; it still does not defeat nonidentifiability under unsupported assumptions. Little and Rubin (2019) give a rigorous treatment.

3.16.1 Why predictive quality is not sufficient to justify imputation

Suppose a model predicts observed household expenditure well. Does that mean its predictions are valid for households that did not respond? Not necessarily. The observed households may differ systematically from the missing group. High cross-validated accuracy among respondents measures prediction on a restricted distribution. It does not verify the distribution of unobserved incomes. A meaningful assessment asks whether the missingness assumptions are credible and how the final estimate changes under alternatives.

A useful report should show at least the fraction missing for each important field, variation by geography and subgroup, the complete-case estimate, the imputed estimate where justified, uncertainty that reflects the imputation process, and a sensitivity analysis where nonignorable missingness is plausible. Reporting only the model's accuracy is inadequate.

3.17 Measurement error can make a real relationship look weaker

Suppose the true relationship between an outcome Y and a predictor X is $Y = \alpha + \beta X + \varepsilon$, with ε mean-zero and independent of X. We do not observe X directly; instead $W = X + U$, where U is classical independent mean-zero measurement error. The slope from a simple regression of Y on W tends, under the usual conditions, toward

$$\beta_W = \beta \frac{\mathrm{Var}(X)}{\mathrm{Var}(X) + \mathrm{Var}(U)}. \tag{3.19}$$

The multiplier is between zero and one, so the estimated slope is attenuated toward zero. One might wrongly conclude that the relationship is weak even when the true relationship is substantial. This is an example of a model that converges statistically to the wrong target as more data are added. For error-in-variables methods and broader cases see Carroll et al. (2006) and Fuller (1987).

For example, if the variance of X is 9 and the error variance is 3, the multiplier is $9/(9+3) = 0.75$. A true slope of 2 produces a limiting naive slope of 1.5. This calculation rests on the classical additive-error assumptions. Differential, correlated, systematic or spatially structured errors may behave differently. The equation is not a general correction formula for every dataset.

3.17.1 Why an independent reference is valuable

If we have repeated measurements or a credible validation subsample, we may estimate measurement-error parameters. Without such data, the decomposition of observed variation into true variation and noise may be underdetermined. This explains why a few expensive but trustworthy reference measurements can be essential even in a project with millions of cheap proxy observations.

The same issue appears when satellite brightness is used to infer activity. Brightness is not a direct reading of firm counts. It is an observation affected by many sources of light, local electrification, land use, atmospheric effects and measurement processing. To argue that it predicts an economic quantity well, we need independent outcome data and a validation design matched to the intended use. To argue that changes in brightness reveal *growth*, we need temporal business data, not merely a satellite time series. No amount of smoothness in the brightness image resolves that distinction by itself.

3.18 Alignment: when two correct datasets disagree

Two datasets can each be internally correct and still be inappropriate to join. An electricity survey may refer to households present during a field period, while an administrative connection register counts meters installed by the end of the calendar year. A satellite product may provide monthly radiance for pixels with variable observation quality. A manufacturing census may count establishments using a definition that changed between rounds. Matching these files on a district name does not make their measurements equivalent.

3.18.1 Temporal alignment as part of study design

Before combining sources, construct a timeline showing the period represented by every variable. Distinguish a measurement date from a reference period. A survey collected in February may ask about income over the previous twelve months; an electricity record may be a stock at 31 December; a satellite composite may average nights within a calendar year. Any temporal mismatch should be stated in the design, with sensitivity checks if the research question makes it important.

A related danger is information leakage. If a model is intended to guide a decision on 1 June, its inputs must be information available by 1 June. A later revised annual statistic may improve a retrospective fit but cannot serve as a genuine prospective input. A split into random training and test records does not protect against this error. We must split according to when the data would actually have become available.

3.18.2 Geography is not just a label

Suppose an administrative boundary changes so that one sub-county becomes two. If we count establishments in one year's boundaries and households in another year's boundaries, the derived rate may be meaningless. A defensible harmonisation chooses a common geography, documents which units were combined or excluded, and checks that extensive quantities, such as counts, have been aggregated appropriately.

Different variables aggregate differently. Counts add across disjoint areas; averages generally require suitable weights; rates require their numerator and denominator to be aggregated before division. For example, combining two areas with 20 businesses in 1,000 residents and 30 businesses in 9,000 residents gives $50/10{,}000 = 5$ businesses per 1,000 residents. Averaging the original rates, $(20 + 3.33)/2 \approx 11.67$, would be wrong for the combined population. This is a mathematical error, not merely a problem of map formatting.

3.19 A data dictionary that can survive a new analyst

A practical data dictionary records more than column names. Each field should identify its concept, unit, type, geographical and temporal reference, allowed values, missing-value codes, collection mechanism, source file, processing rule and any restrictions on sharing. Where the field is derived, retain the formula and the source columns.

These are *illustrative field names*, not a claim that an actual Uganda National Panel Survey file has precisely these columns. Before running the real-data laboratory, inspect the published metadata, codebooks and access terms. The World Bank (2026) documentation provides the survey programme and links to round-specific information. The exact variables and available geographic identifiers must be established from the downloaded files, not guessed from an attractive template.

Table 3.2: Illustrative data-dictionary entries for a household electricity study

Field	Meaning and validation rule	Important caution
`hh_id`	Anonymous analysis identifier; unique within the correct round	Not a public household address
`access`	Defined categorical or binary electricity-access measure	Define source and reference period
`weight`	Documented survey analysis weight	Use the correct round and domain
`psu`	Primary sampling unit code where disclosed	May be restricted or altered for privacy
`wave`	Survey round identifier	Round definitions and samples may change
`region`	Geographic reporting category	Do not assume village resolution

3.20 A defensible audit of the Uganda National Panel Survey

The National Panel Survey is a valuable teaching example because it brings together household and agricultural information with survey design decisions. The published 2019/20 documentation describes the sampling framework and survey instruments, while changes in later surveys require special care in longitudinal comparisons. The World Bank project documents indicate that the UNPS 2019/20 sample contained 3,098 households and that a later refresh affected the sample design; these are design facts from documentation, not results we computed from microdata (World Bank, 2023, 2026).

3.20.1 Start with the actual documentation

Download the specific round's basic information document and questionnaire; identify the unit of observation, sampling stages, strata, weights, repeated-visit structure and geographic levels that may be publicly reported. Make a one-page record of the instrument definition of electricity access. The fact that an item has a familiar name does not guarantee that its meaning is stable across rounds. Check changes in response coding and whether an apparently missing household is a nonresponse, an out-of-scope record or a household no longer followed.

3.20.2 Separate four kinds of output

First, produce a *data inventory*: files, observation counts, key identifiers, variable types and missing-value codes. Second, produce a *weighted descriptive estimate* using the survey's approved design variables, accompanied by an appropriate uncertainty interval. Third, produce a *quality table* showing nonresponse and inconsistencies that could alter interpretation. Fourth, produce a *limitations memo* identifying what cannot be estimated at the desired geographic scale or with the available variables. A refusal to publish an unsupported village-level map is a valid outcome of a careful analysis.

3.20.3 A reproducible notebook skeleton

The following code is a *template*, not an executable analysis of UNPS. It requires actual file paths and variable names from the chosen round's codebook. A notebook that silently guesses names could produce an apparently complete table based on wrong columns.

```
import pandas as pd

# Replace these placeholders only after inspecting the real codebook.
DATAFILE = "path_to_authorized_household_file.csv"
ID = "verified_household_identifier"
WEIGHT = "verified_household_weight"
ACCESS = "verified_access_variable"

# Raw input must remain unchanged; avoid printing identifiers.
df = pd.read_csv(DATAFILE)
assert df[ID].notna().all()
assert df[WEIGHT].notna().all()
assert (df[WEIGHT] > 0).all()
# Before this step, inspect skip codes and recode only documented values.
assert df[ACCESS].dropna().isin([0, 1]).all()

valid = df[ACCESS].notna()
est = (df.loc[valid, WEIGHT] * df.loc[valid, ACCESS]).sum()
est /= df.loc[valid, WEIGHT].sum()
print("Weighted respondent proportion:", est)
# Do not label this a design-valid national estimate until
# response handling, survey strata and primary sampling units
# have been incorporated as the official documentation requires.
```

This skeleton intentionally stops short of reporting a national confidence interval. A full analysis needs survey-aware variance estimation and possibly the correct treatment of panel design, attrition and calibration. It is better to leave a clearly identified computational requirement than to print a polished but unjustified interval.

3.21 How a data error reaches a decision

Most textbook examples stop after identifying a biased measurement. Operational work must ask whether that bias changes what someone does. Suppose two communities are candidates for installing a backup water system. Their true unmet demand is $D_A = 80$ and $D_B = 60$ units, but a reporting failure causes community A's estimate to be $\widehat{D}_A = 50$. If the planner allocates the system to the community with the highest reported demand, the apparent ranking selects B. The error changes the decision, not just a plot.

The decision cost of such an error depends on the capacity of the system, the people served, the alternative arrangements and the consequences of failing to supply water. When data resources are limited, prioritising the measurement whose errors are most likely to change the decision can be more sensible than attempting to perfect every column. This is the bridge back to the value-of-information argument in Chapter 1. The purpose of better data is not simply to obtain a more impressive goodness-of-fit statistic. It is to reduce the chance of a consequential wrong decision.

3.22 Responsible data use is a technical requirement

Public access to a file does not imply permission to publish identifiable records. A survey may release regional categories while withholding fine coordinates to reduce disclosure risk. A hospital may permit analysis of de-identified stock records but not patient-level histories. Good data engineering should therefore use access controls, minimal fields, appropriate aggregation, secure storage and documented retention policies from the start. A technically elegant model built from data obtained without authority is not a successful project.

Reproducibility and confidentiality sometimes pull in different directions. The proper response is to separate what can be made public—code, documentation, synthetic demonstration data, safe aggregates—from what must remain protected. A public notebook can still be independently examined for its calculations even if every restricted observation cannot be shared. Sandve et al. (2013) set out principles for reproducible computational analysis that remain useful here, but reproducibility never overrides the rights of the people represented in the data.

3.22.1 A practical readiness decision

At the end of a dataset audit, we should be willing to write one of four conclusions. The data may be ready for the stated decision under clearly documented limits. They may be ready for descriptive analysis but not for the proposed predictive claim. They may be usable only after targeted additional collection or independent validation. Or they may be inadequate, making the responsible choice to stop or reformulate the question. These outcomes are not grades for the analyst; they are consequences of how much the evidence can support.

A useful readiness record asks: Are the required variables defined? Do we know who and what the sample represents? Are dates and boundaries aligned? Are measurement errors and missingness understood well enough? Can we validate the central claim independently? Does the dataset's permission support the proposed use? If a critical answer is no, the report must say what extra observation would be needed and why.

3.23 Extended laboratory: a data audit before modelling

The laboratory begins with a *constructed teaching dataset*, not a real national survey. Create two groups of households with different probabilities of electricity access and select them at unequal sampling fractions, as in the existing companion program. Record the true finite-population proportion, unweighted sample proportion, weighted estimate, and the fraction missing after imposing two different response mechanisms. Then introduce an independent random measurement error and a shared calibration error into a continuous quantity, confirming equation (3.12) numerically.

The experiment should not end when the numbers match a formula. Explain in ordinary language why the weighted and unweighted estimates differ; which assumption was broken by the imposed response mechanism; why the average of many biased readings is still biased; and what additional data would let the analyst estimate the calibration error. Do not claim that the constructed missingness mechanism can be diagnosed reliably from the completed file alone:

its truth is known to us only because we wrote the simulation.

```
import numpy as np
rng = np.random.default_rng(2026)
true_value = 10.0
shared = rng.normal(0, 1.2) # common calibration error
repeat_error = rng.normal(0, 0.5, 100)
measurements = true_value + shared + repeat_error
print("Mean of repeated readings:", measurements.mean())
print("Known shared error in this simulation:", shared)
```

Run the example with 10, 100 and 10,000 repeated readings, preserving the same randomly sampled shared offset for a fair comparison. The independent error in the mean should shrink; the shared offset remains. The exercise is about the logic of measurement, not about recovering the precise reference value from these observations.

Further exercises. Prove the Horvitz–Thompson total estimator is unbiased under known positive inclusion probabilities. Derive the nonresponse bias identity from $\mathbb{E}[RY]$. Explain why using $\mathrm{Var}(\bar{Y}) = \sigma^2/n$ with correlated readings is unsafe. Construct two latent states that produce the same measured satellite brightness but different establishment counts. Show, using numerator and denominator totals, why an unweighted average of district rates does not generally equal the national rate. Describe a permission-preserving plan for making a restricted-data analysis reproducible.

3.24 Case study: when an apparently complete dataset is not ready

Imagine a district planning office asks for a map of household electricity access to help locate backup power investments. Three files are available: household survey estimates, a list of electricity connections and night-time satellite brightness. At first glance the request seems straightforward. Merge the files by location, fit a model and draw the map. Each step, however, hides a different question about what the data mean.

The household survey may measure whether sampled households report access to an electricity source. Connection records may count meters, some serving several households and others disconnected or inactive. Night-time brightness detects emitted light, much of which may come from roads, shops or public facilities. The three sources are related to electrification, but they do not measure the same thing. A high correlation between two of them is not sufficient proof that either is an unbiased estimate of the target measure.

3.24.1 Start with a question that has a unit

Suppose the policy question is: “Which eligible service areas have the greatest proportion of households without dependable electricity for evening health services?” This already contains several definitions that need work. Is the unit a household or a clinic? Does dependable mean connected, available for a minimum number of hours, or sufficient to power critical equipment?

Are service areas administrative polygons, electricity network zones or a fixed travel distance around facilities? A household connection is not proof that a clinic can operate a cold-chain refrigerator through an outage. The outcome should be defined by what the decision requires.

If the team instead asks, "Which small areas have the lowest proportion of households reporting grid access?" a household survey may help, but only at the spatial resolution supported by its design and confidentiality provisions. The first question may require facility load and outage measurements, while the second may use documented household responses. A model should not silently answer one question while its title promises the other.

3.24.2 The data-inventory meeting

A productive first meeting does not need a computer screen. Place the source documents on a table and write down each dataset's unit, dates, definition, coverage, selection mechanism, spatial precision and permission. For survey estimates, note the sampling strata and weights. For administrative connections, check duplicate meter records and the status date. For satellite radiance, record the compositing period and quality flags. Mark any definition that cannot yet be reconciled.

A preliminary alignment table might read:

Source	Unit recorded	Question it can address directly
Household survey	Sampled household response	Reported access in supported survey domains
Connection register	Meter or customer account	Administrative connections under its status definition
Night-time lights	Georeferenced radiance pixel	Observed light emissions in the selected period

The table is intentionally cautious. It does not claim any of these sources measures reliable clinic electricity. If the planner really needs that outcome, the proper next action may be a targeted meter and outage survey at candidate facilities rather than a more elaborate satellite model.

3.25 An identification calculation before fitting a regression

Write a deliberately simplified brightness model as $L = \alpha H + \beta I + \epsilon$, where H is household electrification, I represents industrial or commercial lighting, and ϵ represents other influences. Even if α and β are known and $\epsilon = 0$, a single observed L cannot uniquely recover two unknown components. For example, with $\alpha = \beta = 1$ and $L = 100$, both $(H, I) = (80, 20)$ and $(50, 50)$ are consistent. If the decision depends on H, the distinction matters; if it depends only on total brightness, it may not.

One independent measure of I or H may allow the other component to be estimated, depending on error and modelling assumptions. Alternatively, multiple measurements that respond differently to the components may help. But adding more measurements of exactly the same

mixture does not necessarily resolve it. The key is independent *distinguishing information*, not sheer volume. This is why measurement design should precede model selection.

3.25.1 A small linear-algebra example

Let $z = (H, I)^\top$ and suppose two measurement instruments produce

$$y = Az, \qquad A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}. \tag{3.20}$$

The determinant is one, so the noiseless measurements identify both components uniquely. From $y_1 = 100$ and $y_2 = 130$ we obtain $I = 30$ and $H = 70$. Now replace the second row by $(2, 2)$. The resulting measurements are linearly dependent; the determinant becomes zero, and the second instrument contributes no new independent equation. Even millions of noiseless repeated observations of these two mixtures cannot separate the components. In practice, nonzero determinant is only the beginning: if the rows are nearly proportional, measurement noise can be strongly amplified. The condition number of A becomes relevant for assessing stability.

This simple example should be read as an argument about information, not as a literal physical model of electricity and satellite light. Real brightness relationships are nonlinear, spatially mixed and affected by many sources. Their complexity strengthens rather than weakens the need to test what the observations can distinguish.

3.26 Three validation designs with different meanings

Suppose the team has an outcome record for a set of areas and wishes to predict it from satellite radiance. A random split of individual household rows into training and testing data can give an optimistic answer when neighbouring households share infrastructure and the same underlying imagery. A district-level holdout asks a more demanding question: can the method predict in districts it did not see during fitting? A time-based holdout asks whether it works in a later period using only information available earlier. These are different tests of generalisation.

A model can perform well under one test and badly under another without any contradiction. The model may have learned persistent features of familiar districts that do not transfer elsewhere. It may rely on relationships that change when electrification expands. Or it may be accurate at the district level while too uncertain for village decisions. The evaluation should match the application, and its spatial units should be stated before model tuning.

3.26.1 A simple leakage thought experiment

Suppose we have ten observations from each of fifty districts. Every observation has a district-specific feature u_j plus individual noise. A flexible model that sees eight observations from each district during training may infer u_j and predict the other two rather well. But when asked about a new district, the same model has never seen its u_j. Its performance may deteriorate sharply. A random-row test answered the question “Can I predict another observation from an

already represented district?" It did not answer "Can I predict a district that has never been represented?" Neither question is inherently wrong; only one may match the planned use.

For a spatial interpolation task, also inspect error against distance from the nearest training measurement. Two held-out areas in the same district can have very different proximity to observed sensors. Grouped validation is therefore necessary but not always sufficient: the design should reflect the geography, timing and operating conditions of the final application.

3.27 When an estimate is too uncertain for a map

Maps can make uncertainty look small. A continuous colour scale fills every polygon, even when some areas have weak evidence. A map with bright colours and precise numbers may communicate confidence that no statistical method actually established. Before publishing, ask how the map was produced, what each pixel or polygon means, and how uncertainty changes with location.

Suppose area A has an estimated electricity-access proportion of 0.70 with a broad uncertainty interval $[0.35, 0.90]$ and area B has an estimate of 0.60 with interval $[0.55, 0.65]$. Ranking A above B by the point estimates alone would hide substantial ambiguity. If the decision depends on a threshold of 0.65, then the uncertainty is consequential. A responsible map may display a separate panel for uncertainty, distinguish unobserved areas, or refuse a precise ranking where the evidence does not support one.

When combining survey estimates with modelled spatial surfaces, do not interpret the resulting fine pixels as independently observed households. A map can display a conditional model prediction at fine resolution; it does not thereby create a fine-resolution household census. A reader should be able to trace every geographic claim to its unit of support.

3.28 A complete data-readiness report

The district team can finish with a useful report even without a final predictive map. The report should identify the decision, list every dataset and its limitations, describe what can be estimated at a permitted resolution, and specify what extra information would resolve the most consequential ambiguity. It should separate three categories of findings: directly observed records, statistical estimates conditional on explicit assumptions, and planning scenarios based on unverified conditions.

A defensible conclusion might be: "The survey supports descriptive estimates at its authorised reporting level. Night-time lights and connection registers provide relevant but non-equivalent information. We cannot yet estimate reliable clinic power availability from these sources. A targeted study of outage duration and facility demand is required before selecting battery capacities." This conclusion may be less visually impressive than a high-resolution map, but it gives the decision-maker a clear next step and explains why it matters.

3.28.1 A final audit exercise

Ask students to prepare two documents. The first is a one-page data dictionary for a published survey field of their choice, using only the codebook and legitimate documentation. The second is a Data Readiness Assessment describing an intended decision, available evidence, three sources of uncertainty, one validation design and one claim they would refuse to make. Require a reference for each definition taken from an external source. Allow the final recommendation to be "do not deploy yet."

The assessment should reward the clarity of the reasoning and the connection between data and decision. A student who identifies a fatal mismatch in measurement units and explains how to correct it has solved an important part of the problem even if no advanced model has been fitted. A student who produces a sophisticated prediction without checking what the outcome means has not.

3.29 Worked investigations: what the data can really support

The exercises below begin with apparently simple calculations and end by asking whether the result has the meaning its author claims. This is the habit we want readers to acquire before they begin fitting elaborate predictive models.

3.29.1 Investigation 1: calculate a survey total with unequal inclusion probabilities

Suppose a tiny population contains four units with target values $(10, 20, 30, 40)$. Their inclusion probabilities are $(1/2, 1/2, 1/4, 1/4)$. The actual population total is 100. Imagine that a probability sample happens to contain units 1 and 3. A naive sum of observed values is $10 + 30 = 40$, which is neither a population total nor a representative estimate of one without a selection argument. The Horvitz–Thompson estimator gives

$$\widehat{T} = \frac{10}{1/2} + \frac{30}{1/4} = 20 + 120 = 140.$$

This particular sample overestimates the true total. There is no contradiction with design unbiasedness: unbiasedness concerns the mean of the estimator over repeated samples from the specified design, not the equality of every sample estimate to the truth. We also need the joint inclusion probabilities to calculate its exact design variance. A sampling weight does not guarantee that every realised weighted estimate is close to the target.

An analyst should not invent weights as the inverse of a convenient demographic percentage. They must come from an understood sampling design and any documented adjustments for nonresponse or calibration. Where the survey provides official weights, use those with its stated domains and replicate or cluster information as appropriate.

3.29.2 Investigation 2: a confidence interval with and without finite-population correction

Under simple random sampling without replacement of n elements from a finite population of N, the sample mean $\bar{y}$ has estimated variance

$$\widehat{\mathrm{Var}}(\bar{y}) = \left(1 - \frac{n}{N}\right)\frac{s^2}{n},$$

where s^2 is the sample variance with denominator $n-1$. Suppose a student samples $n = 100$ of $N = 1{,}000$ households and obtains $s = 12$ units. The estimated standard error is $\sqrt{(1-0.1)(144/100)} = \sqrt{1.296} \approx 1.138$. Ignoring the finite-population correction would give 1.2. The difference is modest here, but it becomes substantial when a large fraction of a small population is sampled.

This calculation is not a valid confidence interval for an actual clustered survey merely because it contains a square root and familiar sample sizes. It assumes simple random sampling without replacement and a suitable approximation for the sampling distribution. In multi-stage or stratified surveys, the variance calculation must respect the actual design. One can use an elegantly implemented formula for the wrong sampling design and obtain an unjustifiably narrow interval.

3.29.3 Investigation 3: why the average of regional rates may be wrong

Two districts record 30 cases among 1,000 people and 30 among 9,000 people. Their rates per thousand are 30 and about 3.33. The unweighted average of district rates is about 16.67 per thousand. The combined area, however, has 60 cases among 10,000 people, or six per thousand. The unweighted average treats the districts as equally large, which is not the target when estimating the population-weighted rate.

The correct combination follows from counts: $r = (30 + 30)/(1000 + 9000) \times 1000 = 6$. An arithmetic mean of area rates could be appropriate for a different question, such as the mean rate across an equally weighted collection of administrative units. The problem is not the arithmetic but failure to state what is being averaged. We should distinguish the mean of district rates from the rate among residents in the combined region.

3.29.4 Investigation 4: missing outcomes and the price of a single number

A sample of one hundred facilities is asked whether an essential medicine was available throughout the previous month. Sixty-five report yes, twenty report no and fifteen do not answer. The observed-respondent proportion is $65/85 \approx 0.765$. But among all one hundred selected facilities, the proportion with availability lies between 0.65 and 0.80 without further assumptions. The upper bound assigns all fifteen nonrespondents a yes; the lower assigns them a no. Neither bound is a claim about the response mechanism.

What extra evidence might narrow the range? A permitted follow-up of nonrespondents, a reliable inventory record, or auxiliary covariates with a credible missing-at-random argument. What would not narrow the range honestly? Reporting three decimal places on 65/85 or fitting

a flexible model that treats imputed answers as observed facts. The missing observations have not become known simply because a numerical procedure produces values for them.

3.29.5 Investigation 5: a large sample from one faulty measuring system

Suppose a sensor has independent random error with standard deviation one and a shared calibration error with standard deviation two. The error variance of the mean of n readings is $4 + 1/n$ rather than $5/n$. With $n = 100$, the variance is 4.01, so the standard deviation of the average's error is about 2.0025. If we mistakenly treated all error as independent, we would calculate $\sqrt{5/100} \approx 0.224$, an enormous understatement. This example captures why thousands of readings from one instrument can offer little protection against shared bias.

The solution is not simply to average harder. A calibration reference, independent instruments or an external validation design may be needed. The number of observations should never be reported without considering how they were generated and which components of error are shared.

3.29.6 Investigation 6: a real-data proposal with a justified refusal

A student proposes to estimate village-level electricity access from the Uganda National Panel Survey and to validate it using night-time satellite imagery. The survey has a national design, and its published domains do not automatically give representative household estimates for every village. Satellite brightness also measures emitted light, not household access as defined in the questionnaire. The validation is therefore ambiguous, both geographically and conceptually.

A stronger plan would first select a target that the survey can support under its documented design, then investigate what the lighting data predict at an appropriate aggregation. If the aim is village-level allocation, the student should describe the additional village-level ground measurements required to validate the estimates rather than simply asserting that a model will be able to resolve them. The correct scientific conclusion may be that the proposed data do not yet support the requested resolution. Being able to explain that limitation is a substantial achievement, not an admission of technical failure.

3.30 Laboratory 3: reproduce the sampling illustration

The code companion produces Figure 3.1 and Table 3.1. Reproduce them before attempting the National Panel Survey laboratory. Change the relative region populations and sample allocations, then explain when equal observation weights would and would not be appropriate. Repeat the synthetic experiment with different seeds to see how random variation and weighting interact.

Next, introduce missingness in the artificial sample. First, remove responses completely at random. Then remove more responses from the higher-access region or from households without access. Compare the complete-case average and population truth. This exercise shows that the pattern of missingness matters; it does not establish which missing-data mechanism operates in any real survey.

The full real-data version is a protocol rather than a completed result in this draft. Once the chosen round's files are obtained legitimately, an independent student should be able to follow the data dictionary, processing script and recorded version to reproduce the descriptive tables. The final publication should include the actual outputs only after the code has been executed and checked on the documented public data.

Exercises

1. A sensor reports a stable reading with small random variation. Explain why collecting one hundred more readings might not remove a fixed calibration error.
2. For the two-region population in equation (3.5), calculate the proportion if Region A contains 60% rather than 75% of households. Explain which quantity changed.
3. From equation (3.6), derive the result when all weights are equal. State the population condition under which equal weights are appropriate for the intended estimator.
4. Ten responses contain six "yes", two "no" and two missing entries. Derive the bounds in equation (3.8), and explain what assumption would be necessary to report only 75%.
5. Give examples of MCAR, MAR and MNAR in a school attendance dataset. Explain why inspecting the observed attendance records alone may not prove the mechanism.
6. A dashboard reports a change in a district's number of firms, but the district boundary changed in the second year. Specify the checks needed before claiming growth.
7. **Project exercise.** Find the documentation for one publicly available survey dataset. Write a Data Readiness Assessment for a narrow descriptive question without performing any analysis that exceeds the dataset's permitted level of disclosure.

What this chapter establishes. Evidence has a history: measurements were produced in a particular place, at a particular time, under particular rules. A defensible solution acknowledges that history and tests its implications. The next part of the book can now introduce prediction and optimisation without pretending that a larger model is automatically a better answer.

Part II

Building Reliable Solutions

Chapter 4

Mathematical Models for Prediction and Understanding

What can we predict from imperfect observations? How certain should we be, and how can we discover when a good-looking prediction will fail in an unfamiliar place? This chapter develops regression, count models, forecasting, spatial prediction and honest validation, then connects them to decisions about air-quality monitoring.

4.1 A prediction is an answer to a particular question

A district officer may ask whether a market is likely to be flooded tomorrow. A health centre may ask how many patients will arrive next Monday. An environmental planner may ask about particulate pollution between monitoring stations. All three questions involve an unknown quantity, but the information available before the answer is needed differs. A rainfall forecast issued tomorrow morning cannot legitimately use tomorrow evening's recorded rainfall; a prediction at a new air-quality station cannot use yesterday's readings from that station if its purpose is to establish how well the method works *without* a monitor there. The time and location of available information are part of the mathematics of the problem, not administrative details.

Write Y for the quantity of interest, X for predictors actually available when a decision is made, and f for a prediction rule. The point prediction is $\widehat{Y} = f(X)$. For squared-error loss, the population-optimal rule is the conditional expectation,

$$f^*(x) = \mathbb{E}[Y \mid X = x]. \tag{4.1}$$

This result needs assumptions: the conditional mean must exist and the squared loss must represent the error we wish to minimise. If underestimating medicine demand is far more costly than overestimating it, the conditional mean need not be the decision-relevant summary. If the officer must choose an action, we ultimately need $\arg\min_a \mathbb{E}[L(a, Y) \mid X]$, as Chapter 1 argued. Prediction is often an input to that calculation, not its final answer.

Proposition 4.1 (Why the conditional mean minimises squared error)**.** *Assume* $\mathbb{E}[Y^2 \mid X =$

$x] < \infty$. *For any real number* c,

$$\mathbb{E}[(Y - c)^2 \mid X = x] = \operatorname{Var}(Y \mid X = x) + (c - \mathbb{E}[Y \mid X = x])^2.$$

Hence the minimum occurs at $c = \mathbb{E}[Y \mid X = x]$.

Proof. Write $m = \mathbb{E}[Y \mid X = x]$ and decompose $Y - c = (Y - m) + (m - c)$. Square both sides and take conditional expectations. The mixed term vanishes because $\mathbb{E}[Y - m \mid X = x] = 0$. What remains is the displayed decomposition, whose second term cannot be negative. □

The proposition is simple, but it exposes an important limitation. Even the best possible forecast carries irreducible uncertainty $\operatorname{Var}(Y \mid X = x)$. A sufficiently clever algorithm cannot recover information that was never observed. The proper question is whether the error that remains is small enough for the intended use. This perspective is consistent with statistical decision theory and the scoring-rule literature (Gneiting and Raftery, 2007; Raiffa, 1968).

4.2 Constructing regression from the residuals

Suppose a clinic has observations (x_i, y_i) where x_i is the number of staffed service hours and y_i is the number of patients processed. A linear description begins with $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$. It is tempting to describe β_1 as the effect of additional staff hours. That would be premature: facilities may schedule more staff precisely when they expect more patients. A fitted slope can describe an association without identifying a causal intervention. Chapter 3's discussion of measurement and selection still applies.

The least-squares estimates minimise

$$Q(\beta) = \sum_{i=1}^{n} (y_i - x_i^{\mathsf{T}} \beta)^2 = \|y - X\beta\|_2^2. \tag{4.2}$$

For a model with an intercept, the first column of X consists of ones. Differentiating gives $\nabla Q(\beta) = 2X^{\mathsf{T}}(X\beta - y)$. A stationary point therefore satisfies the normal equations $X^{\mathsf{T}} X \widehat{\beta} = X^{\mathsf{T}} y$. If X has full column rank, $X^{\mathsf{T}} X$ is positive definite and the minimiser is unique. Otherwise some coefficients are not separately identifiable, even when fitted predictions may be well defined.

Proposition 4.2 (Least-squares geometry)**.** *If the columns of* X *are linearly independent, then the residual* $r = y - X\widehat{\beta}$ *satisfies* $X^{\mathsf{T}} r = 0$. *The vector of fitted values is the orthogonal projection of* y *onto the column space of* X.

Proof. The normal equations imply $X^{\mathsf{T}}(y - X\widehat{\beta}) = 0$. Thus r is orthogonal to every column of X, hence to every vector in their span. Since $X\widehat{\beta}$ is itself in that span, the projection property follows. □

This gives a practical diagnostic. Large residuals may arise from unusual observations or an inadequate model; small residuals do not prove that an omitted cause is irrelevant. The equations also explain why a nearly singular $X^{\mathsf{T}} X$ is dangerous. When two predictors carry almost the same information, substantially different coefficient vectors can produce almost identical predictions. A reader should not mistake numerical sensitivity for evidence that two economic or physical mechanisms have large, opposite effects.

4.2.1 What regularisation changes

Ridge regression replaces Q by

$$Q_\lambda(\beta) = \|y - X\beta\|_2^2 + \lambda\|\beta_{-0}\|_2^2,$$

where the intercept is not usually penalised and $\lambda \geq 0$. After centring the variables, its solution is $(X^\mathsf{T}X + \lambda I)^{-1}X^\mathsf{T}y$. The additional term stabilises directions that would otherwise have small eigenvalues, but introduces bias. It does not repair biased sampling, inaccurate sensors or predictors measured after the decision time. Standardise predictors before penalising coefficients, using the training data only. Otherwise a change of measurement units changes which coefficient is penalised most strongly (Hastie et al., 2009; Bishop, 2006).

Worked example 4.1 (An apparently helpful predictor)**.** *Imagine two weather stations where daily electricity consumption and refrigerator use track each other closely. A regression for peak load uses both. If one column almost duplicates the other, each coefficient may vary greatly when one day is removed, even though total-load predictions barely change. Keep the two roles distinct: prediction can remain acceptable while component-level explanation is unreliable. If the engineering decision is whether to increase generator capacity, prediction of the total may suffice. If the decision is whether to replace refrigerators, the individual effect demands stronger identification.*

4.3 Counts, rates and probabilities are different outcomes

Not every observation is well represented by ordinary least squares. A weekly count of stock-out incidents is nonnegative and discrete. A proportion of late reports lies between zero and one. If Y_i is a count with exposure $E_i > 0$, a Poisson regression can model

$$Y_i \mid X_i \sim \text{Poisson}(\mu_i), \qquad \log \mu_i = \log E_i + \beta_0 + X_i^\mathsf{T}\beta.$$

The offset $\log E_i$ has coefficient fixed at one. It expresses a rate model: doubling exposure with unchanged conditions doubles the expected count. For instance, clinics serving populations of 12,000 and 120,000 cannot be compared fairly using raw event counts without explaining the exposure. The offset assumption itself should be checked; the event rate may not scale exactly with population.

The Poisson variance equals the mean conditional on the predictors. Operational counts often vary more widely, because risk differs across sites or events cluster in time. A negative-binomial model introduces additional dispersion, commonly written $\operatorname{Var}(Y \mid X) = \mu + \alpha\mu^2$ with $\alpha > 0$. It may improve uncertainty assessment, but should not be chosen automatically: excess variance can also reveal omitted structure, inconsistent observation periods or changes in recording. For binary outcomes, logistic regression gives $\Pr(Y = 1 \mid X) = \{1 + \exp(-X^\mathsf{T}\beta)\}^{-1}$; the probability needs calibration against observed frequencies, not only a classification threshold. See McCullagh and Nelder (1989); Agresti (2015) for the broader generalised linear model framework.

Worked example 4.2 (Why exposure matters)**.** *Suppose an illustrative facility serving 20,000 people records ten reports, while another serving 80,000 records twenty. The second has more*

reports but half the rate: 0.5 *versus* 0.25 *per 1,000 residents. Which facility is "worse" depends on what a report means and how reliably it was recorded. A rate adjustment cannot distinguish genuine risk from stronger reporting systems.*

4.4 Forecasting respects the direction of time

Time series are not ordinary shuffled tables. A model fitted with future observations is not a forecast of the past, even if cross-validation software calls it one. A useful baseline is persistence, $\widehat{y}_{t+h|t} = y_t$, while seasonal naive prediction uses the last observation in the corresponding season. These are difficult to beat consistently in sparse settings and must not be omitted merely because a complicated algorithm is available (Hyndman and Athanasopoulos, 2021).

An autoregressive model $y_t = c + \phi y_{t-1} + \epsilon_t$ assumes a particular dependence structure. When $|\phi| < 1$, shocks decay geometrically in expectation and the process can have a finite stationary variance under the usual white-noise assumptions. But structural changes can invalidate a good historical fit. Likewise a state-space representation,

$$z_{t+1} = Fz_t + Gu_t + w_t, \tag{4.3}$$

$$y_t = Hz_t + v_t, \tag{4.4}$$

separates latent system conditions z_t from noisy measurements y_t. Here w_t and v_t may represent distinct uncertainties. This becomes useful when an observed signal has both process variability and sensor error. A Kalman filter is a conditional estimator under linear-Gaussian assumptions; a forecast still depends on knowing those matrices sufficiently well (Kalman, 1960; Durbin and Koopman, 2012).

4.4.1 A rolling-origin evaluation

At decision time t, estimate parameters from data available through t, produce forecasts for $t+1$ through $t+h$, and score only when these outcomes become observable. Repeat for several origins. If hyperparameters are tuned, that operation belongs inside each training period. A later revision to a weather dataset cannot be silently used to represent what was known at the earlier forecast date. Record the data release or observation timestamp separately from the physical measurement date.

4.5 Spatial prediction: a station is not a pixel

Air-quality monitoring offers a natural reason to learn spatial statistics. A sensor measures air at its particular location and time, subject to its device, calibration and sampling conditions. A mapped grid cell is a numerical estimate for an area. Neither is an exact substitute for the other. Let s denote location and write

$$Y(s) = m(s) + Z(s) + \epsilon(s), \tag{4.5}$$

where $m(s)$ is a trend or covariate regression, $Z(s)$ is a spatially related residual process, and $\epsilon(s)$ is measurement error or fine-scale variability. These components are useful only to the extent

that their assumptions can be checked. For example, a spatial Gaussian-process model might specify $\mathrm{Cov}(Z(s), Z(s')) = k_\theta(s, s')$ with a range parameter controlling how quickly similarity decays with separation (Cressie, 1993; Rasmussen and Williams, 2006).

For observed sites $s_1, \ldots, s_n$, assume a known mean m, covariance matrix K for Z, and sensor-error matrix R. For a new site s_*, let k_* be the covariance vector between $Z(s_*)$ and the observed residual process. Under a Gaussian model the conditional mean and variance are

$$\widehat{Y}(s_*) = m(s_*) + k_*^{\mathsf{T}}(K + R)^{-1}(y - m), \tag{4.6}$$

$$V(s_*) = k(s_*, s_*) - k_*^{\mathsf{T}}(K + R)^{-1}k_*. \tag{4.7}$$

If the goal is prediction of a noisy future reading rather than the latent mean, add its measurement-error variance to V. If covariance parameters are estimated, the plug-in variance above usually omits parameter uncertainty. An attractive narrow band based on this formula is not automatically a calibrated prediction interval.

Proposition 4.3 (Why a distant site can remain uncertain)**.** *For a zero-mean Gaussian residual process with positive-definite training covariance $K + R$, suppose the covariance vector $k_*(s)$ tends to zero as a target site moves far from all observed sites. Then the conditional mean correction tends to zero and the conditional variance tends to the prior variance $k(s, s)$, provided the prior marginal variance converges.*

Proof. The correction is $k_*^{\mathsf{T}}(K + R)^{-1}(y - m)$ and the variance reduction is $k_*^{\mathsf{T}}(K + R)^{-1}k_*$. Since the inverse is fixed and finite, both tend to zero as $k_* \to 0$. The remaining variance is the prior marginal variance. □

The statement is not a guarantee that a fitted model reports honest uncertainty at distant sites: a wrong range parameter or missing covariate can undermine it. It explains why the distance to the nearest functioning monitor belongs in an operational error analysis.

4.5.1 Inverse-distance weighting is a baseline, not a physical law

A simple estimator uses weights $w_i(s_*) = d(s_*, s_i)^{-p} / \sum_j d(s_*, s_j)^{-p}$ for $p > 0$ when all distances are positive. Then $\widehat{y}(s_*) = \sum_i w_i(s_*)y(s_i)$. We must define the zero-distance rule and coordinate units. More importantly, proximity does not imply similarity across hills, coastlines, major roads or source regions. Inverse-distance weighting is a benchmark to compare against, not a substitute for testing spatial assumptions.

4.6 Uncertainty statements must have an operational meaning

A 90% prediction interval for a new observation is supposed to cover that observation about nine times out of ten under the relevant repeated-sampling or exchangeability conditions. A confidence interval for a mean answers a different question. Credible intervals under a Bayesian model depend on its prior and likelihood. An interval that is narrow but covers only half the withheld sites is misleading, however appealing its graph.

For a collection of genuinely held-out outcomes y_i and intervals $[\ell_i, u_i]$, empirical coverage is

$$\widehat{C}_{0.9} = \frac{1}{m}\sum_{i=1}^{m} \mathbf{1}\{\ell_i \le y_i \le u_i\}.$$

It should be reported together with average interval width, bias and error. A very wide interval can achieve high coverage without guiding decisions. Proper scoring rules offer a principled way to balance probability accuracy and sharpness (Gneiting and Raftery, 2007).

A clear validation table includes mean absolute error $m^{-1}\sum|y_i - \widehat{y}_i|$, root mean squared error $(m^{-1}\sum(y_i - \widehat{y}_i)^2)^{1/2}$, signed mean error, coverage and interval width. Report the unit (for example $\mu g/m^3$) and the level of aggregation. A model may perform well by hour but poorly during the short pollution episodes at which an alert threshold matters. The proper aggregation depends on the downstream decision.

4.7 Why ordinary random cross-validation can be wrong

Suppose a sensor contributes thousands of hourly records. Randomly splitting individual hours into train and test sets leaves nearly identical readings from the same sensor on both sides. The resulting score largely tests interpolation in time at known sites; it does not establish performance at an unobserved station. The cross-validation design must match the intended deployment: hold out stations for a new-location claim, hold out future periods for forecasting, and hold out sites *and* future periods for the strongest joint claim. Spatial dependence may persist between nearby stations, so also record the distance between held-out and training sites (Roberts et al., 2017).

The procedure is straightforward to state but easy to violate. First choose the held-out stations, without looking at their pollution outcomes. Fit imputations, transformations, spatial covariance parameters and hyperparameters on the remaining stations only. Tune within grouped training folds. Evaluate once on the held-out stations. Report results per station as well as across all observations, because a station with a faster reporting frequency should not necessarily dominate every summary.

4.8 A case study in Kampala: from a map to a defensible claim

The chapter's principal empirical laboratory asks: *How well can we estimate $PM_{2.5}$ concentration at a place without a functioning monitor?* The public measurement candidates are AirQo and OpenAQ. Their APIs, documentation, measurement histories, coverage and permissions must be checked at the time of download. AirQo's public API documentation describes authenticated access to current and historical measurements; access to a particular historical period must not be assumed. OpenAQ is a separate aggregation route with heterogeneous coverage and data provenance. Neither guarantees the site density or covariates required for every proposed model (AirQo, 2026; OpenAQ, 2026).

A minimum data inventory contains station identifier, coordinates, timestamp and timezone, observed $PM_{2.5}$, quality flag, device provenance, data version and units. Add weather, road or

land-use covariates only when they can be aligned with the prediction time. A station with duplicate identifiers or moving coordinates should not silently be treated as stationary. The analyst should publish the data coverage diagram before fitting anything.

Compare, on exactly the same held-out observations, a training-set mean, an inverse-distance-weighted predictor, a generalised additive model, a random forest, a Gaussian process and, where covariates allow, gradient boosting followed by kriging of training residuals. An ordinary gradient-boosting model is a useful separate baseline; residual kriging is not necessarily beneficial if the remaining errors are spatially unstructured. This comparison is motivated by the classes of methods, not by an assumption that the most elaborate method will win (Breiman, 2001; Friedman, 2001; Wood, 2017).

Table 4.1: Required outputs from the measured-data air-quality laboratory. No numerical performance claims are made before data acquisition.

Output	Question it must answer
Site map and date coverage	Which stations and time periods are actually observed?
Held-out station results	How does the prediction behave where the model has seen no outcomes?
Error against nearest-station distance	Does accuracy deteriorate away from measured locations?
Prediction interval coverage and width	Does uncertainty describe observed disagreement, or merely appear precise?
Missing-input and quality-flag sensitivity	Does failure of one instrument or covariate materially change the conclusions?
Runtime, memory and storage	Could an ordinary office computer reproduce and use this approach?

An example of an appropriate conclusion would be: “The model’s error increased beyond the observed station-density range, so we cannot recommend use of these predicted concentrations for site-specific regulatory enforcement there.” An inappropriate conclusion would be: “The map measures air quality in every Ugandan district,” when all stations were in a few urban locations. A spatial map is useful only when its uncertainty, time period and evidence limits travel with it.

4.9 Worked investigation: forecast accuracy is not decision accuracy

Consider an illustrative authority that issues a warning whenever tomorrow’s $\mathrm{PM}_{2.5}$ exceeds a threshold T. A false warning costs $C_{\mathrm{FP}} > 0$ in disruption; a missed exceedance costs $C_{\mathrm{FN}} > 0$ in avoidable harm. With a calibrated predicted exceedance probability p, expected loss if we warn is $(1-p)C_{\mathrm{FP}}$ and if we do not warn is pC_{FN}. The warning is preferred exactly when

$$p > \frac{C_{\mathrm{FP}}}{C_{\mathrm{FP}} + C_{\mathrm{FN}}}. \tag{4.8}$$

This threshold is derived, not guessed. If a missed event is ten times more costly than an unnecessary warning, the probability threshold is $1/11$, not $1/2$. The result assumes those costs are appropriate representations of real consequences and that the forecast probability is

trustworthy. It does not assign an official public-health warning threshold, which belongs to the responsible authority.

Two models can have similar average $PM_{2.5}$ error and very different exceedance probabilities. The second may make better decisions even with a slightly worse RMSE. Conversely, a highly calibrated probability can still be operationally useless if no resources exist to communicate or act on the warning. These issues become the main subject of Chapter 6.

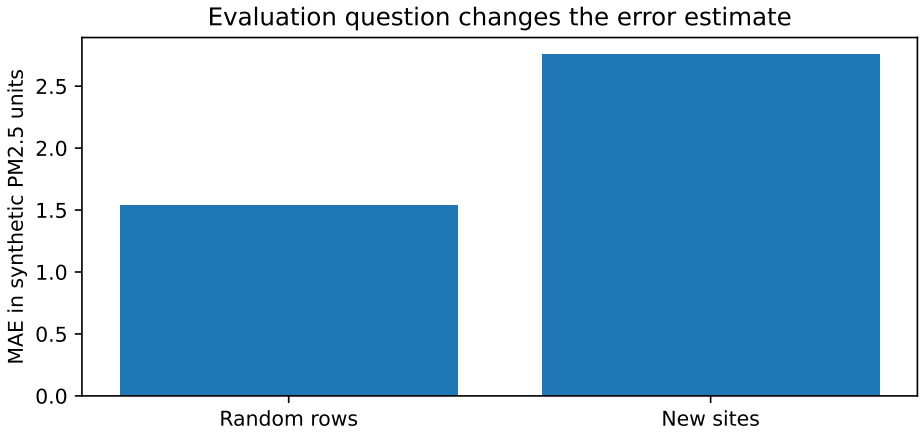


Figure 4.1: An entirely synthetic teaching comparison. The larger held-out-site error in this particular simulation illustrates the risk of row-wise splitting when the goal is prediction at new sites; neither error is an AirQo or Uganda measurement.

4.10 Laboratory 4: an honest spatial-validation exercise

The companion Python programme `code/ch04_spatial_validation.py` creates *explicitly synthetic* station readings and compares ordinary row-wise splitting with grouped held-out-site testing. It reports the grouping and stores a site-level error table. The example is a unit test of evaluation design, not a measured Kampala result. The more important second part of the laboratory asks the reader to replace the synthetic input with an appropriately licensed measured dataset while preserving the same validation protocol. The code intentionally keeps the input schema separate from the model implementation.

Before running the laboratory, predict what may happen when many near-duplicate records belong to a single station. After running it, change the number of stations and the spatial correlation length. Explain why the numerical direction of the gap between random-split and held-out-site error can vary with the data-generating process, even though random splitting is conceptually inappropriate for a claim about unseen sites. Do not force the simulation to demonstrate a predetermined outcome.

4.11 A closer look at predictive uncertainty

The claim that "a prediction has uncertainty" can refer to several distinct things. There may be randomness in the outcome even with complete knowledge of the system, uncertainty about parameters because few observations were collected, variation introduced by an imperfect sensor, or uncertainty about whether the chosen model represents the mechanism. These sources are not generally interchangeable. A model that estimates the conditional variance correctly within a familiar training region may still be wrong under changes in the physical process.

Write the response as $Y = m(X, \theta) + \epsilon$ with $\mathbb{E}(\epsilon \mid X) = 0$. If the parameters θ are treated as uncertain, the law of total variance suggests separating variation within a parameter configuration from variation in the fitted mean:

$$\operatorname{Var}(Y \mid \mathcal{D}, X) = \mathbb{E}_{\theta|\mathcal{D}}[\operatorname{Var}(Y \mid \theta, X)] + \operatorname{Var}_{\theta|\mathcal{D}}[\mathbb{E}(Y \mid \theta, X)].$$

The first term describes remaining outcome variation after conditioning on the parameter, while the second reflects uncertainty in the conditional mean across plausible parameter values. This is a useful accounting device in a probability model; it is not a full description of model misspecification, distribution shift or future data-system failure. Those require additional checks.

4.11.1 Why ordinary bootstrap intervals can understate uncertainty

A common computational technique resamples observations with replacement, refits the model and examines the variation in results. This is useful under appropriate sampling assumptions. But if daily readings within a station are strongly correlated, resampling every hour independently creates artificial information: repeated readings from the same station are treated as though they came from independent monitoring sites. For a claim about new locations, resampling or evaluation should respect the station as a cluster. A cluster bootstrap resamples whole sites; if there are very few independent sites, no bootstrap method can conjure a large effective sample (Efron and Tibshirani, 1993; Roberts et al., 2017).

Suppose m stations each provide T observations and within-station correlation is approximately ρ. A simple variance-inflation approximation for equally sized exchangeable clusters is $1 + (T - 1)\rho$. The nominal row count mT overstates effective information when $\rho > 0$. The calculation is illustrative, but it reveals why a million hourly data points from three stations do not automatically establish generalisation across a country.

4.12 When a model gains accuracy by learning the wrong thing

Imagine that a mobile environmental monitor is moved to higher-risk streets during smoke events. If the analyst fits pollution against street type using the collected observations, high readings may appear associated with the selected routes. The difference may be real, but the site-selection rule helped determine which observations exist. Reweighting or modelling the selection process may be necessary before interpreting results as representative of the whole city. The same logic appears in Chapter 3 for household surveys, but spatial monitoring makes the difficulty particularly visible.

A predictive map can also be misleading when several instruments observe the same general plume. The average squared error on held-out hours can be low because nearby training sensors have almost identical readings at those hours. The result does not prove that a remote settlement shares the same pollution process. The evaluation unit should therefore be chosen by the decision: if a city wishes to fill gaps between known monitors, spatial interpolation within observed coverage may be sufficient; if a national agency wants to place warnings in unmonitored districts, extrapolation evidence is required.

4.12.1 The geometry of a kernel matters

A radial covariance of the form $k(s, s') = \sigma^2 \exp\{-\|s - s'\|/\ell\}$ assumes a relationship based on Euclidean distance and a common correlation length ℓ. This may ignore prevailing winds, terrain and emissions. Distances also require a proper coordinate reference system: a degree of longitude is not a fixed number of kilometres, and treating raw latitude and longitude as planar metres can distort interpolation. The analyst should describe the distance calculation and ideally test sensitivity to its choice. A model whose fitted range is shorter than the gaps between most stations will have little ability to identify local variations at unobserved sites, however persuasive its coloured prediction surface appears.

For the two-station case, the algebra is transparent. Suppose the latent covariance matrix is $\sigma^2 \left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)$ and independent sensor noise has variance τ^2. At a target with covariance vector $k_* = (a, b)^\mathsf{T}$, the predicted residual is $(a, b)(K + \tau^2 I)^{-1} r$. Increasing sensor noise weakens the influence of the observations because the inverse discounts uncertain signals. Increasing correlation ρ without gaining spatial coverage can create redundancy rather than independent information. The formula demonstrates precisely why the number of recorded rows is not an adequate measure of spatial information.

4.13 A useful comparison table before fitting

The comparison should be organised around differences in assumptions rather than a list of fashionable names.

What should be called improvement? A model that gains 0.1 units of MAE at the cost of a fivefold increase in computation has not necessarily improved a decision. A model that predicts the average well but misses every high-risk episode may be unfit for alerts. Improvement should be judged against the predeclared use, with intervals and error decomposition supporting rather than replacing the decision argument.

4.14 Full worked investigation: deriving the value of an observation

Consider a simplified pollution state Z at a proposed unmonitored location, modelled as Gaussian with prior mean m and variance $v > 0$. Suppose an instrument could provide $Y = Z + \epsilon$ with independent Gaussian noise variance τ^2. Under these assumptions the conditional variance

Table 4.2: Methods and the first question to ask before using each one

Method	Essential assumption or vulnerability
Spatial mean	The large-scale average is informative at the target; ignores local variation.
Inverse-distance weighting	Nearby sites are more relevant under the chosen distance and power rule.
Linear or additive regression	Chosen covariates represent meaningful structure; extrapolation remains uncertain.
Gaussian process	Covariance and measurement-error model reasonably describe residual dependence.
Random forest	Training covariate patterns are relevant at prediction sites; outside-range behaviour is limited.
Boosting with residual kriging	Residual spatial dependence remains after the learned trend and can be estimated without leakage.

after a measurement is

$$\operatorname{Var}(Z \mid Y) = \left(\frac{1}{v} + \frac{1}{\tau^2}\right)^{-1} = \frac{v\tau^2}{v+\tau^2}.$$

The reduction in variance is $v - v\tau^2/(v+\tau^2) = v^2/(v+\tau^2)$. This quantity increases with prior uncertainty v and decreases as measurement noise grows. A highly noisy instrument may provide little useful information despite producing thousands of readings. The calculation connects sensor placement to Chapter 1's value-of-information perspective without claiming that variance reduction alone determines the best location.

Derivation. The joint covariance of (Z, Y) is $\left(\begin{smallmatrix} v & v \\ v & v+\tau^2 \end{smallmatrix}\right)$. The conditional Gaussian variance is the upper-left element minus the squared cross-covariance divided by the variance of Y, giving $v - v^2/(v+\tau^2) = v\tau^2/(v+\tau^2)$. □

If the decision is to issue a warning when pollution exceeds a limit, a sensor whose measurements chiefly reduce uncertainty far below that limit may not be the best investment. A more decision-focused value of information compares expected loss before and after receiving a measurement, integrating over possible observed values and future actions. The two criteria can rank sites differently.

4.15 An air-quality investigation that another researcher can reproduce

For each candidate public dataset, the final manuscript should record the retrieval date, source terms, coordinates, station identifiers and quality flags. A compact input table should include a codebook explaining whether the pollutant concentration refers to hourly, daily or another aggregation. A station with many missing hours may produce a biased daily average if outages correlate with high pollution; ignoring that mechanism should be acknowledged. If sensors are calibrated by a model using nearby regulatory instruments, their readings are not necessarily independent of those instruments. The original calibration procedure may matter for validation.

The main performance table should not collapse every withheld site into a single number. Show a row for each site, then a summary across sites. This lets the reader see whether a low pooled MAE is dominated by dense stations. Next show error against distance to the nearest training site and counts in each distance band. Interpret empty bands as a failure of geographic coverage, not as zero error. Repeat the assessment across seasons if the record spans them; a method that works during one dry interval may fail under different weather.

Where actual licensed data are unavailable, the book's computational exercise remains a demonstration of validation machinery. There is nothing shameful about using synthetic data to teach what the code does, provided the book is explicit about what has and has not been measured. The scientific mistake would be to report such a demonstration as evidence that a model works on Ugandan air pollution. Readers should see this distinction repeated in the conclusions of empirical chapters rather than buried in a footnote.

4.16 Extended case study: when an air-quality map deserves to be believed

A field map presents an unusual temptation. It can look complete even when its underlying measurements cover only a handful of streets. The grid is filled with coloured cells, so the reader's eye sees data everywhere. Yet the underlying evidence may come from irregular samples at a few stations. A good environmental map should make both the estimate and the geographic reach of the evidence visible. This worked investigation develops a complete assessment plan for that problem. It deliberately avoids assigning numerical success to any model before an appropriate measured dataset has been obtained and analysed.

4.16.1 First identify the user and the decision

Suppose an urban-planning office wishes to prioritise locations for additional monitoring rather than issue immediate individual health advice. Its decision is where to place a small number of new monitors. The relevant output might be a spatial estimate of likely $PM_{2.5}$ level together with uncertainty and distance from existing instruments. That is different from estimating an individual person's exposure or determining compliance with a legal concentration limit. An official compliance determination may require a specified measurement instrument, averaging period, and quality-control protocol. A map produced by the methods in this chapter cannot substitute for those requirements by default.

A second office may want to choose roads for dust-control maintenance. That task requires information about the part of pollution actually affected by the intervention. A map estimating total $PM_{2.5}$ may predict well while failing to identify the change achievable by road work. The distinction is causal: a predictive relationship between road proximity and observed pollution does not establish the amount by which resurfacing a road will change concentrations. For the first office, prediction and uncertainty are useful inputs. For the second, additional intervention evidence is necessary.

4.16.2 The data inventory should be legible without the code

Before building a dataset, prepare a table naming each source and the question it answers. Observed $PM_{2.5}$ comes from functioning instruments, accompanied by timestamps, units, calibration flags and physical coordinates. Weather covariates describe potentially relevant atmospheric conditions; road density and land-use variables describe possible sources and exposures. A population grid may help identify who could be affected, but population is not itself a pollutant sensor. Each input should have an acquisition date and permitted use. Different variables may describe different years, spatial supports or time resolutions. The analyst needs to make those mismatches explicit.

A point measurement and an area average are mathematically distinct. If $Z(s)$ is a spatial field and A is a grid cell, a cell average is $\overline{Z}_A = |A|^{-1} \int_A Z(s)\, ds$, whereas a monitor measures approximately $Z(s_0)$ plus instrument error for a particular s_0. The two can disagree without either being wrong. If a model is trained on point observations but evaluated against a satellite grid product, the meaning of the comparison changes. The support of each outcome belongs in the evaluation design.

4.16.3 Choose the holdout before making a map

Divide monitors into a training group and an untouched test group. If there are 20 usable sites, an illustrative design might reserve five entire sites for testing. The choice should consider spatial coverage, not merely sample size. If all five are near training sites, the experiment tests interpolation. If several are far from any training site, it also tests more challenging spatial transfer. Publish a site map showing the groups and report the distance from each test location to its nearest training site.

A meaningful error table could have columns for site, observation days, MAE, mean signed error, RMSE, interval coverage and nearest-training-site distance. Do not report a pooled hourly error alone. A sensor that reports every minute can dominate one that reports daily, while the practical decision may concern equal importance across locations. Site-averaged metrics give each held-out station one contribution, but the choice of weighting must itself be described. A statistically principled estimate of citywide accuracy would need information about how the station locations represent the city; equal weighting by station is not automatically population weighting.

4.16.4 Keep selection and fitting separate

Use grouped cross-validation within training stations to choose hyperparameters. For the Gaussian process, estimate the spatial range and noise variance on training information; for forests, choose tree depth, leaf size and feature selection through the training folds. If the final test sites influence these choices, the reported final test error is optimistic because the evaluation has become part of model development. The issue remains even if the analyst did not explicitly copy test outcomes into training; repeated inspection and revision can gradually overfit the test set.

Suppose a boosted model uses both rainfall and station identifier. Station identifier may explain

stable differences between known sites, but the identifier has no meaningful value at a new station. Any preprocessing that one-hot encodes known identifiers without a plan for new ones can produce undefined or misleading behaviour. It is often better to use physical covariates and coordinates whose meanings transfer to new sites, while acknowledging the remaining uncertainty.

4.16.5 Explain residual kriging rather than invoking it by name

Let a trend model $\widehat{m}(s)$ be fitted on the training stations. Its training residuals are $r_i = y_i - \widehat{m}(s_i)$. If there is spatial correlation left in those residuals, fit a covariance model to them using training data alone. At a new site, combine the trend and a kriged residual correction:

$$\widehat{y}(s_*) = \widehat{m}(s_*) + k_*^{\mathsf{T}}(K + R)^{-1} r.$$

This expression is not a guarantee of improvement. If the trend already captures spatial structure, the estimated residual correlation may be weak or unstable. If residual dependence changes with weather, a stationary covariance model can be inadequate. If training residuals are unrealistically small because the trend model overfits, residual kriging may learn little or produce overconfident intervals. Cross-fitting residuals for covariance estimation may reduce this problem, but must also respect the station-group structure.

A useful diagnostic compares residual correlations by separation distance, and then tests whether residual correction improves the same held-out sites as the uncorrected trend. A claimed gain should include uncertainty across stations and a compute-cost comparison. If two methods are effectively indistinguishable on independent sites, prefer the one with clearer assumptions and lower maintenance demands until more evidence is available.

4.16.6 Distance-dependent reliability and a refusal region

Suppose held-out sites near training monitors show small errors, while those beyond 10 kilometres show much larger errors. This suggests a practical decision boundary: do not present site-specific values beyond the demonstrated range as if they have the same reliability as interpolated values. A robust map might display an uncertainty overlay or label regions that have no validation support. The distance at which the method becomes unreliable must be inferred from held-out evidence, not chosen after looking for a convenient result.

There is also an important counterexample. A distant site with the same meteorological and emissions conditions as well-observed areas may be predicted accurately from covariates even when spatial correlation is weak. Conversely, a near site separated by a major emission source may be poorly predicted. Distance alone is a useful diagnostic, not a complete description of distribution shift. Report both distance and covariate novelty where possible.

4.16.7 A concrete uncertainty assessment

Suppose a fitted model returns means $\widehat{y}_i$ and standard deviations $\widehat{\sigma}_i$ for the held-out observations. A Gaussian 90% interval is approximately $\widehat{y}_i \pm 1.645\widehat{\sigma}_i$. Calculate the fraction of held-out values within these bounds and their average width. If coverage is 65%, the intervals are

not empirically calibrated at the nominal level in that test. An interval-widening adjustment might improve coverage but should be fitted on a separate calibration set; fitting it to the final test outcomes invalidates that test. Calibration under exchangeability may be approached through conformal methods, but spatial dependence and distribution shift complicate the usual finite-sample guarantees (Shafer and Vovk, 2008).

The relevant communication to a planner is not merely a confidence percentage. It should say which observed regions supported the interval, how the observation frequency was aggregated, and where data are too limited to justify a precise recommendation. If a predicted concentration is 27 units with a 90% interval from 10 to 55, treating the point estimate as an exact exposure can lead to an unsound intervention decision.

4.16.8 From a map to an additional-monitor decision

Suppose a planning office can afford one more monitoring station. A naive rule chooses the area with the highest predicted concentration; another chooses the location with the greatest model variance. These are different objectives. The first targets possible high exposure, while the second aims to reduce prediction uncertainty. A third approach chooses the site with greatest expected reduction in decision loss, balancing where people live, the need for evidence and monitoring cost. Each choice requires an explicit goal, and the resulting site may be different.

For a Gaussian process with candidate measurement location c, one can calculate the expected reduction in variance at a target s under a simple noise model:

$$\Delta V(s; c) = \frac{\operatorname{Cov}(Z(s), Y(c) \mid \mathcal{D})^2}{\operatorname{Var}(Y(c) \mid \mathcal{D})}.$$

Summing this reduction over weighted planning locations gives an illustrative design criterion. Yet the weights reflect whose areas matter and the covariance model reflects observed spatial structure. A high predicted variance at an inaccessible road junction is not a feasible monitoring proposal. Site selection is therefore an optimisation problem subject to maintenance, power, security, calibration and access constraints.

4.16.9 A genuine publishable result versus an attractive illustration

A scientifically credible manuscript emerging from the laboratory would publish documented measurements, actual station-level holdouts, appropriate baselines, uncertainty calibration, resource measurements and clearly bounded conclusions. It should include a counterexample or a systematic description of where the method fails, not only a map of favourable predictions. A real operational study would go further: place a monitor at a proposed site, measure its readings prospectively and determine whether the resulting information changes planning decisions. That final step cannot be inferred from a retrospective random split.

The chapter's synthetic demonstration is intentionally less ambitious. It checks that grouping changes the evaluation target and that code can produce auditable outputs. The data were invented; the improvement or deterioration of any method in this exercise is not an observation about Kampala. This distinction must remain visible to readers who reuse the code.

4.17 Mathematical workshop: four predictions that look alike but are not

Students often encounter a formula and ask which values to put into it. A more useful habit is to ask what the formula assumes about the observation and what kind of unknown it estimates. We compare four apparently similar tasks: predicting a mean at a known set of conditions, predicting an individual outcome, predicting at an unseen site and predicting a future period. The numerical expressions can be related, but their uncertainty calculations and validation requirements differ.

4.17.1 Mean response versus new observation

In a linear regression $Y = x^\mathsf{T}\beta + \epsilon$ with ϵ independent of the predictors and variance σ^2, the fitted mean at x_* is $\widehat{m}_* = x_*^\mathsf{T}\widehat{\beta}$. Under classical full-rank assumptions with homoscedastic independent errors,

$$\operatorname{Var}(\widehat{m}_* - m_*) = \sigma^2 x_*^\mathsf{T}(X^\mathsf{T}X)^{-1}x_*.$$

A future individual observation has an additional noise term, so its predictive variance is approximately $\sigma^2\{1 + x_*^\mathsf{T}(X^\mathsf{T}X)^{-1}x_*\}$. The second interval is wider because it must accommodate the unpredictability of the new observation itself. A confidence band for the conditional mean does not provide the promised coverage for future individual measurements.

4.17.2 Why this derivation works

Under the standard linear model, $\widehat{\beta} = \beta + (X^\mathsf{T}X)^{-1}X^\mathsf{T}\epsilon$. Its covariance is $\sigma^2(X^\mathsf{T}X)^{-1}$. Taking the linear functional $x_*^\mathsf{T}\widehat{\beta}$ gives the first variance. If the new error ϵ_* is independent of the training errors, its variance adds σ^2 to the variance of the fitted mean, yielding the second expression. The distinction disappears from neither a bigger data set nor an elaborate algorithm; only the reducible part can shrink as more observations are collected.

4.17.3 Predicting an unseen station requires another uncertainty component

If several measurements share a station-specific random effect b_j, an appropriate model might be $Y_{jt} = x_{jt}^\mathsf{T}\beta + b_j + \epsilon_{jt}$ with $\operatorname{Var}(b_j) = \tau_b^2$ and $\operatorname{Var}(\epsilon_{jt}) = \sigma^2$. At a known station, earlier observations inform its particular b_j. At a new station, that effect is unobserved. The prediction distribution may need to include τ_b^2 in addition to daily noise and parameter uncertainty. Randomly withholding an hour at a known station and withholding an entire station do not measure the same task.

If τ_b^2 is large and only three stations are observed, the estimated population distribution of site effects may be poor. Reporting thousands of hourly measurements does not remove the limitation that only three independent site effects were sampled. This is a key reason to examine both between-site and within-site variability before choosing an evaluation scheme.

4.18 Deriving the Poisson regression score

The count model $Y_i \sim \text{Poisson}(\mu_i)$ with $\mu_i = E_i \exp(x_i^\top \beta)$ has log-likelihood, up to constants independent of β,

$$\ell(\beta) = \sum_i \{y_i(\log E_i + x_i^\top \beta) - E_i \exp(x_i^\top \beta)\}.$$

Differentiation gives score

$$\frac{\partial \ell}{\partial \beta} = \sum_i x_i(y_i - \mu_i).$$

Thus the fitted coefficients balance covariate-weighted residuals. The Hessian is $-\sum_i \mu_i x_i x_i^\top$, negative semidefinite; under full-rank conditions it is negative definite. These facts explain why Poisson regression has a convenient convex optimisation structure for the negative log-likelihood. They do not establish that the Poisson conditional variance assumption or the exposure offset is correct for the data.

The exponential link forces positive fitted means, which is appropriate for counts. It is not appropriate to use the raw predicted mean as though it were an actual nonnegative integer; a conditional mean of 3.7 is meaningful even though observed counts are discrete. When planning future capacity, the full predictive distribution is often more informative than rounding the mean to four.

4.19 What a proper score can and cannot guarantee

Consider a binary event with true probability q, and let a model report p. The expected Brier loss is $q(1-p)^2 + (1-q)p^2$. Expand the expression:

$$q - 2qp + p^2 = (p-q)^2 + q(1-q).$$

Its minimum occurs uniquely at $p = q$. This small proof explains why the Brier score is a proper probability score: under the stated data-generating probability, the best expected score comes from reporting the true probability. But a model minimising empirical Brier loss on a biased sample does not necessarily estimate the deployment population's probabilities. Scoring-rule optimality does not correct sample selection.

4.20 An honest prediction interval check with few independent sites

Suppose five withheld stations have site-aggregated observed values, and a model provides a 90% interval for each. If four are covered, empirical coverage is 80%. It would be misleading to assert that the population coverage is exactly 80%: with only five independent sites, that fraction is highly variable. Nor does high hourly coverage establish site-level reliability if hundreds of correlated hours were treated as separate Bernoulli trials. Present numerator and denominator, and when possible provide an interval for the coverage estimate based on the appropriate independent units.

A useful uncertainty report states: “Four of five held-out stations had their aggregated values inside the nominal 90% prediction intervals.” It avoids claiming that national coverage is known precisely. With more stations, subgroup and distance-stratified coverage can become meaningful. With few stations, a careful limitations statement is part of the quantitative result.

4.21 A compact checklist for an empirical prediction paper

Before concluding that one method is better than another, verify that the intended outcome is defined, inputs would be available at prediction time, all methods used comparable training information, test sites were excluded from every fitted transformation, and uncertainty was evaluated against actual observations. Then ask whether the measured accuracy difference is large enough to matter to the user, whether it survives reasonable variations in data quality, and whether the method can be maintained at a tolerable cost.

The reason for such a checklist is mathematical, not bureaucratic. Each missing condition weakens the link between the score computed in the laboratory and the error a user would experience. Prediction is useful only when that link is sufficiently strong for the decision at hand.

4.22 A deeper mathematical view of spatial prediction

The Gaussian-process formula is sometimes taught as a ready-made block of algebra. It is more useful to understand why its pieces appear. Assume jointly Gaussian random vectors Z_* at an unobserved location and Y at observed locations. Their joint mean is $(m_*, m)^\mathsf{T}$ and covariance is

$$\begin{pmatrix} K_{**} & K_{*Y} \\ K_{Y*} & K_{YY} \end{pmatrix}, \qquad K_{YY} \text{ positive definite.}$$

A conditional Gaussian calculation gives

$$\mathbb{E}[Z_* \mid Y = y] = m_* + K_{*Y}K_{YY}^{-1}(y - m), \quad \operatorname{Var}(Z_* \mid Y) = K_{**} - K_{*Y}K_{YY}^{-1}K_{Y*}.$$

The variance reduction is a Schur-complement term. Because covariance matrices are positive semidefinite, this conditional variance is nonnegative. The algebra expresses a simple idea: the prediction corrects the prior mean according to how strongly the unobserved location covaries with measured ones, weighted by the reliability and redundancy of the observed data.

4.22.1 The danger of substituting a fitted covariance as though it were known

The preceding equations assume a covariance structure that is known. In practice the variance, correlation length and nugget parameters are estimated. With only a few monitoring sites, several covariance combinations may explain the observed residuals nearly equally well, yet produce different predictions at distant sites. A plug-in interval based on one parameter estimate may be narrower than the full uncertainty warrants. The analyst should inspect sensitivity to plausible parameter settings, provide an appropriate held-out assessment and avoid claiming certainty about a correlation range that the data cannot identify.

Numerical stability also matters. Forming $(K + R)^{-1}$ explicitly is generally less desirable than solving the linear system $(K+R)z = y-m$. A Cholesky factorisation is efficient when the matrix is symmetric positive definite. If the matrix is nearly singular, numerical difficulties may indicate duplicated coordinates, implausibly tiny measurement noise, or redundant measurements. Adding a small diagonal numerical jitter can stabilise computation, but it changes the model slightly and must be reported if it materially affects predictions.

4.23 Learning a forecast without confusing seasonal variation with change

Environmental or service data may have a seasonal pattern and a gradual trend. A simple decomposition writes $Y_t = T_t + S_t + R_t$ for trend, seasonal and residual components. The decomposition is descriptive unless a process model is specified. A forecast constructed by estimating a seasonal average using future years is not an honest historical forecast. The correct backtest must estimate the pattern from only the records that were available before each prediction date.

Consider a hypothetical monthly series with a recurring high-demand season. If a training period contains only one season, fitting a complex seasonal model may be unstable. A simple prior seasonal average may also be poorly identified. A transparent evaluation should show the available years and how error changes across seasons. If the intended decision is to buy medicine before a seasonal peak, the forecast must be issued early enough to influence the supplier's lead time. Information published after the purchase deadline can improve explanation of the past but not the earlier purchasing decision.

4.24 Why aggregation can hide failure

An agency may report one prediction per district by averaging high-resolution estimates. If predictions are biased high in one part of the district and biased low in another, the mean error can cancel even while local recommendations are wrong. Conversely, small location errors may aggregate into a useful district average. The desired spatial support must be specified. An average of predicted point concentrations is not automatically an estimate of population-weighted exposure. That calculation would require a defensible population surface and assumptions about where and when people are exposed.

Similarly, a model predicting average pollution over a month may be unsuitable for identifying rare daily exceedances. These are different target variables, not alternative visualisations of one objective. When the decision is a daily warning, model evaluation should reflect daily exceedance outcomes. When the decision is long-term monitoring allocation, broader spatial uncertainty may matter more. A single headline RMSE cannot certify all possible uses.

4.25 A direct exercise in evaluation leakage

Take a synthetic dataset with 15 sites and 100 records per site. Assign each site a stable latent level and add small independent daily noise. A predictor that memorises site identifiers will look excellent under random-row cross-validation: each test site is already represented among training rows. Under held-out-site validation, the same identifiers offer no information. This experiment isolates the effect of grouping from differences in model architecture.

Repeat with strong spatial covariates that explain most site differences. Held-out-site prediction should then improve because new-site covariates carry transferable information. The result depends on the synthetic mechanism; the point is not that grouped error must always exceed random error. The point is that evaluation has to ask the question the eventual user needs answered. A good student should be able to state the information-set difference in words before looking at any code output.

4.26 Exercises and investigations

1. Derive the minimiser of expected absolute error and explain why the conditional median, rather than the mean, appears. State a condition under which it is unique.
2. For n observations and a single explanatory variable with intercept, derive the two normal equations explicitly. Explain how a nearly constant predictor affects identification.
3. Derive the ratio interpretation of a Poisson coefficient under a log link. Why is adding population as an ordinary predictor not identical to using it as a fixed offset?
4. A monitoring network has eight stations, six in a city and two in rural districts. Design a held-out-site evaluation that reports rural performance without claiming nationally representative accuracy.
5. Prove that the posterior variance reduction in (4.6) is nonnegative when the covariance matrix is positive definite. Explain why this proof does not establish empirical calibration.
6. Design a figure showing bias against distance to the nearest training station. State how spatial coordinates and duplicate readings would be checked.
7. A forecast has 91% interval coverage but intervals nearly as wide as the entire observed range. Explain what is missing from the evaluation.
8. **Research project.** Obtain documented public measurements under an appropriate licence. Keep one complete station out of training. Produce a transparent data inventory, a simple baseline, one spatial method and a defensible limitations statement.

What we have learned. A prediction is an answer conditioned on particular information. Mathematics can derive a sound estimator under explicit assumptions; validation reveals whether it remains useful when those assumptions meet real observations. The next question is what action to take with the information that survives these tests.

Chapter 5

Optimisation When Resources Are Scarce

Prediction tells us what may happen. Optimisation asks what to do, under actual limits on money, time, people, capacity and risk. We develop linear and integer programming, duality, network flows, facility location, queueing and robust decision-making, paying particular attention to the assumptions that separate an elegant mathematical solution from a practical one.

5.1 A scarce resource creates a choice

A district owns one vehicle and has several health facilities to supply. A cooperative has enough grain to process three orders, but only one machine. A health officer may have the money to improve two clinics but not six. These problems are not solved by predicting everything perfectly. Even with complete information, the decision-maker faces mutually incompatible uses of limited resources. Mathematics helps make the trade-off explicit; it does not decide whose welfare should matter without a public or institutional judgment.

Let x_j be a controllable quantity: deliveries, equipment purchases, patients assigned to a facility or hours of operation. An elementary linear programme is

$$\min_{x \geq 0} c^{\mathsf{T}} x \qquad \text{subject to } Ax \geq b. \tag{5.1}$$

The objective $c^{\mathsf{T}}x$ has a meaning only when the units and prices of each decision are defined. The constraints $Ax \geq b$ state obligations such as minimum service provision. If x counts whole vehicles or facilities, fractional values are not meaningful and an integer restriction must be included. If capacity is unknown, its uncertainty cannot disappear because the solver accepted a numeric parameter.

5.1.1 Build the model backwards from an actual decision

Start with the action someone can approve or carry out. Does the manager choose a route, a dispatch date, a staffing level, a purchase quantity, or a location for new infrastructure? Next

identify feasibility rules that must never be broken, such as an equipment safety rating or a budget ceiling. Only then write an objective. There is no honest way to optimise "development" as a single quantity without explaining whose outcomes are weighted and how. The mathematical difficulty is often less than the work of getting the objective right.

5.2 A fully worked allocation problem

A small service programme has 40 staff-hours available. Service A requires two hours per client and returns an illustrative benefit score of five per client; service B requires four hours and returns a score of eight. At most 12 clients can be served by A and at most eight by B. With $x_A, x_B \geq 0$ the continuous planning problem is

$$\max\ 5x_A + 8x_B, \quad 2x_A + 4x_B \leq 40, \quad x_A \leq 12, \quad x_B \leq 8.$$

These scores are invented to illustrate allocation. They are not values assigned to real patients. Because benefit per staff-hour is $5/2$ for A and $8/4$ for B, A is preferable per hour while both caps are respected. Allocate $x_A = 12$ at a cost of 24 hours, leaving 16 hours for $x_B = 4$. The objective is $5(12) + 8(4) = 92$. In this small problem the optimum can be checked without a solver: any move that replaces two hours of A with two hours of B reduces benefit until the cap on A is reached, and extra B cannot be added without hours.

Proposition 5.1 (Optimality of the illustrative allocation)**.** *For the preceding continuous allocation problem,* $(12, 4)$ *maximises* $5x_A + 8x_B$ *and yields value 92.*

Proof. From the hour constraint, $x_B \leq 10 - x_A/2$. Hence $5x_A + 8x_B \leq 80 + x_A$. Since $x_A \leq 12$, the objective is at most 92. The point $(12, 4)$ is feasible and attains the bound. Thus it is optimal. □

What if the programme's priority is to serve at least five B clients? Then $x_B \geq 5$ is a legitimate new constraint; the optimum changes. Mathematical optimality is always relative to the stated objective and feasible set. It cannot certify that the original objective was ethically appropriate.

5.3 Duality explains the value of an extra unit

For the canonical minimisation form in (5.1), the dual is

$$\max_{y \geq 0}\ b^{\mathsf{T}} y, \qquad A^{\mathsf{T}} y \leq c.$$

Every dual feasible y gives a lower bound on the primal objective because $c^{\mathsf{T}} x \geq y^{\mathsf{T}} A x \geq y^{\mathsf{T}} b$ for feasible x, y. This elementary inequality is *weak duality*. Under the usual finite-dimensional linear-programming feasibility and finiteness conditions, strong duality states that the optimal values coincide (Boyd and Vandenberghe, 2004; Hillier and Lieberman, 2021).

Proposition 5.2 (Weak duality)**.** *If* $x \geq 0$*,* $Ax \geq b$*,* $y \geq 0$ *and* $A^{\mathsf{T}} y \leq c$*, then* $b^{\mathsf{T}} y \leq c^{\mathsf{T}} x$*.*

Proof. Multiply $Ax \geq b$ by the nonnegative vector y to obtain $y^{\mathsf{T}}Ax \geq y^{\mathsf{T}}b$. Multiply $A^{\mathsf{T}}y \leq c$ by $x \geq 0$ to obtain $x^{\mathsf{T}}A^{\mathsf{T}}y \leq x^{\mathsf{T}}c$. The shared middle expression yields the claim. □

A dual multiplier may be interpreted locally as a *shadow price*: the improvement in the optimised objective for a small relaxation of a constraint, while the current active set remains stable. It is not a market price, a promise of gain from a real investment, or a valid prediction for arbitrarily large changes. In the staff-hour example, the hour resource has local value two benefit units per hour while x_A remains capped and x_B has spare capacity. An additional hour can serve a quarter of a B client in the continuous relaxation, increasing value by two. If clients are indivisible, the marginal gain becomes discontinuous: one extra hour may change nothing until enough additional hours accumulate.

5.3.1 Integer decisions change what is feasible

For a facility-opening variable $y_j \in \{0, 1\}$, a continuous relaxation allows $y_j = 0.3$ and may produce an unrealistically attractive bound. The relaxation is valuable computationally because it supplies a lower bound for a minimisation model; it is not a deployable answer. Integer programmes require binary or integer variables, and the solver must report optimality gaps or time limits honestly. A feasible solution with a 4% gap is not a proof of global optimality unless the gap is zero within appropriate numerical tolerances.

5.4 Road networks: distance is not travel time

An urban delivery network can be represented by a directed graph $G = (V, E)$. Vertices stand for intersections or facilities; edges stand for navigable links. A shortest-path algorithm such as Dijkstra's algorithm finds the least-cost route when edge costs are nonnegative. What counts as cost is the modelling decision. Edge distance and edge travel time are not interchangeable. If one road floods, the best route may change even when its geometric length does not. Road geometry from OpenStreetMap gives useful network structure but cannot, by itself, establish actual wet-season passability or hourly travel speed (Dijkstra, 1959; Geofabrik, 2026).

For edge flows $f_e \geq 0$, a single-commodity network model imposes flow conservation at each node v:

$$\sum_{e \in \delta^+(v)} f_e - \sum_{e \in \delta^-(v)} f_e = b_v,$$

where b_v is net supply. Capacity bounds $0 \leq f_e \leq u_e$ protect links or conveyances. A water distribution or medicine-delivery system with losses, limited vehicles or time windows needs more constraints. The diagram is not the network: a missing bridge on the map or an overestimated carrying capacity can completely invalidate the proposed schedule.

5.5 The difference between locating a facility and assigning users

Suppose populations indexed by i must be served by candidate facilities j. Let p_i denote the size of population group i, d_{ij} a route-based travel cost, x_{ij} the fraction assigned and y_j whether a facility is opened. A capacitated location–allocation formulation is

$$\min_{x,y} \quad \sum_{i,j} p_i d_{ij} x_{ij} + \sum_j f_j y_j \tag{5.2}$$

$$\text{subject to} \quad \sum_j x_{ij} = 1 \quad (\forall i), \tag{5.3}$$

$$\sum_i p_i x_{ij} \le K_j y_j \quad (\forall j), \tag{5.4}$$

$$0 \le x_{ij} \le y_j, \quad y_j \in \{0,1\}, \quad \sum_j f_j y_j \le B. \tag{5.5}$$

If opening costs f_j appear in both the objective and budget, they are counted differently for different reasons: the objective includes a cost trade-off, while the budget imposes an affordability limit. Avoid inadvertently charging the same cost twice in a monetary total. If the policy fixes the number of new facilities rather than a monetary budget, use $\sum_j y_j \le k$.

Here K_j is service capacity in people, not an abstract score. The model treats p_i as divisible among facilities; if populations cannot realistically be divided, or the service requires exclusive assignment, binary x_{ij} may be preferable. A mathematical optimum may be inequitable if a few remote groups bear very large travel burdens. A constraint such as $\sum_{j:d_{ij}\le T} x_{ij} \ge \alpha_i$ for specified groups can represent a minimum fraction served within a travel limit T. The values T and α_i are policy choices that should be discussed with affected communities rather than presented as facts discovered by an optimiser.

Worked example 5.1 (Two clinics and one mobile team)**.** *For illustration, imagine three villages with populations* $(300, 200, 100)$ *and possible mobile-team sites* A, B*. Their travel minutes are* $(20, 40, 60)$ *to* A *and* $(50, 20, 15)$ *to* B*. If only one team can be based at a single site, the population-weighted travel costs are* $300(20) + 200(40) + 100(60) = 20{,}000$ *minutes for* A *and* $300(50) + 200(20) + 100(15) = 20{,}500$ *for* B*. Under this objective* A *wins by 500 weighted minutes. But village 3 faces a 60-minute journey to* A *rather than 15 minutes to* B*. Whether this is acceptable depends on travel-time equity and the service's clinical urgency; the weighted mean alone cannot answer it.*

5.6 Queueing and service capacity

An assignment model that sends more people to a clinic must consider the waiting time produced by that load. Suppose arrivals at a single service point follow a Poisson process at rate λ and independent service times are exponential at rate $\mu > \lambda$. In the elementary M/M/1 model, utilisation is $\rho = \lambda/\mu$ and mean time in the system is

$$W = \frac{1}{\mu - \lambda}.$$

As $\lambda \uparrow \mu$, the mean waiting time diverges. The formula makes the cost of operating near full utilisation vivid; it also depends on assumptions that may not fit scheduled appointments, interruptions or heterogeneous clinical tasks (Gross et al., 2008; Kleinrock, 1975). If the arrival process is highly variable or the staff member performs several duties, a more suitable queueing model or a simulation should be justified.

A practical facility-location problem is therefore joint: opening a new clinic may reduce travel time yet fail to reduce total time if staffing is insufficient. Optimising only road distance can shift the burden from transport to queues. More complex models couple route assignments and service capacity, but the additional variables demand data that an existing clinic register may not contain. Start with a defensible baseline and expose the missing information before expanding the model.

5.7 Planning when demand and infrastructure are uncertain

A single demand estimate, solar forecast or travel-speed assumption can produce a brittle plan. A scenario set $\mathcal{S}$ supplies plausible alternatives s with objective $C(x, s)$. Expected-cost optimisation uses probabilities π_s and minimises $\sum_s \pi_s C(x, s)$, while a worst-case plan minimises $\max_{s \in \mathcal{S}} C(x, s)$. These are distinct notions of risk. An expected-cost solution can have rare catastrophic outcomes; a worst-case plan can be expensive if the uncertainty set contains impossible combinations. Neither should be called "robust" without defining the set of circumstances and how they were selected (Bertsimas and Sim, 2004; Ben-Tal et al., 2009).

A chance constraint $\Pr\{g(x, \xi) \leq 0\} \geq 1 - \alpha$ requires a probability model for ξ. A scenario approximation based on ten invented disruptions does not establish a 99% reliability guarantee. If historical road closures or supplier delays are limited, consider transparent sensitivity analysis and state the range of assumptions rather than attaching unwarranted probabilities.

Proposition 5.3 (The nominal optimum can be fragile)**.** *Consider two actions with costs* $C(A, 1) = 1$, $C(A, 2) = 100$, $C(B, 1) = 10$, *and* $C(B, 2) = 11$. *Under scenario 1 alone, A is optimal, but under a worst-case criterion over scenarios 1 and 2, B is optimal.*

Proof. Under scenario 1, $1 < 10$. Under worst-case evaluation, $\max\{1, 100\} = 100$ and $\max\{10, 11\} = 11$. Therefore B has the lower worst-case cost. □

This invented counterexample is intentionally tiny. It shows that a model can be solved flawlessly and still give a poor recommendation if uncertainty has been suppressed. It does not tell us whether the worst-case criterion is the right policy in any actual district.

5.8 An honest health-facility access laboratory

The book's main location exercise uses geographic candidates from Uganda's OpenStreetMap extract, contributed facility records such as Healthsites.io and an explicitly dated population surface such as WorldPop (Geofabrik, 2026; Healthsites.io, 2026; WorldPop, 2026). The dataset must state which facilities are in the source and which are verified operational. A reported health-site point is not evidence that staff or essential services are available there. Population

rasters are model-based estimates, not household enumeration at each pixel. Travel-speed attributes need checking; otherwise define several labelled scenarios.

The first stage is deliberately modest: find the nearest reachable candidate site for each population cell over the observed road graph, comparing weighted straight-line distance with network distance. The second stage imposes a limit on new facilities and plausible capacities. The third tests whether the decision survives changes in travel speeds, missing road connections and candidate closures. This order reveals exactly what each layer of complexity contributes.

Table 5.1: Questions to answer before claiming a facility-location result

Check	Why it matters
Are facilities operational?	A mapped building may not offer the intended service.
Are populations and facilities time-aligned?	Settlement patterns and capacities change.
Is route travel time measured?	Distance and assumed speeds do not establish observed journeys.
Are referral and urgency rules included?	Nearest is not always clinically appropriate.
Are candidate sites legally and physically feasible?	A site can be mathematically optimal but impossible to acquire.
Are worst-served communities reported?	Means can conceal concentrated harm.

5.9 Worked investigation: verify an optimiser without trusting it blindly

For a small problem, exhaustive enumeration can establish a reference answer. Consider candidate sites A, B, C and a budget for one. Calculate the total assignment cost for each candidate with the same cost table and constraints. The minimum is an exact discrete optimum. Then solve with a mixed-integer optimiser and verify the selected site's value against that table. Next increase the budget to two facilities and enumerate all three two-site subsets. A solver returning a lower cost than exhaustive enumeration indicates a mistake in the model, input data or comparison procedure. This is software verification, not validation of the road speeds.

The companion programme `code/ch05_allocation.py` reproduces the continuous staff-hour example, exhaustively evaluates a small facility-location illustration and checks every returned decision against the capacity and selection constraints. It intentionally requires no commercial solver. For larger real networks, the exercise can be extended using a documented open-source mixed-integer solver, but the book's claims should remain limited to the problem actually solved and checked.

5.10 The decision record is part of the result

A planner should not receive only a map and a number. A useful decision record identifies the recommended action, the objective, hard constraints, alternatives considered, uncertainty

assumptions, feasibility checks, required data and the consequences of being wrong. A local planner may reject a mathematically preferred location because community land agreements, referral patterns or financing make it infeasible. Such a rejection is information about the model's boundary, not evidence that the mathematics was useless.

5.11 Feasibility comes before optimality

In practice, one of the first questions about an optimisation problem is whether any feasible decision exists. A solver that reports infeasibility is not necessarily malfunctioning. It may be telling us that the stated requirements are mutually incompatible. Suppose a district requires at least 300 visits from a team that can conduct only 200 in the available working days. If all assumptions are true, no clever allocation can meet the target. The useful result may be identifying the smallest change to staffing, time or service expectations that restores feasibility.

A simple diagnostic introduces nonnegative slack variables s_i and minimises their weighted sum subject to relaxed constraints. For example, if $Ax \geq b$ cannot be satisfied, solve $\min_{x,s\geq 0} \sum_i w_i s_i$ subject to $Ax + s \geq b$. The weights encode priorities: relaxing a non-negotiable clinical safety condition should not be treated as equivalent to delaying a routine administrative task. The diagnostic provides a transparent account of incompatibility, but it is not itself approval to violate a requirement.

5.11.1 A feasible decision can still be numerically fragile

Optimisers rely on floating-point arithmetic. If a capacity constraint is satisfied within a solver's tolerance, the result may still be marginally infeasible when rounded to whole staff shifts or medicine packs. For any proposed x, independently calculate residuals $r = Ax - b$ and compare them with tolerances appropriate to physical units. A violation of 10^{-6} litres may be irrelevant in a reservoir model; a violation of one full vehicle or a medicine cold-storage threshold may not be. The checking programme should not use the same mistaken conversion routine that created the original optimisation model.

5.12 Why dual prices are local descriptions, not permanent promises

Consider a simple facility investment with two binding constraints: staff-hours and vehicle capacity. If one additional vehicle has no use without extra staff, the shadow price for vehicle capacity may be zero. After staff capacity increases, vehicle capacity may become valuable. This does not contradict duality; it reveals a change in which constraints are active. A common misuse of optimisation is to multiply a local shadow price by a huge proposed budget expansion and claim that amount as the gain. The local approximation need not remain valid when another constraint becomes active.

For a minimisation programme with right-hand side b, write its optimal value as $V(b)$. Under suitable regularity and an unchanged optimal basis, the dual vector y^* gives a local directional sensitivity: $V(b + \Delta b) \approx V(b) + y^{*\mathsf{T}} \Delta b$ (with signs chosen consistently for the constraint

orientation). A full calculation should reoptimise after a meaningful change. The mathematics indicates where a new resource might matter, but the real cost of supplying that resource must be added before recommending action.

5.13 A richer road-network counterexample

Imagine a driver choosing between roads A and B between the same two sites. Road A is 12 kilometres and is assumed to permit 20 kilometres per hour. Road B is 18 kilometres and permits 45 kilometres per hour. Their nominal travel times are 36 and 24 minutes, respectively. The shorter road is slower. If heavy rain reduces the safe speed on road B to 15 kilometres per hour, its time becomes 72 minutes and the preferred route reverses. These values are invented, but the reversal is a reliable consequence of $t = d/v$ under the stated scenario.

Now suppose the rainfall forecast is uncertain. If B is slower only during severe rain, one needs the frequency and operational consequences of that condition to choose an expected-time route. A route that minimises average travel time may be unreliable for urgent medical supplies, where the worst delays matter more than small reductions in the mean. An appropriate objective might be the 90th percentile of delivery time or expected cost plus a penalty for late arrival. Such a decision can be implemented only if the relevant risk is measured or if scenario assumptions are openly presented.

5.14 Comparing location decisions through a sequence of models

Rather than building the full location problem in one step, an instructor can show students how assumptions change results. In a first model, assign each village to the nearest functioning facility by verified road distance. In a second, use travel time and plausible road speeds. In a third, add facility capacities. In a fourth, allow opening one additional site. In a fifth, add an explicit equity requirement for remote populations. Keep the same candidate set and data for every comparison so that the effect of each change is visible.

At every stage ask what new information was required. Switching from distance to travel time requires speeds, surface types or observations. Adding capacities requires service-specific facility information, not simply building areas. Adding a new site requires a catalogue of physically feasible candidates. Adding equity requires a public choice about acceptable disadvantage. A model is richer when it answers a more important question under defensible assumptions, not merely when it has more variables.

The resulting report can be organised around four maps: existing accessibility, predicted access after nearest-site assignment, proposed allocation under capacity limits, and locations where the choice changes under plausible disruptions. The report should include people for whom travel time increases, not only people for whom it decreases. A numerical objective without a distribution of effects can hide an intervention’s unequal burden.

5.15 An allocation problem with uncertain capacity

Suppose two vehicles nominally have capacities 50 and 40 boxes, while total demand is 85 boxes. The nominal plan is feasible if no capacity is lost. If wet-road conditions reduce the second vehicle's usable capacity to 25, total capacity becomes 75 and the plan is infeasible. An optimisation model that assumes the nominal capacity in all periods cannot guarantee delivery of all 85 boxes. It can make a defensible plan only by including extra trips, reserve capacity, alternate transport or accepted unmet delivery in its constraints and objective.

One possible two-stage formulation chooses initial resource decisions x before knowing the disruption and recourse decisions y_s after observing scenario s:

$$\min_{x,\{y_s\}} \; c^{\mathsf{T}}x + \sum_{s\in\mathcal{S}} \pi_s q_s^{\mathsf{T}} y_s, \quad T_s x + W_s y_s \geq h_s \quad (\forall s).$$

Here x may represent advance vehicle booking while y_s represents emergency rescheduling. Recourse is meaningful only if a second-stage action can actually occur when disruption is known. A road closure discovered after the delivery deadline provides no useful opportunity to hire extra transport. Timing belongs in the feasible set.

5.16 Computation and explanation for the final decision

For a large integer programme, record solver name, version, tolerances, elapsed time, thread count and an optimality gap. Test a small instance by exhaustive enumeration, then perform a constraint audit on the large solution. The number of nodes considered or iterations completed is not a measure of policy benefit. Report both the mathematical objective and the operational outcome; a model that reduces travel minutes but increases missed urgent deliveries has not necessarily improved the service.

Where possible, show a transparent baseline next to the optimised recommendation. The baseline may be the current allocation, nearest-site rule or a simple greedy heuristic. If the advanced optimisation improves little over the baseline, that finding is valuable. It may show that there is no need to maintain a complex tool. If it improves substantially only under one travel-speed assumption, the next investment may be better measurement rather than new software.

5.17 Extended case study: one district, a limited service budget

Consider a district planning exercise in which a mobile service team must visit settlements separated by difficult roads. We imagine the planner has a basic road graph, a list of candidate service points and approximate population totals. The budget permits only one new service point. This is a plausible type of decision, but all numerical values in this investigation are illustrative; no actual site is being recommended.

The planner first needs to distinguish the purpose of a visit from the number of visitors. A routine vaccination session, supply delivery and emergency referral have different constraints. A location suitable for an occasional mobile clinic may be unsuitable for an ambulance service,

even if the same road graph is used. Define the service, the minimum operating standard and the time horizon before calculating travel cost. “Closest hospital” is not necessarily meaningful without specifying which services are available at each facility.

5.17.1 Stage one: build an auditable baseline

Create a table of settlements with stable identifiers, population counts, coordinates and geographic source dates. Build another table of operational facilities, with location and verified services. Construct the road network and calculate route costs with a documented algorithm. If a settlement is disconnected in the mapped road graph, record that fact rather than silently assigning straight-line travel distance. Where transport modes vary, declare a separate cost model for each mode.

The most basic decision is to assign every settlement to a reachable existing facility. Let $d_i^{(0)} = \min_{j \in F_0} d_{ij}$ denote travel cost from settlement i to the nearest functioning facility in the baseline set F_0. An average accessibility score is $\overline{d} = \sum_i p_i d_i^{(0)} / \sum_i p_i$. This is population-weighted travel cost, not an observed patient journey time. It depends on the assumptions used to turn roads into travel times. The baseline should also report the largest distances and the number of people beyond a chosen threshold.

5.17.2 Stage two: test a single proposed location

For each candidate j not already open, calculate the new minimum travel cost $d_i^{(j)} = \min\{d_i^{(0)}, d_{ij}\}$. The weighted improvement is $\Delta_j = \sum_i p_i (d_i^{(0)} - d_i^{(j)})$. The best site under this one-dimensional objective maximises Δ_j, subject to feasibility. The calculation can be completed by enumerating every candidate without sophisticated software. This is a useful exact benchmark against which more elaborate location models should be checked.

Now add an operational cost f_j. If the budget permits only one new site with $f_j \leq B$, the feasible candidates may shrink. Next consider capacity. An apparently well-located site that can serve 40 clients per week cannot automatically absorb 600 clients simply because it is closest. If demand is concentrated in one settlement, a capacity-limited assignment may send some clients farther away. The site that maximises travel savings without capacity can cease to be optimal when staffing is considered.

5.17.3 Stage three: introduce equity without hiding the trade-off

An average travel-time objective can favour dense central settlements. To see why, imagine one village of 1,000 residents and a remote village of 40 residents. Reducing the large village’s travel time by ten minutes yields 10,000 weighted minutes saved. Reducing the remote village’s time by 100 minutes yields 4,000. The mean-based objective favours the large village, even though the remote residents face a much greater journey. This preference follows from the chosen objective, not from an empirical finding that the second group’s burden is acceptable.

A planner can represent equity explicitly. One option minimises the maximum travel time; another imposes $d_i \leq T$ for settlements with designated priority. A third uses a weighted

objective with additional priority weights, but those weights require open discussion. A mathematically optimal plan can be sensitive to the chosen representation of fairness. The result should present the alternatives rather than declare one universal definition of an equitable allocation.

5.17.4 Stage four: explain infeasibility to a decision-maker

Suppose the district has only one ambulance and policy requires all emergency patients to reach a referral centre within 30 minutes. The road network shows a settlement whose minimum travel time is 80 minutes even with the best available candidate site. No assignment algorithm can satisfy the stated target. The correct result is infeasibility, together with options for what additional resources or infrastructure would be required. One might need a nearer emergency service, an additional vehicle, improved road access or a revised referral strategy. Choosing a different solver is not a solution.

5.17.5 Stage five: account for disruption

Assume one bridge becomes impassable during some periods. Remove its edge from the network and recompute the assignments. Repeat for other plausible closures or speed changes. A robust decision could minimise maximum travel cost over the designated scenarios or satisfy minimum access in each scenario. That decision may differ from the nominal optimum. The price of robustness is the additional cost or lost nominal performance accepted to reduce vulnerability. It should be reported, not hidden behind a label such as “climate-smart.”

The scenario exercise is useful even if historical closure probabilities are unavailable. It shows which decisions depend critically on a single uncertain road link. It does not establish how often that link fails. If better observations would alter the preferred plan, the project may justify collecting road-speed and passability records before investing in a new facility.

5.18 Mathematical detail: a small exact facility-location method

Suppose a set J has m candidates and exactly k may be opened. The brute-force method evaluates $\binom{m}{k}$ candidate subsets. For each subset S, compute $C(S) = \sum_i p_i \min_{j \in S} d_{ij}$. Choose an admissible subset attaining the minimum. This method is computationally expensive for large m, but it is exact for a small benchmark. A student should understand what the enumeration proves before accepting an integer solver’s certificate.

The code companion enumerates subsets of three candidate sites for one-site and two-site plans, reporting every objective. It is important that the objective table includes *all* admissible subsets, not only the winning one. This makes the result independently checkable. Extending the code to capacities requires changing the assignment procedure; merely adding capacity values to a printed table does not enforce them.

5.19 Network errors and their decision consequences

If a bridge is missing from the mapped graph, a route may appear to require a long detour. If a river crossing is incorrectly mapped as a driveable road, travel time may be underestimated. These errors have directional consequences for investment decisions. It can be useful to classify uncertain edges by how often they are used in optimal solutions. An uncertain bridge that changes the selected facility under plausible interpretations deserves higher-priority verification than a road segment never used by any feasible route.

This suggests a practical, decision-oriented data-audit procedure: perturb uncertain map attributes within justified ranges, recompute the decision, and rank the information gaps by whether they change the chosen action. If the optimal location is stable across all considered assumptions, elaborate additional surveying may not be immediately necessary for that decision. If the recommendation changes repeatedly, more data could have substantial value. This is a stronger argument for data collection than counting map features with missing labels.

5.20 A policy briefing that exposes uncertainty

A final briefing should show the current access pattern, the proposed choice, its resource requirements and the people who do and do not benefit. It should include nominal and disrupted-route results, operational caveats, and what site verification remains before investment. Claims such as “travel time reduced by 28%” must identify whether the time was measured from journeys or modelled from road geometry and assumed speeds. The numerical precision of a solver does not determine the precision of those inputs.

A planner who decides not to invest immediately may still have benefited from the analysis. The work could reveal that a proposed facility is infeasible because of staffing limits, or that a cheap road verification survey is more valuable than another optimisation run. A sound engineering book should teach readers to recognise that outcome as a legitimate solution to the current problem.

5.21 Mathematical workshop: reading a constrained solution

Optimisation models can feel like collections of inequalities until the reader learns to interpret each constraint as a real operational promise. Consider a two-variable allocation with unit costs c_1, c_2 and resource limits. The feasible region is the intersection of half-spaces. In two dimensions it can be drawn, but in larger problems its geometry is encoded by the matrix A. The model is not complete until every coefficient has a unit: hours per client, shillings per item, litres per day or kilometres per trip. A coefficient that adds hours to litres does not become meaningful because a computer accepted the matrix.

5.21.1 A dual certificate of optimality

Return to the illustrative programme with $\max 5x_A + 8x_B$ subject to $2x_A + 4x_B \le 40$, $x_A \le 12$, $x_B \le 8$, $x \ge 0$. A simple upper bound arises from the hours constraint:

$$8x_B \le 80 - 4x_A, \qquad 5x_A + 8x_B \le 80 + x_A \le 92.$$

The candidate $(12, 4)$ attains 92, so the bound certifies optimality without reliance on solver output. This kind of certificate is important pedagogically. A numerical programme should be checked against mathematics on small instances so that students understand what it means to trust an optimisation result.

The bounding argument also reveals shadow prices. The objective bound was constructed by assigning price two to the hour resource and price one to the cap on service A. Indeed, $2(2x_A+4x_B)+1(x_A) = 5x_A+8x_B$, and the corresponding resource values give $2(40)+1(12) = 92$. These nonnegative prices form a dual feasible certificate. Each is a local marginal interpretation. If the hour allocation increases enough for the B cap to bind, these dual prices need not remain valid.

5.21.2 Integrality creates a different kind of difficulty

If clients are counted as indivisible people, x_A and x_B should be integers. In the example, the continuous optimum happens to use integers, so no difference arises. That coincidence should not be assumed generally. Consider $\max x$ subject to $2x \le 3$ and $x \ge 0$. The continuous optimum is $x = 1.5$. If x counts vehicles, the integer optimum is $x = 1$. The relaxation's value 1.5 remains an upper bound for the integer maximisation, but it is not a feasible operating plan.

For facility-location models, this gap can be substantial. An integer solver may use branch-and-bound or cutting planes to prove that no undiscovered integer solution can improve the incumbent beyond a stated bound. A solution marked "feasible" has a different status from one marked "optimal." Teaching students to read that distinction is as important as teaching them how to call the solver.

5.22 A short network-flow derivation

Let B be the node-edge incidence matrix of a directed network, with one $+1$ at the tail and one -1 at the head of each directed edge. A flow f must satisfy $Bf = b$ where b records net injection at each node. Summing all node equations gives $\mathbf{1}^{\mathsf{T}}Bf = \mathbf{1}^{\mathsf{T}}b$. Because every edge contributes one $+1$ and one -1, $\mathbf{1}^{\mathsf{T}}B = 0$ and therefore $\sum_v b_v = 0$ is necessary for feasibility. Total supply must equal total demand in the simplified lossless network.

Proposition 5.4 (Aggregate conservation)**.** *A flow satisfying $Bf = b$ can exist only if $\sum_v b_v = 0$.*

Proof. Multiplying $Bf = b$ by $\mathbf{1}^{\mathsf{T}}$ gives $\mathbf{1}^{\mathsf{T}}Bf = \mathbf{1}^{\mathsf{T}}b$. Every column of B sums to zero, so the left side is zero. Hence the right side must be zero. □

In a water-distribution model with leakage, the loss must be represented as an additional sink or accounted for in the balance. Otherwise a solver may report infeasibility for a system that is

physically possible, or report a mathematically feasible plan that ignores lost water. Physical conservation is a structural property that can reveal modelling mistakes early.

5.23 Why shortest path does not solve vehicle routing

A shortest-path problem asks for one low-cost path from a source to a destination. A vehicle-routing problem may require visiting several customers, observing vehicle capacity, returning to a depot and respecting deadlines. Even when every pairwise shortest path is known, finding the best visiting order is a separate combinatorial problem. A small delivery cooperative may save substantial time by changing its visit sequence, but those savings cannot be inferred from a point-to-point route alone.

An elementary illustration has depot O and deliveries A, B, C. Suppose symmetric travel times are $d(O, A) = 5$, $d(O, B) = 6$, $d(O, C) = 7$, $d(A, B) = 3$, $d(A, C) = 8$, $d(B, C) = 4$. The route $O \to A \to B \to C \to O$ takes $5 + 3 + 4 + 7 = 19$ time units. The route $O \to C \to A \to B \to O$ takes $7 + 8 + 3 + 6 = 24$. Enumerating every tour establishes the best route in this tiny instance. For realistic networks, more efficient algorithms and carefully evaluated heuristics are necessary.

The model also needs demand quantities and loading rules. A vehicle that can carry only 20 boxes cannot necessarily serve three customers needing 15 each on one tour. Its route may require returning for more stock, using another vehicle or changing delivery dates. Distance optimisation without capacity and operational scheduling gives incomplete advice.

5.24 Robustness as a family of tests, not a single algorithm

An uncertain parameter may be continuous, such as demand between 80 and 120, or discrete, such as a bridge being open or closed. Robust optimisation typically seeks decisions that remain feasible over a specified uncertainty set. The size and shape of this set matter as much as the solver. If the set is too small, the solution may fail under plausible conditions. If too large, it may be excessively costly. An uncertainty set should be based on evidence or transparently selected planning scenarios rather than invented after the optimisation to produce a preferred result.

For a constraint $a^{\mathsf{T}} x \le b$ with uncertain a lying in a box $\{\bar{a} + u : |u_j| \le \delta_j\}$ and nonnegative x, robust feasibility requires

$$\bar{a}^{\mathsf{T}} x + \sum_{j} \delta_j x_j \le b.$$

Why? The worst permitted perturbation chooses $u_j = \delta_j$ for each nonnegative decision component. The derivation is elementary, yet it makes the conservatism visible: the model protects against every coefficient taking its worst value at once. If such simultaneous extremes are physically implausible, a less conservative uncertainty set may be justified (Bertsimas and Sim, 2004; Ben-Tal et al., 2009).

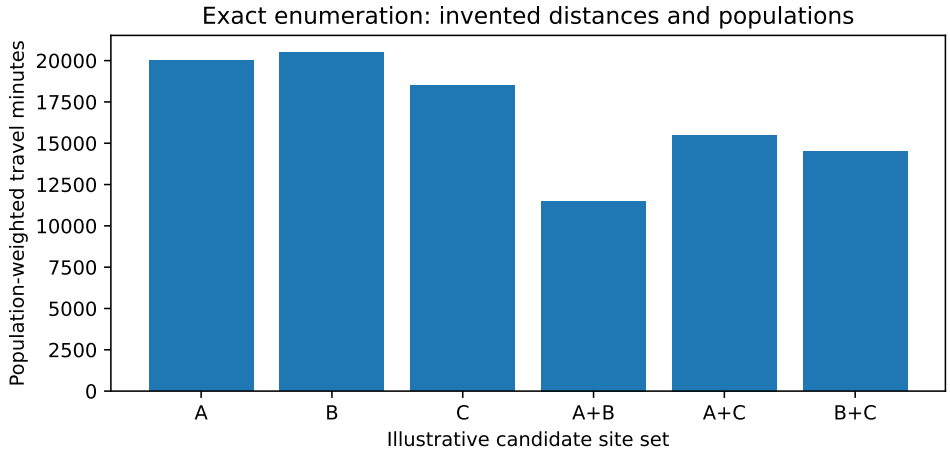


Figure 5.1: An exhaustive comparison on an invented three-candidate problem. The visual shows how the cost falls when two sites may be opened; it is not a cost estimate for an actual Ugandan district.

5.25 How to present an optimisation result responsibly

The final decision note should give the objective in interpretable units, the key constraints, a feasible baseline, the optimised choice, the best available bound, and the sensitivity of the result to uncertain parameters. It should report the communities or clients experiencing worse outcomes under the plan as well as those who benefit. It should separate quantities observed in the world from those supplied as policy assumptions. If the preferred location changes when a single unverified road-speed assumption changes, the conclusion should explain this weakness openly.

The strongest practical recommendation may therefore be an information-gathering action: check whether a bridge is passable, confirm a clinic's actual capacity, or measure vehicle loading times. The optimisation may show that this missing observation is the pivotal factor. That is a rigorous and useful outcome, even before a new road, vehicle or facility is purchased.

5.26 Laboratory extension: building a trustworthy transport-cost matrix

In a real location model, the travel-cost matrix $D = (d_{ij})$ is as important as the optimisation algorithm. It encodes assumptions about roads, speeds and access that determine which communities appear close to a service. Begin with a dated road graph and a dated set of population or village points. Confirm that both use compatible coordinates. Snap each population point and candidate facility to appropriate reachable road nodes; record the snapping distance. A village several kilometres from the nearest mapped road should be flagged, not assigned an unrealistically short route simply because its coordinate was rounded to the road.

For each population point i, calculate travel cost to each candidate j. If no route exists in the supplied network, mark $d_{ij} = \infty$ or record the pair as unreachable; do not invent a straight-line

fallback without explicitly treating it as a different scenario. Review a sample of routes on the ground with a partner. Geometry may contain roads that cannot be traversed by the intended vehicle, especially after heavy rain.

Once the matrix is constructed, check a few entries by hand. Does a 10-kilometre segment at 20 km/h contribute 30 minutes? Are one-way roads respected? Does an apparent shortcut cross a river without a bridge? Are walking paths being treated as vehicle routes? Small systematic errors in these calculations can dominate the supposed gain from choosing one optimisation method over another.

5.26.1 Why optimistic road speeds can distort facility choices

Suppose one candidate site is close to most villages by rough tracks, while another is farther away by well-maintained roads. If all roads are assigned the same speed, the first site may look better. Under vehicle-specific speeds, the second may have lower travel time and more reliable access. Conduct at least three sensitivity scenarios: documented conservative speeds, plausible nominal speeds and disruptive conditions where selected links are unusable. Report when the selected site changes.

The result should be a stability table with one row per scenario, preferred sites, weighted travel time, worst-served group and feasibility violations. If the selected site changes in every scenario, the recommendation is contingent on information that needs verification. A robust location analysis is not merely a solver producing an answer despite uncertainty; it identifies which uncertainty matters to the decision.

5.27 A mathematical comparison of mean access and worst-case access

For a selected facility set S, let $d_i(S)$ denote the travel time for settlement i. A mean-cost objective is $M(S) = \sum_i p_i d_i(S) / \sum_i p_i$. A worst-access objective is $W(S) = \max_i d_i(S)$. These criteria generally order alternatives differently. The mean criterion reflects population totals, while the maximum gives the most remote settlement a dominant role. Neither is a universally correct measure of equity.

Consider two possible plans serving three equally weighted villages. Plan A produces travel times $(10, 10, 60)$ minutes, so mean time is $80/3 \approx 26.7$ and maximum is 60. Plan B produces $(30, 30, 30)$ minutes, giving mean 30 and maximum 30. A mean objective prefers A, while the worst-case objective prefers B. The example shows how an implicit ethical choice appears as a mathematical objective. A public policy report should make this difference visible rather than announce a single mathematically optimal site.

5.27.1 A compromise with an explicit maximum travel constraint

A planner may minimise population-weighted mean time subject to $d_i(S) \le T$ for every priority village. This combines efficiency with an access guarantee. Its feasibility depends on the threshold T and candidate network. If no candidate set meets the threshold within budget, the

model should return infeasible and identify which areas fail. The honest response may be to seek funds for a second mobile team or improve a critical road link.

A further refinement is to report the fraction of population within 30, 60 and 90 minutes rather than optimise one arbitrary threshold. These descriptive curves let decision-makers see how an intervention affects different access standards. The thresholds should follow service needs and policy guidance where available; they are not learned from the route algorithm.

5.28 Cost accounting beyond a single objective

A facility may have construction, staffing, transport, maintenance and energy costs. The cheapest site to build may have high recurring operating costs, and a remote location may require expensive backup power. A multiyear present-value calculation can represent recurring costs under a specified discount rate r:

$$C_{\mathrm{PV}} = C_0 + \sum_{t=1}^{T} \frac{C_t}{(1+r)^t}.$$

The discount rate, time horizon and price assumptions must be declared. This calculation does not automatically value improvements in health or access; those outcomes may be reported alongside costs rather than reduced to a disputed monetary equivalent.

If the planning office is financially constrained today but expects future funding, a present-value optimum may violate the immediate cash budget. Include annual expenditure constraints as well as a long-run cost objective when necessary. The mathematical model should represent how the organisation actually receives and spends resources, not how a hypothetical unconstrained investor would act.

5.29 Field questions that mathematics cannot answer alone

A map may show an optimal facility location on private land. The chosen site may lack clean water or have unsuitable ground conditions. Residents may prefer a different location for reasons not represented by travel time. Health workers may need accommodation or security. These are not “soft” details to dismiss after optimisation; they determine feasibility and acceptance. A practical planning process uses mathematics to compare declared possibilities, then returns to affected users and specialists to verify what the model could not know.

5.30 Exercises and investigations

1. Derive the dual of a two-variable minimisation programme of your own design. Verify weak duality for one feasible primal-dual pair.
2. Show how the optimal staff-hour allocation changes if the maximum number of A clients is reduced to ten. Explain when the continuous relaxation ceases to be a meaningful implementation.

3. Construct a four-node directed network in which the shortest-distance path has greater travel time than a longer path. State all edge costs and verify both paths by hand.
4. Explain why $\sum_j x_{ij} = 1$ is insufficient to represent a capacitated facility network. Supply the missing constraints and units.
5. In the location example, introduce an equity rule limiting each village's travel time to 45 minutes. Identify whether either single-site solution remains feasible.
6. Derive the M/M/1 waiting time when arrival rate is 7 per hour and service rate is 10 per hour. What assumption is broken when arrival rate reaches 10?
7. Compare expected-cost, worst-case and chance-constrained decisions for two invented road-closure scenarios. Explain what additional evidence would justify assigning scenario probabilities.
8. **Field investigation.** Obtain an approved road dataset and a documented facility list. Reconcile the geographic coordinate systems, flag disconnected sites, and write a one-page explanation of what remains unverified.

What this chapter establishes. Optimisation can clarify trade-offs and reveal resource bottlenecks. Its guarantees refer to the mathematical model and stated constraints. Whether the recommended action works depends on data, implementation, people and uncertain conditions. The next chapter asks when learning algorithms add enough value to justify their cost.

Chapter 6

Machine Learning and AI That Earn Their Place

When should a learnt model replace a simpler rule? The answer is not "when its accuracy is higher" without qualifications. We develop statistical learning, honest evaluation, probability calibration, decision-sensitive comparison, distribution shift and retrieval-based language assistance. The central test is whether added complexity produces a useful, defensible improvement.

6.1 Start with what the system must decide

A programme manager may be offered an AI system that predicts medicine shortages or daily pollution. The proposed system may require a computer, connectivity, software updates, labelled data and staff who know when to question its outputs. A simpler alternative may be a published threshold rule or a transparent forecast. It is not enough to ask which model has a lower reported error. We must ask what information each consumes, what failures it creates and whether the decision made with it is better than the decision made without it.

Let (X_i, Y_i) denote training observations, let $f \in \mathcal{F}$ be a prediction function and let ℓ be a loss. Empirical risk minimisation chooses

$$\widehat{f} \in \arg\min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} \ell(f(X_i), Y_i) + \lambda \Omega(f). \tag{6.1}$$

The first term measures fit to the observed sample, the second penalises complexity, and the parameter λ controls the trade-off. A good training score is not a statement about future performance. Generalisation requires a relationship between the training observations and the situations in which predictions will be used (Vapnik, 1998; Hastie et al., 2009; Bishop, 2006).

6.2 Training error can be misleading

Imagine a model that memorises the average $PM_{2.5}$ value for each monitoring station. If the training and testing sets contain different hours from the same stations, its test performance

may be impressive. Ask it to predict at a newly installed station, and its stored averages are useless. The problem is not simply "overfitting" in the narrow sense of memorising individual rows. The assessment tested a different task from the one the model was proposed to perform.

Similarly, an algorithm may learn the future through a feature that was recorded only after an event. Suppose the outcome is whether an electricity fault will occur by Friday, but one predictor is the repair ticket created on Saturday. The data table may make this predictor statistically powerful, but it is unavailable at the time of action. Leakage is therefore best defined relative to the actual information set $\mathcal{I}_t$ available at decision time t, not just a rule about columns named "target".

A sensible evaluation begins with the simplest useful benchmark. For regression this may be a constant mean, persistence forecast, linear model or inverse-distance weighted estimate. For classification it may be a published rule or a constant prevalence-based probability. If a sophisticated model does not consistently beat these benchmarks on independent observations and a decision-relevant outcome, further complexity is difficult to justify.

6.2.1 Validation belongs inside the design

A model comparison requires data for fitting, data for choosing hyperparameters, and data that remain untouched until final assessment. It is acceptable to split by groups or time rather than by arbitrary rows. Every operation that learns from data, including normalisation, imputation, feature selection, threshold tuning and interval calibration, must be fitted on the appropriate training portion only. For spatial observations, station-blocked validation is critical; for repeated households, split by household; for forecast operations, preserve time order (Roberts et al., 2017; Sculley et al., 2015).

A common error is to report the best score among dozens of models tried repeatedly on the same test set. Once the test set has been used to select a model, it is no longer an independent final evaluation. A reproducible project documents the model-search procedure and reserves a final assessment that does not influence design. When the available dataset is too small to support all these divisions, be explicit about the uncertainty rather than claiming to have solved it.

6.3 From regression to trees, forests and boosted models

The simplest parametric predictor specifies a functional form, such as a linear relationship between rainfall, land use and an outcome. A decision tree instead partitions the predictor space and fits simple values in each region. A tree can discover threshold-like interactions, but a deep single tree is often unstable: a small change to the training data can substantially alter its branches.

Random forests train many randomised trees and average their predictions. Their appeal is variance reduction and flexibility, not a guarantee of correct extrapolation outside the training range (Breiman, 2001). Gradient boosting builds an additive model in stages, each stage improving a chosen loss relative to the current predictor. Modern implementations can be powerful on structured data, but hyperparameter search, missing-value handling, calibration

and feature leakage remain necessary concerns. Foundational accounts include Friedman (2001) and Chen and Guestrin (2016).

A generalised additive model takes an intermediate route:

$$g\{\mathbb{E}(Y \mid X)\} = \beta_0 + f_1(X_1) + \cdots + f_p(X_p),$$

where each f_j is a smooth function and g is an appropriate link. It can capture nonlinear associations while leaving a more inspectable decomposition than a large ensemble (Wood, 2017). But inspectability is not causal identification: a smooth curve can still summarise a confounded relationship.

6.4 Prediction uncertainty and the limits of probability labels

A probability reported as 0.8 is useful only if similar cases occur with the predicted frequency in the population where decisions are made. A classifier can rank patients or locations well while producing badly calibrated probabilities. Calibration can be assessed by grouping predictions and comparing mean forecast probabilities with observed event rates. Grouping is a visual approximation; sample size, uncertainty and the choice of bins matter.

Worked example 6.1 (A confident classifier that should not be trusted)**.** *Suppose, in an explicitly illustrative holdout sample, a model assigns a probability near* 0.9 *to 100 cases but only 60 cases experience the event. Its predictions are too confident in that group. A warning system using probabilities as if they were calibrated will misstate expected false-alarm and missed-event costs. Perfect ranking on that sample would not repair the problem. Probability calibration must be checked separately from discrimination (Guo et al., 2017).*

Post-hoc calibration, such as a fitted logistic mapping or isotonic regression, must be trained on calibration data that were not used to fit the underlying model. Even then, calibration learned in one setting can fail after a change in the sensor network, weather distribution, reporting rules or population. Ovadia et al. (2019) provides a systematic study of uncertainty under dataset shift. A trustworthy system should monitor change and have a procedure for withholding or qualifying outputs when its evaluation no longer applies.

6.5 The mathematical distinction between error and consequence

Consider a binary decision $a \in \{0, 1\}$, where $a = 1$ issues a warning and $Y = 1$ denotes an actual adverse event. Let false warnings cost $c_{10} > 0$ and missed events cost $c_{01} > 0$, with zero loss for correct decisions. Given a calibrated event probability p, the expected losses are

$$R(1 \mid p) = (1 - p)c_{10}, \qquad R(0 \mid p) = pc_{01}.$$

Choose a warning when $p > c_{10}/(c_{10} + c_{01})$. This reproduces the threshold derived in Chapter 4, but we now use it to compare algorithms. Two prediction systems should be assessed using the same predeclared consequences. The system with the lower average cross-entropy need not have the lower operating loss at a particular threshold, and the reverse can also occur.

Proposition 6.1 (Threshold sensitivity)**.** *Let $\tau = c_{10}/(c_{10} + c_{01})$. If two calibrated predictors produce probabilities p_1, p_2 for a case and both lie on the same side of τ, they recommend the same action under this loss, even if their numerical probabilities differ. If they lie on opposite sides, they recommend different actions.*

Proof. The decision rule depends only on the sign of $p - \tau$, obtained by subtracting the two conditional expected losses. The result follows immediately. □

This seemingly modest fact has a practical implication. Improving accuracy on cases far from an action threshold may change a reported average metric without changing a single decision. Errors close to the threshold can be much more consequential. It is therefore sensible to examine the costs and calibration of decisions near the threshold, provided the threshold and costs themselves are defensible.

6.5.1 Selective prediction and human review

A model need not force a decision for every input. It can abstain and refer uncertain cases to a qualified person. Let $a = \perp$ denote deferral with cost $c_\perp$; compare that with the expected losses of automatic actions. The optimal decision among the three minimises $\{pc_{01}, (1-p)c_{10}, c_\perp\}$. Real deferral is possible only if someone is available and capable of responding. A system that "refers" hundreds of warnings to one busy worker has moved its bottleneck rather than removed it.

6.6 Dataset shift, fairness and where the model stops working

A model trained in central Kampala may encounter different emission sources, road geometry, sensor types, weather patterns and building densities elsewhere. Formally the joint distribution $P_{\text{train}}(X, Y)$ may differ from $P_{\text{use}}(X, Y)$. Under *covariate shift*, $P(X)$ changes while $P(Y \mid X)$ stays constant; under *concept shift*, the conditional relationship changes. These definitions are useful distinctions, not conditions that can be verified from input covariates alone. If rural ground truth is unavailable, an observed change in $P(X)$ is a warning, not a calibrated rural error estimate.

Subgroup evaluation matters because good average results may conceal poor performance for areas with fewer sensors, unreliable electricity or different conditions. Report subgroup sample sizes and uncertainty. Do not create an apparently precise fairness score from three observations. The appropriate protected or operational subgroups depend on the application; environmental risk may require separation by monitoring density, distance from training monitors and pollution regime rather than importing group labels that do not relate to the decision.

6.7 Complexity must be measured in the intended environment

Software cost includes training time, inference latency, memory, storage, electricity if measured, data acquisition, update requirements and the time skilled staff need to diagnose errors. For a

device expected to operate during network outages, online server latency is only one dimension. A simple model that can be checked and rerun offline may be more useful than a complex model whose marginal accuracy improvement depends on a cloud service that is frequently unavailable.

A transparent comparison includes the number of variables, training observations, compute hardware, software versions, run repetitions and measurement procedure. If energy consumption is not metered or estimated through a validated tool, do not label a wall-clock time comparison as an energy study. The hidden costs of maintaining learnt systems are real engineering concerns, discussed in Sculley et al. (2015) and Kleppmann (2017).

6.8 A worked comparison: adding learning complexity to air-quality prediction

Return to the $PM_{2.5}$ problem in Chapter 4. Keep exactly the same held-out stations, time coverage, quality rules and outcome aggregation. Begin with a station-independent mean and inverse-distance weighting. Add a generalised additive model. Then consider random forests, boosted trees and residual kriging only when the necessary inputs exist. Each method must see the same permissible information, though methods may express that information differently.

Before training, write the protocol. The primary measure may be station-averaged MAE at withheld sites; secondary measures include RMSE, bias, interval coverage, distance-stratified error, prediction time and disk footprint. The preferred method should not be chosen solely by the smallest one-decimal-place improvement in RMSE, especially if it is slower, poorly calibrated or sensitive to missing covariates. Declare what size of improvement would actually change a planning decision.

When the outcome is a threshold exceedance, construct the warning decisions from each model's probability or predictive distribution and compare expected decision losses. A regression point forecast that exceeds the threshold is not equivalent to a calibrated exceedance probability. The analysis should show both prediction and decision performance, keeping these claims separate.

6.9 Generative AI: a small public-service information system

The book should not introduce language models with an unconstrained chatbot that dispenses health, agricultural or legal instructions. A safer and more teachable case begins with a finite collection of official public documents and a narrow purpose: retrieve the correct paragraph that answers a known information question, or state clearly that the documents do not contain an answer.

Compare four approaches on a common set of questions: keyword retrieval; semantic retrieval; a language model that answers from retrieved passages; and a language model without access to the documents. The last approach helps reveal why retrieved evidence matters, not why one should deploy a source-free system. Questions must include deliberately unanswerable cases. Construct a source-verified answer key before evaluating any method. Track whether a citation actually supports the sentence it accompanies.

Retrieval-augmented generation can be represented schematically as $q \rightarrow R(q, D) \rightarrow$

$G(q, R(q, D))$, where q is the user question, D the vetted document collection, R a retrieval rule, and G a generator. Each arrow can fail. Retrieval may miss the correct document; the generator may misread an accurate passage; a source may be outdated; a fluent answer may combine unsupported details. It is a mistake to infer that attaching citations makes every sentence true.

6.9.1 Evaluation before deployment

For each question, store the approved source, whether an answer exists, the expected content and what an appropriate refusal or referral looks like. Count factual accuracy, source support, invalid citations, unsupported assertions, abstention for missing answers, latency and running cost. Assess questions in the languages and reading styles the service will actually encounter. A human review of the first deployment is necessary because an apparently high accuracy rate may conceal dangerous errors concentrated in unusual questions.

Documentation such as dataset descriptions and model cards makes these assumptions visible to users and later maintainers (Gebru et al., 2021; Mitchell et al., 2019). An explanation method such as a local surrogate can help an analyst inspect a prediction, but an appealing explanation does not itself guarantee that the model is fair, causal or reliable (Ribeiro et al., 2016).

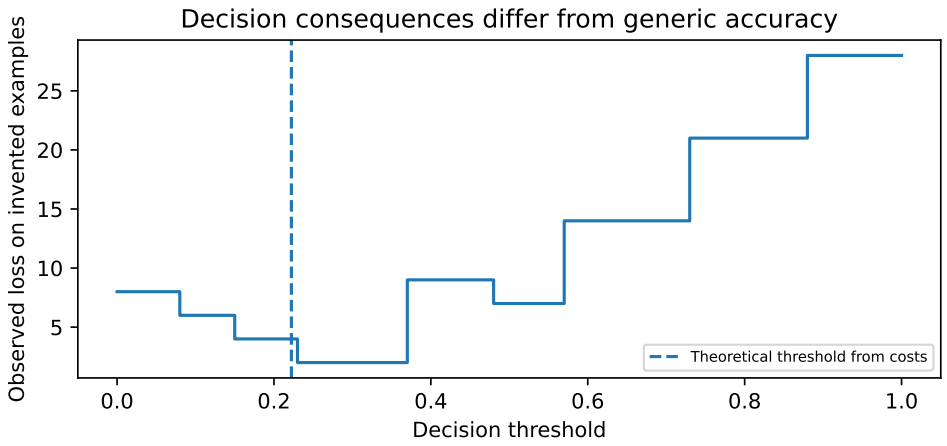


Figure 6.1: Observed losses for thresholds applied to eight invented binary outcomes. The dashed line is the cost-derived threshold, which need not minimise the realised loss on such a small sample; the mathematical rule minimises conditional expected loss for calibrated probabilities and specified costs.

6.10 The model audit the reader should be able to perform

A complete audit contains an intended-use statement, disallowed uses, training-data provenance, missing-data rules, leakage check, baseline, split method, uncertainty assessment, subgroup performance, compute cost, release version, update policy and incident pathway. The questions are not decorative compliance language. Each one corresponds to a failure that could change what an organisation should do. A model card written before a model has been evaluated is

only a plan; it must not report imagined performance numbers.

The companion `code/ch06_decision_threshold.py` illustrates threshold decisions with deliberately invented probabilities and outcomes. It computes false-positive and false-negative counts, expected loss and an abstention option. The programme makes no claim about measured medical or pollution outcomes. Its purpose is to let readers test the mathematical rule before adapting it to an approved real application.

6.11 Why high predictive accuracy can hide a harmful decision rule

Think of a programme that uses a model to choose which water pumps receive preventive maintenance. The model predicts the number of failures per pump, but the number of people affected depends strongly on the pump. Predicting one failure at a pump serving ten households and one failure at a pump serving five thousand households produces the same count error if either prediction is wrong. Yet missing the second failure may have much greater consequences. A model chosen purely for unweighted prediction accuracy may therefore direct attention to the wrong sites.

A decision-sensitive objective can include severity weights w_i or, better, use a complete cost model. For actions $a_i \in \{0, 1\}$, maintenance budget B, maintenance costs c_i and expected avoided disruption v_i, a simple allocation is $\max \sum_i v_i a_i$ subject to $\sum_i c_i a_i \leq B$. The predictive model enters through estimates of v_i, which combine event probability and consequence. A programme with two independent models—one predicting failures and another estimating affected users—may still be unreliable if either target is measured badly.

This is the natural junction between learning and operations research. The mathematical comparison should not obscure values: the importance assigned to a school, a health centre or a remote community is an institutional or ethical choice. It should not be smuggled into a coefficient fitted from past service records.

6.12 An explicit example of a ranking and threshold disagreement

Consider four illustrative situations with true event outcomes $(1, 1, 0, 0)$. Model A predicts probabilities $(0.51, 0.50, 0.49, 0.48)$; Model B predicts $(0.95, 0.20, 0.10, 0.05)$. Under a decision threshold of 0.5 with the convention $p \geq 0.5$ triggers action, A warns for both events and no non-events. Model B warns for only the first event. But B assigns much higher probability to that first event and substantially lower probabilities to non-events. Depending on the score used, the models can be ranked differently. Neither model has a universal claim to superiority; the decision threshold, costs and calibration quality determine which differences matter.

More generally, a proper scoring rule evaluates the full probability forecast rather than just binary actions. Logarithmic loss for independent binary observations is

$$-\frac{1}{n}\sum_i \{y_i \log p_i + (1 - y_i)\log(1 - p_i)\}.$$

It strongly penalises confident mistakes. Brier loss averages $(y_i - p_i)^2$. Both are useful for probability evaluation, but neither substitutes for specifying the costs of an actual warning decision (Gneiting and Raftery, 2007). A model can have excellent average probability scores and still fail on the rare events a small institution most needs to anticipate.

6.13 Learning under rare events and small samples

When adverse outcomes are uncommon, a classifier can obtain high overall accuracy by predicting that nothing will happen. If 98 of 100 days have no breakdown, an always-negative classifier achieves 98% accuracy while detecting no breakdowns. Precision and recall expose different errors, but their interpretation depends on event prevalence, recording completeness and the proposed intervention. A model trained on reported incidents may learn which events get documented rather than which events occur.

Small samples create two additional risks. First, estimates for rare subgroups can be highly uncertain; second, a flexible model can find patterns that are accidental. Under small data conditions, a simple model with transparent uncertainty can be preferable to a large ensemble even when the latter has a slightly lower fitted error. Before building the larger model, ask whether obtaining a modest number of better-quality observations would do more for the decision than adding parameters. This is a value-of-information question.

6.14 A critical view of explanation tools

A feature-importance chart identifies relationships inside a fitted model, not necessarily causes in the world. If a transport-delay model relies on a district code, the code may proxy road quality, distance, reporting differences or management practices. Changing the code in software does not change the road or the institution. Local explanation methods can identify which features influenced the prediction in a neighbourhood of a particular input, but the meaning of such a neighbourhood can be questionable for strongly dependent or categorical variables (Ribeiro et al., 2016).

One reliable exercise is to compare model explanations with deliberately known data-generating structure in a synthetic example. A second exercise tests whether explanations are stable when correlated predictors are exchanged. Neither demonstrates that explanations will have causal meaning for an observational real dataset. Documenting uncertainty about explanations can be more responsible than presenting a visually tidy ranking of “drivers.”

6.15 A retrieval system should know what it does not know

Suppose a community information service indexes verified public pages about procedures for applying for a service. The correct answer to a question may depend on a particular regulation version. If the relevant document is absent, the system should refrain from inventing a procedure. We can assess this behaviour mathematically by introducing an abstention category. Let Q_A be answerable questions and Q_U unanswerable questions under the indexed corpus. Report

accuracy on Q_A, abstention on Q_U, and the rate of unsupported answers separately. Overall accuracy may look high if the test contains very few unanswerable questions.

A useful test collection contains paraphrases, spelling variations and questions whose answers differ between old and current documents. The retrieval corpus should record document version and effective date. If a policy changes, storing both versions without distinguishing their valid periods can create apparently supported but outdated answers. The same provenance discipline introduced for datasets in Chapter 3 applies to text.

For multilingual use, literal translation is not enough. Speakers may use locally common terms, abbreviations and mixed languages. Evaluation should use questions that actual intended users can understand, with informed consent where user data are collected. A model that performs well on standard English may not be reliable when it receives a short Luganda phrase or local usage unfamiliar to its training corpus. Such a gap should be measured, not assumed away.

6.16 A model-selection decision record

When recommending a more complex model, write a short decision record explaining its measurable advantage over the baseline, its additional data requirements, costs, observed failure cases, expected maintenance burden and who will monitor it. Include the possibility of deciding not to deploy. The record should make it possible for a later researcher to see whether a genuine improvement was established or whether the team merely followed a technology trend.

If a model's accuracy improves but its decision loss does not, report that as a result. If the apparent advantage disappears under station-blocked validation, the test design has revealed a problem before the system was entrusted with a service. If the model requires data the organisation cannot maintain, the design should be revised before deployment. This is how machine learning earns a place in engineering: by giving the organisation something it can use more reliably than its current alternative.

6.17 Extended case study: comparing learning systems on one decision

A useful evaluation should place every candidate method in the same operational situation. Imagine a municipality that considers issuing a temporary notice when predicted pollution exceeds a threshold. A conventional model produces a daily concentration forecast. A second model produces an exceedance probability from weather and traffic features. A third uses a nonlinear ensemble. Which one is useful? A fair comparison needs the same information cutoff, the same observed outcomes and a common loss structure. A lower test MAE does not by itself settle the question.

6.17.1 Begin with the no-model policy

A baseline need not be a statistical estimator. The current office might issue notices only when a monitor records a specified concentration. That policy may not cover unmonitored

locations but is easy to inspect. Another baseline uses the last measured concentration. A third uses a historical seasonal mean. Record the actual rules and who follows them. A model that outperforms a constant predictor but offers no advantage over the policy already used has a weak deployment case.

The model's information set is determined by the decision time. If notices must be issued by 7 a.m., the model may use observations and forecasts available before then. It may not use noon's realised meteorological conditions. A retrospective dataset should preserve the timestamp at which each input became available, including data publication delays. Without this information, the evaluation may inadvertently permit future knowledge.

6.17.2 Compare probability forecasts, not merely labels

Suppose the three methods provide probabilities p_{it} for events at site i and day t. Calibrate them on a validation period that does not overlap the final test. Then evaluate log loss or Brier score, observed calibration against predicted probability, and the action recommended by the stated cost threshold. A threshold chosen after seeing final test outcomes would not provide independent decision assessment.

For false-notice cost C_{FP} and missed-event cost C_{FN}, the expected per-case loss of action a is $R(a \mid p)$. The model's empirical decision loss is an average of observed costs on held-out cases, while conditional expected loss is computed from predicted probabilities. Agreement between these summaries is informative but not automatic. If event costs are uncertain, repeat the comparison across a plausible range rather than using a single arbitrary ratio.

6.17.3 Understand the cost of a false sense of certainty

Imagine the nonlinear model frequently predicts exceedance probabilities of 0.99 but is wrong in a meaningful fraction of those cases. A more cautious model may have slightly poorer ranking yet better calibrated probabilities. For a decision threshold near 0.9, the distinction can change many notices. A model should therefore be evaluated at the action thresholds its users will actually employ, not only through aggregate discrimination. This is particularly important when the outcome is rare and an unnecessary action has substantial cost.

6.17.4 A selective system with human review

Allow the model to abstain when its predictions are insufficiently supported. One possible rule is to refer cases whose predicted probability is close to the decision threshold. Another uses a validated uncertainty estimate. This can improve safety, but referral itself uses capacity. If a municipality has one analyst available for fifteen cases per day, the deferral policy must respect that workload. Otherwise the theoretical option $a = \bot$ is not feasible.

The analyst's review can also be wrong. A rigorous evaluation of human-in-the-loop performance should document the information available to reviewers, their response time and the final action. A study of a model alone cannot establish the quality of a combined human-machine workflow. Deference to a human is not a magic error correction; it is an operational process requiring adequate staffing and accountability.

6.17.5 Model updating is a decision with its own risks

A model may lose accuracy when equipment changes, a major road opens or reporting practices are revised. A maintenance plan should specify which signals justify investigation, how new outcomes are collected and when retraining is permitted. Retraining every week can introduce instability; never retraining can allow persistent deterioration. A versioned evaluation pipeline should compare the old and new models on agreed reference cases and recent independently assessed observations.

A new model must not be promoted simply because it performs better on the sample used to develop it. In a small organisation, even a simple approval protocol can improve reliability: record the reason for the update, the validation results, a rollback plan and the person authorised to approve deployment. Model governance is one of the points at which rigorous methodology becomes practical engineering.

6.18 Extended generative-AI investigation: from a document to a supported answer

A language model can generate plausible text even when its source does not contain the answer. Consider a public-service document collection with eligibility rules, office hours, procedures and dates. A citizen asks, “Can I receive this service without travelling to the district office?” The system should identify the relevant source and either answer from it or acknowledge that the available documents do not establish the answer. It should not invent a telephone number or an online service merely because such facilities exist elsewhere.

6.18.1 Create a test set before trying models

Choose a small, verified collection of public documents that may be legally redistributed for teaching. Record each document’s title, issuing agency, effective date, location and source URL. Prepare 40 questions: perhaps 25 with directly supported answers, five requiring information from two documents, five whose answers are absent, and five designed to test an outdated source against a newer one. These proportions are illustrative; in a real study the test distribution should represent intended use as well as deliberately challenging cases.

The independent answer key should identify the exact passage supporting each answer, what must not be inferred, and when a referral is appropriate. Then compare the same questions across keyword search, semantic retrieval, a retrieval-augmented generator and an unsupported generator. The first three may all fail for different reasons: poor wording, mismatched embeddings, missing documents or unsupported synthesis. Reporting only the percentage of answers that sound reasonable is not enough.

6.18.2 Citation validity is not a cosmetic measure

A generated answer can display a real URL that does not contain the alleged fact. The evaluation should therefore verify proposition-level support. If a response contains three factual

claims but the cited passage supports only two, count the unsupported claim. Distinguish an absent citation from an invalid citation and from a citation to an outdated but formerly correct document. The reader should be trained to inspect support, not merely the appearance of scholarly apparatus.

Another failure is over-answering. Suppose the document lists required identity documents but says nothing about processing times. A suitable answer might report the requirements and state that no processing time is specified. Adding an invented "typically three working days" can be more harmful than refusing that part of the question. A good system can supply a partially supported answer while clearly separating what remains unknown.

6.18.3 How to measure abstention

Let A be the set of answerable test questions and U the set of unanswerable ones. Calculate supported-answer accuracy on A and the correct-abstention rate on U. In addition, report the rate of unsupported factual claims among all answered questions. A system that abstains on every question achieves perfect avoidance of unsupported claims but provides no service, so answer coverage must be reported. A system that answers everything may maximise coverage while making unsupported assertions. The useful operating point depends on the consequences of misinformation and the possibility of human referral.

6.18.4 Local-language and low-bandwidth testing

A service intended for people using mixed English and local-language expressions should be evaluated with those expressions, not only with translated benchmark questions. Consent and privacy are necessary when real user utterances are collected. If the system relies on a remote server, test actual response time over the target network and identify what happens when connectivity fails. An offline keyword index with verified passages may be a better operational solution for a narrow service than a large cloud model. If a language model genuinely adds value, the evidence should show which questions benefit and what additional cost is incurred.

6.19 What the two investigations teach together

The pollution notice and the document question-answering service look technically different, but the evaluation problem is similar. Both require a clearly defined output, suitable ground truth, a simple baseline, independent evaluation, calibrated uncertainty or abstention, and a plan for error recovery. Both can be harmed by using data unavailable at the time of action. Both benefit from specifying where the system should not be trusted. The success of a learned model is therefore not its ability to produce output on every input; it is its ability to improve a defined human decision under demonstrable conditions.

6.20 Mathematical workshop: why a training score is not a guarantee

Suppose a team tries M candidate prediction rules on a sample of n independent observations, and each rule has a bounded loss in $[0, 1]$. For any fixed rule, Hoeffding's inequality provides a simple concentration bound for its empirical average loss $\widehat{R}(f)$ around its population risk $R(f)$:

$$\Pr\{|\widehat{R}(f) - R(f)| > \epsilon\} \leq 2\exp(-2n\epsilon^2).$$

A union bound over a *fixed* finite family $\mathcal{F}$ of M rules gives

$$\Pr\left\{\sup_{f\in\mathcal{F}} |\widehat{R}(f) - R(f)| > \epsilon\right\} \leq 2M\exp(-2n\epsilon^2).$$

Therefore, with probability at least $1 - \delta$, all the rules have deviations no larger than $\sqrt{\log(2M/\delta)/(2n)}$ under these assumptions.

This inequality makes an important idea explicit. Choosing the lowest empirical loss among many candidates creates an opportunity to select a rule that benefited from chance. The bound grows with the logarithm of the candidate-family size. It does not imply that a complex model necessarily generalises badly; it illustrates why model selection and evaluation cannot be treated as independent when they use the same outcomes. It also does not apply without modification when observations are clustered, temporally dependent or adaptively reused to invent new candidates after examining results.

6.20.1 A worked numerical interpretation

Take $n = 200$, $M = 20$ and $\delta = 0.05$. The bound is approximately $\sqrt{\log(800)/400} \approx 0.129$. This is a loose theoretical upper bound on the largest deviation among the fixed candidate rules under the stated assumptions, not an estimate of actual forecast error. If the data contain only ten independent stations with many repeated readings, $n = 200$ cannot be inserted as though all observations were independent for a new-station claim.

The practical lesson is not to replace validation with a theorem. It is to understand why the same dataset cannot be used indefinitely for trying choices and then advertised as an untouched confirmation. Separate data roles, grouped evaluation and transparent reporting are the empirical safeguards.

6.21 The Bayes decision depends on costs, not only probabilities

For a binary event with probability p, false action cost c_{10} and missed action cost c_{01}, the Bayes rule chooses action 1 when $(1 - p)c_{10} < pc_{01}$. Algebra gives $p > \tau = c_{10}/(c_{10} + c_{01})$. This rule is optimal for the stated conditional losses if p is the true conditional probability and no other constraints apply. Its reliability depends on calibration and on whether the cost assumptions capture real consequences.

If a model gives $\widehat{p}$ rather than p, making the wrong decision is possible only when $\widehat{p}$ and p lie on opposite sides of τ. This clarifies where probability accuracy is operationally critical. A

small estimation error far from the threshold may not affect the action, whereas a small error near the threshold can reverse it. The importance of this observation is easily lost in a global average accuracy measure.

A second limitation arises when multiple actions share a budget. If a district can respond to only five of fifty alerts, an independent threshold rule may select more than five. The optimisation problem becomes choosing the set of actions with the greatest expected avoided loss subject to resource constraints. Good classification probabilities are then inputs to a constrained decision problem, as Chapter 5 developed.

6.22 A model can be predictable but not interpretable in the causal sense

Suppose past service delays are correlated with the number of staff on duty. A fitted model may conclude that delays are higher when more staff are present. One possibility is that extra staff are deployed on the busiest days. Predicting the delay from staffing and arrivals may still be useful, but interpreting the positive staffing coefficient as evidence that hiring causes delays would be false. To estimate the effect of staffing interventions, a causal comparison must address how staffing decisions were made and what other factors changed.

This distinction is central in a book about useful solutions. A municipality may need a good forecast of demand to plan the next day, and that forecast need not identify the causal effect of each covariate. But when the intervention itself changes the covariate distribution, a purely predictive model may no longer describe the new system. A successful prospective implementation should therefore be monitored for changes induced by the action.

6.23 Auditing the information available at the decision point

For each feature, write its source and the earliest time it is known. For an environmental alert issued at 7 a.m., yesterday's verified sensor readings are available, but today's midday observations are not. For medicine procurement placed on Monday, an inventory count recorded the preceding Friday may be available, but a Tuesday stock reconciliation is not. A feature table that lists availability times can identify leakage before fitting any model.

The audit should also ask whether deployment will provide each input at the same quality as the historical dataset. A research dataset may be carefully curated after the fact, while real-time systems face missing records, delayed updates and device failures. Training on retrospectively cleaned data without testing real-time missingness can produce optimistic results. A deployment evaluation should therefore replay information as it arrived, not as it appeared in a perfect archive months later.

6.24 When the right model is a rulebook rather than AI

Consider a service desk that must answer a finite set of approved procedural questions. If the approved answers are stable, a searchable rulebook may be more dependable and cheaper

than a generative model. A language model may still help interpret varied wording or identify the relevant document, but that value should be demonstrated on realistic test questions. If every question requires authoritative approval before an answer is sent, the cost of the review workflow belongs in the comparison.

The strongest evaluation does not attempt to show that AI wins. It asks which method provides the needed result at acceptable risk and cost. In some tasks the answer will be a conventional rule, a statistical model or a retrieval system. That is a positive conclusion of applied computational science, not a failure of ambition.

6.25 Laboratory extension: testing when learning complexity is worth paying for

A proper machine-learning laboratory should begin with a predeclared comparison rather than a search for the method that produces the most impressive graph. Consider a small tabular dataset with a numerical outcome and geographic groups. Choose a simple linear or additive baseline, then one nonlinear alternative such as a random forest. The same groups must be withheld from both models. All preprocessing, hyperparameter tuning and missing-value handling occur inside training folds. Compute the test error and resource use for both. Then ask whether the difference would change an operational decision.

6.25.1 A learning curve with honest group counts

Repeat training with an increasing number of complete groups, not an increasing number of randomly sampled correlated rows. Plot held-out-group error against the number of independent training groups. This helps distinguish an underfit method from a data-limited problem. If error remains high as the number of rows grows but the number of stations stays fixed, the extra rows may be providing less new geographic information than expected. Report the number of groups and observations at each point.

The curve is not necessarily monotone on a finite test set. Additional noisy groups can change fitted parameters and produce temporary increases in measured error. Averaging over several predeclared splits can reduce such fluctuations, but very few independent groups make fine-grained comparisons unreliable. Avoid drawing conclusions about a universal scaling law from one small dataset.

6.25.2 A resource-cost curve

For each candidate model, measure training wall-clock time, peak memory when practical, saved model size and time to produce predictions on a fixed workload. Use the same hardware and repeat timing trials. If a method requires GPU acceleration, distinguish the resource condition from a CPU baseline. If the target institution owns only an ordinary office laptop, a method that is convenient on a large research server may not be a realistic deployment option.

An appropriate result table could contain test MAE, interval coverage, inference time, model size and a short failure description. A low-error model with severe performance deterioration

when a sensor feature is missing may be less useful than a slightly less accurate model with a documented fallback. Resource and reliability results belong in the same comparison, not in separate promotional sections.

6.26 A test of missing inputs

A model deployed in a field organisation will occasionally encounter missing weather, delayed laboratory results or unavailable network features. Build a controlled test in which one predictor is removed at a time from held-out observations. For each removal, apply the missing-input procedure that would actually be used in deployment. A method that silently imputes from the full test dataset is using information unavailable to the system.

Record how the prediction, interval width and operational decision change. A good system may produce a less precise estimate, abstain or fall back to a simpler method. The aim is not to make the model answer every possible input. It is to make degraded performance visible and prevent unsupported confidence during foreseeable failures.

6.27 A short experiment on dataset shift

Construct a teaching example where training covariates occupy $X \in [0, 1]$ while deployment inputs include $X \in [2, 3]$. A tree ensemble may produce a prediction close to its boundary training values because trees partition observed regions rather than extrapolate a physical trend. A linear model may extrapolate, but that extrapolation is not automatically correct either. Neither should be trusted without evidence about the new regime.

Now introduce a physical or operational constraint. If the outcome must be nonnegative, a fitted line predicting negative values is invalid as a physical prediction. A log-link or constrained model may enforce nonnegativity but still be inaccurate outside the observed range. This illustrates why plausibility constraints, distribution-shift detection and independent validation are complementary.

6.28 A practical definition of model improvement

The reader should end this laboratory with a comparison record, not a trophy for a particular algorithm. It states the intended decision, held-out design, test sample, performance difference, uncertainty, computation cost, sensitivity to missing inputs and conditions under which the model should be withheld. If a nonlinear model's advantage disappears outside its training region, the honest conclusion is conditional usefulness, not universal superiority.

6.29 Exercises and investigations

1. Explain, with a concrete example, why a station-blocked test estimates a different kind of generalisation from a randomly sampled hourly-record test.

2. For a logistic predictor, explain the distinction between discrimination and calibration. Construct a small dataset in which ranking is good but probability estimates are miscalibrated.
3. Derive the optimal warning threshold when false positives cost two units and false negatives cost seven. Explain how the answer changes when the costs become equal.
4. A model's MAE improves by 2%, but its memory use increases tenfold. State what additional facts would be needed before recommending deployment on a shared low-cost computer.
5. Explain why selecting the best of 30 models on the final test dataset invalidates the independence of that test.
6. Design a source-verification test with six answerable and four unanswerable questions for a small government-document retrieval system.
7. Construct a model card for a proposed air-quality predictor without inventing results. Separate fields that can be completed now from those requiring held-out-site analysis.
8. **Research project.** Reuse a measured dataset from Chapter 4 and compare one interpretable baseline with one nonlinear alternative. Report changes in operational decisions, not just regression errors.

What this chapter establishes. Learning can be valuable where relationships are complex and trustworthy measurements exist. Its value must be earned through independent tests and consequences that matter to users. Chapter 7 turns to the less glamorous but equally important question of whether the resulting software survives the conditions in which it must operate.

Chapter 7

Building for the Machine and Infrastructure People Actually Have

A correct algorithm is not yet a usable application. This chapter develops a small offline-first system that tolerates power loss, interrupted connections, duplicate submissions and limited storage. We treat transactions, synchronisation, security, recovery and maintainability as part of scientific reliability, not as details to be postponed after a model has been built.

7.1 When a connection is a luxury rather than an assumption

Imagine a field worker recording a water-quality measurement at a rural treatment point. The worker has a low-cost device, enough battery for an hour, and uncertain network coverage. A programme designed for uninterrupted cloud access may lose the record when the connection fails after the worker presses Submit. A programme that saves locally first can continue collecting data, then synchronise later. But this introduces other questions: what if the same record is uploaded twice, the phone is lost, two workers edit the same visit or the electricity fails in the middle of writing a file?

These are engineering questions with precise answers. The goal is not to replicate the infrastructure of a large technology company. It is to determine which failures can reasonably occur and establish what the software will do when they occur. Foundational treatments of reliable data systems and transactions include Kleppmann (2017); Gray and Reuter (1992). Their lesson is especially relevant here: a system's promises should be expressed as properties that can be tested, not as reassuring adjectives.

7.2 Specify an operational contract before drawing screens

For each data record, specify required fields, allowed ranges, units, a local identifier and the meaning of a saved state. An illustrative water-quality record might contain `record_id`, `site_id`, `sampled_at`, `measurement`, `unit`, `device_id` and `revision`. If a measurement is outside a plausible instrument range, the application should flag it for review rather than secretly replace it with a typical value. If the timestamp comes from a device whose clock may

be wrong, that risk belongs in metadata. No choice of programming framework fixes a defective measurement process.

A small system can use states such as `draft`, `validated`, `queued`, `acknowledged` and `conflict`. A record moves through those states under stated conditions. For example, a valid local record should be committed to persistent storage before the interface tells the worker it has been saved. A successful network request is not enough to claim remote persistence until the server confirms a durable result. If acknowledgement is lost after the server commits, retrying the same operation should not create a second observation.

7.3 SQLite transactions and the meaning of atomicity

A database transaction groups changes that must be treated as one logical operation. Suppose a new measurement is inserted and a synchronisation queue entry is created. If the application inserts the measurement but crashes before creating the queue entry, the observation may never be uploaded. If both changes occur in one transaction, the system can ensure that either both are committed or neither is committed, subject to the database's documented guarantees and correct configuration.

The essential logical operation is

$$\texttt{BEGIN} \rightarrow \{\texttt{INSERT measurement}, \texttt{INSERT outbox}\} \rightarrow \texttt{COMMIT}.$$

This protects a local invariant: every committed record that requires synchronisation has a corresponding pending outbox event. Transaction properties do not protect against a damaged storage device, loss of encryption keys or an application that never performs a backup. Reliable systems need more than a correct database primitive.

Proposition 7.1 (An atomic outbox invariant)**.** *Assume a transaction contains exactly one insertion into a measurement table and one insertion into an outbox table, and that neither insertion occurs elsewhere without appropriate controls. If the database guarantees atomic commit, a crash cannot leave a committed measurement from that transaction without its committed outbox event.*

Proof. Atomicity permits only two persistent outcomes: all transaction effects are committed, or none are. A state with only one of the two insertions is excluded. The claim relies on the stated transactional assumptions and does not address later manual modification or database corruption. □

The proof establishes a software property under assumptions, not that all real-world data will be delivered. The worker may have misidentified the sampling site or the device may be stolen before the local backup is made.

7.4 Delivery at least once, effect at most once

Network requests can fail in an ambiguous way. Imagine the server receives and stores a record, but the reply never reaches the client. The client cannot know whether the upload succeeded.

If it retries, the server may receive the same logical operation twice. We therefore design a unique *idempotency key*, typically the stable record identifier and revision, so that repeated delivery has the same stored effect as one delivery.

Let T_k denote the server-side application of a message with key k. An idempotent write should satisfy

$$T_k(T_k(S)) = T_k(S)$$

for all valid server states S. This is a statement about the *effect*, not a guarantee that the message arrived exactly once. A retry policy with exponential backoff improves behaviour under temporary outages but should have a limit and should not silently abandon pending data. The system needs an accessible queue status and an escalation path when a record remains unsynchronised.

Worked example 7.1 (The duplicate that looks like new evidence)**.** *A field worker submits a water measurement of 35 units. The server saves it, but the connection drops before acknowledgement. A naive retry inserts a second row with the same value and location. The analyst later sees two measurements and may infer that the site was sampled twice. With a stable operation key enforced by a unique database constraint, the second submission returns the existing result instead of creating a new observation. The original physical measurement still needs quality checks; idempotency only prevents duplication of a logical record.*

7.5 Conflicts are decisions, not exceptions to hide

If two devices update the same record while offline, the server may receive incompatible revisions. A “last write wins” rule is simple but can discard a valid correction without informing anyone. For certain noncritical fields it may be acceptable, but for a measurement record or approval it may be indefensible. A safer approach is to store the conflicting versions and require an authorised reviewer to resolve them, with a complete audit trail.

Use revision numbers or version vectors appropriate to the system. A simple optimistic concurrency rule accepts an edit only if its declared base revision equals the current revision. Otherwise the server reports conflict and preserves both pieces of information. This prevents accidental overwriting; it does not establish which record is scientifically correct. When people collect actual observations, a conflict-management screen is part of data quality.

7.6 Power loss, partial writes and recovery

We should distinguish a process crash from loss of electricity, sudden device restart, disk failure and intentional deletion. Each has different effects. A process crash during a database transaction is often recoverable by the database’s journal or write-ahead log. A damaged storage medium may require restoration from a backup. An application-level bug can commit the wrong values perfectly; transaction mechanisms do not fix its logic.

Design tests that kill the process between operations. Verify that a committed record survives restart, a partially written transaction is rolled back, and retrying an acknowledged upload does not duplicate it. Then disconnect the network while a series of records are being entered,

reconnect it, and inspect the remote counts. A successful demonstration consists of an input log, observed outputs, checks of invariants and clearly stated software versions. It is not merely a video showing that a screen continues to respond.

7.7 Measuring the actual cost of reliability

Resource-constrained deployment requires measurements, not assumptions that a small codebase uses little energy. Important quantities include storage per record, memory footprint, time to open and save a form, synchronisation bytes per record, time to catch up after an outage, and battery use where a defensible measurement process exists. Measure representative workloads repeatedly. Distinguish startup costs from steady-state behaviour, and avoid comparing two applications run under different device conditions.

Suppose an offline application stores n records of average payload size b bytes, plus metadata and index overhead $h(n)$. Its storage requirement is approximately $S(n) = nb + h(n)$. This elementary formula is useful for capacity planning, but real overhead depends on database pages, indexes, compression and old versions. If the application accumulates records faster than they can be synchronised, its queue may grow without bound. Let incoming records average λ per day and successful uploads average μ per connected day. A necessary long-run stability condition is that effective service capacity exceeds arrival volume under the actual connectivity schedule; simply choosing $\mu > \lambda$ on a fully connected day is insufficient.

7.8 A bounded resource policy

A deployment should specify what happens as storage approaches its limit. Never silently delete unacknowledged field measurements to keep the interface responsive. The system may warn the worker, disable nonessential media capture, compress approved attachments or arrange safe manual export. A hard stop on new records can also have consequences; users must understand the trade-off. Design the policy with the institution that carries the operational responsibility, not solely with the software developer.

A device-local cache is not a backup. A backup should be restorable, access-controlled and periodically tested. Where sensitive records exist, protect them in transit and at rest according to the actual threat model, rather than assuming that adding a password to the interface makes a stolen device safe. Established security principles include least privilege and minimisation of exposed information (Saltzer and Schroeder, 1975; NIST, 2020).

7.9 A threat model for a modest field system

Start by listing what is worth protecting: participant identities, facility records, credentials, location history and the integrity of measurements. Then identify plausible threats: a lost phone, an unauthorised person using an unlocked workstation, a malicious attachment, a compromised synchronisation endpoint or an accidental spreadsheet export to a public folder. For each threat, state its consequence and a practical mitigation.

Table 7.1: Examples of engineering controls matched to specific failure modes

Failure mode	Design response and verification
Network unavailable	Local commit before remote send; test queued entries after restart.
Acknowledgement lost	Stable idempotency key; retransmit and confirm one server record.
Device clock inaccurate	Preserve device and received timestamps; flag large disagreements.
Duplicate record	Unique key with explicit merge or rejection rule.
Two offline edits conflict	Revision check; retain history and request resolution.
Storage nearly full	User-visible warning and supported safe-export plan.
Device lost	Authentication, access control, appropriate encryption and revocation.
Power interrupted	Crash tests and documented restore procedure.

No control is effective without testing it. For instance, storing data in encrypted form but leaving a permanently logged-in device unattended may offer little protection against ordinary unauthorised access. Conversely, an elaborate security architecture that prevents a nurse from completing routine tasks may be bypassed. Security and usability are part of the same operational design problem.

7.10 Architecture of a minimal offline-first application

A deliberately small application has four components: a form that validates data, a local transactional store, an outbox of pending operations and a synchronisation worker that sends records when possible. The server applies idempotent writes and returns a stable acknowledgement. The user interface never invents a server status; it distinguishes "saved on this device" from "received by the server."

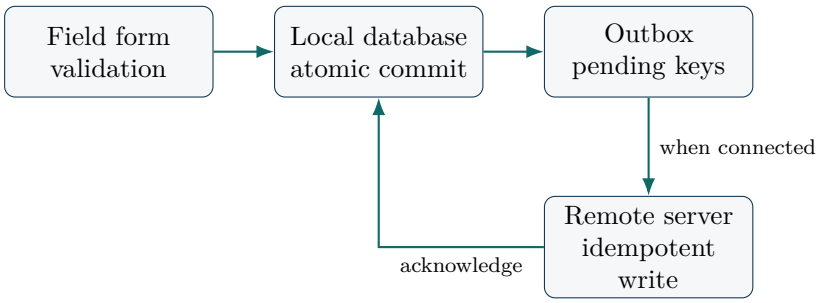


Figure 7.1: Logical components of an illustrative offline-first system. Network failure must not erase a committed local observation. The arrow from server to local store represents acknowledged synchronisation, not a physical connection assumed to be continuously available.

7.11 Complete implementation exercise

The companion `code/ch07_offline_store.py` implements a local SQLite measurement table and outbox, inserts both within a transaction, and simulates remote idempotent delivery using

a second SQLite database. The first run creates several invented records, including a retry of an already delivered identifier. Assertions check that the local measurement and outbox counts agree at creation and that retries do not increase the remote unique-record count. It is a teaching core, not a production-ready application: it does not implement user authentication, encryption, key management, real network transport or an approved privacy policy.

Students should modify the programme to simulate a crash after beginning but before committing a transaction, test what survives, and document the result. Then add an intentionally incorrect implementation that inserts into two tables using separate commits. Demonstrate that a crash between commits can violate the intended invariant. Finally, introduce two conflicting revisions and design a safe path for human review. This is a more convincing exercise than merely writing an offline user interface.

7.12 Maintenance is a design outcome

A student-led system often ceases to work when its original author leaves. Reproducible setup instructions, tested dependencies, audit logs, data export formats and named institutional responsibilities should therefore be considered part of the output. A programme requiring frequent remote software changes may be unsuitable where staff cannot install updates and cannot verify compatibility with older devices.

An operational handover records who can create and revoke accounts, who owns data, who authorises software changes, how incidents are reported, and who pays for connectivity, backup and replacement equipment. It should state the minimum hardware on which the application has actually been tested. Claiming “works on low-cost devices” without reporting the devices and measurements is not an engineering result.

7.13 An offline application as a small distributed system

Even a two-device deployment becomes a distributed system once both devices exchange changes with a server. The devices can have different information at the same moment. One may know that a record has been locally saved while another knows only the last acknowledged version. This is not automatically an error; it is the unavoidable consequence of intermittent communication. The application must therefore describe what users are allowed to infer from a displayed state.

A local record marked “saved” should mean the transaction has committed on that device. A record marked “sent” may mean only that a network request was attempted. A record marked “received” should require the server’s acknowledgement. A record marked “verified” should mean something more: that an authorised check of its scientific or operational content has occurred. Conflating these four meanings creates misleading dashboards and can have consequences for downstream decisions.

The names of these states are not matters of visual design alone. A state transition diagram can be translated into unit tests. For instance, a transition from `queued` to `acknowledged` should be impossible without a valid server response. A retry should preserve the same idempotency

key. Editing an acknowledged record should create a new revision rather than silently changing the data that a previous report already used.

7.14 Versioned changes and the problem of concurrent edits

Suppose two workers begin from revision 4 of the same observation. Worker A changes a measurement unit while Worker B corrects the location identifier. Both changes may be valid, but applying them in the wrong order can overwrite information. An optimistic concurrency mechanism requires each update to declare its base revision. The server accepts an update only if it matches the current revision; otherwise it reports a conflict. A human or explicitly tested merge procedure can then decide whether the changes are compatible.

A useful audit table records `record_id`, `revision`, `changed_at`, `author_role`, and separate fields for earlier and updated values, along with a reason where necessary. It should not store secret authentication information in plain text. An audit trail is useful because the correct data value can depend on why a field was changed, whether a replacement sample was taken and who authorised the correction.

7.15 Reproducibility when packages and networks are unavailable

A development laptop may have current packages and reliable internet access, while a deployed device cannot download updates. Record dependency versions and build a reproducible installation path that works without fetching remote components at the point of use. Establish whether a security update can be delivered through a controlled offline mechanism. A deployment plan that requires downloading a new environment every morning is not an offline-first plan.

Software results should contain a version identifier and a timestamp. If a numerical decision rule is updated, previously produced reports may need to indicate which version generated them. Otherwise an auditor may attempt to reproduce a past calculation with a newer implementation and obtain a different result without understanding why. This is the software counterpart of recording data releases and survey documentation.

For a reliable scientific application, inputs and settings matter as much as executable code. An analysis should save the dataset version, model parameters, configuration file and checksums where appropriate. If a random procedure is involved, record the seed and algorithm version. A seed alone cannot guarantee identical output across incompatible libraries or hardware. Reproducibility is a property of the complete execution environment, not a single command (Sandve et al., 2013).

7.16 Designing failure experiments as engineering evidence

A useful failure experiment has a hypothesis and a measurable criterion. For example: “If power is interrupted before transaction commit, neither the measurement nor the corresponding outbox row remains visible after restart.” The test then deliberately terminates the process

at a controlled point, reopens the database and queries both tables. A second hypothesis may concern a connection dropped after the server stores a record but before the client receives acknowledgement. The test retransmits and asserts that the server contains one logical measurement.

Results should report the number of attempts, software and database versions, operating system and where the interruption was introduced. A laboratory that never actually interrupts a write cannot support a claim about crash recovery. Likewise, a network-disconnection test performed only by adding a long delay is not equivalent to physically dropping the connection during a request. Start with small deterministic faults for which expected behaviour is clear; add more realistic fault injection only after the basic invariants pass.

7.17 A bounded deployment readiness exercise

Choose a representative low-cost machine and attempt the complete workflow: install without privileged internet access, create 200 records offline, restart after a simulated crash, synchronise after reconnection, inspect data integrity and recover from an exported backup. Measure how long each step takes, how much local storage is used and what help users need. If a requirement is not met, record it as a failure and revise the design. A failed test is evidence that improves the engineering process when its cause and conditions are documented.

The final question is organisational. Who will answer the phone when staff cannot synchronise a week's records? Who may reset credentials? Who decides which device is authoritative? Can the system be repaired without its original student developer? These questions are neither an afterthought nor a reason to abandon mathematical work. They are the point at which a mathematical solution becomes a dependable service.

7.18 Extended implementation study: from a record to an auditable database

A reader should be able to build one small, dependable component rather than only admire an architecture diagram. We therefore develop a practical protocol for a field-data application. The example stores invented water-quality readings in SQLite. It is small enough to inspect completely, yet it encounters the same logical difficulties as a larger system: validation, atomicity, duplicate handling, synchronisation state and recovery. The implementation is deliberately not a finished mobile application.

7.18.1 Write an input specification

The first table is a measurement record. Assign a stable identifier when a new observation is created, not each time it is uploaded. Require a site identifier, timestamp, measurement value, unit and observer or device context as permitted by the application. Reject records with missing required fields. Distinguish a missing reading from zero concentration. A zero may be an observed value; an absent reading should not be converted to zero simply to avoid a database null.

The identifier should not be based solely on device time, because clocks can be inaccurate and two readings can occur at the same instant. A randomly generated identifier or a well-designed device-specific sequence can serve the purpose, but its collision risks and data-handling implications should be understood. For public teaching examples, use arbitrary non-identifying record keys. Real deployments should not encode patient identity or confidential location information directly into identifiers that will be transmitted widely.

7.18.2 Keep the database schema small and enforceable

The local database needs a measurements table keyed by record identifier and an outbox table keyed by the same identifier. The outbox indicates which logical records require a remote operation. A unique constraint enforces the property that two uploads with the same logical key do not create two measurement records. Additional checks can restrict nonnegative values, valid units and allowed status transitions. For a small system, the ability to state and test these invariants is more valuable than adopting an elaborate database framework.

One insertion routine should validate all fields, start a transaction, insert the record, create the pending outbox operation and commit. The interface then reports that the record is locally saved. It should not report that the central database has received the observation. A separate synchronisation routine selects pending operations, sends them, and marks each acknowledged only when a server response establishes durable acceptance.

7.18.3 Simulate the lost acknowledgement

Run the server-side receiver twice with the same operation key. In the correct implementation, the first call creates a row and the second returns the already stored logical result. The count of distinct remote records remains one. Next deliberately remove the uniqueness constraint in a temporary test database and run the same operation. The duplicate appears. The comparison shows that idempotency is an enforceable property rather than a label in the design document.

One subtlety matters: “ignore duplicate insert” is not a safe universal rule. If two submissions use the same key but different payloads, silently ignoring the second may conceal an inconsistency. A production design should detect payload mismatch and report conflict. The teaching programme demonstrates the narrow case of identical retries; extending it to conflict handling is a necessary next step before real deployment.

7.18.4 Restore after interruption

Create three records, confirm the local database count, terminate the programme and reopen the database file. The records should remain. Then arrange a controlled failure inside a transaction before commit and verify that neither partially inserted table change appears. Finally, export a backup, restore it into a new file and rerun a set of consistency checks. A backup that has never been restored is only a claim that recovery may be possible.

The result table should include the test name, triggering condition, expected invariant, observed state and verdict. For example, the duplicate-upload test expects one remote record after two attempts; the atomicity test expects either two related rows or none; the restart test expects

committed rows to persist. These are deterministic tests and can support strong claims about the tested software configuration. They do not support claims about every possible hardware failure.

7.19 A complete failure matrix for a small field pilot

Several failures deserve separate experiments. Network refusal before upload tests queuing; network loss after server commit tests ambiguous acknowledgement; process termination during a local transaction tests atomicity; a full disk tests graceful storage failure; a wrong device clock tests timestamp warnings; simultaneous edits test conflict resolution; incorrect credentials test access control. The designer should decide which failures are safety critical and which are merely inconvenient.

Under each condition, document both the technical state and the user's experience. A technically correct system that shows a spinning icon for ten minutes without explanation may lead the user to repeat an operation or abandon it. A technically correct rejection with an incomprehensible error message may also be unusable. Good systems reveal relevant state in ordinary language: "Saved on this device; upload pending" is more useful than a generic green success symbol.

7.19.1 Unreliable connectivity changes workflow design

Suppose synchronisation succeeds for only fifteen minutes each evening. The application must retain a local queue throughout the day. If the queue receives 100 small records per day but each successful connection can upload 300, the backlog may clear regularly. If uploads are slow because records contain large photographs, the queue could grow despite the same record count. Measure bytes and retry overhead, not merely the number of requests.

A field worker may need to confirm that a submitted observation can be viewed by a central supervisor. The interface should expose the last confirmed synchronisation time and unresolved conflicts. The system should not clear local data as soon as it begins uploading. A safe retention policy needs a verified acknowledgement and an appropriate backup procedure. How long data remain on the device is a policy question balancing recovery, device storage and privacy risk.

7.20 The cost of an update

A program installed on dozens of inexpensive devices will eventually need an update. If the update changes the database schema, the transition must preserve existing records and remain recoverable if power fails. Test schema migration on a copy of a real-sized *synthetic* database before distributing it. If an update modifies the units of a measurement column, store the original units and explicit conversion; changing numeric values without provenance can invalidate historical analyses.

There is a broader lesson here. A low-cost application is not necessarily cheap to own. Training users, repairing devices, replacing batteries, documenting changes and providing support can dominate the initial development effort. The book should teach readers to calculate a multiyear maintenance budget, not only the time taken to write the first version of a programme.

7.21 A responsible field readiness review

Before a live pilot, an organisation should confirm its authority to collect data, the retention and privacy rules, the workflow that will use the results and a named maintenance owner. Test with users under the expected network and electricity conditions. If a tool is intended for clinical or public-service decisions, identify what happens when its recommended action is wrong. A data-entry prototype can demonstrate that observations are saved and synchronised; it cannot prove that a health service, water supply or field programme will improve until the relevant operational outcomes are evaluated.

The final project deliverable is therefore a small working application component, a complete test suite, a threat model and a documented decision on whether further deployment is justified. It is acceptable for that decision to be "not yet" when a critical failure remains.

7.22 Mathematical workshop: safety and progress in a synchronisation system

A reliable synchronisation system needs at least two kinds of claims. A *safety* property says that something bad never happens: for instance, a message retry must not create a second logical measurement. A *liveness* property says that something good eventually happens: for instance, a queued record will eventually be delivered when communication and server availability permit progress. Safety can often be demonstrated with finite invariant checks. Liveness requires assumptions about the future environment. Confusing the two is a common source of exaggerated reliability claims.

Let Q_t be the number of pending records at the start of day t, A_t the arrivals of new records and S_t the successfully acknowledged records. Then

$$Q_{t+1} = Q_t + A_t - S_t, \qquad 0 \le S_t \le Q_t + A_t.$$

This is a conservation equation, like the stock balances used in Chapter 2 and Chapter 8. It does not imply that the queue will ever empty. If average arrival volume exceeds effective upload capacity over long periods, backlog grows. If connectivity is unavailable for a week, a system should preserve queued records, show the backlog and avoid claiming that they are centrally available.

7.22.1 A queue stability calculation

Suppose 100 records arrive each day. A connection succeeds on two days per week, with capacity to upload 400 records on each connected day. Weekly arrivals total 700 and nominal weekly upload capacity is 800, so the capacity condition permits long-run backlog clearance. But timing matters: immediately before a connection, the queue may contain several days of records. If storage is sufficient for only 300 records, the system can fail before the weekly throughput calculation becomes relevant. Capacity planning must consider the maximum expected backlog as well as average rates.

Now suppose each retry consumes time and network capacity. The theoretical maximum of 400 records per session may not be attained when packets are repeatedly lost, or when each record contains large image attachments. A real deployment needs measured upload throughput and a conservative storage buffer. The book's simple queue calculation is a planning tool, not an empirical guarantee about a mobile network.

7.23 What exactly must be idempotent?

It is tempting to say that "the API is idempotent" without specifying which action. A record creation request, a stock deduction and a notification can have different meanings. Repeating the request to save one observation should not create a duplicate measurement. Repeating an instruction to deduct five medicine packs should not deduct ten packs. Repeating a notification may be harmless in one application and unacceptable in another. The operation's business meaning determines the idempotency key and its scope.

For a stock transaction, let a unique transaction identifier k represent one physical issue event. The server stores $(k, \text{medicine}, \text{quantity})$ with a uniqueness constraint on k. A retry with the same identifier and identical payload returns the prior result. A retry with a different payload under the same key should be rejected as a conflict, not silently accepted. Otherwise a correction could be lost, or a malicious client might exploit the ambiguity.

7.24 A useful consistency invariant

Suppose the local database has a table of committed observations and a table of outbox operations. Let R be the set of record IDs that need synchronisation and O the set of record IDs in the outbox. An invariant at creation time is $R \subseteq O$. Atomic insertion can preserve this condition, provided all relevant code paths obey the same rule. A program that allows direct insertion into R without creating an outbox row breaks the proof. In production, database constraints, application permissions and tests should collectively enforce the intended data pathways.

After acknowledgement, entries may be removed from the pending queue or marked acknowledged, so the invariant must be reformulated in terms of the complete outbox history rather than only pending entries. A good specification makes the lifecycle explicit. The fact that a database has transactions does not tell us what the application should do with its states; the developer must state and implement the appropriate invariant.

7.25 Security trade-offs with an offline device

If a device is stolen, encrypted local storage may protect confidentiality, but encryption keys must be managed. If the device remains unlocked, an unauthorised person may still access the application. If data cannot be decrypted after a key is lost, a backup and recovery plan is essential. These risks cannot all be solved with a single mechanism. Begin with the sensitive fields, likely adversaries, device support and practical constraints. Then choose controls

proportional to the consequences of disclosure or tampering.

A particularly important engineering distinction is between confidentiality and integrity. Encryption can help prevent unauthorised reading, but a dataset may still be changed through an authorised account used incorrectly or compromised. Audit trails, checksums, access control and review processes serve different functions. The design should state the threat against which each control is intended to work and test its failure path.

7.26 A reproducible stress-test protocol

A well-written stress test begins with a fixed local database state, a known collection of invented input records and a deterministic failure schedule. For each stage, record the expected number of local measurements, pending operations and unique server records. Introduce one fault at a time: crash after beginning a transaction, lost acknowledgement after server commit, and conflicting second revision. Check the resulting state against the expected invariant before adding another fault.

Once these tests pass, introduce randomised faults and repeat under different seeds. Do not infer a probability of real-world data loss from a few simulated faults. A field test is still required to assess actual storage reliability, device behaviour and network conditions. The value of the deterministic tests is that they establish the software's intended response to precise failure cases, reducing the chance that a real outage reveals an entirely unexamined design flaw.

7.27 What to put in the technical handover

A handover package should contain a complete system diagram, a data dictionary, database schema, setup instructions, dependency versions, backup and restoration instructions, user roles, error codes, known limitations and a test log. It should identify the person or office responsible for restoring service and the criteria for escalating unresolved data conflicts. If a deployment cannot be handed over, it is a prototype dependent on its developer rather than an institutionally sustainable solution.

The best measure of success is whether trained users can enter a record, recover from an interruption, identify a pending upload, reconcile a conflict and produce the intended data output without calling the original developer. Such a test combines technical correctness with human usability and is far more persuasive than measuring the number of lines of software.

7.28 Laboratory extension: measuring a realistic offline workload

A working application needs more than correct database transactions. The user must be able to complete an ordinary task without excessive delay or repeated errors. Define a representative workflow: open the application, choose a site, enter a measurement, validate it, save it locally, and check whether synchronisation is pending. Repeat the same workflow with a fixed number

of records under connected and disconnected conditions. The protocol should report median and worst-observed completion times, not only a best-case result.

7.28.1 Workload size and storage

Create databases with 100, 1,000 and 10,000 invented records. Measure actual file size after insertion, then after marking records acknowledged and applying an appropriate database-maintenance routine. The apparent bytes per record may change because of indexes, pages, journal files and free space. Do not infer that storage scales exactly linearly from one small experiment. Repeat with a representative distribution of record lengths and attachment sizes if the real application includes text notes or images.

7.28.2 Concurrency and delayed synchronisation

Simulate two devices collecting records independently while disconnected. Give each device a distinct collection of stable identifiers, with one deliberately conflicting edit to an existing record. After reconnection, synchronise both and inspect the server state. The desired outcome is not necessarily automatic merging. It is preservation of valid independent observations and explicit recording of the conflict. A conflict that goes undetected is worse than one that requires manual resolution.

7.28.3 Energy and connectivity costs

If devices are battery powered, a serious energy claim requires measurement with a suitable instrument or carefully documented platform counters. Wall-clock time alone does not establish joules consumed. Compare network bytes, number of retries, screen-on time and synchronisation duration; these are useful engineering measures even when energy itself is not directly recorded. If the programme requires uploading large attachments, consider whether locally processing or compressing them reduces communication without destroying the evidence the organisation needs.

7.29 Reliable failure messages

A field worker should not be confronted with a technical traceback as the only explanation of failure. Error messages can distinguish an invalid measurement, lack of network access, expired credentials, server rejection and an unresolved conflict. Each needs a different user action. For example, an invalid unit can be corrected locally, while a duplicate key with mismatched content may require supervisor review. The system should avoid encouraging repeated submission when the server state is ambiguous.

7.30 A staged deployment decision

The final lab asks the team to answer four questions in order: Does the core database preserve data under expected interruptions? Does the synchronisation protocol avoid duplicate effects? Can intended users understand and complete the workflow? Is there an institution prepared to own and maintain it? Passing the first two is necessary but not sufficient for deployment. An honest answer may be that the software is ready for a supervised pilot but not for use with sensitive real records.

7.31 Exercises and investigations

1. Define atomicity and explain why saving a measurement and its outbox event in different transactions can cause silent loss of synchronisation.
2. Distinguish at-least-once message delivery from exactly-once effect. Show how a stable idempotency key changes the second property without promising the first.
3. Sketch a state machine for a locally saved observation. Identify which transitions require a user action, a local commit or a server acknowledgement.
4. A field device stores 600 records per week but can successfully upload only 400. Calculate queue growth over ten weeks, ignoring deletion. Explain which assumptions the calculation omits.
5. Design a crash test that distinguishes rollback of an incomplete transaction from loss of a committed transaction following storage failure.
6. Give a policy for detecting and resolving two offline edits to the same observation. Why may “latest timestamp wins” be unsafe?
7. Prepare a short threat model for a tablet holding confidential facility data. Identify one plausible risk that encryption alone cannot remove.
8. **Implementation project.** Extend the companion SQLite example with a testable conflict policy, restoration test and a report of disk use for 100, 1,000 and 10,000 invented records.

What this chapter establishes. Reliability is a collection of specific properties: saved records survive expected failures, repeated messages do not duplicate observations, people can understand system states, and maintainers can repair problems. Chapter 8 applies these lessons to a real operational problem: keeping essential medicines available without tying up resources the health service cannot afford.

Chapter 8

Reliable Health Services and Essential Supplies

A medicine stock-out is not simply a forecasting error. It may involve uncertain demand, delayed procurement, incomplete records, finite storage and decisions made under a limited budget. This chapter develops an inventory model that distinguishes observed dispensing from true need, derives reorder policies and shows how to evaluate them without confusing a retrospective simulation with demonstrated improvement in patient care.

8.1 What a stock-out reveals—and what it conceals

A patient arrives at a health centre needing an essential medicine. The medicine is not available. The register records no tablet dispensed. If an analyst treats recorded dispensing as demand, the day's demand appears to be zero. A forecasting model trained on that ledger may therefore learn that fewer medicines are needed precisely during the periods when shortages are worst. This is more than an inconvenient missing value. It is a measurement process in which stock availability constrains what the records can reveal.

Let D_t denote units requested or clinically indicated in period t, I_t inventory available, and Y_t units actually issued. Without substitutions or other complications, $Y_t = \min(D_t, I_t)$, so $Y_t \leq D_t$. When $I_t < D_t$, the observed issue Y_t is a censored version of demand. A model that explains Y_t well can still underestimate what the facility needs to avoid future stock-outs. This observation should be central to the chapter because it connects Chapter 3's measurement warnings with Chapter 6's machine-learning caution and Chapter 5's optimisation.

8.1.1 The consequences have multiple owners

Patients experience delays, return visits, private purchases or interrupted treatment. Staff experience frustration and emergency ordering. The facility may waste supplies through overordering or expiry, while the district budget carries the financial cost of tying up too much working capital. These outcomes need not move together. A policy that minimises average stock value can cause frequent shortages; a policy that keeps every item at a very high level

can be unaffordable and increase expiry losses. The objective must state the trade-off and distinguish non-negotiable clinical safety requirements from costs that can be traded in a model.

The WHO Service Availability and Readiness Assessment provides an established framework for examining facility readiness and tracer commodities, but it is not a longitudinal transaction ledger of medicine issues, orders and lead times at a particular Ugandan clinic (World Health Organization, 2015). The main measured-data study therefore requires authorised, de-identified operational records. Such records are not included with this book; their availability must not be implied.

8.2 The inventory balance is a conservation equation

For a single medicine, define I_t as usable stock just before period t begins, R_t as receipts during the period, Y_t as units actually dispensed, and W_t as losses from expiry or damage. The recorded balance is

$$I_{t+1} = I_t + R_t - Y_t - W_t. \tag{8.1}$$

Every term has units of medicine units, tablets or packs, with conversion factors recorded explicitly. The equality assumes no unrecorded transfer, theft, counting error or retrospective correction. If those events occur, the balance discrepancy should be investigated rather than hidden in a generic noise term. The right-hand side must satisfy $I_{t+1} \geq 0$ under a feasible dispensing process. Conservation is a necessary quality check, not evidence that the demand record is complete.

Suppose a facility starts a day with 30 packs, receives ten, dispenses 28 and writes off two expired packs. Its closing stock should be $30 + 10 - 28 - 2 = 10$ packs. If the register reports 14, at least one recorded component, unit conversion or timing convention requires reconciliation. A sophisticated forecast of tomorrow's stock cannot repair today's unexplained four-pack discrepancy.

8.3 On-hand stock, on-order stock and backorders

A reorder policy should not depend only on what is physically present. Suppose 20 packs are on hand and 100 are already ordered but will arrive next week. Ordering another 100 today may be unnecessary if the next delivery is reliable. Define inventory position as

$$P_t = I_t + O_t - B_t,$$

where O_t is outstanding ordered stock and B_t is backordered demand, if the system permits backorders. This is a bookkeeping identity under the chosen convention; lost demand that never enters a backorder ledger is not represented by B_t. In healthcare, certain unmet needs cannot safely be treated as backorders that can simply wait. Each product requires a clinically informed representation of what happens when it is unavailable.

Order lead time L is the interval from placing an order until the stock becomes usable. It may include approval, supplier preparation, transport and inspection. A single average lead time does not describe repeated delays or long-tail disruption. An effective policy must consider both the typical lead time and its uncertainty.

8.4 Deriving a reorder point rather than guessing one

Assume daily demand D_t is independent and identically distributed with mean μ_D and variance σ_D^2, and lead time is a fixed integer L. Demand accumulated during lead time is

$$D^{(L)} = \sum_{k=1}^{L} D_{t+k}, \qquad \mathbb{E}[D^{(L)}] = L\mu_D, \qquad \operatorname{Var}(D^{(L)}) = L\sigma_D^2.$$

The final variance equality requires independence; with autocorrelation, covariance terms must be added. If a normal approximation is defensible and a cycle-service probability of $1 - \alpha$ is desired, an illustrative reorder point is

$$r = L\mu_D + z_{1-\alpha}\sigma_D\sqrt{L}. \tag{8.2}$$

The first term covers expected lead-time demand; the second is a safety-stock allowance. The standard normal quantile $z_{1-\alpha}$ is not an ethical statement about acceptable clinical shortages. Its value must be selected with the service and checked against actual demand behaviour. Skewed, intermittent and nonstationary demand may make the normal approximation poor (Silver et al., 2017; Zipkin, 2000).

Worked example 8.1 (Illustrative lead-time calculation)**.** *Suppose observed and verified daily demand averages 12 packs with standard deviation four packs, lead time is five independent days, and a one-sided 95% normal service target is used. Then expected lead-time demand is* 60*, and the standard deviation is* $4\sqrt{5} \approx 8.94$*. With* $z_{0.95} \approx 1.645$*, the reorder point is* $60 + 1.645(8.94) \approx 74.7$*, rounded to 75 packs. If demand is underestimated because days with zero stock are recorded as zero demand, this calculation can give false reassurance. The arithmetic is correct only under the demand and lead-time assumptions.*

8.4.1 When lead time itself is random

Let L be a positive integer-valued random lead time independent of the daily demands, with mean $\mathbb{E}[L]$ and variance $\operatorname{Var}(L)$. The law of total variance gives

$$\operatorname{Var}(D^{(L)}) = \sigma_D^2\mathbb{E}[L] + \mu_D^2 \operatorname{Var}(L). \tag{8.3}$$

Proof. Conditional on L, the accumulated demand has mean $L\mu_D$ and variance $L\sigma_D^2$. Therefore $\operatorname{Var}(D^{(L)}) = \mathbb{E}[\operatorname{Var}(D^{(L)} \mid L)] + \operatorname{Var}(\mathbb{E}[D^{(L)} \mid L])$, yielding the formula. □

The second term can dominate when supplier delays vary greatly. Replacing a volatile lead time with its mean erases a risk that a predictive demand model cannot correct. If demand and lead time are dependent, the displayed formula needs modification and should not be used uncritically.

8.5 Service levels are not the same as stock-out days

A cycle-service level is the probability that demand is met during a replenishment cycle without a stock-out. A fill rate is the fraction of demanded units supplied without delay. The percentage

of days with any stock-out is a third measure. These are not interchangeable. A facility may have only one stock-out day but fail to meet a large fraction of demand on that day. Alternatively it may experience frequent short gaps with relatively few missed units. Where clinical severity differs across medicines, a common service target may also be inappropriate.

For a horizon of T periods, define total unmet demand $U = \sum_t (D_t - Y_t)$. The empirical unit fill rate, when D_t is genuinely observed, is $1 - U/\sum_t D_t$. If actual demand is unknown during shortages, the denominator is unidentified and the ratio is not calculable from the issues ledger alone. Reporting a high fill rate based only on recorded dispensing then becomes circular. An honest report may supply lower bounds, uncertainty ranges or explicitly measured patient requests from an additional source.

8.6 The order-up-to policy and its limitations

A periodic review policy places an order intended to raise inventory position to a target S. The nominal order at review is $Q_t = \max\{0, S - P_t\}$. Under deterministic lead time and known demand the target can be chosen to cover the review interval plus lead time. With uncertain demand, the target incorporates a protection level. Rounding to supplier pack sizes, minimum order quantities, delivery limits and cold-chain storage requirements can make the idealised formula infeasible.

Proposition 8.1 (Monotonicity of the simple order rule)**.** *For fixed S, $Q(P) = \max\{0, S - P\}$ is nonincreasing in inventory position P. Increasing available or on-order stock cannot increase the suggested order under this rule.*

Proof. Both the function $S - P$ and its maximum with zero are nonincreasing in P. Thus $P_2 > P_1$ implies $Q(P_2) \le Q(P_1)$. □

The proposition is a small property worth testing in software. If a programmer implements a rule that increases suggested orders when on-order inventory increases, the implementation may have reversed a sign or misunderstood the inventory position. Formal invariants are useful even for simple tools.

8.7 Choosing a forecasting baseline for intermittent demand

Some supplies are used almost daily; others are requested irregularly. A moving average may be adequate for frequently issued products but poor for an item with many periods of zero use and occasional large requests. Intermittent-demand methods seek to separate the size of nonzero demand from the interval between occurrences. Croston's original approach is historically important, but later work examined its bias and modifications (Croston, 1972; Syntetos and Boylan, 2005). No forecasting family should be chosen solely because a medicine appears "specialised". The observed zeros must first be classified as genuine zero need, non-reporting, stock-out censoring or another cause.

A practical comparison includes a documented current ordering rule, a fixed minimum-stock rule, a moving-average forecast, an intermittent-demand baseline when appropriate, and a

probabilistic forecast if data justify it. A machine-learning model may be added, but should not be presumed superior. If the historical data cover only a few orders, an elaborate model may learn the supplier's past mistakes rather than underlying need.

8.8 Optimisation across multiple medicines

A facility may need to allocate scarce budget among several medicines j. Let q_j be order quantity, c_j unit cost, v_j storage volume and B available purchasing budget. At a minimum,

$$\sum_j c_j q_j \le B, \qquad \sum_j v_j q_j \le V, \qquad q_j \ge 0.$$

Products may be restricted to integer pack counts. Additional conditions can represent minimum clinical stocks, expiry horizons, temperature-sensitive storage and supplier pack sizes. One formulation minimises expected shortage and holding costs:

$$\min_q \sum_j \mathbb{E}\Big[h_j(q_j - D_j)^+ + p_j(D_j - q_j)^+\Big] \quad \text{subject to budget and capacity constraints.} \tag{8.4}$$

The notation $x^+ = \max\{x, 0\}$ separates surplus stock from shortage. The coefficients h_j, p_j represent modelled consequences; for essential medicines, a shortage penalty is not a substitute for explicit clinical safety constraints or ethical oversight. The model can rank feasible plans under stated assumptions, but cannot infer those values from routine spreadsheets alone.

For a single product without constraints, the classical newsvendor model yields a critical fractile: if excess stock costs C_o per unit and unmet demand costs C_u per unit, order to the smallest q for which $F_D(q) \ge C_u/(C_o + C_u)$, under the standard continuous-demand assumptions (Zipkin, 2000; Silver et al., 2017). This formula illustrates why a forecast and a cost assessment must be considered together. It does not justify assigning arbitrary large monetary values to patient harm.

8.9 Evaluating policies in the correct order

A responsible study first checks that the baseline ledger reconciles. It then establishes which demand observations are reliable and what happened during stock-outs. Next it documents the current rule, lead-time distribution and cost assumptions. Only then does it compare alternative policies. Historical replay can be useful, but if orders would have changed under the new policy, some counterfactual receipts, stock-outs and hence recorded demand would also have changed. A retrospective simulation needs an explicit demand model and should be labelled as such.

The minimum evaluation uses a rolling time split: choose a past cutoff, fit forecasts only on information then available, simulate or assess decisions over a subsequent period, and repeat. Compare stock-out days, unmet demand where identifiable, expiry, average inventory, emergency orders, cost and required working capital. Report sensitivity to delayed deliveries and sudden demand increases. A policy that wins only under a narrow assumed lead-time distribution is not robust evidence for deployment.

8.9.1 The problem of demand not recorded during shortages

When only $Y_t = \min(D_t, I_t)$ is observed, D_t is known exactly on days when stock is sufficient, but on a stock-out day we know only $D_t \geq Y_t$ without further observations. The demand distribution may therefore be unidentifiable from the issue ledger alone. Patient requests, prescriptions, referral records or replenishment patterns might provide additional information, but each has its own missingness and operational bias.

Proposition 8.2 (Non-identifiability from censored issues alone)**.** *Fix a day with available inventory $I > 0$ and observed issues $Y = I$. Both latent demands $D = I$ and $D = I + 100$ produce the same observed issue $Y = \min(D, I)$. Therefore this single issue observation cannot identify the amount of unmet demand.*

Proof. For both candidate demands, the minimum with I equals I. The observation is compatible with at least two distinct latent demand quantities, so no unique determination follows. □

This proof is intentionally elementary. It prevents a sophisticated modelling technique from making a claim the available measurements cannot support. Where sufficient additional records do not exist, report bounds and scenario results instead of fabricated ground truth.

8.10 A simulation with transparent assumptions

The companion programme `code/ch08_inventory.py` creates an explicitly artificial daily demand sequence, a supplier lead-time scenario and two simple replenishment policies. It preserves daily demand separately from actual dispensing so that unmet need can be computed honestly in the teaching simulation. It records receipts, available stock, dispensed units, shortages and orders, then checks conservation. Readers can change the lead time and demand pattern, compare policies and see how conclusions depend on assumptions.

A valuable exercise deliberately hides the artificial demand on stock-out days and recomputes a forecast using only dispensing. Because the simulated ground truth is known to the programme, the reader can measure the resulting bias. In a real facility, the ground truth is usually not available, so this exercise must not be presented as validation of an empirical correction method. The distinction between the teaching simulation and an authorised operational study should appear in every table and caption.

8.11 Designing the human-facing stock-planning tool

An acceptable prototype should produce a suggested order and an explanation: stock on hand, outstanding orders, verified lead-time assumptions, demand estimate, safety allowance, budget effect and the possibility that demand is censored. Staff should be able to inspect and override the recommendation, with the reason recorded. If the tool's input is missing or contradictory, withholding an automated recommendation may be safer than outputting a confident number.

An institution needs a procedure for inventory counts, corrections, back-ups, monthly review and software updates. Chapter 7's offline-first ideas apply immediately: dispensing should

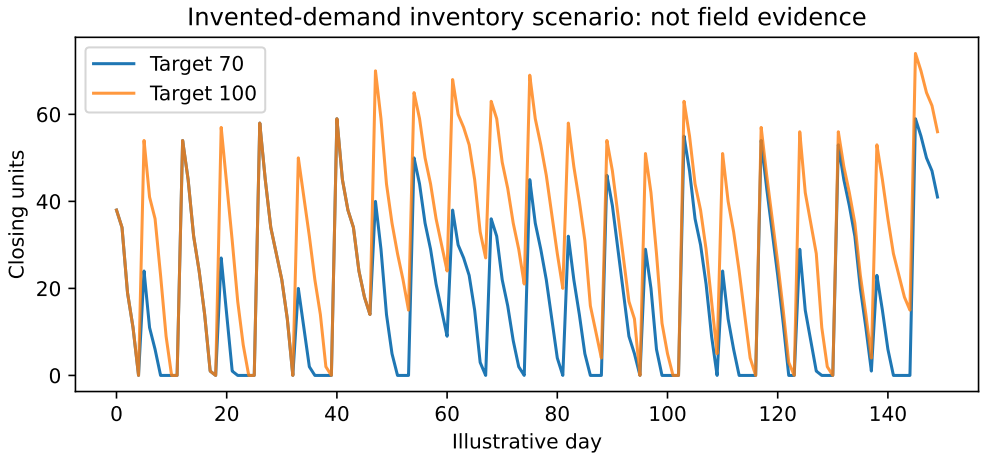


Figure 8.1: Invented demand and fixed review/lead-time assumptions; each policy sees the same artificial demand sequence. This comparison checks the mechanics of a policy simulator, not changes in a real health-centre stock-out rate.

remain recordable when the network is unavailable; duplicate synchronisation should not double the stock deduction; each adjustment must have an audit trail. For controlled or sensitive medication records, access and privacy requirements are operational obligations, not optional additions after a pilot.

8.12 From a promising simulation to a measured health benefit

A policy improvement in simulation is not proof of fewer real stock-outs. A prospective study would record the pre-existing process, implementation fidelity, staff time, supplier behaviour, stock balances, patient consequences and any unintended workload. Depending on the number of facilities and feasibility, a staged pilot, a carefully designed comparison or another appropriate evaluation may be possible. The study would need ethics and institutional permissions where patient information or care delivery is involved. Even if average stock-outs fall after deployment, concurrent changes in budgets or procurement schedules may explain part of the change (Hernán and Robins, 2020; Oberkampf and Roy, 2010).

The public demonstration should use de-identified or synthetic information. Identifiable patient-level records cannot be published merely because they would make the book's example appear more realistic. A validated prototype with a partner is Level B evidence under the book's convention; demonstrated benefits require evaluation appropriate to the actual operational outcome.

8.13 A deeper look at stock-out censoring and statistical identification

The data problem in inventory control deserves more careful treatment than simply filling in missing rows. On an in-stock day, an observed issue may be a reasonable proxy for requested quantity, subject to dispensing practice. On a stock-out day, observed issues provide a lower bound on demand. The missing amount is not absent by random accident: it is missing because a shortage prevented it from being observed through the dispensing channel. This is a form of informative censoring that can make ordinary complete-case estimation unreliable (Little and Rubin, 2019).

Suppose the observed pair is (Y_t, I_t) and assume $Y_t = \min(D_t, I_t)$. If $Y_t < I_t$, the observation is consistent with $D_t = Y_t$ under the simplified model. If $Y_t = I_t$, then $D_t \geq I_t$. Thus a likelihood contribution for a parametric demand distribution with density f_θ and distribution function F_θ may take the form

$$\mathcal{L}_t(\theta) = \begin{cases} f_\theta(Y_t), & Y_t < I_t, \\ 1 - F_\theta(I_t^-), & Y_t = I_t, \end{cases}$$

with appropriate adaptations for discrete quantities and the adopted observation convention. The second case is a tail probability, not the density at the observed issued quantity. Treating censored observations as exact demand values distorts the likelihood.

A likelihood method still needs assumptions about the demand distribution, the timing of available stock and the completeness of records. If stock-outs are systematically correlated with epidemics, supplier failures or reporting practice, a simple independent-demand model may be badly wrong. Additional records about patient requests or clinical orders can be much more valuable than a more elaborate fitting algorithm.

8.14 Ordering under nonstationary demand

Demand may vary with rainfall, seasonal disease patterns, treatment guidelines or changes in clinic attendance. A moving average reacts slowly to abrupt changes, while a highly adaptive forecast can overreact to measurement noise. The choice is therefore a bias-variance and decision-cost question. A safe evaluation uses rolling historical origins and compares whether orders based on each forecast would have been feasible under the observed procurement rules.

One can represent a changing mean demand by $D_t = \mu_t + \epsilon_t$ with μ_t itself evolving over time. State-space or regression models can incorporate known predictors, but such predictors must be available at the order date. If a weekly disease bulletin is published after orders were placed, its realised values cannot legitimately enter those orders in retrospective evaluation. The same information-timing discipline appears in Chapter 4, but inventory decisions make the consequences directly visible.

8.15 Why a zero demand period may mean several things

Consider a specialist medicine with issues $(0, 0, 0, 8, 0, 0, 3, 0, 0)$. One explanation is intermittent patient need; another is that the medicine was unavailable for several periods; a third is that issues were recorded weekly while the timeline shows days. Each mechanism suggests a different forecast and policy. A data dictionary, stock-on-hand ledger, issue timestamps and reporting calendar are therefore prerequisites for a trustworthy intermittent-demand analysis.

A model of zero-inflated demand may distinguish the occurrence of a demand event from its size. But its apparent fit to issue data cannot validate the latent demand process if stock-outs were not recorded. The most useful improvement may be a simple new field in the register: “patient requests not filled.” Recording it accurately and consistently can transform the identifiability of a later analysis, though privacy and staff workload must be evaluated before adding any field.

8.16 Comparing stock rules with a consistent simulator

A simulation should define the order of events in each day. For example: (1) deliveries scheduled for the morning arrive, (2) expiring packs are removed, (3) patient demand occurs, (4) available stock is issued, and (5) orders are placed for future delivery. If another simulator places deliveries after demand, it may report different stock-outs even with identical random numbers and parameters. The sequence is part of the model and belongs in the published code.

Let $A_t = I_t + R_t - W_t$ be stock available after receipts and write-offs. Then $Y_t = \min\{D_t, A_t\}$ and unmet demand is $U_t = (D_t - A_t)^+$. The closing inventory is $I_{t+1} = A_t - Y_t$. An invariant is $I_{t+1} \geq 0$, and conservation requires $I_{t+1} + Y_t + W_t = I_t + R_t$. Test these properties for every simulated period. A simulation that occasionally creates negative inventory or discards deliveries cannot support a comparison of reorder policies.

An illustrative baseline may order a fixed number of packs every week; a second policy may order up to an inventory-position target whenever the position falls below a threshold. The comparison should use the same generated demand and lead-time scenarios for both policies to reduce arbitrary differences caused by simulation randomness. If a method appears superior only because it was assigned a more favourable sequence of disruptions, the comparison is not fair.

8.17 Economic cost and clinical consequence should not be conflated

Inventory holding cost may include storage, cold-chain electricity, financing and expiry. Emergency procurement may add transport and administrative expenses. Those are quantities that can often be priced. Clinical harm from a stock-out is more complicated and cannot responsibly be reduced to a convenient invented shilling value without justification. A multi-criteria report may therefore present financial cost alongside stock-out days, unmet units, number of affected patients and staff workload, with clinical safety requirements specified separately.

The point is not to avoid optimisation. It is to prevent the model from concealing ethical choices

inside an opaque cost function. A transparent constrained formulation may minimise ordinary inventory expenditure subject to a minimum service level for essential items. Another context may call for explicit high shortage penalties. In either case, the decision-maker must know what assumptions produced the recommendation.

8.18 What evidence would support an actual operational improvement?

Before a field pilot, assemble a historical dataset that reconciles stock balances and documents lead times and stock-outs. Record how staff currently decide what to order. A useful prototype can then produce side-by-side recommendations and be reviewed without automatically changing clinical practice. Record whether staff understand the output, what corrections they make and the time required to operate the tool. This is a legitimate feasibility study even if no measured stock-out reduction has yet been established.

If an authorised pilot changes ordering, prospectively monitor relevant outcomes and possible confounders such as supplier disruptions, funding changes or seasonal demand. A comparison design is preferable when practical and ethical, but a small pilot may support only feasibility and descriptive improvement. The reported claim should match the evaluation actually performed. A mathematical demonstration is Level A; a validated operational prototype may be Level B; demonstrated benefit to service delivery requires a suitable Level C evaluation.

8.19 Extended case study: the medicine that disappears from the record

Consider a fictional district health centre that dispenses an essential medicine in packs. The manager has a monthly ledger showing openings, receipts, issues and closing stock. The immediate request is to predict next month's issues and recommend an order. A student may be tempted to import the spreadsheet, fit a forecasting model and minimise RMSE. That procedure could be computationally correct and operationally dangerous if the medicine was unavailable during several weeks. The full investigation begins by reconstructing how the records were produced.

8.19.1 Step one: reconcile the stock ledger

For each period, calculate opening stock plus receipts minus recorded issues and write-offs. Compare the result with closing stock. If the difference is not zero, investigate transfers, damaged stock, returns, unit conversions and delayed entries. A negative recorded balance is a warning about the ledger or event order. The analyst should preserve original records and add documented correction fields; replacing raw figures without an audit trail prevents later verification.

The numerical exercise can begin with an invented five-day sequence. Initial inventory is 20 packs; receipts are $(0, 10, 0, 0, 15)$; recorded dispensing is $(8, 7, 6, 9, 4)$; write-offs are zero. Under

a simple receipt-before-dispensing convention, successive closing stocks are $12, 15, 9, 0, 11$. The fourth day has zero closing stock. That alone does not prove unmet demand occurred: the clinic might have dispensed exactly its available stock. To determine whether any patient was turned away, further information is needed.

8.19.2 Step two: distinguish issues from need

Suppose patients requested twelve packs on day four, while only nine were available. The ledger records nine dispensed. The remaining three are unfilled demand. On a day with no stock and five requests, the ledger would record zero dispensed despite five units of demand. These two facts demonstrate why an issue forecast is not automatically a demand forecast.

If patient request records do not exist, the magnitude of missing demand is unidentified without assumptions. A model might infer latent demand from days when stock was available, but it should present uncertainty and acknowledge the unverifiable extrapolation to stock-out days. A credible data-improvement proposal may be to document unmet requests and clinically justified substitutes. The new recording practice should be tested for staff burden and privacy, since a data field that cannot be reliably maintained does not solve the measurement problem.

8.19.3 Step three: choose the simplest ordering rule that can be evaluated

Begin with the facility's present ordering rule. For example, it might request 60 packs every second Monday, subject to budget approval. Then compare a fixed minimum-stock rule and a periodic order-up-to target. Both need an explicit convention for lead time, outstanding orders and receipt timing. The analyst should write the policy in sentences before coding it. Otherwise two implementations that share the same name may take different actions when a delivery is delayed.

For a policy with target S, let inventory position before review be P_t. The proposed order $Q_t = \max\{0, S - P_t\}$ is not necessarily the quantity finally purchased. The supplier may require complete cartons, the budget may cap purchases, or several medicines may compete for one transport trip. A deployment tool should distinguish *recommended* from *approved* and *received* quantities. A simulated stock balance that assumes every recommendation arrives on time is rarely a defensible model of a constrained supply chain.

8.19.4 Step four: compare under exactly the same circumstances

To compare policies fairly, use a common sequence of underlying demand and supplier disruptions. The policies may make different orders and therefore experience different stock-outs, but they must face equivalent externally generated conditions. Record stock-out days, unfilled demand, average stock, expiry and emergency orders. Include procurement cost and any storage restriction. Do not count a policy's avoided shortage as proven patient benefit unless the connection to patient outcomes has actually been measured.

A meaningful sensitivity analysis changes lead time, shortage severity, seasonality and budget. A policy that wins when suppliers deliver in five days might lose when delivery frequently takes fifteen. A policy requiring a large inventory buffer may be impossible when the facility has

insufficient secure storage. The final recommendation should identify the assumptions under which it remains feasible and the measurements that would most reduce uncertainty.

8.20 A teaching calculation linking order decisions to uncertainty

Assume an illustrative medicine has daily demand D with mean ten packs and standard deviation three, and supplier lead time is seven days. Under independent daily demand, expected lead-time demand is 70 packs and its standard deviation is $3\sqrt{7} \approx 7.94$. An approximately normal 90% one-sided reorder point is $70 + 1.282(7.94) \approx 80.2$, so one might select 81 packs before considering pack sizes. If lead time doubles to fourteen days, expected demand becomes 140 and standard deviation becomes $3\sqrt{14} \approx 11.2$, giving a reorder point around 155. The change arises largely from the longer period that must be covered, not merely from the safety term.

If supplier lead time is itself uncertain, use equation (8.3) only when its independence assumptions are plausible. For example, disruptions caused by heavy rainfall may coincide with higher disease demand; then lead time and demand are dependent. Using a formula that assumes independence can understate risk. A scenario model with explicitly defined joint disruptions may be more honest until enough operational data exist to estimate dependence.

8.21 When a sophisticated forecast can make the plan worse

Suppose a machine-learning model fits historical dispensing with exceptional accuracy by learning that reported issues fall during periods when stock is zero. It then predicts low demand for the same season next year. An optimisation model treats those predictions as true demand and recommends smaller orders. The resulting plan may perpetuate the stock-out. The problem is not repaired by lower cross-validation error against historical issues: the target variable was the wrong quantity.

This is a useful teaching counterexample because it demonstrates the limits of model complexity. The correct first intervention may be improving demand observation, separating stock-out periods and setting a conservative policy for essential supplies. Only after those issues have been addressed does it make sense to test whether more flexible forecasting improves decisions.

8.22 A prospective evaluation protocol

The next stage requires a willing facility and a documented baseline. Record stock counts, issues, unmet requests where feasible, lead times, budgets and emergency purchases for a sufficiently long period to describe the relevant variation. Allow staff to review the tool's suggestions without automatically accepting them. Collect usability information: time needed to enter data, frequency of corrections, trust in the explanation and the cost of maintaining the system.

If the pilot proceeds to changing orders, the evaluation should specify its primary outcome and how unrelated changes will be handled. A decline in stock-outs might arise from higher procurement funding, improved supplier availability or reduced disease demand rather than

the software. If comparison facilities or staggered adoption are feasible, they may strengthen inference, but a small pilot may still be best described as a feasibility assessment. A serious book should teach readers how to report a limited result accurately rather than pressure every project to claim impact.

8.23 A tool the facility could keep using

The desired deliverable is not an unreviewable prediction dashboard. It is a modest planning system that displays stock on hand, open orders, recent verified need, delivery-time assumptions, proposed quantity, expected budget use and the reason for any warning. It should flag contradictory data and retain authorised overrides. If electricity or internet connectivity fails, the local transactions should preserve records and synchronise without duplication once the service returns.

The system's ongoing cost includes time to reconcile stock, training staff, backup, equipment maintenance and the administrative work of ordering. A method whose calculations are free but whose data requirements overwhelm staff is not an affordable solution. The right measure of success is whether the process helps people make dependable supply decisions over time, not whether an algorithm produces an impressive graph at the end of a student project.

8.24 Mathematical workshop: the newsvendor rule from first principles

A basic one-period inventory problem captures a central trade-off. Before demand D becomes known, the manager chooses stock quantity $q \geq 0$. If $q > D$, the excess incurs holding or expiry cost C_o per unit. If $D > q$, the shortage incurs cost C_u per unfilled unit. Expected cost is

$$G(q) = C_o \mathbb{E}[(q - D)^+] + C_u \mathbb{E}[(D - q)^+].$$

Assume continuous demand distribution function F with sufficient regularity. Differentiating under the expectation gives

$$G'(q) = C_o F(q) - C_u\{1 - F(q)\}.$$

Setting the derivative to zero yields $F(q^*) = C_u/(C_o + C_u)$. The derivative is nondecreasing because F is nondecreasing, so a quantile satisfying the condition minimises expected cost. The result is not a universal target for essential medicines: it assumes a one-period decision, a meaningful shortage cost and a demand distribution that adequately describes the future. Constraints on cash, suppliers and service levels can alter the feasible optimum.

8.24.1 A numerical illustration

Suppose demand is uniformly distributed between 0 and 100 packs, overstock costs one unit and shortage costs four units. The critical fractile is $4/(4 + 1) = 0.8$, so the optimal unconstrained order is 80 packs. If shortage cost is increased to nine, the target rises to the 90th percentile,

90 packs. This is a mathematically valid sensitivity result under the illustrative distribution. It does not tell a real pharmacy that a missed treatment is worth four or nine monetary units; such consequences need separate and responsible assessment.

8.25 Service levels under discrete demand

Medicine packs are discrete. If D takes integer values, the optimal order under the one-period model is the smallest integer q satisfying $F(q) \geq C_u/(C_o + C_u)$, with possible ties. A normal approximation to demand can be useful for sufficiently regular high-volume items, but discrete or intermittent demand may require direct empirical distributions or suitable count models. Replacing every demand distribution with a Gaussian can produce negative simulated demand, an obvious physical contradiction that should be handled explicitly rather than silently clipped without explanation.

For an item with many true zero-demand days, the distribution may have a point mass at zero and a long right tail. A mean forecast alone hides both features. An order chosen as the mean demand can perform poorly when shortage consequences are asymmetric. This is why probabilistic forecasting and decision-oriented evaluation often matter more than small changes in point-prediction error.

8.26 The inventory position and the order pipeline

The inventory position $P = I + O - B$ contains outstanding orders that are not yet physically available. Suppose on-hand stock is 30, outstanding confirmed orders total 50, and there are no backorders. With an order-up-to target of 100, suggested additional order is $100 - 80 = 20$, not 70. A naive programme using only on-hand stock would overorder by 50. If outstanding orders are delayed or unreliable, the decision becomes more complicated: the planner must represent possible delivery failure rather than pretend that the existing order is certain to arrive.

For daily updating, use a pipeline indexed by expected receipt day. A new order Q_t scheduled with lead time L_t adds Q_t to the receipt queue at $t + L_t$. At each day, receipts are drawn from the scheduled queue before demand is served under the chosen event order. The implementation should test that every placed order arrives at most once in the simulation. If deliveries may be partial or lost, the model must represent those outcomes explicitly.

8.27 A worked policy stress test

Consider a pharmacy whose normal demand is ten packs daily. A fixed order of 70 packs every seven days matches mean usage in a perfectly reliable seven-day cycle, but supplies no buffer for lead-time delays or demand variation. If the delivery scheduled for day seven arrives on day ten, the facility may experience shortage before the next order is received. An order-up-to policy with a higher target can reduce shortages, but it also increases average inventory. Compare the alternatives under a common sequence of demand and delayed receipts, calculating every day's stock, dispensed units and unmet need.

The result should report at least two competing outcomes. If target 100 reduces unfilled units relative to target 70 but raises mean closing stock, neither policy is universally "better" without a cost or service requirement. If a medicine has a short expiry horizon, higher average stock can translate into waste. If the cold-chain capacity is limited, target 100 may be infeasible. A sound comparison never announces victory on one metric while hiding the trade-off in the other.

8.28 Reconciliation as a software property

Suppose opening stock is I_t, receipts R_t, issues Y_t, write-offs W_t, and closing stock I_{t+1}. Every valid inventory transaction should satisfy

$$I_t + R_t = Y_t + W_t + I_{t+1}.$$

A unit test can generate nonnegative feasible events and verify the identity to machine precision for numeric models, or exactly for integer pack counts. If the test fails, the code or bookkeeping is wrong. The test does not establish that the opening balance or receipts were measured accurately; that requires physical counts and authorised records. The distinction mirrors Chapter 2's separation between verifying equations and validating their representation of the world.

8.29 How a clinician and a programmer should work together

A mathematically consistent reorder rule might suggest substituting one medication for another when stock is low. That is not a safe decision for a generic optimisation algorithm. Clinical substitution depends on indications, dosage, contraindications and professional authority. Such choices must be specified by qualified clinicians and regulatory requirements, not learned solely from historical stock records. The software should support human-approved workflows rather than invent clinical equivalences.

The same applies to prioritising scarce medicines. A shortage penalty in a mathematical objective may summarise a planning preference, but cannot represent all consequences for patients. For essential items, hard minimum-stock or escalation rules may be more appropriate. The role of the book's mathematics is to make these decisions clearer and more accountable, not to replace professional judgment.

8.30 Reporting the result of the computational laboratory

The companion simulation writes daily CSV ledgers for two illustrative order-up-to policies and a summary of unmet units, stock-out days and mean stock. These outputs are generated from invented demand. A researcher can reproduce the calculations exactly under the stored random seed, inspect the event order and challenge the assumptions. If the results were collected from a real facility, the report would need provenance, permissions, quality checks and independent validation. The word "simulation" should appear in every table and figure that depends on the invented demand model.

An excellent student result may be the discovery that ranking of policies reverses when lead time becomes uncertain or when expiry costs are increased. That finding identifies a measurement need: supplier-delay records or item-specific shelf life. A practical project can be successful when it establishes what information is missing, even before it produces a deployable recommendation.

8.31 Laboratory extension: designing the evidence needed for a real pharmacy tool

The hardest step in an inventory project may be agreeing on what constitutes demand. Dispensing records are often easier to obtain than patient requests, but the two are not equivalent during stock-outs. Before building software, ask staff how an unavailable medicine is recorded, whether substitutions are made, and whether partial issues are documented. Draw the information pathway from a patient's need to a prescription, an approved issue and an eventual stock deduction. Each transition may lose information.

8.31.1 A minimum operational dataset

For each medicine, collect a documented item identifier, unit and pack conversion, date, opening balance, receipts, issues, returns, transfers, adjustments, expiry and closing balance. A separate order table contains order date, supplier, quantity requested, quantity approved, expected delivery and actual receipt dates. If permitted and feasible, an unmet-request table records requests not filled without exposing unnecessary patient-identifiable details. These records should share stable identifiers, but patient privacy must remain a primary design constraint.

Do not assume that joining two tables on medicine name will work reliably. Different spelling, strengths, formulations and pack sizes can create incorrect matches. A medicine identifier should include the attributes necessary to distinguish genuinely different products. If two brands are clinically substitutable, that relationship requires approved professional input and cannot be inferred from identical stock codes.

8.31.2 Physical counts as a reality check

At defined intervals, staff compare the recorded closing stock with a physical count. Discrepancies should be logged, explained where possible and corrected with an audit trail. A modelling team should not overwrite the historical balance to make the conservation equation look perfect. The difference may reveal losses, timing errors or unit confusion that also affect the demand forecasts. A data-cleaning pipeline that removes these cases without explanation may delete the evidence most relevant to the operational problem.

8.32 A complete policy comparison protocol

Define at least three strategies: the current practice, a simple fixed minimum-stock rule and a proposed probabilistic or order-up-to policy. Specify review days, lead times, order limits, budget, storage and emergency purchase conditions. Run all strategies against the same

permitted historical periods and, for unobserved counterfactual demand, the same clearly declared scenario assumptions. Hold out the final period from model selection and report both service and financial consequences.

The primary output can be unfilled units or stock-out days, depending on what can be measured reliably. Secondary outputs include average inventory, expiry, emergency purchase frequency, ordering workload and working capital. Report results across plausible supplier-delay scenarios. A complicated method should not be preferred when its advantage disappears under a modest change in an unverified assumption.

8.32.1 A realistic stopping criterion

A prototype should stop and request human review when required inputs are missing, stock balances are irreconcilable, the forecast is based mainly on censored periods or the proposed order violates a budget or clinical constraint. An automated recommendation is not automatically better than an explicit warning that information is insufficient. The ability to state why a recommendation was withheld is part of reliable design.

8.33 Interpretation and responsible handover

The final report should identify exactly what the programme established: the arithmetic of an inventory policy, its results under stated simulated scenarios, and any performance measured in an authorised pilot. It should separately identify what was not established: true unmet demand when no request records exist, causal reduction in real stock-outs, or clinical benefit. The handover plan names staff who maintain the records, review recommendations, approve orders, and diagnose system failures after the original developer leaves.

8.34 Exercises and investigations

1. Derive the stock balance when emergency purchases and transfers between facilities are recorded separately. Identify all terms and units.

2. Explain why observed dispensing on a day with zero stock does not measure demand. Give two different demand histories consistent with the same issue record.

3. Starting from independent daily demand, derive the variance of total demand during five days of fixed lead time. State why the calculation fails with correlated days.

4. Use (8.3) to explain the contribution of lead-time uncertainty when average daily demand is large. Which term is missed by replacing random lead time with its mean?

5. Compare a cycle-service level with a unit fill rate and percentage of stock-out days. Construct a small example in which the measures rank two policies differently.

6. Verify the monotonicity of the order-up-to rule and write a software unit test expressing it.

7. Design a rolling historical evaluation comparing an existing reorder rule with a moving-average rule. Identify every piece of information that must be available before each simulated order.
8. **Partner study design.** Draft a data-request and governance plan for a willing clinic: inventory counts, receipts, issues, expired stock, delays, orders and stock-out dates. Specify how you would avoid making patient-identifiable data public.

What this chapter establishes. Better forecasts alone do not make essential medicines available. The decision is a coupled problem of measurement, supplier behaviour, budgets, inventory conservation, robust policy and accountable people. That is precisely why the book began with the problem rather than with an algorithm. The following chapters extend the same method of reasoning to energy, agriculture and infrastructure.

Part III

From Prototypes to Lasting Impact

Chapter 9

Knowing Whether a Solution Actually Works

A hospital can have accurate forecasts, correctly functioning software and unchanged patient outcomes. This chapter asks a harder question than whether an algorithm runs: did introducing the proposed practice cause an improvement that matters? We learn to formulate that question, build a defensible comparison, expose the assumptions behind causal estimates, and report costs and uncertainty without claiming more than the evidence supports.

9.1 The decision that remains after the software works

Imagine the district health office that has followed the first eight chapters. Staff have checked the stock ledger, reconstructed plausible demand, compared ordering policies and built an application that works offline. Its simulation predicts fewer shortages. The district now asks a reasonable question: *If we adopt it, will patients actually find essential medicines when they need them?*

The question is not answered by a smaller forecast error. Suppliers may miss deliveries; clinical staff may continue ordering informally; or a cost-saving policy may increase shortages of a medicine that is particularly important. Sometimes an intervention helps for reasons other than the model. Extra supervision during the pilot can improve record keeping, giving the appearance that the algorithm is responsible. To claim improvement we need to ask what would have happened to the same service without the intervention.

These claims should be evaluated separately. A favourable answer in one row does not imply a favourable answer in the next. This separation is essential to implementation research (Proctor et al., 2011; Donabedian, 1988) and the logic of impact evaluation (Gertler et al., 2016; Hernán and Robins, 2020).

Table 9.1: Four claims that require different evidence.

Claim	Suitable question	Typical evidence
Mathematics is valid	Do the deductions follow from stated assumptions?	Proofs, counterexamples, dimensional checks
Code implements the rule	Does the program compute the intended quantities?	Tests, code review, replay, reproducible runs
System can be used	Does the tool survive the actual workflow?	Field usability, failure drills, process records
Service improves	Is the outcome better *because* of the intervention?	Credible comparison, measured outcomes, sensitivity analysis

9.2 Start from a decision, not a statistical test

For the medicine-ordering example, the decision maker needs at least five statements in writing: which facilities are eligible, which practice changes, which outcomes matter, how long the assessment will run, and what size of benefit would justify the extra cost. These statements should be agreed before seeing the post-intervention outcomes.

A useful primary outcome might be the proportion of scheduled days on which a specified essential medicine was available. Other outcomes include unfilled prescriptions, expiry, emergency procurement, staff work time and total resource cost. Define numerator, denominator and recording rule. A facility that closes for two weeks must not quietly disappear from the denominator merely because the new software recorded no transactions there.

An illustrative estimand. Let $Y_i(1)$ denote facility i's average medicine-availability rate over a fixed follow-up period under the new ordering practice, and $Y_i(0)$ its rate under the existing practice. The average treatment effect for an eligible population of n facilities is

$$\tau = \frac{1}{n}\sum_{i=1}^{n}\{Y_i(1) - Y_i(0)\}. \tag{9.1}$$

If we instead care about the effect among facilities that actually adopt the tool, we need an estimand conditional on adoption. This is not the same as (9.1) if adoption depends on local leadership or resources. The distinction between an intention-to-treat effect and a treatment-on-the-treated effect is similarly important in a randomised rollout (Angrist and Pischke, 2009; Gertler et al., 2016).

The counterfactual problem is immediate: a facility cannot simultaneously follow both ordering practices at the same time under the same circumstances. We observe one of $Y_i(1)$ and $Y_i(0)$, not both. That is the fundamental obstacle that experimental and observational designs address (Hernán and Robins, 2020).

9.3 What a controlled comparison can identify

Suppose eligible facilities are randomly allocated to immediate implementation ($Z_i = 1$) or continued usual practice ($Z_i = 0$), and assume complete outcome measurements. Define $p = \Pr(Z_i = 1)$, with $0 < p < 1$. Under random assignment, Z_i is independent of the pair of potential outcomes. Then

$$\mathbb{E}(Y_i \mid Z_i = 1) = \mathbb{E}\{Y_i(1)\}, \qquad \mathbb{E}(Y_i \mid Z_i = 0) = \mathbb{E}\{Y_i(0)\}.$$

Consequently, the difference of group means targets the population average treatment effect. The argument is simple but its assumptions are substantial: the groups must truly be assigned by a valid mechanism, the outcome must mean the same thing in both groups, and the intervention used for one facility must not silently alter the counterfactual for another.

Proposition 9.1 (Unbiased difference under complete randomisation)**.** *For a fixed finite population with potential outcomes $Y_i(1)$ and $Y_i(0)$, let exactly n_1 of n units be assigned to treatment uniformly among all assignments of that size, where $0 < n_1 < n$. Under consistency and no interference, the estimator $\widehat{\tau} = \overline{Y}_{Z=1} - \overline{Y}_{Z=0}$ satisfies $\mathbb{E}_Z(\widehat{\tau}) = n^{-1} \sum_i \{Y_i(1) - Y_i(0)\}$.*

Proof. Each unit has inclusion probability n_1/n. Linearity of expectation yields

$$\mathbb{E}_Z\left[\frac{1}{n_1}\sum_i Z_i Y_i(1)\right] = \frac{1}{n}\sum_i Y_i(1).$$

Similarly, because each unit is in control with probability $(n - n_1)/n$, the expected control mean is $n^{-1}\sum_i Y_i(0)$. Subtracting proves the statement. □

Randomisation does not guarantee that one small trial will look balanced, or that the reported estimate is precise. A small programme can have a valid experiment with an uninformative confidence interval. Moreover, the unbiasedness proof does not rescue a trial in which outcomes are selectively missing or a control facility uses the intervention by another route. The evaluation must document implementation, measurement, and deviations.

9.3.1 Unit of assignment versus unit of observation

A district may assign ten health facilities but collect thousands of patient-level transactions. The resulting study does not have thousands of independent assignments. Patients within a facility share staff, procurement and stock records. If assignment is at facility level, uncertainty must reflect that clustering; treating every transaction as independent produces artificially narrow intervals. Where a simple model of equal-sized clusters is reasonable, the design effect is approximately

$$\mathrm{DEFF} = 1 + (m - 1)\rho, \tag{9.2}$$

where m is the number of observations per cluster and ρ is within-cluster correlation. This formula is a planning approximation, not a replacement for cluster-aware analysis (Cochran, 1977; Lohr, 2019).

If a pilot includes six facilities, no amount of transaction volume turns it into a high-powered study of effects across facilities. This is an especially common temptation when data collection is inexpensive but participating institutions are few.

9.4 When random assignment cannot be used

An observational comparison can still be valuable, but the burden of explanation grows. The programme may be installed first in facilities with the most motivated managers; these facilities might have improved under either policy. Routine data may also change when the new application begins to record shortages that previously went unreported. Such changes can produce spurious differences even when the arithmetic is correct.

A causal diagram can help identify candidate confounders: procurement budgets, supplier reliability, geographical accessibility, staffing and management. Adjusting for a variable measured *after* the intervention that is itself affected by the intervention may remove part of the effect or create bias. Adjustment should follow a causal model rather than a convenient collection of columns (Hernán and Robins, 2020).

9.4.1 Difference-in-differences: derive before trusting

Assume we observe an outcome for a treatment group $G = 1$ and a comparison group $G = 0$, in periods $t = 0$ before and $t = 1$ after rollout. Write $\mu_{gt} = \mathbb{E}(Y \mid G = g, t)$ and define

$$\widehat{\tau}_{\mathrm{DiD}} = (\overline{Y}_{11} - \overline{Y}_{10}) - (\overline{Y}_{01} - \overline{Y}_{00}). \tag{9.3}$$

The estimator removes a permanent difference between groups and a common period shift. To interpret it causally, we need a *parallel trends* assumption: absent treatment, average outcomes in the two groups would have changed by the same amount. This is an assumption about an *unobserved* counterfactual trend, not a property that the formula automatically creates (Angrist and Pischke, 2009; Gertler et al., 2016).

Proposition 9.2 (Identification with parallel counterfactual trends)**.** *Suppose only group $G = 1$ receives treatment at $t = 1$. Assume consistency, no anticipation, no spillovers, stable measurement and*

$$\mathbb{E}[Y_1(0) - Y_0(0) \mid G = 1] = \mathbb{E}[Y_1(0) - Y_0(0) \mid G = 0].$$

Then the population counterpart of (9.3) *equals $\mathbb{E}[Y_1(1) - Y_1(0) \mid G = 1]$, the average effect in the treated group after implementation.*

Proof. The treated post-period mean is $\mathbb{E}[Y_1(1) \mid G = 1]$. Every other cell is observed under no treatment. Add and subtract $\mathbb{E}[Y_1(0) \mid G = 1]$ in the first difference. The remaining untreated trend difference is zero by the stated assumption; what remains is the average treated post-period effect. □

A diagram of pre-intervention trends is valuable: divergence before implementation is strong evidence against a simple parallel-trends interpretation. Similar pre-trends, however, are not proof of the assumption. Report alternative comparison groups, possible concurrent policy changes, changes in recording and sensitivity to the observation window. With staggered treatment dates and heterogeneous effects, conventional two-way fixed-effects regressions can behave unexpectedly; methods designed for staggered adoption should be considered (Callaway and Sant'Anna, 2021).

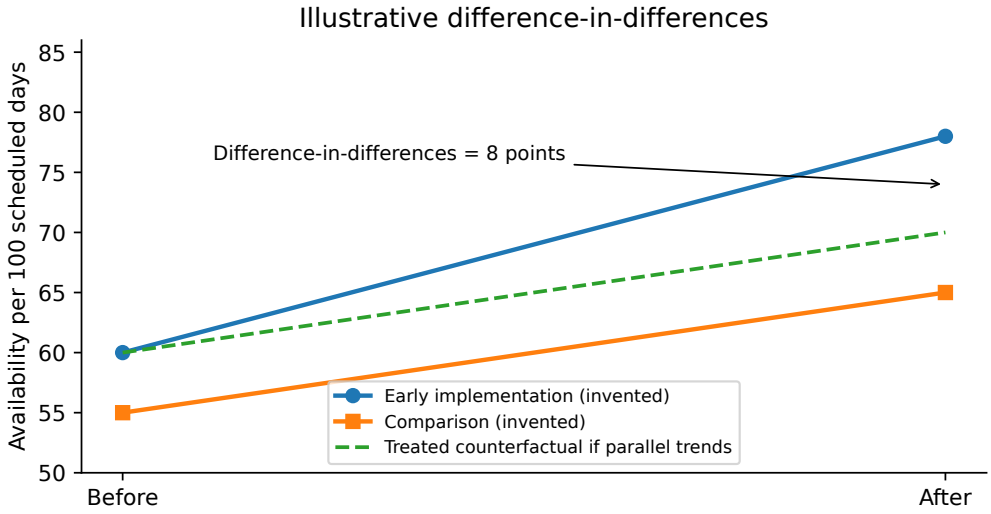


Figure 9.1: Invented before-after outcomes. The dashed line is a hypothetical untreated path for the treated group under parallel trends, not an observed counterfactual.

Worked example 9.1 (The deceptively successful rollout)**.** *In an* invented *four-cell example, treated facilities improve from 60 to 78 days of availability per 100 scheduled days, while comparison facilities improve from 55 to 65. Their before-after changes are 18 and 10 points, so the difference-in-differences estimate is* 8 *percentage points. The naive treated before-after improvement,* 18 *points, assigns the entire time trend to the intervention. Neither number is automatically causal: if the treated facilities also received a special medicine shipment, the* 8*-point estimate remains confounded.*

9.4.2 Matching, regression discontinuity and interrupted time series

Matching can help construct more comparable groups on *measured* baseline covariates, but it cannot correct unmeasured differences on its own. In a regression discontinuity design, a programme assigned by a known cutoff may identify a local effect near that cutoff when potential-outcome regression functions are continuous there and assignment cannot be strategically manipulated. The estimate generally describes people or sites near the threshold, not the whole country. An interrupted time series uses repeated observations to model baseline levels and trends; changes to case definition, reporting, seasonality or concurrent policy still require scrutiny. None of these design names is a certificate of validity; each works only when its own assumptions can be defended (Shadish et al., 2002; Angrist and Pischke, 2009).

9.5 An evaluation that can be carried out with few resources

Evaluations do not have to begin with expensive technology. A facility can record daily stock on hand, scheduled service days, patients unable to receive an indicated item, emergency orders, costs and delivery delays. The difficult part is reliable definitions and a defensible counterfactual.

A practical design should be proportionate to the importance and reversibility of the decision.

1. **Write the causal question.** Name the treatment, comparison, population and follow-up window.
2. **Register outcomes and decisions.** Agree the primary outcome and meaningful minimum effect with practitioners before viewing the results.
3. **Map the assignment mechanism.** Who chooses the first facility? Can selection be random? What nonrandom factors influence uptake?
4. **Audit baseline records.** Investigate missing periods, reporting changes and seasonality.
5. **Preserve the comparison.** Record supply-chain policy changes, extra funding and changes in supervision for both groups.
6. **Analyse at the right level.** Use cluster-aware inference for facility-assigned interventions and report uncertainty honestly.
7. **Check unintended effects.** A policy could improve one medicine's availability while diverting funds from another.
8. **Decide what happens next.** Continue, modify, extend the pilot, or withdraw according to evidence and operational judgment.

A pilot whose confidence interval includes both a valuable benefit and meaningful harm has not proved the solution works. It may still have taught the institution something important about feasibility, data collection and what additional measurement is needed.

9.6 Missing outcomes and changed measurement

Suppose stock availability is observed in 80 of 100 scheduled facility-days and 60 observed days show stock present. The recorded rate among observed days is $60/80 = 0.75$. Without assumptions on the 20 missing days, the rate across all 100 scheduled days lies between $60/100 = 0.60$ and $(60 + 20)/100 = 0.80$. The seemingly precise 75% is conditional on observation; it cannot represent the full period without a missingness assumption.

Proposition 9.3 (Elementary missing-outcome bound)**.** *Let $Y_i \in [0, 1]$ for n scheduled observations, and suppose r outcomes are observed with sum S. With no assumptions on the unobserved outcomes, the population mean lies in*

$$\left[\frac{S}{n}, \frac{S + n - r}{n}\right]. \tag{9.4}$$

These bounds are sharp when the missing outcomes can attain zero and one.

Proof. Each of the $n - r$ missing contributions is between zero and one. Adding their smallest and largest possible totals to S and dividing by n gives (9.4). Setting all missing outcomes to zero or one attains the endpoints. □

Repeated missingness may depend on stock-outs, power loss or staff workload. Those mechanisms should be investigated. For a causal comparison, bounds or sensitivity analysis should be applied separately to treatment and comparison groups before differences are interpreted. Multiple imputation is a method that relies on its own assumptions, not a way of creating observed facts (Little and Rubin, 2019; Rubin, 1976).

9.7 What uncertainty intervals mean here

A confidence interval describes uncertainty arising under a specified sampling or randomisation model; it is not the probability that a fixed parameter lies in the reported interval. In a randomised comparison, finite-sample randomisation inference may be appropriate when the assignment mechanism is known. In a sufficiently large observational study, robust uncertainty estimates can account for clustering and serial dependence where the design supports them. Reporting a narrow interval around a biased estimate is not progress.

For the simple two-group case with independent observations within groups, a conventional standard error is

$$\widehat{\mathrm{se}}(\widehat{\tau}) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}},$$

but the formula is inappropriate for naive use when assignments are clustered or outcomes are serially correlated. An evaluation report must say what constitutes an independent unit and which uncertainty model it uses.

9.8 Costs must be attached to outcomes that matter

A model can reduce measured shortages and still be unaffordable. Count device costs, training, internet, maintenance, batteries, staff work, replacement and record verification. Classify one-time costs separately from recurring costs. State the institutional perspective: a change may save district money while increasing unpaid work for patients.

For two alternatives, an incremental cost-effectiveness ratio is

$$\mathrm{ICER} = \frac{C_1 - C_0}{E_1 - E_0}, \tag{9.5}$$

provided $E_1 \neq E_0$ and a meaningful common effectiveness scale exists. The ratio becomes misleading when effects are close to zero or of opposite sign; present costs and outcomes separately and inspect dominance. Where a defensible willingness-to-pay value λ exists, incremental net benefit is $\lambda(E_1 - E_0) - (C_1 - C_0)$. No threshold should be imported uncritically from a different health system or funding context.

Worked example 9.2 (A transparent, illustrative budget-impact calculation)**.** *Suppose the current service costs 1,200,000 local currency units over the assessment period and provides 8,000 available medicine-days. A proposed service costs 1,320,000 and provides 8,400 days, under* explicit hypothetical assumptions*. The extra 400 medicine-days cost 120,000 units, or 300 units per additional day. This ratio does not estimate value to patients and says nothing*

about the distribution of the 400 days across essential medicines. A district would need to examine opportunity cost: which other service loses the 120,000 units?

Uncertainty about costs and effects should be evaluated jointly. It is poor practice to use the most optimistic effectiveness estimate and the most optimistic cost estimate from unrelated scenarios.

9.9 Reproducible laboratory: an explicit counterfactual trap

The file `code/ch09_impact_evaluation.py` generates an *invented* facility-period dataset with two time periods, a common improvement, a group-specific change and a stated intervention effect. It produces four-cell means and a difference-in-differences summary. Changing the untreated trend for treated facilities illustrates how an estimator can be computed correctly and still fail to identify the effect. Randomised noise in the script serves teaching only; no output is claimed to represent Ugandan hospital records.

The exercise has three stages. First, check the four-cell calculation by hand. Second, alter the group-specific untreated trend and record the bias relative to the known simulation parameter. Third, design a feasible real pilot that could distinguish a stock-policy effect from a procurement-budget change. A technically proficient reader should be able to explain why code verification and causal validity are separate checks.

9.10 The evaluation protocol and evidence record

Before an institution begins, create a short protocol containing the following fields:

Table 9.2: Minimum impact-evaluation record, to be completed before the pilot.

Field	What the team must specify
Decision and owner	The adoption decision, responsible institution, and who may stop the intervention
Eligibility and assignment	Included facilities, selection rule, timing, contamination risk
Treatment	Exact changes to ordering, staffing, training and software
Comparison	Existing practice or alternative, with evidence for its credibility
Primary outcome	Definition, units, denominator, frequency and missingness rule
Other outcomes	Cost, equity, workload, safety and unintended consequences
Analysis	Estimand, estimator, adjustment, clustering, sensitivity tests
Ethics and protection	Permissions, data minimisation, consent where needed
Reporting	Who receives negative results, evidence limits and next decision

A report should distinguish *was deployed*, *was used*, *operated reliably*, and *caused an improvement*. Each is important, none is synonymous with the others. For institutions with very limited resources, good evaluation may be the decision *not* to scale: it prevents a costly idea from being repeated simply because the first demonstration looked attractive.

9.11 Deeper method: choosing an evaluation design before a pilot begins

A decision maker may be willing to test a change but unwilling to deny it permanently to some facilities. This does not rule out a controlled study. The evaluator can sometimes randomise the order in which eligible facilities receive the change. Such a rollout can be ethically and operationally attractive when the institution could not implement the policy everywhere immediately anyway. The comparison is then between facilities already offered the intervention and those not yet offered it, over precisely defined periods.

The rollout order, however, must not be chosen after inspecting the early outcomes. If managers move successful sites to the front of the schedule or change the intervention in response to mid-study results, the analysis must account for those decisions. A transparent assignment record matters more than a sophisticated regression written months later.

9.11.1 Cluster assignment in a practical setting

When the intervention changes an entire facility's ordering rules, facility-level assignment is generally more sensible than randomising individual patients. It would be difficult and unsafe to apply two incompatible purchasing rules to patients within one shared stockroom. The facility, rather than an individual prescription, becomes the experimental unit. The analysis may measure outcomes daily, but uncertainty about treatment effects must respect the number of independently assigned facilities.

Before randomising, group facilities into relevant strata if doing so improves comparability and is justified in the design: for example, comparable size categories or geographical service types. Document the strata and randomly assign within them. Do not create many strata with only one eligible facility per condition, leaving little randomisation. The published result should report the actual assignment procedure, not simply say "randomly selected."

The design may need to consider shared procurement depots. If treated facilities draw from a common, capped medicine supply, their extra orders can deprive comparison facilities of stock. The treatment and comparison groups are then coupled through the resource system. A valid evaluation may require assignment at the level of the shared depot, or explicit measurement of spillovers. This is a substantive problem, not a technical nuisance solved by cluster-robust standard errors.

9.11.2 The intention-to-treat question

Suppose the randomised rollout offers a tool to ten facilities, but only eight actually use it. The effect of assignment measures what happens when an institution offers the programme under realistic implementation conditions. It includes the possibility of nonadoption. This intention-to-treat effect can be a policy-relevant quantity: a ministry considering rollout may care about the effect of the real programme it can deliver, not the idealised effect among only perfect adopters.

A comparison based on actual use rather than assigned offer can be confounded: motivated staff

may be more likely both to use the tool and to maintain complete stock records. Estimating an effect among compliers requires additional assumptions or a design such as an instrumental-variable analysis. The simple difference in outcomes between users and nonusers is not generally that effect (Angrist and Pischke, 2009; Gertler et al., 2016).

9.12 Regression discontinuity with an explicit limit

Suppose scarce service grants are assigned to facilities whose pre-policy readiness score X is at least cutoff c. Facilities just below and just above the cutoff might be comparable, aside from receiving the grant. A sharp regression discontinuity design studies the jump in the conditional expectation of an outcome at c:

$$\tau_{\mathrm{RD}} = \lim_{x \downarrow c} \mathbb{E}[Y \mid X = x] - \lim_{x \uparrow c} \mathbb{E}[Y \mid X = x]. \tag{9.6}$$

The limits have a causal interpretation near the cutoff only if, in the absence of treatment, the expected potential outcomes evolve continuously through c, along with other relevant assumptions. The design is vulnerable if facilities can manipulate the readiness score to pass the threshold or if another policy changes at the same score. A difference near the cutoff need not describe what would happen to facilities with very low readiness scores.

Worked example 9.3 (A local effect is not a national effect)**.** *Suppose a hypothetical subsidy begins at score 70. Facilities scoring 69 and 71 differ little in observed readiness, and an appropriately designed analysis suggests a benefit near 70. A facility scoring 25 may have very different staff capacity and record quality; the estimated local effect does not establish that the same subsidy will help it. Transport from near-threshold facilities to all facilities is a new question. It needs the reasoning of Chapter 11.*

Researchers should report the bandwidth, functional form, data density around the cutoff, covariate balance checks, and sensitivity to reasonable analytical choices. A graph with a sharp-looking discontinuity is helpful but not a substitute for identifying which programmes changed at the cutoff (Shadish et al., 2002; Angrist and Pischke, 2009).

9.13 Interrupted time series and the meaning of a baseline trend

Suppose a district deploys an improved appointment system throughout all clinics on one day, leaving no contemporaneous comparison district. If adequate observations exist both before and after, an interrupted time series can compare post-intervention behaviour with an extrapolated pre-intervention pattern. A simple segmented model is

$$Y_t = \beta_0 + \beta_1 t + \beta_2 \mathbf{1}\{t \geq t_0\} + \beta_3 (t - t_0)\mathbf{1}\{t \geq t_0\} + \varepsilon_t. \tag{9.7}$$

The coefficient β_2 represents an immediate level change at implementation in the model, while β_3 represents a change in slope. But the model assumes the baseline trend would otherwise have continued, after accounting for factors such as seasonality and autocorrelation. A vaccine shortage or a revised reporting form at t_0 may cause a change unrelated to the appointment system.

A credible analysis should examine pre-trend fit, residual dependence, outliers, seasonality, policy history and alternative explanations. If the data consist of only three before and three after observations, a flexible segmented model may be too fragile to support a strong claim. A longer series is not automatically sufficient either; it must contain comparable measurements across the relevant time periods.

9.14 Stepped implementation: design and ethics together

A staged rollout can reduce the practical tension between wanting an evaluation and wanting all eligible sites to receive a promising service. Yet it does not remove the ethical obligation to consider risk. If the intervention is known to be safer and delaying it would cause unacceptable harm, an evaluation design that intentionally withholds it may not be appropriate. Conversely, if efficacy and safety are genuinely uncertain, rapid unmeasured rollout can harm many more people than a carefully monitored pilot.

The ethical decision belongs to responsible institutions and relevant review bodies, informed by patients and staff where appropriate. A book can explain design options; it cannot authorise a clinical or public-service experiment.

9.15 An evaluative model of service quality

The ordering algorithm may be only one contributor to medicine availability. In a stylised structural model,

$$Y_t = f(D_t, L_t, B_t, S_t, M_t, U_t),$$

where D_t is demand, L_t delivery delay, B_t budget, S_t staffing, M_t order-management practice and U_t unmeasured influences. A policy changing M_t may have a large effect when L_t is short and almost no effect when the supply chain has collapsed. The average effect across facilities depends on the distribution of these modifiers. This is one reason impact evaluation should report heterogeneous outcomes rather than a single mean when sufficient evidence exists.

The model is not identified merely by writing f. Its purpose is to guide what should be measured and why a plausible counterfactual might fail. Chapter 2 made the same distinction for physical models: equations clarify assumptions; observations determine whether those assumptions describe the service.

9.16 Negative results and publication responsibility

A study in which the new method provides no demonstrated benefit can be valuable if it identifies a reliable baseline, an unforeseen barrier or a data problem. Repeatedly changing the outcome or evaluation window until a favourable number appears undermines inference. Ideally, primary outcomes, analysis choices and major exclusions should be documented before examining the intervention results. Changes may be necessary, but the report should distinguish planned and exploratory analyses.

When manuscripts or project reports describe a multi-stage evaluation, include the dates of design, data collection, intervention and analysis. An institution should be able to see when an outcome was selected relative to the observed results. This discipline helps prevent a computer-generated table from becoming an unsupported institutional claim.

9.17 Extended investigation: why a medicine pilot can mislead

Consider a district with fourteen facilities. Six receive a new stock-ordering protocol at the beginning of a financial quarter. The other eight keep the old procedure. Before the change, the intervention facilities tend to be larger, closer to the procurement depot and managed by staff who have previously participated in training. An analyst receives monthly medicine-availability records and is asked to assess the new protocol.

The first danger is selection. A facilities list is not an assignment mechanism. The health authority may have chosen the six facilities because they were ready, because they had poor outcomes, or because their managers requested the intervention. Each reason suggests a different potential bias. If the selected sites had unusually low availability in the final pre-period, regression toward more typical levels could create an apparent gain even without the protocol. If the selected sites had better leadership, their outcomes might have improved more rapidly under normal practice. Neither source of bias is repaired by adding many individual dispensing records to the model.

The analyst should reconstruct the decision that assigned facilities, not merely the dates when software appeared. This may require interviews and written programme records. A carefully documented nonrandom pilot can still be informative, but it should not be described as though assignment had been random.

9.17.1 The outcome is produced by a recording system

Assume the initial ledger reports the number of stock-out days for each item. Before implementation, staff record a stock-out only when a monthly register is reviewed. After implementation, the tool records a stock-out automatically whenever a transaction cannot be fulfilled. The new records are more complete. A naive before-after analysis might conclude that shortages increased even if the actual physical availability improved. A second analyst might incorrectly delete the new flagged records as “outliers” and conclude that the project produced large gains.

A defensible study checks whether the numerator and denominator represent the same event under both systems. One possibility is a repeated independent physical count for a randomly selected set of medicines and dates in both groups. This will cost staff time, but it produces evidence about measurement. If comparable physical counts cannot be obtained, the evaluation should present the remaining uncertainty rather than pretend that the administrative indicators are interchangeable.

9.17.2 A fully worked, invented panel

Let $\overline{Y}_{gt}$ be the average percentage of scheduled medicine-days with stock available. The following numbers are chosen only to teach the arithmetic.

Group	Before	After	Change
Early implementation	61	79	+18
Comparison	57	68	+11
Between-group difference	+4	+11	+7

The DiD estimate is $18 - 11 = 7$ percentage points. Suppose, however, that records show the intervention group also received a temporary logistics grant, while comparison facilities did not. The grant could improve availability independently of the software. The untreated trend for the treated group is no longer credible without additional adjustment or a design that isolates the effects. The calculation is still seven; the causal interpretation is not established.

A useful sensitivity exercise imposes a hypothetical untreated-trend advantage δ for the treated group relative to the comparison. Under the simple additive sensitivity model, the attributed intervention effect becomes $7 - \delta$. For $\delta = 0, 3, 7, 10$, the values are $7, 4, 0, -3$ percentage points. The numbers do not estimate a probability distribution for bias. They reveal how much an unobserved differential trend would have to matter before the decision changed.

9.17.3 Why pre-trends are informative but not decisive

Suppose we can recover six pre-intervention months. Plot the treated-minus-comparison gap for every month, not just the last month. If the gap had been widening consistently before implementation, a two-period parallel-trends assumption is difficult to defend. If the gap had been stable, a differential post-intervention procurement shock could still invalidate it. Pre-trend checks test some violations, not every possible violation. When group sizes are small, lack of a statistically significant pretrend can simply reflect low power.

To improve the design, consider several comparison candidates, matched on pre-period availability and documented supply-chain characteristics. Record each choice before viewing the post-period outcomes. Where the programme's rollout order is not yet fixed, a transparent lottery among eligible sites can be a better use of limited administrative flexibility than extensive after-the-fact model adjustment. If the institution cannot randomise, careful staggered-adoption methods may be available, but they too require clearly described assumptions (Callaway and Sant'Anna, 2021).

9.18 Extended mathematics: individual versus policy effects

Not every intervention has an effect that can be captured by an individual treatment indicator. A stock-ordering policy can affect suppliers' allocation decisions. If one facility orders more, another facility may receive less from the same constrained depot. Then facility i's outcome may depend on the treatment vector $\mathbf{Z}$ rather than only Z_i. Potential outcomes must be

written $Y_i(\mathbf{Z})$ if such interference is material. Proposition 9.1 relies on no interference, so its interpretation can fail when the intervention changes the shared supply system.

For a district-wide policy, it may be more appropriate to assign districts, evaluate system-level outcomes, or explicitly measure spillovers. Those alternatives are not automatically feasible; the right design depends on the actual decision and size of the service system. They illustrate a general rule: when a model assumes the units do not affect one another, ask whether resources, information or human behaviour couple them.

9.18.1 A decision threshold for continuing a pilot

Imagine the district's decision is to expand only if the new protocol improves medicine availability by at least five percentage points, without causing meaningful harm to remote facilities or exceeding the agreed budget. Let τ denote the population effect and write the substantive target as $\tau \geq 0.05$ when availability is measured on a zero-to-one scale. A confidence interval $[-0.01, 0.11]$ does not establish that the target has been reached. Nor does a point estimate of 0.07 by itself satisfy a safety criterion. The decision record should say whether the evidence supports expansion, continued limited evaluation or redesign.

This avoids a common mistake: treating a small p-value or an impressive graph as sufficient justification for a complex operational decision. Statistical significance is one part of evidence. Magnitude, practical relevance, uncertainty, distribution of benefits, affordability and durability also matter.

9.19 Laboratory notebook: a deliberately broken design

The Chapter 9 script deliberately separates the known simulated intervention effect from an optional group-specific untreated trend. With default arguments it generates 40 invented treated and 40 invented comparison facilities, two periods per facility, a shared time trend, an effect of eight outcome units and random measurement noise. Its estimated DiD is close to, but not exactly, eight because the sample includes random noise.

Now run the script with a nonzero `-trend-difference` parameter. The estimated DiD moves approximately by that amount, even though the coded intervention effect remains eight. This is a transparent counterexample: perfect execution of the estimator does not imply identification when parallel trends fails. A robust report must include the generating mechanism in the teaching case and the uncertainties about that mechanism in the field case.

9.19.1 A field instrument that exposes missingness

For a selected medicine and day, collect: whether the facility was scheduled to open, whether it actually opened, whether the item was physically counted, stock on hand at a specified time, any patient request that could not be filled, whether substitution was possible, and which staff member verified the record. Record separate reason codes for unavailable count, absent staff, lost paper record, and facility closure. A blank entry should not automatically mean zero stock, zero demand or normal operation.

This instrument should be developed with clinical staff so its definitions make sense. It may need translation and testing in the actual workflow. If collecting identifiable patient information is not necessary, avoid doing so. If it is necessary for a justified study, obtain institutional and ethical permissions and implement suitable data protection. A pedagogical chapter cannot confer permission to collect private clinical records.

9.20 The district's final evidence statement

A transparent statement after a limited pilot could read: "The ordering tool was installed in six facilities and staff completed the required transaction logs. We observed a reduction in recorded stock-outs relative to eight comparison facilities, but recording changed at rollout and concurrent procurement support cannot yet be separated from the tool's effect. Operational benefit is therefore uncertain. We recommend improving common measurement and continuing evaluation before district-wide expansion."

That statement does not claim failure. It distinguishes installation, adoption, records and causation. It is useful because the next decision follows from evidence rather than promotion. Another report, supported by a strong design and consistent measurement, could justifiably reach a more positive conclusion. In both cases the obligation is the same: show how the claim is earned.

9.21 Mathematical workshop: selection, missingness and value of information

The preceding examples show why an outcome difference may not be an intervention effect. We now work through three further mathematical tools. Their purpose is to make assumptions inspectable; they do not make observational data automatically causal.

9.21.1 Selection probabilities and inverse-probability weighting

Suppose a programme assigns intervention with a known probability $e(X_i)$ conditional on pre-treatment covariates X_i, with $0 < e(X_i) < 1$. Under conditional exchangeability, consistency and suitable sampling assumptions, a population average treatment effect can be represented as

$$\tau = \mathbb{E}\left[\frac{ZY}{e(X)} - \frac{(1-Z)Y}{1-e(X)}\right]. \tag{9.8}$$

To understand the expression, condition the treated term on X and the two potential outcomes. If Z is conditionally independent of them, then

$$\mathbb{E}\left[\frac{ZY}{e(X)}\middle| X, Y(0), Y(1)\right] = \frac{\mathbb{E}[Z \mid X, Y(0), Y(1)]}{e(X)} Y(1) = Y(1).$$

The control term similarly recovers $Y(0)$. The unconditional difference therefore yields the target effect. The calculation can be valid in a randomised programme with known assignment

probabilities. In an observational study, the estimated propensity $e(X)$ does not establish exchangeability: unmeasured selection factors can still influence treatment and outcome (Hernán and Robins, 2020; Angrist and Pischke, 2009).

Suppose eligibility favours large facilities, with intervention probabilities of 0.8 for large sites and 0.2 for small sites. If an analyst compares all treated and untreated facilities without accounting for their different size distributions, group composition can produce an outcome difference unrelated to the programme. The weights $1/e(X)$ and $1/[1 - e(X)]$ correct observed group composition under the assumptions. But a small site with $e = 0.01$ receives weight 100 if treated; a single unusual outcome can then dominate an estimate. The mathematics suggests an operational action: examine overlap before trusting the weighted comparison.

9.21.2 When lack of overlap prevents a useful answer

Suppose every well-equipped facility received the intervention and every poorly equipped one did not. No data identify what the intervention would have done in a poorly equipped facility under those rules. A statistical model might extrapolate the effect, but the result depends on unverifiable modelling assumptions. Report the population for which there is overlap, or alter the evaluation design to include eligible sites from both groups. The appropriate target estimand may need to be narrowed and declared, not silently extrapolated.

A particularly revealing plot is intervention probability against baseline facility characteristics, with treatment and comparison observations distinguished. If the groups barely overlap, elaborate matching cannot create new examples where none were observed. The book's general lesson about identifiability, first developed in Chapter 3, applies directly to programme evaluation.

9.21.3 A placebo intervention date

If several pre-intervention periods are available, choose a date before the true rollout and compute the same before-after or difference-in-differences estimator as though treatment had started then. A large apparent "effect" before the treatment existed indicates that the design may be responding to unequal trends, seasonality or measurement changes. A small placebo estimate does not prove the true analysis is valid; it is one diagnostic among others.

For the invented panel with treated and comparison trends, a student can create an additional pre-period, apply (9.3) to two pre-periods and explain why the result should have expectation zero if the counterfactual trends are truly parallel. Repeat with a deliberately added treated-group trend. Report both the simulated true effect and estimated placebo effect. This is a test of a known teaching mechanism, not evidence that a real parallel-trends assumption holds.

9.21.4 Baseline adjustment: why precision and identification differ

In a valid randomised trial, baseline covariate adjustment can improve precision by explaining predictable outcome differences. It is not required to create exchangeability; randomisation is the source of identification. In an observational design, including baseline covariates may reduce

confounding if the necessary exchangeability assumption holds, but a regression coefficient cannot defend that assumption on its own.

A simple adjusted outcome model is $Y_i = \beta_0 + \tau Z_i + \beta^\top X_i + \varepsilon_i$. If the conditional outcome structure is appropriate, the parameter τ has a useful interpretation. If treatment effects vary with X, the model should either include interactions or define an averaging procedure. If X is measured after intervention and affected by it, the coefficient may omit mediated effects or be biased by collider relationships. The correct set of variables is determined by the causal question.

9.22 Small pilots: power, precision and honest limits

A proposed pilot with four treated and four comparison facilities may be attractive because it is affordable. Yet between-facility variability can be large. A study can be designed to establish whether staff can operate the new system and whether data collection is feasible without having enough independent units to identify a small average benefit.

For planning a simple two-arm independent comparison, a rough equal-sized sample formula for detecting a difference Δ in group means under normal approximations is

$$n_{\text{per arm}} \approx \frac{2\sigma^2(z_{1-\alpha/2} + z_{1-\beta})^2}{\Delta^2}, \tag{9.9}$$

where σ^2 is the assumed outcome variance, α the two-sided type-I error rate and $1-\beta$ the desired power. This is an initial approximation for independent units, not a plug-and-play calculation for a cluster-randomised trial. In facility assignment, cluster sizes, between-facility variance, intracluster correlation, attrition and the number of clusters matter (Cochran, 1977; Gertler et al., 2016).

Worked example 9.4 (Why small target effects need more evidence)**.** *Suppose an illustrative outcome has standard deviation $\sigma = 15$ points across independent facilities, and the policy target is $\Delta = 5$ points. With $z_{0.975} \approx 1.96$ and $z_{0.8} \approx 0.84$, the approximation is $n \approx 2(225)(2.8)^2/25 \approx 141.1$ facilities per arm. Under those assumptions, a pilot of ten facilities cannot reliably detect a five-point difference with the specified power. The precise requirement depends on the study design, but the calculation makes the scale issue explicit.*

An underpowered pilot can still provide valuable estimates of operational feasibility and variance to inform a larger design. The mistake is to report "no impact" when the evidence is compatible with a wide range of effects. A confidence interval that includes meaningful benefit and meaningful harm should be described that way.

9.23 Evaluation when the outcome is a rate

Medicine availability is often represented as a percentage, but not every percentage has the same precision. A facility with three observed days and two in-stock days has an estimated availability of two-thirds; a facility with 300 observed days and 200 in-stock days has the same proportion but a different sampling picture. The days may be serially correlated, so a simple

independent-binomial model is frequently unrealistic. Before choosing an estimator, decide whether the unit of analysis is a facility, medicine, day, patient or district and whether repeated measurements represent independent information.

A weighted facility average can differ from the unweighted average. Suppose a large hospital has 90% availability over 1,000 medicine-days, and a small clinic has 30% over 100 medicine-days. The unweighted facility mean is 60%; the pooled medicine-day rate is $(900+30)/(1000+100) \approx 84.5\%$. Neither is intrinsically wrong. The first answers how an average facility performs; the second answers how an average recorded medicine-day performs. Policy conclusions can differ when large facilities dominate the denominator. State the estimand before calculating.

9.24 Economic evaluation with consequences outside the ledger

The economic evaluation must identify whose costs and benefits count. From the district's perspective, the tool costs money and staff time. From a household's perspective, fewer unfilled prescriptions may avoid repeat travel, missed work and additional consultations. A solution that shifts cost from the district to patients may look efficient under a narrow accounting perspective and harmful under a broader one. No universal spreadsheet automatically chooses the correct perspective.

For an evaluation horizon of several years, define a consistent valuation date and discount both costs and benefits if appropriate. Record the source and uncertainty of each parameter. Do not treat volunteer labour as zero resource use merely because no salary line exists; note the opportunity cost. Conversely, avoid assigning speculative monetary prices to outcomes whose valuation is contested. Present them separately when necessary.

Worked example 9.5 (One decision, three perspectives)**.** *Imagine an invented intervention costing a district an extra 120,000 units, saving 200 patient journeys and reducing 30 emergency purchases. The district may compare procurement and staffing costs; patients may value journeys differently depending on travel time and missed earnings; a national health authority may assess system-wide continuity of care. The intervention's conclusion depends on which consequences enter the objective. The right report shows a disaggregated ledger before presenting an aggregated summary.*

9.25 An evidence-grading exercise

Give students the following statements and require them to assign an evidence category, identify what is missing and rewrite the claim accurately.

1. "The programme exactly reproduced 10,000 simulated ordering decisions." This is computational verification under synthetic inputs, not proof of improved patient care.

2. "The software was used for three months at a clinic without a recorded crash." This is some operational evidence, but it depends on what was monitored and whether relevant failures could be unrecorded.

3. “Availability increased after rollout.” This is a before-after observation, not necessarily a causal effect.
4. “A randomised rollout reduced verified stock-out days.” The design may support a causal claim if assignment, measurement, interference and missingness are properly handled; magnitude and uncertainty remain necessary.
5. “The same software runs on a second country’s devices.” This establishes a component of technical transfer, not local clinical benefit.

This exercise is not busywork. The distinction between mathematical, computational, observational and causal claims is what allows a practitioner to act confidently without pretending to know more than the data reveal.

9.26 Exercises, derivations and field investigations

1. Explain in your own words why the same facility cannot reveal both potential outcomes under the same conditions. Provide a nonmedical example.
2. Prove Proposition 9.1 when treatment probability is p under independent Bernoulli assignment, using inverse-probability weighted means with fixed denominators. What if treatment probabilities vary by unit?
3. Compute a DiD estimate from treated means $(0.62, 0.79)$ and comparison means $(0.58, 0.69)$. Name two reasons it might not be causal.
4. Create a counterexample to parallel trends in which DiD produces a positive estimate even though the intervention effect is zero. Show every number.
5. A district has eight assigned facilities with approximately 200 patient visits per facility and intraclass correlation 0.08. Calculate the design-effect approximation and explain why the number of independent facilities still matters.
6. Derive the bound (9.4) for an outcome bounded between a and b. Discuss its relevance to missed stock-count days.
7. Design an outcome-definition sheet for a stock-out study that cannot be improved merely by hiding or deleting stock-out days.
8. Explain when an interrupted time series may be preferable to a simple before-after comparison, and identify a concurrent event that could still invalidate its causal interpretation.
9. Develop a sensitivity analysis for unknown unmet demand on stock-out days. Show how the range of plausible values changes the estimated programme benefit.
10. Write a one-page pre-pilot protocol for a water, school or clinic service. Name the ethical review or administrative permissions that would have to be checked locally.
11. Run the companion code after changing the group-specific trend from zero to five units. Explain the numerical difference between the estimator, its sampling variability and its causal bias.

12. Evaluate a policy with both better average availability and worse availability at the most remote facilities. What decision rule could make the trade-off explicit?

What this chapter establishes. A trustworthy evaluation is an argument about an unobserved alternative, not merely a statistical calculation. It has an estimand, a credible comparison, measured outcomes, defensible assumptions, uncertainty and a decision. Chapter 10 asks what an institution must put in place if that evidence is strong enough to justify a continuing service.

Chapter 10

Deployment, Governance and Long-Term Maintenance

A correct prototype can fail after the people who built it leave. Deployment means more than installing code. Someone must own the service, repair failures, protect records, train colleagues and pay for the system next year. This chapter turns a promising prototype into an accountable operational plan, connecting software reliability to human responsibilities and institutional budgets.

10.1 The handover on a Monday morning

The district stores team has completed a promising pilot of an offline inventory application. On Monday the person who wrote the program is away. A nurse enters a correction to yesterday's stock count. The internet disappears. At midday a second device uploads the same order twice. Later a manager asks for the history of who changed the figure. None of these situations is unusual. Each should have been considered before the application became part of routine work.

Chapter 8 dealt with the ordering decision; Chapter 9 dealt with whether changing the policy improves outcomes; Chapter 10 deals with the institution that must continue producing those outcomes. An unreliable institution can defeat a sound mathematical model. Conversely, even an excellent institution cannot compensate for a system whose data are systematically corrupted by duplicate writes.

A deployment plan begins with a basic question: *Who is responsible when the system stops working?* If the answer is "the developer" and the developer is not employed or contracted to provide support, the institution does not yet have an operational service.

10.2 Distinguish building software from operating a service

Chapter 7 introduced transactions, offline data storage, sync failures and recovery procedures. We use those mechanisms here, but our concern has changed. An engineering test checks whether a duplicate request increments stock twice; a governance review checks who may correct

an error, what evidence is retained and who is told if a patient may have been affected. The first is necessary for the second, not a substitute.

Table 10.1: A minimal operational responsibility map for a small service.

Role	Decision authority	Evidence that the responsibility is functioning
Service owner	Continue, change, fund or withdraw the service	Signed ownership and escalation process
Operational lead	Daily workflow, staff cover, corrected records	Review logs and reconciliation sign-off
Technical custodian	Updates, backups, security and recovery	Version log and tested recovery procedure
Data steward	Access, quality, retention and permitted use	Data register and access review
Safety or domain reviewer	Clinical or other high-stakes decisions	Written rules for override and escalation

A very small organisation may assign several of these roles to one person, but it should not erase the distinctions. A shared device requires particularly careful access controls: a record made while an account was left logged in cannot be regarded as a reliable attribution of who acted. Where accountability matters, users need separate credentials and the ability to correct rather than silently overwrite past transactions.

10.3 Follow a single record through the institution

Consider an inventory adjustment of -12 units. The correct technical implementation inserts a transaction with a unique identifier, authorised user, timestamp, medicine identifier, reason and an auditable link to the previous stock balance. A reconciliation process then checks that

$$I_{t+1} = I_t + R_t + T_t^{\text{in}} - Y_t - W_t - T_t^{\text{out}} + A_t, \tag{10.1}$$

where receipts R_t, issued units Y_t, waste W_t, transfers $T_t^{\text{in/out}}$ and authorised adjustments A_t are separately recorded. The equation is a conservation check, not a guarantee that each reported event is true. A dishonest or mistaken entry can satisfy arithmetic conservation while misrepresenting the physical medicine.

The workflow therefore includes periodic physical counts, independent review of material discrepancies and a rule for reporting unresolved differences. Transaction idempotence means retrying a committed operation does not repeat its effect. The institution should test this property with the actual synchronisation workflow and store-and-forward devices; see Chapter 7 and Kleppmann (2017); Gray and Reuter (1992).

10.4 Reliability can be measured, not merely described

Saying that software is “robust” does not tell an operator how long a service may be unavailable. A useful first measure is *availability*: fraction of scheduled service time during which essential

functions are usable. Outage frequency and repair time are separately important. A tool that fails once a year for a month can have a similar average unavailability to one that fails briefly every day, while creating very different consequences.

Under an idealised alternating renewal model, let independent operating durations have finite mean U and repair durations have finite mean D. The long-run fraction of time in the operating state is

$$A = \frac{\mathbb{E}U}{\mathbb{E}U + \mathbb{E}D}. \tag{10.2}$$

If mean time between failures (measured in operating time) is M and mean time to restore operation is R, this is often written $A = M/(M + R)$ (Ross, 2014). This formula is a planning model. A process with long correlated outages, partial functionality or inconsistent definitions requires a more suitable analysis.

Proposition 10.1 (Elementary availability sensitivity)**.** *For $M, R > 0$, define $A(M, R) = M/(M + R)$. Then*

$$\frac{\partial A}{\partial M} = \frac{R}{(M + R)^2} > 0, \qquad \frac{\partial A}{\partial R} = -\frac{M}{(M + R)^2} < 0.$$

At fixed M, reducing restoration time always increases modelled availability.

Proof. Differentiate the rational function with respect to each variable. Both denominators are positive, establishing the signs. □

Worked example 10.1 (Why local repair capacity matters)**.** *Take a deliberately illustrative system with an average of 240 operating hours between failures. If restoration takes 12 hours on average,* (10.2) *gives $A = 240/252 \approx 0.9524$. If training and local spare equipment reduce restoration to 3 hours, modelled availability rises to $240/243 \approx 0.9877$. The point is not that either value describes a real clinic. It is that a repair process can be more consequential than another improvement in predictive accuracy.*

Average availability does not measure patient harm. A half-hour failure during a vaccination clinic may be more important than a four-hour outage after closing. For a schedule $w(t)$ that gives operational consequence per unit of downtime, a simple weighted loss is

$$L = \int_0^T w(t)\, \mathbf{1}\{\text{service unavailable at } t\}\, dt. \tag{10.3}$$

The weights must be defined with domain experts; the formula is not a licence to assign money values to patient safety without evidence.

10.5 Backup, restoration and recovery objectives

A backup file is not evidence of recoverability. Someone must actually restore it on a different device and verify the result. Specify a recovery point objective (RPO), the maximum tolerable loss of recent data expressed as a time window, and a recovery time objective (RTO), the targeted time to restore essential service. These objectives depend on criticality and resources;

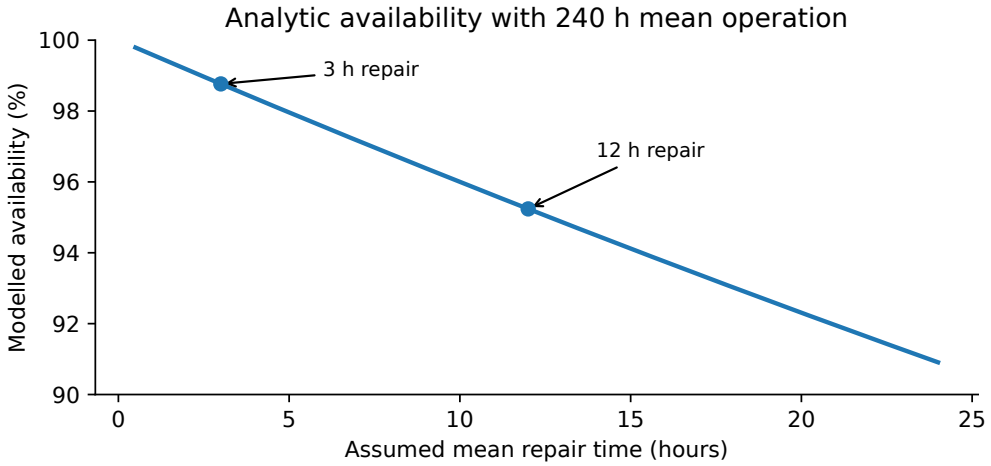


Figure 10.1: Analytic availability $M/(M+R)$ as assumed mean repair time R varies for fixed $M = 240$ operating hours. The figure is a model illustration, not observed uptime.

writing "zero loss, zero downtime" without architecture and funding that can deliver it is not a plan.

A low-bandwidth service may keep local transaction logs and periodically export protected copies to a separate medium. On recovery, the operator replays validated transactions, rejects duplicate identifiers and reconciles totals. Every recovery exercise should record backup date, software version, number of records restored, missing-record count and who signed off.

Proposition 10.2 (Idempotent replay under unique transaction identifiers)**.** *Suppose a ledger applies a transaction's stock effect at most once for every unique identifier k and stores applied identifiers durably in the same atomic transaction as the stock update. Replaying an already committed transaction with identifier k leaves the stock balance unchanged.*

Proof. After the first successful atomic commit, the identifier belongs to the durable set of applied identifiers. On replay, the guard sees k in that set and performs no additional stock update. Atomicity prevents a committed stock update without its associated identifier record. □

The guarantee requires the transactional assumptions. If the stock update and identifier are written in separate, interruptible steps, a power failure between them can invalidate the claim. The difference between an application-level check and a transaction-level guarantee should be visible in the code and tests.

10.6 Security begins with a threat model

No amount of encryption is sufficient if shared passwords permit anyone to edit sensitive records without attribution. A threat model asks what must be protected, from whom, by what means, and which failures are most consequential. In the district example, assets may include patient-linked data, procurement records, stock balances, staff credentials and backups. Plausible

threats include lost laptops, accidental email disclosure, malicious changes, unauthorised exports, old unpatched devices and privilege misuse.

The classic principles of least privilege, fail-safe defaults and separation of duties give a useful starting point (Saltzer and Schroeder, 1975; NIST, 2020). Apply them proportionately: a nurse entering dispensing data should not automatically be able to change supplier payment details; an analyst studying aggregate shortages does not necessarily need patient identities. The required law and policy depend on jurisdiction and institution, so an implementation must obtain local legal review rather than copy a foreign compliance checklist.

10.6.1 Data minimisation and a retention decision

The most secure unnecessary sensitive record is one never collected. Before adding a new field, ask whether a decision truly requires it. A stock-planning model may need aggregate issues by item, date and facility, not individual names. Records should have specified retention periods and lawful access rules. Where disclosure could be harmful, the institution needs a breach-response procedure with named authorities and reporting thresholds that are appropriate to the applicable regime.

An audit trail must itself be protected against unauthorised editing, yet still support legitimate corrections. Record corrections as new events that preserve the earlier entry and the reason for the change. This is operationally more informative than silently replacing yesterday's value.

10.7 People are part of the system design

A service that only its developer understands is fragile. The operator's manual should use language and examples familiar to the staff, including a short procedure for when no computer or internet is available. Training is not complete when a demonstration has been attended. Ask a staff member who did not build the tool to perform a normal task, make a common mistake, restore an interrupted workflow and explain when the recommendation must be overridden.

Human oversight must be real. A screen asking users to click "approve" without showing the data, uncertainty, consequences and available alternative is not meaningful control. People need time, authority and documentation to question an output. If a pharmacist rejects a suggested reorder quantity because an emergency outreach campaign is coming, the system should support that decision and record the reason. It should not turn a useful override into an opaque deletion of evidence.

10.8 Total cost of ownership and the cost of dependence

A donated computer does not make a system free. Write the lifecycle cost over planning horizon H as

$$\mathrm{PV}(C) = C_0 + \sum_{t=1}^{H} \frac{C_{\mathrm{staff},t} + C_{\mathrm{energy},t} + C_{\mathrm{data},t} + C_{\mathrm{maintenance},t} + C_{\mathrm{replacement},t}}{(1+r)^t}, \tag{10.4}$$

where r is the explicitly chosen discount rate and the amounts have consistent price-year units. This equation does not capture every consequence; training time, procurement delays, vendor migration and opportunity costs may be difficult to measure. Record the exclusions, and include a stress scenario for exchange-rate changes or unexpected equipment replacement.

Worked example 10.2 (A reproducible maintenance decision)**.** *Consider two invented configurations for three years. Option A uses a 500,000-unit device and requires 280,000 units of annual operations; option B uses a 900,000-unit device but only 150,000 units annually. With a hypothetical annual discount rate of* 10%*, the present values are*

$$C_A = 500{,}000 + 280{,}000 \sum_{t=1}^{3} (1.1)^{-t}, \quad C_B = 900{,}000 + 150{,}000 \sum_{t=1}^{3} (1.1)^{-t}.$$

The three-year annuity factor is approximately 2.4869*, giving* $C_A \approx 1{,}196{,}320$ *and* $C_B \approx 1{,}273{,}035$*. Option A is cheaper in the specified model, despite the higher annual operations budget. If A's repair costs rise substantially, the comparison may reverse. No claim about actual procurement prices follows from invented numbers.*

Costs should be compared under the same service requirements. The cheapest application that frequently loses records may not be an acceptable option. An apparently free proprietary platform that stores the only usable copy of institutional data creates migration costs and bargaining dependence. Prefer documented formats, export rights, clear licences, versioned interfaces and a tested exit path when feasible. Procurement should make ownership of data and responsibility for maintenance explicit.

10.9 Adoption and implementation outcomes

An application can be adopted by administrators but rejected in practice. Implementation research distinguishes acceptability, adoption, appropriateness, feasibility, fidelity, cost, penetration and sustainability (Proctor et al., 2011). These dimensions are not interchangeable with health or service outcomes. For instance, high adoption does not prove improved medication availability; low adoption may explain why a technically effective intervention had little measured effect.

Each indicator needs a denominator and a process for preventing the measurement itself from being gamed. A programme should not report "100% uptime" because it simply stopped logging during power loss.

10.10 Preparing for incidents, not hoping they never happen

An incident is an event that threatens availability, integrity, confidentiality or safe decisions. Classify incidents by consequences and urgency rather than by how embarrassing they are. A useful response procedure is to detect and record the problem, preserve relevant logs, protect people and data, contain the failure, restore verified functionality, communicate to affected parties as required, and review why the event occurred.

Table 10.2: Practical measures for the first year of operation.

Measure	Operational definition to choose locally
Availability	Scheduled service hours usable / scheduled service hours
Restoration time	Time from confirmed outage to verified return of essential service
Data reconciliation	Proportion of planned stock-count checks completed without unresolved discrepancy
Training resilience	Percentage of shifts with at least one staff member able to operate and recover the tool
Adoption	Eligible facilities recording specified core transactions in the evaluation window
Exception handling	Time and authority used to resolve flagged records or unsafe recommendations
Sustainability	Approved next-year budget and identified responsible support staff

There must also be a *fallback*: paper stock cards, an authorised manual reorder procedure, or a safe pause in automated recommendations. Staff should know which tasks can continue when the network, database or model is unavailable. An intervention whose failure prevents any service at all may be worse than the less sophisticated system it replaced.

10.10.1 A controlled change process

When updating code or a forecasting model, record the version, tests, expected data schema and rollback procedure. A model trained with a changed medicine identifier convention could produce plausible but incorrect reorder quantities. For high-consequence rules, a limited release with side-by-side checking can expose mistakes before full use. Do not introduce a new model merely because it scores better on the old training dataset; evaluate the new decision rule against the current workflow, costs and safety conditions (Sculley et al., 2015).

10.11 A first-year deployment plan

The service owner can work through four phases. Before rollout: identify the owner, data rights, required clinical approvals, baseline workflow, budget and stop criteria. During a limited launch: train a small number of staff, test failed sync and restoration, reconcile daily balances and record actual workload. During monitored expansion: check whether the intervention still serves its intended decision across facilities, including those with weaker infrastructure. At handover: identify a local custodian, rehearse disaster recovery, secure funding and retain documentation that does not depend on one expert.

The transition from one phase to another should be decided using a written readiness record, not an attractive dashboard. A severe unresolved data-loss defect is a reason to stop expansion even when prediction accuracy looks excellent.

10.12 Reproducible laboratory: uptime and restoration

The programme `code/ch10_service_reliability.py` calculates the analytic availability (10.2), runs a seeded illustrative failure-and-repair simulation, evaluates lifecycle budgets and writes a small CSV report. It requires no connection to a real clinic. The script also checks the ledger-replay invariant in a toy transaction list: duplicate identifiers must not change the final balance. An output label describes every result as an assumed or simulated quantity.

A disciplined laboratory report states what was modelled, which model assumptions were tested, which measurements would be required to estimate failure and repair distributions, and what happens when outages are correlated or concentrated during busy hours. Students must not claim that a simulated availability of 98% proves the application will be available 98% of the time in use.

10.13 A deployment-readiness record

A one-page readiness sheet should have an answer and a named signatory for each question:

1. Is there an identified service owner, technical custodian and domain reviewer?
2. Have normal and exceptional workflows been observed with actual users?
3. Have offline writes, duplicate uploads, account recovery and power interruption been tested on intended hardware?
4. Can a backup be restored without the original programmer, and has it been demonstrated?
5. Are data permissions, retention, exports and breach procedures documented?
6. Is there an agreed maintenance budget, including staff time and equipment replacement?
7. Do users have a workable manual fallback and authority to override unsafe recommendations?
8. Are the baseline and post-deployment outcomes measurable for Chapter 9's evaluation?
9. Are the conditions for stopping, redesigning or replacing the service explicit?

Not all services need identical controls. A library checkout counter and a medicine-dispensing system have different consequences if unavailable. The reasoning is transferable: define the harm, evidence and cost of a possible failure; then choose controls proportionate to it.

10.14 Deeper method: a service as a chain of dependent components

An inventory service depends on more than the laptop. The local power supply, database, trained operator, communication link, backup, supplier and clinical process each contribute to whether an order is completed safely. A reliability claim must therefore specify the function

that is meant to remain available. “The app opened” is not the same as “a verified order could be placed and reconciled.”

A simple series system with independent components is operational only when every required component is available. If component j is available with probability a_j over a specified observation instant, then under the stated independence assumption,

$$A_{\text{system}} = \prod_{j=1}^{m} a_j. \tag{10.5}$$

For three independent components with availabilities 0.99, 0.95 and 0.98, this gives approximately 0.9217. A highly reliable laptop does not guarantee a highly reliable end-to-end service. The calculation also warns against casually multiplying field measurements: shared power, weather and operational policies often cause correlated failures. The independent-component formula is a model to validate, not a licence to invent precision.

10.14.1 Redundancy has conditions

If two independent alternative devices can each deliver the required function and the service is down only when both fail, a simple parallel formula is

$$A_{\text{parallel}} = 1 - (1 - a_1)(1 - a_2).$$

Two devices each available 95% of the time would give 99.75% in this idealised model. But devices in the same locked room may both be inaccessible when the key is lost, and two laptops using the same exhausted battery charging arrangement may both fail during a prolonged outage. Independence and actual failover are essential. A second device that cannot restore current data is not necessarily redundancy for the complete service.

10.14.2 A fallback changes what failure means

Suppose the network is down but local transactions can continue safely and be reconciled later. If the essential function is “record a legitimate stock issue with recoverable integrity,” network failure may not count as service outage. If the essential function is “consult live stock balances from every facility,” the network may be required. One application therefore has different availability measures for different functions. Service definitions should be written from user needs rather than only from the architecture diagram.

10.15 Data recovery and the arithmetic of snapshots

A facility may take snapshots every h hours. Under a simplistic model where failures occur uniformly relative to the snapshot schedule, the average age of the most recent snapshot at failure is $h/2$. That is the expected amount of transaction history requiring replay or potentially lost if no durable log exists. The worst-case snapshot age is almost h. The result is a planning approximation; real failures may be correlated with operating schedules or power conditions.

If a durable write-ahead log records each committed transaction between snapshots, the amount of recoverable data may be much greater than the snapshot alone. Yet the log must itself survive a power interruption and be restored in the correct order with duplicate protection. A backup strategy should specify both the physical storage locations and the logical recovery sequence.

Worked example 10.3 (Two backup policies)**.** *Under an illustrative 12-hour operating day, a once-daily snapshot made at closing can leave nearly the entire day's transactions outside the latest snapshot before the next close, if no separate log exists. Hourly snapshots shorten that window but consume more operator time and storage. The correct choice depends on tolerated data loss, recovery time, device cost and whether a reliable transaction log is available. The answer cannot be inferred solely from the price of a memory card.*

10.16 Governance as an explicit state machine

An institutional service can be described in a small number of states: *planned*, *tested*, *approved for pilot*, *limited operation*, *routine operation*, *suspended* and *retired*. Moving between states should require evidence. For instance, a tool may move from tested to approved pilot only when the required risk review, local permissions and operator training are complete. It may move from limited to routine operation only after the institution has reviewed usability, reconciliation and observed service outcomes. A serious integrity incident may force suspension regardless of headline accuracy.

This state description is not a novel scientific theory. It is a way to make decision authority and evidence requirements legible. It prevents "the software has passed its unit tests" from being mistaken for "the institution has authorised its use for clinical decisions."

10.16.1 A sample transition rule

Suppose a pilot can expand only if three conditions hold: the ledger's reconciliation failure count is zero in the accepted verification period; at least two trained staff members can operate the system without the developer; and the responsible safety reviewer has authorised the workflow. Write the gate as $G = G_1 \wedge G_2 \wedge G_3$. The point is that failure of any necessary condition prevents expansion. Replacing the conjunction with an average score could hide a severe defect behind good performance elsewhere.

The actual thresholds and evidence requirements must be chosen by the institution with domain expertise. A book can teach how to express and audit a gate; it cannot choose the safe threshold for every service.

10.17 An accountability puzzle: the algorithm recommends, the person decides

Consider a stock-order recommendation that appears inadequate to a senior nurse because a vaccination outreach is scheduled next week. If the tool has no field for anticipated events, its

recommendation is based on incomplete information. The nurse's override may be the correct decision. The institutional process should permit a documented adjustment and later review. It should not classify every override as a user failure merely because the model did not predict it.

On the other hand, repeated overrides might reveal a poorly calibrated model, weak training, missing features or external pressure to over-order. The service owner should analyse patterns rather than simply suppress them. The same applies to manual corrections and failed synchronisation: exceptions are signals about how the system meets the world.

10.18 Ownership of an open-source tool

Open-source software can reduce licensing barriers and improve inspectability, but it does not eliminate maintenance. Dependencies may have security vulnerabilities; device interfaces and operating systems change; and a public repository may not include training or local support. An institution choosing open-source tools should verify that it can build, test, install, update and recover the exact version it depends on. It should know who is responsible for reviewing upstream changes and how long records must remain readable.

A proprietary service may supply reliable support but introduce vendor dependence and data export risks. Neither category is universally superior. The correct comparison includes lifecycle cost, security, support, migration and the institution's capacity to understand its own operation.

10.19 Training evidence instead of training attendance

Training records often count who attended an orientation session. Attendance is not competence. A better assessment asks a trainee to perform essential tasks while an observer checks accuracy and recovery: create a legitimate record, recognise a malformed value, correct a mistaken entry transparently, respond to loss of connectivity, and explain when to stop using an automated recommendation.

The assessment should be fair to staff with different prior computing experience. Test the system with users who have the least reliable access to technology, not only with the development team. Where necessary, adapt interfaces for local languages and accessible interaction. An effective handbook must teach technology design around people rather than expect people to adapt to every poorly designed interface.

10.20 Adoption is not a single event

Diffusion research emphasises that innovations move through social and institutional networks, with perceived advantages, compatibility, trial opportunities and communication affecting adoption (Rogers, 2003). A public-sector deployment also depends on procurement cycles, leadership transitions and staff turnover. A positive pilot may be forgotten after a manager moves. A tool may spread rapidly because a donor funds devices but become unused when subscriptions or consumables are due. Those patterns should be monitored as part of sustainability rather than treated as unrelated social issues.

The service owner should ask what happens after the project's grant ends. A strong answer identifies local funding authority, support capacity, documented data ownership and evidence that the service remains worth maintaining. If those cannot be established, maintaining a limited prototype rather than promising national scale may be the rational decision.

10.21 Extended investigation: what an offline clinic system must survive

A small clinic keeps inventory records on two inexpensive laptops. One is used near the dispensing counter and the other in the storeroom. Each laptop can work without internet. Once each day, staff attempt to synchronise with a district server. This is not a hypothetical excuse for poor software quality; it is a real design setting in which local action must continue while communication is uncertain. The test data below are invented, and the design must be assessed against actual hardware before adoption.

Consider an order event $e = (k, t, u, m, q)$, where k is a unique event identifier, t a local time, u an authorised user, m a medicine identifier and q a signed quantity. An event log can reconstruct the current balance by applying valid events in an agreed order. But the word "valid" matters. Two devices may assign the same local identifier if their schemes are poorly designed. The clocks may differ. One event may correct another; an item may have been transferred before the correction reaches the second device. These problems require explicit semantics rather than a rule that simply keeps the most recently uploaded row.

10.21.1 A simple conflict that cannot be resolved by timestamps

Suppose two offline workers both read a stock balance of 20. The first dispenses 12 units; the second dispenses 10. If each device writes the new balance as an absolute value, one stores eight and the other ten. A last-write-wins synchronisation rule may keep either eight or ten, although the signed events imply a net balance of $20 - 12 - 10 = -2$, indicating that the two offline decisions were not jointly feasible. The correct response is not to accept a negative physical stock silently. The system must expose the over-allocation, reconcile the actual dispensed quantities and decide what the application is allowed to do while disconnected.

If the application requires a single stock-holding authority for each physical store, synchronisation can be simplified by assigning each stock movement to a unique store and imposing rules for offline dispensing. If several devices legitimately update one store at the same time, additional coordination or constrained transaction semantics are necessary. A database may provide local atomicity without providing global serialisability across disconnected devices. The software architect should acknowledge the gap (Kleppmann, 2017; Gray and Reuter, 1992).

10.21.2 Write an acceptance test for the risky situation

A credible test should perform the following sequence, with the application installed on the intended devices: record a receipt, interrupt power during a write, restart, replay the same upload twice, perform concurrent offline issues, and reconcile the resulting transactions with

a signed physical count. Inspect the state after each interruption. Check that no committed record disappears, that duplicate identifiers cannot change the balance, and that unresolved conflicts remain visible until reviewed.

The test should name its limitations. If two staff members physically dispense the same stock to different patients while offline, software cannot retroactively prevent the physical shortage. It may only limit permitted operations, reserve stock, require manual verification, or record and escalate the conflict. Engineering cannot remove the need for an operational rule.

10.22 What a service agreement needs to say

A maintenance agreement should not consist only of “contact IT when there is a problem”. It should identify essential functions, support hours, tolerated data loss, escalation contacts and restoration expectations. For a service that can work on paper during outages, the essential availability objective may refer to the ability to dispense safely and later reconcile, not the uptime of the central dashboard. For an emergency system with no safe manual fallback, substantially stricter objectives may be necessary.

A useful service-level record specifies the time window for measurement, what counts as an outage, planned maintenance exemptions, incident severity, who acknowledges an incident and what happens if support is unavailable. Beware of a metric that excludes every difficult interval. A system that never logs failures during blackouts should not be allowed to report perfect availability simply because no data were uploaded.

10.22.1 Reliability is about more than the average

Equation (10.2) is a long-run fraction under a simple renewal model. In reality, repair times may be heavy-tailed: most faults are fixed in minutes, but a rare motherboard failure can disable a device for days. A single mean then conceals relevant risk. Report, where data permit, the frequency of outages, median and upper-tail restoration times, longest outage, proportion of transactions needing reconciliation and fraction of scheduled service time lost.

The modelled expected downtime over an operating window T is $T(1 - A)$ when a stationary availability approximation is appropriate. For $T = 2{,}000$ scheduled hours and $A = 0.95$, the expected downtime is 100 hours. This arithmetic is not a promise about any particular year. The distribution and timing of outages determine whether those hours cause moderate inconvenience or severe disruption.

10.22.2 Repair stock and staff capability as design variables

Suppose a device failure can be restored in three hours if a spare battery or second laptop is available, but in two weeks if replacement hardware must be ordered from another city. The institutional decision is partly a spare-equipment allocation problem. Let c_s be the annualised cost of holding a spare and let p denote the probability of a major failure in the planning period. If having the spare reduces the loss from L_0 to L_1, a simplistic expected-value calculation favours the spare when $c_s < p(L_0 - L_1)$. This rule presumes the losses are measured on a

defensible common scale. If patient safety or essential service continuity is involved, there may be a mandatory minimum control independent of the monetary comparison.

Training another staff member can produce a similar benefit when failures are human rather than mechanical. A service whose only trained user is frequently absent does not become dependable by purchasing more powerful computers.

10.23 A useful procurement evaluation

Procurement teams often compare purchase prices but neglect software lifetime. A more complete evaluation should request the total price for installing and sustaining the service for an agreed period, including device replacement, training, data migration, security updates, helpdesk support, periodic audit and exit. Require a working export of key data to a documented open format, even when proprietary software is chosen. Check whether records can be read after a contract ends.

10.23.1 Four offers that sound similar but are not

Imagine four invented offers: a hosted platform requiring steady connectivity; a single-device offline tool with no synchronisation; an offline-first multi-device tool with expensive integration; and a spreadsheet maintained by one trained clerk. The correct choice depends on data sharing needs, local skills, failure consequences and lifecycle cost. A hosted platform can be excellent where the connection is dependable; a spreadsheet can be sufficient for a very small service; an offline-first design may justify its complexity where disconnected multi-device operation is essential. No product category should be selected simply because it sounds advanced.

The team should run the same practical acceptance scenario for all plausible options. For a stock system, it may require recording a receipt, correcting an error, generating an order, losing internet for a day, restoring power, exporting a backup and reconstructing balances. Compare both the amount of work and the quality of the resulting data.

10.24 The annual review is an engineering activity

Long-term sustainability requires a schedule for checking assumptions, not only replacing worn equipment. Medicine usage changes, a new procurement platform changes identifiers, staff join and leave, and an old tablet may no longer receive security patches. An annual review should ask whether the original problem still exists, whether the data retain their meaning, whether the decision rule remains appropriate, and whether the cost is justified by evidence of service performance.

The review can end with one of four decisions: retain with routine maintenance; retain with specified corrections; redesign or revalidate; or decommission with an orderly migration. Decommissioning is not necessarily failure. Sometimes the institution has outgrown the original tool, or the task has become simpler. Responsible design includes making retirement safe.

10.24.1 An accountable withdrawal

If the service must be stopped, export records under the applicable retention rules; document who has received them; revoke credentials; remove unneeded sensitive data from returned devices using appropriate secure methods; and verify that the previous workflow or replacement can perform essential transactions. Inform operators of the change and record where future corrections should go. The aim is to avoid the common outcome in which a “pilot” quietly becomes the only repository of institutional records, then disappears with its developer.

10.25 A practical handover demonstration

The final practical exercise in this chapter should be performed by someone other than the original programmer. Ask that person to install the documented application on a clean machine, restore an example dataset, complete five ordinary transactions, recover from a simulated interruption, identify a deliberately corrupted record and export an auditable report. Time the process and record where the manual fails.

A handover succeeds only when the receiving team can demonstrate the specified functions. A signed checklist without an observed restoration is weak evidence. If the software cannot be installed and used without a live call to the developer, the team has uncovered an important dependency before it became a service outage.

10.26 Mathematical workshop: reliability, queues and scarce support

Small systems rarely have unlimited technicians or replacement equipment. The right reliability question is therefore partly an allocation problem: how should the institution spend a limited support budget to prevent the most damaging interruptions? A simple model can clarify the trade-offs, provided it does not hide safety requirements.

10.26.1 A repair-time distribution matters

Consider two service options with the same mean restoration time. In the first, repair always takes four hours. In the second, most problems are fixed almost immediately but a small fraction last several days. Their mean can agree while the chance of an extended outage differs sharply. If an essential service cannot tolerate a single multi-day failure, average restoration time is not a sufficient design criterion. A maximum acceptable outage duration, a high quantile of restoration time or a consequence-weighted downtime measure may be more useful.

Define an outage loss function $g(D)$ for outage duration D. If damage grows faster than duration, for example $g(D) = cD^2$ in a clearly stated hypothetical model, then

$$\mathbb{E}[g(D)] = c\,\mathbb{E}[D^2] = c\{(\mathbb{E}D)^2 + \operatorname{Var}(D)\}.$$

Two repair procedures with equal mean but different variances therefore have different expected damage under this model. This does not establish that actual patient harm is quadratic in

outage duration. It shows why a service planner must understand the consequence function, not just report mean downtime.

10.26.2 Preventive maintenance as an optimisation decision

Suppose a facility can pay c_m for preventive maintenance that changes expected failure count from λ_0 to λ_1 over the planning period, where $\lambda_1 < \lambda_0$. If each failure costs an expected L in disruption under a defensible loss model, maintenance reduces modelled expected failure loss by $(\lambda_0 - \lambda_1)L$. In a simple monetary comparison it is justified when

$$c_m < (\lambda_0 - \lambda_1)L.$$

The assumption that failures are independent of service workload and that their expected consequence is the same is often unrealistic. Use this calculation as a starting point for sensitivity analysis, not a universal procurement rule.

When budgets are scarce, there may be several maintenance actions. Let $z_j \in \{0, 1\}$ indicate whether action j is purchased, with cost c_j and estimated risk reduction b_j. A basic selection problem is

$$\max_{z_j \in \{0,1\}} \sum_j b_j z_j \quad \text{subject to} \quad \sum_j c_j z_j \leq B.$$

This is a knapsack-style abstraction. Its additive benefit model neglects interactions: a spare battery and a solar charger may be more valuable together than separately, while two independent backups may have diminishing benefit. Use domain tests to decide whether an additive approximation is suitable.

10.27 Capacity and support arrival rates

Suppose an institution has one technician serving several sites. Requests arrive while others are being handled. Under the highly idealised $M/M/1$ queue with Poisson request arrivals at rate λ, exponential service times with rate μ, and $\lambda < \mu$, the expected time in the system is

$$W = \frac{1}{\mu - \lambda}, \tag{10.6}$$

including waiting and service. This model is not necessarily realistic for district support, where travel, priority and spare-part delays complicate the process (Gross et al., 2008; Kleinrock, 1975). It nevertheless demonstrates why a nearly saturated technician can cause disproportionately long delays.

Worked example 10.4 (A support team at the edge of capacity)**.** *In an invented service model, support requests arrive at $\lambda = 3$ per working day and the technician can finish an average of $\mu = 4$ per day. Formula* (10.6) *gives one day of expected time in system. If demand rises to* 3.8 *requests a day without changing service capacity, the idealised waiting time becomes five days. The dramatic change follows from operating near capacity, not from a mysterious failure of the application. Actual repair durations require measured data and an appropriate queueing model.*

This insight can motivate local training: if staff can resolve safe, routine problems without central support, the central technician sees fewer requests. But it would be dangerous to transfer security-sensitive tasks to untrained staff merely to improve a queueing indicator.

10.28 A bounded-risk approach to deployment

A useful deployment decision does not rely only on expected cost. For high-consequence functions, require explicit protection against certain events: no silent deletion of committed financial transactions, no unauthorised disclosure of patient identities, no dispensing recommendation when required safety information is missing. These requirements define a feasible set, not just a penalty in an objective function.

A decision model may then minimise lifecycle cost among designs that pass the necessary safety and reliability conditions:

$$\min_{a \in \mathcal{A}} C(a) \quad \text{subject to} \quad \Pr(\text{unacceptable service loss under } a) \leq \epsilon, \quad a \in \mathcal{A}_{\text{governance}}.$$

The probability bound is a chance constraint that requires credible failure data or conservative justification. For rare events, a small demonstration may not support a precise estimate of an extremely low probability. Report the evidence behind the constraint, do not fill it with invented decimals.

10.29 Recovery exercise with a measurable acceptance test

A small service can demonstrate reliability without an expensive testing platform. Prepare an example database containing a known opening balance, receipts, issues and corrections. Record its checksum and expected reconciliation totals. Run the system on the actual low-cost machine, then deliberately interrupt power during the prescribed test operations. On restart, restore a previously recorded backup, replay the authorised outstanding events and compare the final state to the independently calculated expected state.

A suitable acceptance report records the number of deliberate interruptions, records correctly recovered, duplicates rejected, unresolved conflicts, time to safe service, observer and software version. It must say whether encryption, user authentication and access permissions were also tested. Passing a laboratory test is stronger evidence than merely asserting “offline-ready”, but it is not the same as years of operational reliability.

10.30 The service’s security boundary

Sensitive systems fail when nobody can state where trust starts and ends. Draw a data-flow diagram showing the field device, local database, sync server, backups, user credentials, exported reports and any third-party service. For each connection, ask who can read, modify, replay or delete data. A system may encrypt network traffic but leave an unencrypted laptop backup in an unlocked cupboard. The threat model should include such simple failures, not merely advanced attackers.

For a stock-management system, the institution might separate permissions to record an issue, approve a correction, change a supplier and export reports. Privileges should be reviewed when staff leave. Backup copies should be protected with the same care as active data. An offline system is not automatically secure because it is disconnected; a stolen device or malicious removable drive can still expose or corrupt records (NIST, 2020; Saltzer and Schroeder, 1975).

10.31 A field-ready incident log

A concise incident record includes discovery time, affected function, whether patients or service users were affected, data-integrity status, temporary fallback, responsible person, restoration time, evidence preserved and corrective action. A later review should distinguish immediate cause from contributing conditions. "User error" alone is not a sufficient root-cause analysis if the interface invited a predictable mistake and no recovery path was documented.

A recurring near-miss is valuable evidence. For example, several staff members may almost submit the same order twice after poor connectivity. Even if the idempotence guard prevents duplicate stock reductions, the repeated confusion is a usability problem worth addressing. Reliability includes making correct behaviour easy under actual working conditions.

10.32 An operational sustainability experiment

Ask a team that did not develop the program to use its documentation for an entire simulated service day. Give them an intermittent connection, one missing stock count, a supplier correction and a device shutdown. Observe where the written instructions are insufficient and whether the manual fallback preserves essential tasks. Repeat after training and compare the error count and time required. These are genuine measurements of the training exercise if undertaken and recorded, but they remain distinct from actual patient-outcome effects.

Finally, require the service owner to explain the annual budget and how the application would be retired without data loss. If nobody can do so, the technical demonstration is not yet an institutional handover. That conclusion may be uncomfortable, but it prevents a fragile prototype from becoming indispensable without support.

10.33 Exercises, failure drills and institutional investigations

1. Evaluate (10.2) for mean operating durations of 90 and 180 hours, with repair time six hours. Explain whether doubling operating time or halving repair time offers a larger modelled benefit.

2. Explain why an application can report high availability but still be unusable when patients arrive. Construct a time-dependent downtime penalty and compare two outages of equal duration.

3. Prove Proposition 10.2 fails if an identifier is recorded only after the stock balance update. Give an exact failure sequence.

4. Design an immutable correction ledger for a wrong receipt: list the minimum fields and identify which roles may perform the correction.
5. Build a backup and restoration drill for a facility without stable internet. Name what must be verified after restoration, rather than merely checking that the backup file exists.
6. Compare the present values of two service options under annual rates 0%, 5% and 15%. State which conclusions are sensitive to the discount assumption.
7. Write a threat model for a shared laptop containing aggregate clinic stock data. Include accidental loss, unauthorised exports and denial of service.
8. Draft a training exercise in which a staff member must reject a model recommendation with an incomplete demand record.
9. Design a procurement clause that guarantees an authorised institution can export its records and documentation when a vendor contract ends.
10. Use the companion code to change mean repair duration and document how simulation variation compares with the analytic expected availability.
11. With a service manager, develop a maintenance plan with annual costs, named responsibilities and a manual fallback. Identify which elements require local verification and which can be taught through simulation.
12. Decide whether to scale a tool with good model accuracy but frequent untracked manual overrides. What new evidence would change your decision?

What this chapter establishes. Successful deployment is a durable agreement between people, processes, mathematics and software. A model does not have a maintenance budget or legal responsibility; an institution does. Chapter 11 tests another limit: will the same working arrangement survive a different country, region or service environment?

Chapter 11

Adapting Solutions Across Countries and Contexts

The same equations can be useful in Uganda, Nepal or Peru without producing the same policy recommendation. This chapter distinguishes what can travel—a mathematical argument, an algorithm, a data schema—from what must be learned again: local demand, geography, regulations, costs and institutions. We derive a simple transport formula, show when it fails, and construct a practical adaptation record for new settings.

11.1 Why a successful pilot is not a universal answer

Suppose a health district introduces a reliable stock-planning tool and observes improved availability under a credible evaluation. A neighbouring district asks for a copy. The same code may run, but the medicines, lead times, budgets, staffing, data definitions and patients are not identical. There are at least three separate questions. Does the *software* run on the available devices? Does the *prediction* remain accurate with the new population? Does the *intervention* improve the outcome after local implementation? Passing the first test implies neither of the other two.

The challenge becomes more visible across countries. A water-service model fitted using urban demand data may be applied to rural settlements with different dry-season patterns. Distances computed on a road map may hide travel routes impassable after rainfall. A clinic may record prescriptions, while another records only items dispensed. The same variable name can describe different measurements. Good adaptation begins with those differences, not by changing the country name on the cover of a report.

We should also resist a misleading division between “advanced” and “low-resource” countries. Contextual limits exist everywhere. A rural district in a high-income country may lack reliable transport, while a large hospital in a lower-income country may have sophisticated infrastructure. The analytical question is which assumptions change, and whether the proposed decision remains justified.

11.2 An inventory of what can and cannot transfer

The basic conservation identity in Chapter 8 remains valid wherever physical stock is counted consistently. The empirical distribution of lead times is local. The same binary decision variable used in Chapter 5 can express whether a facility is opened, but the candidate sites, staffing and affordability require new evidence. The offline transaction requirements of Chapter 7 may travel almost directly; local-language labels and record-retention rules do not.

Table 11.1: Separating portable structure from context-specific content.

Object	Often portable	Must be checked locally
Physical model	Conservation law and units	Boundary, loss terms, measurement method
Statistical model	Estimator, validation procedure	Sampling frame, conditional relationships, calibration
Optimisation	Mathematical formulation and solver	Costs, feasible sites, capacities, constraints
Software	Transaction and backup patterns	Devices, language, legal permissions, support
Intervention	Logic of the causal evaluation	Actual adoption, comparison, population and effects

A *Country Adaptation Record* should describe the target institution, outcome definitions, differences from the source setting, new data requirements, feasibility tests, recalibration plan, legal checks and decision authority. A "transfer complete" label should be withheld until its specified acceptance tests pass.

11.3 Three distinct sorts of shift

Let X denote recorded predictors, Y the outcome of interest and S the setting. A prediction system trained in a source setting $S = s$ is commonly evaluated in a target setting $S = t$. Three situations deserve distinct names.

Covariate shift. The distribution of recorded inputs changes, $p_t(x) \neq p_s(x)$, but the conditional outcome relationship remains invariant: $p_t(y \mid x) = p_s(y \mid x)$. An example would be a different mix of rural and urban facilities, with the same outcome mechanism within each properly measured group. This is an assumption requiring evidence, not an automatic property of a map.

Conditional or concept shift. The relationship $p_t(y \mid x)$ itself changes. A lead-time predictor may change because procurement contracts, roads or delivery schedules differ even after controlling for recorded inputs. Reweighting the source data cannot generally correct this problem.

Measurement shift. The recorded variable changes its meaning or error mechanism. One site counts demand as prescriptions, another counts actual issues, and a third counts received orders. Even if the underlying physical process were similar, the observed Y is not. A model can look surprisingly good against a flawed reference outcome while making poor decisions.

The distinctions help choose a response. Reweighting, recalibration, refitting, revised measurement or a complete redesign solve different problems. The covariate-shift literature gives one precise set of tools and limitations (Sugiyama and Kawanabe, 2012; Ovadia et al., 2019).

11.4 Derive the transport identity

Let $L(f(X), Y)$ be the loss for fixed predictor f, with p_s and p_t the source and target joint distributions. Assume the target input distribution is absolutely continuous with respect to the source input distribution: wherever $p_t(x) > 0$, the source density is positive as well. Assume further that $p_t(y \mid x) = p_s(y \mid x)$. Then

$$R_t(f) = \mathbb{E}_t[L(f(X), Y)] = \mathbb{E}_s\left[\frac{p_t(X)}{p_s(X)} L(f(X), Y)\right]. \tag{11.1}$$

The ratio $w(x) = p_t(x)/p_s(x)$ is an importance weight. Equation (11.1) is an identity under stated assumptions, not proof that weights are estimable from a tiny dataset (Sugiyama and Kawanabe, 2012).

Proposition 11.1 (Risk identity under covariate shift)**.** *Suppose the source and target admit joint densities, $p_t(x) \ll p_s(x)$, $p_t(y \mid x) = p_s(y \mid x)$ almost everywhere on the target support, and the weighted loss is integrable. Then* (11.1) *holds.*

Proof. Starting from the source expectation, substitute the factorisation $p_s(x, y) = p_s(y \mid x)p_s(x)$:

$$\begin{aligned}\mathbb{E}_s[w(X)L] &= \int\int \frac{p_t(x)}{p_s(x)} L(f(x), y) p_s(y \mid x) p_s(x)\, dy\, dx \\ &= \int\int L(f(x), y) p_t(x) p_t(y \mid x)\, dy\, dx = R_t(f).\end{aligned}$$

Support inclusion makes the ratio well-defined where it contributes; integrability permits the stated expectations. □

Worked example 11.1 (A change in the mix of facilities)**.** *Suppose a binary error occurs at rate* 0.10 *in urban facilities and* 0.30 *in rural facilities. In the invented source sample,* 80% *of facilities are urban, giving source risk* $0.8(0.10) + 0.2(0.30) = 0.14$. *In the target region only* 30% *are urban, giving target risk* $0.3(0.10) + 0.7(0.30) = 0.24$. *The model has not changed; the population has. The importance weights are* 0.3/0.8 *for urban units and* 0.7/0.2 *for rural units. Weighting the source conditional errors recovers the target risk* only if *the conditional error rates really remain the same.*

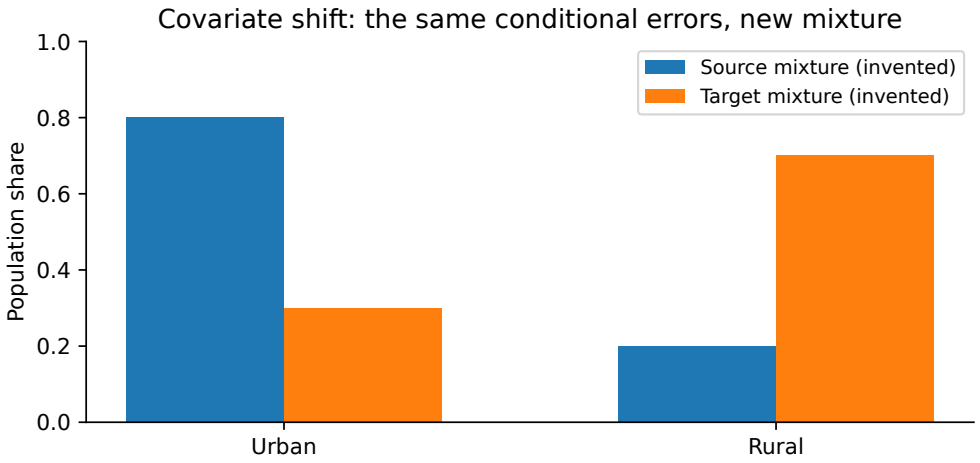


Figure 11.1: Invented source and target compositions for the covariate-shift example. The conditional error rates are assumed invariant, not empirically established.

11.5 Why weights do not create information

The transport identity can be useless in practice if a target population contains categories absent from the source, or if a few source observations receive enormous weights. For nonnegative sample weights $w_1, \dots, w_n$, a common descriptive effective-sample-size diagnostic is

$$n_{\text{eff}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}. \tag{11.2}$$

It equals n for equal positive weights and approaches one when nearly all weight concentrates on a single observation. This is not an exact number of independent observations; it signals potential instability. The weight estimates also have uncertainty, which should not be hidden by evaluating a plug-in weighted risk as though the weights were known.

Proposition 11.2 (Bounds on the effective-sample-size diagnostic)**.** *For $n \geq 1$ and nonnegative weights not all zero, $1 \leq n_{\text{eff}} \leq n$.*

Proof. Since $(\sum_i w_i)^2 = \sum_i w_i^2 + 2\sum_{i<j} w_i w_j \geq \sum_i w_i^2$, the lower bound follows. Cauchy–Schwarz gives $(\sum_i w_i)^2 \leq n \sum_i w_i^2$, yielding the upper bound. □

Clipping weights reduces variance but generally changes the estimand by introducing bias. If such regularisation is used, report the threshold and examine sensitivity. A target population with important unrepresented groups needs new measurements, not more aggressive mathematics.

11.6 Transporting an intervention is harder than transporting prediction

Equation (11.1) concerns predictive loss. A credible causal effect in one place does not automatically transfer when clinicians, suppliers or institutions act differently. Under additional

assumptions, the target effect can be written

$$\tau_t = \int \{\mathbb{E}_s[Y(1) - Y(0) \mid X = x]\}\, p_t(x)\, dx. \tag{11.3}$$

But this requires transportability of conditional causal effects, suitable overlap and valid identification of those effects in the source. If the new setting has different implementation intensity or an unmeasured effect modifier, the formula is not justified. Causal selection diagrams offer a more general language for expressing which mechanisms differ (Pearl and Bareinboim, 2014; Hernán and Robins, 2020).

Worked example 11.2 (Same algorithm, different service process)**.** *In one district, forecasting is the binding problem: orders are generally delivered on time, but quantities are often too low. A better forecast could reduce shortages. Elsewhere, quantities are adequate but deliveries are delayed for weeks. Reusing the forecast model may improve prediction scores without improving medicine availability. The active causal mechanism has changed. The second district may need procurement reform or a buffer-stock policy, not a more complex prediction model.*

This example also shows why a headline such as "the model improves access to medicines" is incomplete. It may be true for one operating system and false in another. The report must identify the intervention and the surrounding mechanism.

11.7 Local data can be sparse without being ignored

Adapting models often requires balancing information borrowed from the source with a small target sample. A hierarchical formulation might describe a site-specific parameter as

$$\theta_j \sim N(\mu, \sigma_\theta^2), \qquad \widehat{\theta}_j \mid \theta_j \sim N(\theta_j, s_j^2).$$

Under this explicitly Gaussian model, the posterior mean of θ_j for fixed hyperparameters is a precision-weighted average:

$$\mathbb{E}[\theta_j \mid \widehat{\theta}_j] = \frac{\sigma_\theta^{-2}\mu + s_j^{-2}\widehat{\theta}_j}{\sigma_\theta^{-2} + s_j^{-2}}. \tag{11.4}$$

No real service parameter is guaranteed to follow the Gaussian model. Its value is explanatory: noisy local estimates can borrow strength without being treated as identical. Measurement mismatch or severe nonexchangeability can make such pooling misleading (Gelman and Hill, 2007).

Before fitting a sophisticated hierarchical model, ask whether the new setting has enough reliable observations to check its assumptions. A small, well-audited local dataset may be worth more than millions of poorly aligned records from another institution.

11.8 Transfer also changes optimisation and fairness

Prediction is only one input to a decision. In location planning, demand estimates, travel-time matrices, candidate sites, clinic capacity and cost limits may all change. A solution a_s^* that

minimises source cost need not be feasible in the target setting:

$$a_s^* \in \mathcal{F}_s \quad \not\Rightarrow \quad a_s^* \in \mathcal{F}_t.$$

Even if the same problem formulation is appropriate, the target optimum is

$$a_t^* \in \arg\min_{a \in \mathcal{F}_t} \mathbb{E}_t\{L(a, \xi)\}. \tag{11.5}$$

The target feasibility set $\mathcal{F}_t$ should incorporate genuine local limits: staffing, road accessibility, budget, legislation and protected service standards. Replacing source prices with target prices without checking the rest of the constraints is not a complete adaptation.

Proposition 11.3 (A simple stability bound for a transferred decision)**.** *Suppose source and target share a feasible set $\mathcal{F}$, and their objective functions satisfy $\sup_{a \in \mathcal{F}} |J_t(a) - J_s(a)| \le \varepsilon$. If a_s minimises J_s and a_t minimises J_t, then*

$$0 \le J_t(a_s) - J_t(a_t) \le 2\varepsilon.$$

Proof. Insert $J_s(a_s)$ and $J_s(a_t)$ between the two target values:

$$\begin{aligned} J_t(a_s) - J_t(a_t) = [J_t(a_s) - J_s(a_s)] + [J_s(a_s) - J_s(a_t)] \\ + [J_s(a_t) - J_t(a_t)] \le \varepsilon + 0 + \varepsilon. \end{aligned}$$

Nonnegativity follows because a_t is target-optimal. □

The proposition gives a useful *conditional* guarantee: if every decision's expected cost changes by at most ε and feasibility does not change, source optimisation incurs at most 2ε regret in the target. It does not say that ε is small. The most valuable transfer task may be to measure or bound the shift. If the feasible set changes, the proof cannot be applied as written.

11.9 A reproducible cross-context teaching investigation

Our central teaching case considers water-service access in four *illustrative* district settings labelled Uganda, Kenya, Nepal and Peru. The labels signal intended contexts for later data collection; the numbers inside the supplied script are intentionally invented and should never be reported as country statistics. In each setting we vary settlement distribution, a resource limit and an assumed unit travel burden. The same facility-allocation logic is used, but we re-estimate demands and re-solve the decision rather than blindly copying the original site's solution.

For real work, appropriate starting points include the WHO/UNICEF Joint Monitoring Programme, whose datasets and documented application programming interface provide WASH estimates at several geographic levels where available (WHO/UNICEF JMP, 2026); national statistical services; geospatial and road data with licence and completeness checks; and locally verified infrastructure and cost records. National WASH percentages alone do not identify travel times, village-to-tap distances or facility construction costs. Joining those quantities without measuring their definitions creates spurious precision.

11.9.1 Using a public dataset responsibly

For each country, record the particular indicator, reference year, geographic level, source method, missingness and access date. Preserve the original downloaded file with a checksum. When comparing countries, check whether an indicator is defined and estimated consistently. A nationally representative estimate cannot automatically be used as a village-level input. If disaggregated estimates are unavailable, say so and choose a defensible aggregate analysis rather than inventing local measurements.

The companion programme `code/ch11_transport_audit.py` demonstrates the weighted-risk identity, effective sample size and failure when the conditional outcome mechanism changes. It also writes an invented cross-context allocation table. It contains no claimed measurements from the four countries. An extension instructs the reader to replace the invented data only after securing documented sources and preserving the original field definitions.

11.10 A decision-oriented adaptation protocol

Before moving a model, complete the following sequence.

1. **Name what is being transported.** Mathematical model, software, predictive relationship, decision rule or claimed causal effect?
2. **Define the new population.** Which users, institutions, seasons and sites are included? Who is excluded?
3. **Reconcile measurements.** Confirm units, event definitions, missingness and survey design.
4. **Check overlap.** Are important target conditions present in the source? If not, collect new information.
5. **Check mechanisms.** Which physical, social and institutional processes may have changed?
6. **Update constraints.** Recalculate costs, capacity, access, legal obligations and service priorities.
7. **Validate independently.** Use local held-out sites or times appropriate to the deployment question.
8. **Review equity.** Disaggregate by relevant geographic or service groups, with appropriate privacy safeguards.
9. **Test technical operation.** Confirm offline operation, accessibility, support and failure recovery.
10. **Decide and document.** Adopt, recalibrate, retrain, redesign, test further or reject the transfer.

A country adaptation record is especially important when a donor-funded project is reproduced in several locations. The transferable achievement may be the disciplined method of inquiry

rather than identical code or identical performance.

11.11 What constitutes evidence of successful transfer?

A successful installation proves that the software executes on the target platform. Acceptable target prediction errors establish empirical performance for the measured population and evaluation window. A feasible optimisation result establishes feasibility under specified constraints. A measured improvement against a credible comparison provides evidence of target impact. Maintaining those gains over time is yet another question. Each claim requires new evidence at the target location.

If an institution only possesses national aggregates, the appropriate conclusion may be modest: a proposed model is *plausible and testable*, but decisions at individual service sites are not yet supported. Readers should not confuse this restraint with pessimism. It is what prevents research from disguising missing facts as precision.

11.12 Deeper method: transfer decisions, not only models

A predictive model can achieve similar error in two places and still lead to different decisions. Suppose a shortage-probability model gives a facility risk $p = 0.3$ in both settings. In the first setting, an emergency order costs little and usually arrives the next morning. In the second, an emergency shipment is slow and expensive. Even with identical probability estimates, optimal precautionary stock may differ. The decision depends on local losses, constraints and response options, not simply the distribution of predictions.

Let action $a \in \mathcal{A}$ incur loss $L_s(a, Y)$ in setting s, and let $p_s(Y \mid X)$ describe local uncertainty. The optimal local action for observed covariates x is

$$a_s^*(x) \in \arg\min_{a \in \mathcal{A}_s} \mathbb{E}_s[L_s(a, Y) \mid X = x].$$

A transferred model may correctly estimate $p_s(Y \mid X)$ and still select a poor action if it retains an outdated loss function or feasible set. This is particularly important when monetary costs, clinical consequences or regulatory permissions differ. A good transfer audit must therefore validate the *decision rule* as well as the predictive component.

11.12.1 A two-action example

Suppose action $a = 1$ orders a protective reserve and incurs known additional cost c, while $a = 0$ leaves the reserve unpurchased. If a shortage occurs with probability p, the loss from not ordering is h and the reserve prevents that particular shortage under the simplified assumptions. The expected losses are $C(1) = c$ and $C(0) = ph$, so the reserve is preferred when $p > c/h$. A change in c or h changes the decision threshold even with the same model.

If the source has $c/h = 0.2$ and the target has $c/h = 0.5$, a predicted risk of 0.3 triggers the reserve in the source but not in the target. Blindly transferring the source threshold creates a policy mismatch. For essential services, hard safety constraints may override this cost

calculation; the point is the role of local consequences, not a universal threshold for medicine stocks.

11.13 Robust transfer when information is scarce

Sometimes a target district has so little reliable data that a fitted local distribution is not credible. In that situation a robust decision may be more defensible than a speculative precise optimum. Define an uncertainty set $\mathcal{P}_t$ of plausible target distributions and choose

$$a^* \in \arg\min_{a \in \mathcal{F}_t} \sup_{P \in \mathcal{P}_t} \mathbb{E}_P L(a, \xi). \tag{11.6}$$

This minimax formulation asks which action has the best worst-case expected performance across the declared distributions. Its strength depends entirely on whether the uncertainty set captures relevant possibilities. If $\mathcal{P}_t$ is unrealistically narrow, the result will still be fragile. If it is too broad, the method may produce excessive conservatism. Robust optimisation is a framework for representing uncertainty, not a substitute for measuring local conditions (Ben-Tal et al., 2009; Bertsimas and Sim, 2004).

11.13.1 An example in water allocation

Suppose a district can add capacity to one of two sites. Under normal seasonal demand, expanding site A costs less travel. During a dry-season shift, expanding site B serves a settlement that otherwise has no nearby functioning water source. A robust model may select B, but only if the dry-season scenario is considered plausible and the resulting cost is acceptable. A useful report displays the demand scenarios and the action preferred in each, rather than presenting the robust recommendation as an unquestionable scientific conclusion.

11.14 Source data quality and target performance

A common assumption in transfer studies is that more source data must be better. Not if their labels measure the wrong quantity. For a stock-out model, a source dataset with millions of dispensing entries may provide less information about latent demand than a small target dataset containing unmet requests during shortages. The source and target observations should be compared by their measurement process before any sample-size argument.

One way to make this concrete is to describe the observation equation in each setting. Let true requested demand be D, observed issues $Y = \min(D, I)$, and reported issues $Y^{\text{rep}} = Y + \epsilon$ with recording error ϵ. If source facilities have ample stock, issues approximate demand. If target facilities are often out of stock, issues are systematically censored. A model of Y^{rep} in the source cannot be assumed to identify D in the target. This is a measurement and identifiability problem, not merely covariate shift.

11.14.1 The danger of silent imputation

When a target field is unavailable, replacing it with the source average may seem practical. But it can erase exactly the difference that matters. If local supplier lead time is unknown, assigning a source mean of three days can create a serious shortage risk in a location where deliveries routinely require two weeks. If the decision depends strongly on that parameter, the correct response is to collect it, use a sensitivity range or withhold the recommendation until it is verified.

11.15 A practical hierarchy of transfer evidence

A sensible transfer project progresses through distinct levels:

1. **Conceptual portability.** Can the decision and mathematical model be defined coherently in the new setting?
2. **Data compatibility.** Do target measurements exist with comparable definitions and adequate quality?
3. **Empirical adequacy.** Does the method pass independent target validation for the task it will perform?
4. **Operational feasibility.** Can people operate it on their devices, under their budgets and legal duties?
5. **Demonstrated benefit.** Does a credible target evaluation show the expected improvement in actual outcomes?
6. **Sustained benefit.** Can the institution maintain the service after initial project support ends?

No later level is implied by an earlier one. A technically portable algorithm has achieved something useful, but its international benefit remains unproven until the later evidence exists.

11.16 A cross-country research collaboration that respects local expertise

A responsible international project should avoid treating local partners merely as providers of labels or devices. Their knowledge is required to define outcomes, identify missing constraints, interpret unusual measurements and understand institutional authority. Local experts may know that a route closes seasonally, that a record field changed after a procurement reform, or that users will not adopt a design requiring continuous internet. Those facts can determine the mathematical solution.

An equitable collaboration should agree in advance on data ownership, authorship or credit, capacity building, publication permissions, local validation and who will maintain the result. A study should not extract records, publish a favourable cross-country comparison and leave

participating institutions without access to usable tools. The book's emphasis on reliable decisions includes the responsibilities of the people conducting research.

11.17 Why success in one setting may require a different model elsewhere

The same research question can call for different mathematical instruments. Where water demand is stable and source yield is the limiting factor, a conservation model and deterministic capacity optimisation may be sufficient. Where demand is strongly seasonal, a time-series model and robust stock or storage decision may matter. Where data collection is weak, careful sampling and measurement could be the most valuable technical contribution. Where governance is the main bottleneck, the priority may be institutional redesign rather than computation.

A good international handbook should therefore teach readers to *choose* among methods, not to deploy one method everywhere. That is the sense in which its ideas can be world-class while remaining grounded in resource constraints: they are precise about what must be known and honest about when the available information cannot support a claim.

11.18 Extended investigation: a country label is not a dataset

Imagine four teams preparing water-service proposals for Uganda, Kenya, Nepal and Peru. The first team has a nationally reported drinking-water indicator. The second has a district road network and settlement estimates. The third has records from a small household survey. The fourth has costs supplied by a regional engineering office. A careless joint report could place the numbers in one table and compare them as though they measured the same thing. Such a table would be attractive and largely uninterpretable.

The teams should begin by defining a common target: for example, the proportion of residents of a specified district who can reach a functioning source meeting an agreed water-quality standard within a stated one-way travel time in a given season. A national survey estimate, a geospatial route model and administrative records may provide parts of the evidence, but none automatically gives this target quantity. Some sources measure access categories rather than routes. Some survey responses describe a household's primary source rather than its actual time under bad weather. Some administrative records count installed infrastructure without checking whether it functions.

11.18.1 A measurement compatibility table

For every source, write a row with unit, population frame, time period, observation process and likely errors. Examples include:

- a survey percentage with a weighted national population denominator and documented uncertainty;
- estimated road distance from a mapped settlement centroid to the nearest listed water point;

- service records showing hours that a particular pump was operational;
- a water-quality test with its date, tested contaminant and measurement uncertainty;
- the present-year local cost of installing and maintaining a particular type of point.

These quantities are complementary, not interchangeable. A rigorous cross-country comparison may therefore present different levels of evidence rather than a falsely harmonised ranking. It may also select one narrower question that can be answered consistently.

11.19 When it is sensible to borrow evidence

Consider estimating a regression parameter for demand response to seasonal rainfall. A source country has many observations; a target district has only a few. Borrowing strength may reduce estimation variance if physical mechanisms and measurement are similar. It can increase bias if the target uses a different water tariff, irrigation practice, or reporting protocol. Hierarchical modelling is not a guarantee of transferability. It formalises a set of assumptions about how source and target parameters relate (Gelman and Hill, 2007).

A transparent analysis might compare three predictors: a target-only baseline, a fully transferred source model, and a partially pooled model. Evaluate all three on a temporally separated local target set, report computational and data costs, and identify whether differences matter for the final decision. If the target sample is too small for a reliable comparison, the result should be an uncertainty statement and a measurement plan, not a declaration that the most sophisticated model "generalises".

11.19.1 A simple mathematical sensitivity bound

Suppose a policy loss is bounded in $[0, L_{\max}]$ and source and target joint distributions differ. If their total-variation distance is at most δ under the convention $\mathrm{TV}(P, Q) = \sup_A |P(A) - Q(A)|$, then for every fixed decision a,

$$|\mathbb{E}_t L(a, \xi) - \mathbb{E}_s L(a, \xi)| \leq L_{\max} \delta. \tag{11.7}$$

The inequality follows by scaling a bounded nonnegative loss to $[0, 1]$ and using the characterisation of total variation as the largest expectation difference over such functions. If the feasible sets are identical, Proposition 11.3 then bounds the target regret of a source-optimal decision by $2L_{\max}\delta$. This is a mathematically valid conditional statement, not an estimator of δ. Without enough comparable observations to establish a useful bound, the theorem may provide no practical assurance.

When losses are unbounded or the feasible set changes, this shortcut no longer applies. That is one reason physical access requirements and safety constraints are often more important to check than average loss alone.

11.20 A transfer audit for a prediction model

A workable audit has four layers. First, evaluate whether input fields still describe the same variables: height above sea level, travel time, patient age or daily demand. Second, compare source and target input distributions, including missingness and relevant subgroups. Third, measure target prediction performance on held-out observations that reflect actual deployment. Fourth, compute decisions and compare their costs, feasibility and distribution of consequences under local conditions.

A source model might maintain excellent aggregate accuracy while failing for remote facilities that were rare in the training sample. The audit must therefore choose meaningful groups before seeing the errors. Reporting only the average can conceal the very contexts that justified the adaptation project.

11.20.1 Calibration is distinct from ranking

For a risk predictor, discrimination assesses whether higher-risk cases tend to receive higher scores; calibration asks whether predictions have the intended frequency interpretation. A model trained where shortages are common may rank facilities sensibly elsewhere but systematically overstate absolute shortage probability. Local calibration may help if sufficient target outcomes are available and the ranking mechanism is reasonably stable. It does not repair a model built on predictors that are absent or newly defined. In a decision rule, recalibration may alter which cases cross a threshold, changing staffing and cost even if ranking remains unchanged (Guo et al., 2017; Ovadia et al., 2019).

11.21 Environmental and institutional stress tests

Cross-country adaptation should deliberately examine stresses that vary across places: seasonal road closure, power outages, cost of data connectivity, staff turnover, storage conditions, language, procurement bottlenecks and regulatory constraints. Some are measurable numerical parameters; others require structured interviews or practical tests. Their presence should not be reduced to a generic “country effect” without attempting to identify the mechanism.

For an offline application, a short test may be more informative than a long theoretical argument: install it on an intended device, disconnect the internet, enter a record, interrupt power, restart, and restore an encrypted backup. For a facility plan, verify a small set of travel-time predictions through field journeys at the times and seasons relevant to the decision. This converts otherwise invisible assumptions into evidence.

11.22 How to write a responsible multi-country result

A cross-country study should clearly separate *common method* from *common evidence*. One chapter can use the same optimisation model in several settings and show how different inputs change the recommended decision. That demonstrates a reusable computational procedure. It

does not establish that the recommendations are correct in each country unless the required local measurements and checks were performed.

A suitable report therefore contains a source inventory, local adjustments, validation results by setting, uncertainty, computational resource use, ethical and legal limitations, and a list of decisions that remain unsupported. If only one country has measured data, describe the others as scenario studies. This is still a valuable book example, provided that the boundary between observation and illustration is unmistakable.

11.22.1 A simple nontransferability counterexample

Let a source population contain only two types of sites, with predictors $X = 0$ or $X = 1$. Suppose the outcome rule is $Y = X$ in the source. A perfectly fitted predictor $f(X) = X$ has zero source error. In the target, if the outcome instead follows $Y = 1 - X$, the same predictor has error one under absolute loss at every site, even if the target and source distributions of X are exactly equal. Reweighting cannot help, because the conditional relationship changed. This elementary example is worth more than a claim that a certain algorithm is "domain invariant" without supporting assumptions.

11.23 Putting a Country Adaptation Record into use

The record should list source and target sites; a one-sentence decision; intended users; data-field definitions; documented differences in population, environment, service process and constraints; methods for identifying missing values; evidence on support and shift; independent target validation; subgroup and safety checks; governance permissions; and the final adoption decision. Include a table with an "unknown" column. The team should not be pressured to fill every unknown with a guess so that the transfer can be declared complete.

A capable researcher may conclude that the most useful next investment is local measurement rather than another model. This is consistent with the book's guiding question: what information and computation are sufficient to make this particular decision reliably? Transferability is not achieved by refusing to change a model; it is achieved by understanding which parts of the solution must change.

11.24 Mathematical workshop: which claims survive distribution shift?

When a model moves, a seemingly minor change in data collection can invalidate the formula used to transfer its results. We examine the conditions behind a few common procedures, then turn the mathematics into observable checks.

11.24.1 Standardisation for a target population

Suppose a programme's treatment effect differs between rural and urban facilities. In the source evaluation, the conditional effects are $\tau(\text{urban}) = 2$ and $\tau(\text{rural}) = 6$ outcome units. If urban facilities make up 80% of the source, the source average is $0.8(2) + 0.2(6) = 2.8$. If only 30% of target facilities are urban, the standardised target effect would be

$$\tau_t = 0.3(2) + 0.7(6) = 4.8.$$

The calculation is elementary. Its interpretation depends on a strong substantive assumption: the conditional treatment effects for urban and rural facilities would be the same in both settings after accounting for relevant effect modifiers. If rural target facilities have different roads, procurement contracts or implementation support, the effect of the intervention may change even within the rural category. Standardisation cannot correct omitted effect modifiers (Pearl and Bareinboim, 2014; Hernán and Robins, 2020).

11.24.2 Uncertainty in standardisation

If the conditional effects are estimated rather than known, their errors matter. Under a highly simplified model in which two conditional estimators are independent with variances v_u and v_r, the variance of a weighted target estimate with fixed weights p and $(1 - p)$ is

$$\operatorname{Var}(p\widehat{\tau}_u + (1-p)\widehat{\tau}_r) = p^2 v_u + (1-p)^2 v_r.$$

If they are correlated, include the covariance term $2p(1-p)\operatorname{Cov}(\widehat{\tau}_u, \widehat{\tau}_r)$. If the target proportion p is estimated from a survey, its uncertainty must also be considered. An exercise that treats every input as known may understate confidence even when the central formula is correct.

11.25 A prediction transfer audit with two stages

Suppose a binary shortage model was trained in a source region and will be used for new sites. Stage one examines whether the same information will be available at the time of decision: previous verified stock levels, approved orders, supplier delays and facility characteristics. A variable based on next month's deliveries cannot be used for an order made today. This is leakage, not a country-specific technical issue.

Stage two evaluates held-out target observations that resemble deployment: future times, different sites and routine rather than retrospectively corrected records. Report discrimination where relevant, calibration, subgroup error, uncertainty, and the operational loss of decisions under an explicitly chosen threshold. If the model predicts probabilities, compare its predicted and observed frequencies within suitably chosen groups, keeping the sample size of each group visible. Small samples can make apparent differences unreliable.

If the model's only advantage is a slight improvement on a random-row split, while a site-held-out test shows no benefit, do not justify a costly deployment. It may be more useful to collect new local data or choose a robust decision rule that requires less information. This repeats the book's central principle in a new setting: computation should earn its cost in relation to the decision.

11.26 When reweighting can make a result worse

In importance weighting, a source observation from a rare-but-important target subgroup may acquire a large weight. The estimated target risk can become highly variable. For example, suppose 99 source observations have weight one and a single observation has weight 99. Equation (11.2) gives

$$n_{\text{eff}} = \frac{(99+99)^2}{99+99^2} = \frac{39204}{9900} \approx 3.96,$$

although the raw dataset contains 100 observations. The computation shows severe concentration; it does not imply that the statistical uncertainty is exactly the same as having four independent observations. The correct response may be targeted data collection in the underrepresented group.

Proposition 11.4 (When source and target risk differ by at most a bounded shift)**.** *Let P and Q be probability distributions on the same measurable space and let g take values in $[0, M]$. Under $\mathrm{TV}(P,Q) = \sup_A |P(A) - Q(A)|$, one has*

$$|\mathbb{E}_P g - \mathbb{E}_Q g| \le M\,\mathrm{TV}(P,Q).$$

Proof. For $M > 0$, write $g/M \in [0,1]$. The layer-cake representation gives $g/M = \int_0^1 \mathbf{1}\{g/M > u\}\,du$. Fubini's theorem and the definition of total variation yield

$$|\mathbb{E}_P(g/M) - \mathbb{E}_Q(g/M)| \le \int_0^1 |P(g/M > u) - Q(g/M > u)|\,du \le \mathrm{TV}(P,Q).$$

Multiply by M. The case $M = 0$ is immediate. □

This is a bound on expectations under a known distributional distance. It is not evidence that the distance is small. If a target site has states absent from the source, or if local observations cannot establish an informative discrepancy, the formula can be numerically correct and operationally useless.

11.27 Data provenance exercise with international indicators

The WHO/UNICEF JMP publishes documented country and other WASH estimates for specified settings (WHO/UNICEF JMP, 2026). These are valuable for international context, but the operational task must match the indicator. An estimate of safely managed drinking-water service in households is not an estimate of a particular project's pump reliability; a school sanitation indicator is not a clinic water-quality measurement. A sound exercise begins by selecting one service type and one clearly defined indicator, rather than collecting all available figures and looking for attractive plots.

A reproducible student workflow is as follows. Visit the official API documentation; record the endpoint and field definitions; select country codes and a reference year; download the unmodified response to a named file; save the retrieval date and checksum; inspect missing values and uncertainty fields; construct a table using only comparable fields; and write an

interpretation that does not exceed the geographic and temporal resolution of the observations. If the interface or coverage changes, document the change and update the exercise. Nothing in this book asserts numerical national WASH results that have not actually been retrieved and checked.

11.27.1 Why national percentages cannot choose a village site

A national proportion is an aggregate over many households. Facility siting requires local locations, service needs and travel constraints. Two districts can have the same national contribution to a reported indicator while differing greatly in the distribution of underserved settlements. Inferring local access from a country mean is an ecological inference error. Even a high-quality national estimate cannot replace local field data for site-specific engineering decisions.

11.28 A transfer decision with three possible outcomes

After a local validation, the team might discover one of three situations. First, the model transfers with adequate performance and feasible operating costs: adopt with monitoring and local accountability. Second, the model remains potentially useful but needs recalibration, retraining or a changed decision threshold: perform the adaptation and validate again. Third, the model lacks required inputs, violates local constraints or creates unacceptable distributional harm: withdraw the transfer recommendation and reconsider the problem definition.

The team's credibility does not depend on always obtaining the first outcome. A transferable book of methods must teach readers when to stop or reformulate an innovation. The Country Adaptation Record should preserve every such decision and the evidence behind it so that future teams do not repeat an avoidable failure.

11.29 Exercises and cross-country investigations

1. Prove Proposition 11.1 for discrete predictors with two categories. State where the proof breaks if the target has a category never observed in the source.
2. In the urban/rural example, calculate the two importance weights and use them to recover target error 0.24. Compute the unweighted source error and explain the gap.
3. Derive the ESS diagnostic for $n - 1$ weights equal to one and one weight equal to M. Describe the limit as $M \to \infty$.
4. Construct two settings with identical input distributions but different conditional outcomes. Explain why importance weighting cannot repair transfer in this example.
5. Derive equation (11.4) by completing the square in a normal likelihood times a normal prior.
6. Give a setting in which the same inventory conservation equation transfers correctly but the same reorder point is dangerously low.

7. Recalculate a simple facility-location problem when one previously available road becomes impassable. Identify why source optimality no longer guarantees target feasibility.
8. Explain every assumption behind the regret guarantee of Proposition 11.3. Find a counterexample if the feasible sets differ.
9. Use the JMP documentation to choose one precisely defined WASH indicator. Write down its unit, reference year, population, geography and limitations before downloading any data.
10. Draft a Country Adaptation Record for transferring a school attendance or clinic appointment system from one district to another. Include legal and language differences.
11. Change the target covariate mix and conditional error rates in the companion code. Explain which experiment satisfies covariate shift and which does not.
12. Design an independent target validation that withholds entire sites rather than random records within sites. Explain when this distinction matters.

What this chapter establishes. A reusable method is not the same as a universally valid prediction or universal intervention effect. Transfer requires a map of what is invariant, new measurements of what changes, independent validation and a fresh decision. The final chapter brings those habits together into one complete project from first problem statement to handover.

Chapter 12

From Problems to Working Solutions: An Integrated Field Handbook

This final chapter is a working journey, not a summary of twelve topics. We begin with a community service problem and carry one transparent set of assumptions through problem definition, data checking, mathematics, prediction, optimisation, software, testing, evaluation and handover. The reader finishes with a completed illustrative solution dossier and an exact template for developing a real one.

12.1 A question that deserves an answer

Four settlements report difficult access to safe water. A local authority has a limited investment budget and can operate no more than two new distribution points. Some households are distant from the nearest existing source; others live close to an unreliable one. A sponsor proposes a mobile application that predicts where water will be needed. Yet the first decision may be more basic: *which new points, if any, should be built, and how will we know they improve access?*

We do not begin by developing the mobile application. We begin by visiting the service process conceptually: where does water come from, how is it stored, how do people reach it, who has the right to make siting decisions, and which records describe demand? Chapter 1 taught this discipline. The people who collect water may care about distance, cost, queuing, safety and hours of availability, while administrators also face capital budgets, land permissions and maintenance liabilities. The mathematical model should make these differences visible.

The complete case in this chapter is a *teaching example with invented numbers*. It is not a report about actual settlements or measured improvements. All quantities that would require field measurement are marked as assumptions. That convention is part of the work, not an apology for it.

12.2 Stage 1: write the problem definition record

A compact problem definition includes the decision owner, affected population, alternatives, outcome, feasibility conditions and risks. For the invented water-service case, the decision maker is the local authority, the primary decision is a set of distribution points, and the primary planning outcomes are accessibility and capital expense. Whether the built points deliver safe water reliably is a separate operational outcome that requires real monitoring.

Table 12.1: A completed *illustrative* problem definition record.

Field	Defined for this teaching case
Population	Four invented settlement groups A–D
Decision	Select at most two candidate water points and allocate planned demand
Primary planning criterion	Minimise total weighted travel distance for demand served
Capital constraint	Maximum 85 budget units (invented, no currency conversion)
Other constraints	Installed site capacities, at most two facilities, full planned coverage
Affected groups	Residents, maintenance workers, users requiring accessible routes
Evidence status	All numerical inputs invented; feasibility is mathematical only
Field evidence still required	Survey demand, walkable routes, source yield, water quality, land rights, financing

This record prevents the problem from changing unnoticed during model development. In a real study the stakeholders may discover that meeting a maximum travel-time standard matters more than minimising average travel distance. That is not a nuisance to suppress: it is a reason to revise the objective and document who authorised the change.

12.3 Stage 2: understand the system and audit the observations

A facility may serve water continuously or only at particular hours. Demand may mean population count, litres per day or observed visits, and these are not equivalent. Chapter 2 introduced dimensional reasoning: a daily water volume cannot be substituted for a head count in a capacity constraint without an explicit conversion. In this case our simplified demand unit is *one planning service unit per represented resident*, and capacities are measured in the same planning units. No claim about litres or hydraulic delivery follows without a corresponding physical model.

A field dataset should record the population denominator, observation date, season, source, units, missing values, coordinate accuracy, road conditions and consent or governance requirements. To avoid misleading travel distances, compare mapped travel paths with field observations, especially where rivers, slopes or seasonal routes matter. If population figures come from a gridded model rather than a census count, note the modelling and uncertainty. For service siting, uncertainty concentrated in remote settlements may change the decision even when the total population estimate is accurate.

12.4 Stage 3: formulate the mathematical choice

Let I index settlement groups, J candidate sites, d_i planned service demand, c_j the opening cost, K_j the site capacity, and ℓ_{ij} the travel distance for settlement i to site j. Introduce $z_j \in \{0,1\}$ for opening a site and $x_{ij} \geq 0$ for the amount of settlement i's demand allocated to site j. A basic capacitated facility-location model is

$$\min_{x,z} \quad \sum_{i\in I}\sum_{j\in J} d_{ij}^{\text{served}}\ell_{ij} \quad \text{with } d_{ij}^{\text{served}} := x_{ij}, \tag{12.1}$$

$$\text{subject to} \quad \sum_{j\in J} x_{ij} = d_i \qquad (i \in I), \tag{12.2}$$

$$\sum_{i\in I} x_{ij} \leq K_j z_j \qquad (j \in J), \tag{12.3}$$

$$\sum_{j\in J} c_j z_j \leq B, \qquad \sum_{j\in J} z_j \leq 2, \tag{12.4}$$

$$x_{ij} \geq 0, \qquad z_j \in \{0,1\}. \tag{12.5}$$

The first constraint serves all assumed demand; the second forbids sending demand to closed sites and limits capacity. The cost and site-count constraints express explicit institutional limits. Fractional allocations are interpreted as dividing aggregate demand across sites; if each settlement must be assigned to exactly one site, the model needs binary assignment variables instead. Chapter 5 supplies the wider optimisation background (Hillier and Lieberman, 2021; Boyd and Vandenberghe, 2004).

The objective is total person-distance under the teaching convention. Minimising it is not equivalent to minimising maximum distance, nor to guaranteeing water quality. When fairness requires a maximum acceptable distance $L_{\max}$, a route whose $\ell_{ij} > L_{\max}$ must be forbidden for relevant users, or the constraint must be formulated at an appropriate individual level. A policy decision is hidden whenever an analyst describes average distance as if it were every household's experience.

12.5 Stage 4: put all illustrative assumptions on one page

Table 12.2: *Invented teaching inputs* for the capstone. Distances are illustrative kilometres.

Settlement	Assumed demand	Distance to S1	Distance to S2	Distance to S3
A	30	2	6	12
B	25	3	3	10
C	20	11	2	4
D	15	12	6	2
Candidate capacity		45	50	55
Candidate opening cost		48	41	35

The overall assumed demand is $30 + 25 + 20 + 15 = 90$ service units. No single site's capacity covers it. Opening S1 and S2 costs 89 units, above the budget $B = 85$. The remaining two-site

combinations, S1+S3 at cost 83 and S2+S3 at cost 76, have enough total capacity. These facts can be verified before using a solver; such simple checks catch mistakes in code and assumptions.

12.5.1 One explicitly feasible allocation

For S1+S3, allocate 30 units from A and 15 from B to S1, and the remaining 10 from B plus all of C and D to S3. Its site loads are 45 and 45, and weighted travel is

$$30(2) + 15(3) + 10(10) + 20(4) + 15(2) = 315.$$

This is a feasible plan, *not automatically the global optimum*; a correct algorithm must check all feasible allocations. In this two-site case we can verify its optimality by examining the difference in cost between routing each settlement to S1 and S3: A saves 10 per unit at S1, B saves 7, while C and D are cheaper at S3. Since S1 has capacity 45, we first send all A's 30 units and the next 15 units of B to S1. That exactly gives the plan above. For S2+S3, the same savings argument must be repeated under S2's capacity.

Proposition 12.1 (Optimal allocation between two open sites)**.** *Suppose exactly two sites j, k are open, demand can be split continuously, their combined capacity covers total demand, and every unit must be served. Define savings $s_i = \ell_{ik} - \ell_{ij}$ from assigning settlement i to site j rather than k. Any feasible optimum can be found by allocating site j's available capacity to the largest positive savings first, while respecting the minimum total allocation to j imposed by site k's capacity.*

Proof. Begin with all units provisionally assigned to k. Moving u_i units of group i to j changes cost by $-s_i u_i$. The problem is to maximise $\sum_i s_i u_i$ subject to $0 \le u_i \le d_i$, $\sum_i u_i \le K_j$ and $\sum_i u_i \ge \sum_i d_i - K_k$. For two groups a, b with $s_a > s_b$, if $u_a < d_a$ and $u_b > 0$, shifting a feasible small amount δ from b to a increases the objective by $\delta(s_a - s_b) > 0$. Repeating exchanges sorts assignments by decreasing savings; the total allocation is chosen to satisfy the stated bounds. This proves the greedy characterisation for this divisible, two-site special case. □

The proposition is not valid for arbitrary nondivisible household assignments or general multi-site integer optimisation. The script `code/ch12_integrated_capstone.py` enumerates feasible opening pairs and optimises the continuous two-site allocations in this small example. It also asserts conservation, budgets and capacities as automatic tests.

12.6 Stage 5: make competing objectives visible

A plan that minimises person-distance may still leave an unacceptable burden on some individuals. Record not just the mean but the maximum assigned route, weighted share above a chosen limit and the worst-served subgroup. Here the decision maker could impose $\ell_{ij} \le 10$ for every route that carries positive demand. The proposed S1+S3 allocation meets that threshold only at the limit for B's diverted demand. A stricter maximum may render both currently affordable pairs infeasible under the assumed capacities; the correct output is then *infeasible*, accompanied by an explanation, rather than a fabricated compromise.

One can formulate a two-objective problem: minimise $(C(x,z), T(x,z))$, with capital opening cost C and travel burden T, under specified service constraints. A Pareto-efficient solution has no alternative that is at least as good in both criteria and strictly better in one. The pair of affordable opening options illustrates why the authority should inspect both costs and accessibility rather than hide the choice behind a single arbitrary weighted sum. If a weighting is used, it must express a decision-maker preference in consistent units.

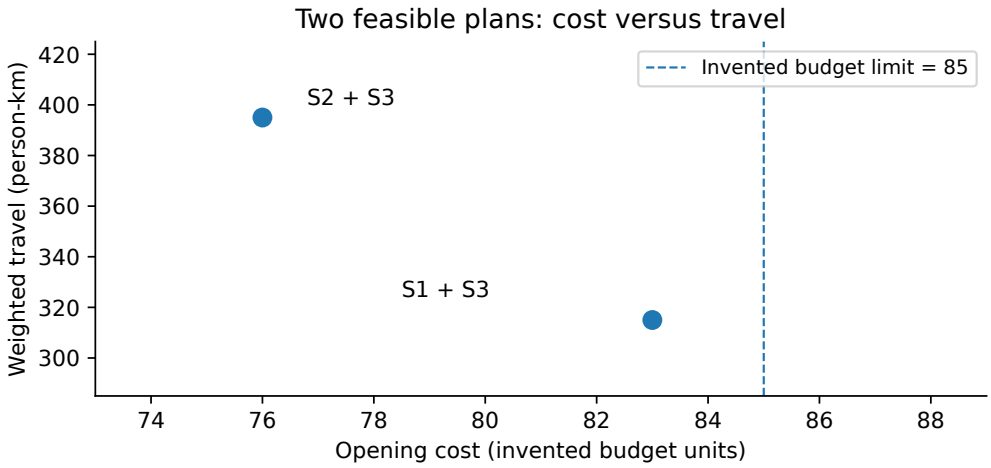


Figure 12.1: Cost and travel burden of the two feasible opening pairs in the fully invented water-service example. No real locations or measured travel times are used.

12.7 Stage 6: prediction only when it changes the decision

We have treated demand as known. Real demand varies by season, opening hours, population movement and source reliability. To forecast the next period, start with a transparent baseline such as a rolling mean, then compare against more complex models using an appropriate temporal and spatial hold-out. A prediction method earns its place only if it improves the quality of a decision enough to justify the extra data and maintenance requirements (Hyndman and Athanasopoulos, 2021; Roberts et al., 2017).

Let D_i denote uncertain future demand. A simple robust version of (12.2) may require enough capacity for all D in a declared uncertainty set $\mathcal{U}$. The larger the set, the more expensive the required plan may become. A stochastic version instead evaluates expected cost under a fitted and validated distribution. Neither is better by definition. Use the approach whose assumptions and consequences are defensible (Ben-Tal et al., 2009; Bertsimas and Sim, 2004).

Worked example 12.1 (Sensitivity that a decision maker can understand)**.** *Suppose future demand at settlement A could rise from 30 to 40 service units without a change elsewhere. S1's capacity is already fully allocated in the illustrated S1+S3 plan, but S3 has 10 spare units. The extra A demand could be routed to S3 only at a long assumed travel distance. A model that reports merely "capacity sufficient" ignores the access consequence. If service to A at that distance is unacceptable, additional capacity or a revised location choice is required.*

This is why sensitivity analysis belongs inside a planning decision. The purpose is not to produce dozens of plots but to determine which uncertain inputs could reverse the chosen action.

12.8 Stage 7: build only the application that is needed

For this project, a small offline tool could store settlement definitions, verified site capacities, approved cost estimates, planned demand, route measurements and each version of the allocation decision. A spreadsheet with transparent formulas may be sufficient initially. If several field officers must collect data independently, an offline-first application with unique record identifiers and transactional reconciliation may be justified, using Chapter 7's engineering principles (Kleppmann, 2017).

The tool should display a rejected recommendation as clearly as an accepted one. For example, "S1+S2 rejected: capital cost 89 exceeds limit 85" is more useful than an empty output. It should also report the date and source of demand assumptions and distinguish verified data from illustrative placeholders. The software must not silently substitute a default zero when a village's demand estimate is missing.

12.9 Stage 8: verification is a sequence of tests

The book's companion capstone script checks these properties, which should later become automated tests for any operational implementation:

1. **Arithmetic.** Total allocated demand equals total declared demand, within numerical tolerance.
2. **Capacity.** No site serves more than its declared capacity, and closed sites serve zero.
3. **Budget.** Opening cost respects the approved limit.
4. **Units.** Inputs and outputs are labelled consistently; costs and travel burdens are not added as if they have the same units.
5. **Monotonicity check.** Removing budget or capacity cannot enlarge the mathematical feasible set.
6. **Failure behaviour.** Missing data, impossible budgets and negative capacities raise explicit errors rather than plausible numbers.
7. **Reproduction.** A second reader obtains the same decision table with the same input and code version.

Proposition 12.2 (Feasibility under tightened resource constraints)**.** *For a fixed optimisation model, if the only change is to reduce available budget from B_2 to B_1 with $B_1 \leq B_2$, then $\mathcal{F}(B_1) \subseteq \mathcal{F}(B_2)$. Likewise, decreasing a capacity upper bound while fixing all other assumptions cannot add a feasible allocation.*

Proof. Every solution satisfying $\sum_j c_j z_j \le B_1$ also satisfies $\sum_j c_j z_j \le B_2$. The rest of its constraints are unchanged. The capacity statement follows from the same inclusion argument. □

The proposition is elementary but powerful as a software test. If the optimiser reports a new feasible plan only after the budget is reduced, check whether the underlying input, solver tolerance or other constraint changed unnoticed.

12.10 Stage 9: from a mathematical plan to a real pilot

The decision to build a distribution point requires more than a feasible allocation. The authority must verify water yield, safety, land agreements, maintenance funding and the actual routes used by residents. A pilot could then compare access, queuing time, service reliability and cost under a defensible evaluation design. If a comparison settlement is used, define why its trend is a credible counterfactual; if sites are randomised, analyse by the assignment unit. Chapter 9 provides the logic.

A plausible outcome is median minutes spent collecting a specified quantity of water over a defined observation period. Record water quality and availability separately. A shorter walk to an unsafe or frequently dry source is not a successful solution. Report who benefits and who bears extra travel or maintenance work. If only a feasibility plan has been produced, the manuscript should say exactly that.

12.11 Stage 10: decide whether the service can survive

The mathematical solution is a recommendation, not an operating institution. Staff must know who cleans, maintains, tests and repairs the water system, who pays, and what happens if a source fails. The service requires budgets and a route for complaints, physical inspections, backup supplies and a clear owner. Chapter 10 showed how to express those obligations and verify that they exist. Software reports can support the process, but cannot replace the infrastructure or decision authority.

A deployment readiness record therefore includes technical acceptance tests, institutional permissions, funding commitments, procurement rules, training, water-quality monitoring and a stop procedure. The absence of a reliable water-quality monitoring mechanism may be a reason not to commission the service, regardless of predicted travel savings.

12.12 Stage 11: what changes in another country?

A version of the project could be developed in any country where service-access problems exist, but the data schema alone is portable. Demand units, route measurements, climate, maintenance capacity, land permissions, tariffs and service rights may differ. An adaptation exercise should redo the problem definition and use documented local data rather than copying our invented example. Chapter 11 provides the transport and re-optimisation tools. Public

WHO/UNICEF WASH estimates can provide contextual information, but they cannot supply missing village-level routes and engineering costs (WHO/UNICEF JMP, 2026).

12.13 Stage 12: the solution dossier

A completed project should leave a structured record that another person can inspect or contest. The *Solution Dossier* developed through the book contains twelve elements corresponding to the intellectual journey, not a claim to have invented a new scientific method.

Table 12.3: Solution Dossier: the book's integrated set of deliverables.

Chapter	Evidence or artefact	Question answered
1	Problem definition record	Whose decision and what consequence?
2	Mathematical model and assumptions	Which system and units are represented?
3	Data provenance and quality audit	Can the measurements be trusted?
4	Prediction validation report	What can be predicted, for whom?
5	Feasibility and optimisation record	What action fits the constraints?
6	Model comparison and cost audit	Is AI worth using here?
7	Software tests and recovery plan	Does the tool work under actual conditions?
8	Domain safety and service analysis	What consequences matter in practice?
9	Impact evaluation protocol or evidence	Did the intervention cause an improvement?
10	Ownership and maintenance plan	Who keeps it operating and funded?
11	Adaptation record	What must be remeasured elsewhere?
12	Integrated acceptance and handover	Can another team verify and continue the work?

The case's dossier is complete as a *teaching prototype*: the inputs and objective are explicit, the feasible assignments are checkable, the code reproduces the comparison, the field measurements and validation still needed are named, and no improvement in real access is asserted. A production dossier cannot earn the same acceptance until field evidence and governance records exist.

12.14 Three further capstones

A book of methods should not dictate only one field of application. Each proposed project below uses the same reasoning but a different practical domain.

12.14.1 Essential medicines across a district

Define a measurable medicine-availability outcome; audit issue and stock records for censored demand; develop a ledger-conserving inventory model; compare simple and advanced demand

forecasts; optimise orders under budgets, delivery time and expiry; implement a transparent offline tool; plan a defensible pilot; document training and maintenance; and assess adaptation to another supply-chain structure. The critical task is distinguishing improved simulation results from observed patient benefit.

12.14.2 School resource allocation

Begin with a specific education decision: how to distribute scarce laboratory equipment or teaching time fairly across schools. Model student counts, class size, distance, timetables and staffing; check whether data correspond to the same school year; optimise allocations while protecting minimum service; develop a small auditable allocation tool; then evaluate whether actual access to lessons improves. The project must account for qualitative classroom constraints rather than infer educational value solely from equipment counts.

12.14.3 Flood-warning decisions

Choose a limited geographic area with available hydrological observations and agreed local warning responsibilities. Compare simple physical or threshold rules against predictive models using time-aware validation. Optimise a warning threshold against missed-event and false-alarm consequences, assess the cost and power needs of communication, test failure procedures, and design a prospective evaluation. Do not claim disaster-risk reduction from a retrospective prediction metric alone. This is a domain in which false confidence is particularly dangerous.

12.15 A practical assessment rubric

A capstone should reward disciplined reasoning and responsible evidence, not the use of fashionable algorithms. The following suggested marks total 100 points and can be adapted to course level.

Table 12.4: Suggested marking rubric for a final-year project.

Criterion	Marks
Clear problem and decision ownership; affected population and ethical limits	10
Model correctness, derivations, units, assumptions and failure cases	15
Data provenance, quality audit and honest evidence classification	15
Independent prediction/decision validation and appropriate baselines	15
Feasibility, resource cost and equity implications	10
Working, reproducible software with tests and failure handling	15
Credible impact-evaluation plan or appropriately qualified field evidence	10
Deployment, maintenance, transfer and quality of communication	10
Total	100

A project may score highly as a rigorous prototype despite having no measured deployment impact, provided it labels that limitation and provides a credible pathway to a field study. A

project presenting invented observations as real data should not receive credit for apparent impact, whatever the sophistication of its model.

12.16 Instructor and practitioner guide

For an introductory course, assign Chapters 1–3 to a problem-definition and data-audit exercise, followed by one chosen application from Chapters 4–8. For an advanced undergraduate course, require mathematical justification, computational tests and an explicit baseline comparison. For a master's or professional training programme, add Chapters 9–11 and require a written evaluation and deployment protocol in collaboration with a domain specialist. All levels can complete a version of the final dossier; expectations differ in the evidence and mathematical depth required.

The companion code is designed for a modest computer and an offline Python environment after installation. It does not require paid cloud services or a GPU. A university may distribute the synthetic fixtures with the exercises, but third-party observed datasets must be obtained and used under their own documented licences and protections. A reproducible result is useful only when another reader can discover what was actually run (Sandve et al., 2013; Oberkampf and Roy, 2010).

12.17 A second integrated capstone: essential medicine ordering

The water case teaches where infrastructure should be placed. A medicine-ordering project teaches another lesson: even when infrastructure exists, the institution may not observe the quantity it needs to predict. We return to the health-service problem of Chapter 8 and use it to show how the twelve chapters cooperate rather than simply sit next to one another.

A district clinic reports that a particular medicine is frequently unavailable. Staff propose training a neural network to forecast monthly issues. The project team first asks whether the ledger records actual demand. Suppose dispensing is capped by stock: if 40 patients need the item but only 25 units are available, the ledger records 25 issued units. A forecast fitted only to issues may predict 25 and appear accurate, while the unobserved need remains 40. This is the central measurement problem. The right next step may be a record of unfilled requests, not a neural network.

12.17.1 The problem record

The decision is an order quantity for specified medicines at a stated review time. The primary service aim is to reduce verified stock-out days without violating budget, storage, expiry and clinical safety constraints. The decision owner is the responsible pharmacy or supply-chain team, not an autonomous algorithm. The target population includes people served by the clinic, with attention to vulnerable groups who may face the greatest harm from shortages.

A full problem definition also identifies the existing policy and alternatives. These might include fixed reorder points, order-up-to levels, a simple moving-average forecast and a more complex predictive method. The intervention is not simply the introduction of software; it is a specified

change to the ordering practice, training and record review. Its consequences may be affected by supplier delays and procurement cycles outside the facility's control.

12.17.2 An illustrative stock equation

Let I_t be physical stock immediately before the daily transaction window, R_t receipts, Y_t issues, W_t wastage and A_t authorised corrections. A simplified balance is

$$I_{t+1} = I_t + R_t - Y_t - W_t + A_t.$$

An observed issue Y_t is not necessarily latent clinical demand D_t. When dispensing is stock-constrained, $Y_t \leq D_t$, and uncertainty about unmet demand must be retained. The model also needs the ordering pipeline: physical on-hand stock differs from inventory position, which includes outstanding confirmed orders and possibly commitments. A simple order rule that ignores these quantities can cause over-ordering or false reassurance.

12.17.3 A fully labelled example

For an illustration only, assume item A has mean demand 30 units per week and a two-week replenishment delay; item B has mean demand 20 per week and a three-week delay. Under a deterministic-demand simplification, lead-time demand is 60 and 60. Suppose the team adds hypothetical safety allowances of 15 and 10 units. The resulting illustrative reorder points are 75 for A and 70 for B. The safety allowances here are assumptions, not clinically validated quantities or statistical estimates.

If item A has 55 units on hand and 10 confirmed in transit, its inventory position is 65, below 75. Item B has 65 on hand and none in transit, also below its reorder point of 70. Both might trigger review. Deciding quantities requires the target order-up-to levels, pack sizes, budget, expiry and supplier rules. A simple threshold comparison is not a complete procurement optimisation.

Suppose proposed purchases are 40 units of A at four cost units each and 20 units of B at six units each. The illustrative cost is $40(4) + 20(6) = 280$, within a budget of 300. That arithmetic alone does not establish that the plan is safe, optimal or appropriately balanced. A budget-feasible purchase could still run out before the next delivery or waste stock through expiry. The clinic's actual clinical decision requires verified demand, lead-time variability, approved medicine lists and professional oversight.

12.17.4 From forecast to tested policy

To compare forecasting methods, hold out future periods rather than randomly shuffling daily records. Compare their predictive accuracy and the outcomes of the order rule when fed those forecasts. If outcomes are evaluated by retrospective policy simulation, account for the fact that changing the order policy changes stock availability and therefore what dispensing records would have been generated. Simply replaying observed issues under a new policy may understate unmet demand.

A sound report might compare a fixed threshold, a moving-average policy and an enhanced forecasting policy. For each, report assumptions on demand during stock-outs, delivery delays,

expiry, budget and emergency orders. Then compare the resulting stock-out metrics, working capital and sensitivity to adverse scenarios. Do not claim a simulated reduction in unfilled requests is an observed reduction in patients denied medication.

12.17.5 From policy simulator to offline service

The clinic needs a usable tool that records stock events, reconciles physical counts, rejects duplicate submissions, explains each recommendation and survives power loss. A correction must remain auditable. Staff must be able to override unsafe recommendations, and the system should state when required information is missing. These are the engineering duties developed in Chapter 7 and the institutional duties in Chapter 10.

Before actual use, the team must test recovery on intended hardware and establish data permissions, training, maintenance and a manual fallback. If a system is intended to influence clinical care, local regulatory and ethical requirements must be checked. The textbook's invented numbers cannot be used as a medicine-ordering policy.

12.17.6 From rollout to causal evidence

A reasonable pilot could compare the changed ordering practice with a credible alternative in suitable facilities, using independent verification of physical availability and carefully defined reporting periods. The design should account for supplier changes, concurrent budgets and shared procurement depots. Its evaluation would examine service outcomes and patient consequences, not merely the number of orders generated or the time the application remained open.

The team should expect the result to be uncertain at first. A pilot may find that staff can operate the system but that improvement in availability is not yet established. This can still justify a new measurement protocol or a revised design. Chapter 9's causal reasoning tells the institution what its evidence does and does not permit it to claim.

12.18 The mathematics of a solution portfolio

An institution may face several possible innovations at once: an inventory tool, a water-point improvement, a school timetable, and a maintenance programme. It cannot finance every project. The book's methods can help compare projects, but only after their outcomes are expressed responsibly. A simple portfolio abstraction is

$$\max_{z_j \in \{0,1\}} \sum_{j=1}^{m} b_j z_j \quad \text{subject to} \quad \sum_{j=1}^{m} c_j z_j \leq B, \tag{12.6}$$

where b_j is the documented benefit on a comparable scale, c_j total resource requirement and B the available budget. This expression is not a universal formula for social value. Water safety, patient care and educational opportunity cannot necessarily be converted into one invented score without institutional and ethical decisions.

The portfolio may instead use mandatory standards, a multi-criteria assessment and staged investments. One project can collect information needed before another is justified. A pilot may have value not only through its immediate service outcome but through reducing uncertainty about future decisions. The model should distinguish demonstrated benefit from hoped-for impact and avoid using optimistic scores merely to win funding.

12.19 The value of learning before spending

Suppose a proposed intervention costs much more than a targeted data audit. The team is uncertain whether shortages arise primarily from demand unpredictability or supplier delay. A short measurement study of requests, issued units and delivery dates may distinguish the two mechanisms. If the main problem is slow delivery, a forecasting system might offer little benefit; if the main problem is systematic under-ordering, a better policy could help. Measuring the actual mechanism before buying software may have high decision value.

Formally, for uncertain state θ, loss $L(a, \theta)$ and current information $\mathcal{I}$, the minimum posterior expected loss is

$$V(\mathcal{I}) = \min_a \mathbb{E}[L(a, \theta) \mid \mathcal{I}].$$

Additional observation Z can reduce expected optimised loss under standard decision-theoretic assumptions. The expected value of sample information can be written

$$\mathrm{EVSI} = V(\mathcal{I}) - \mathbb{E}_Z[V(\mathcal{I}, Z) \mid \mathcal{I}]. \tag{12.7}$$

This is nonnegative when the decision maker can ignore Z at no cost: after seeing Z, they can always choose the previous action. The argument does not imply that data collection is free or that every survey is worthwhile. The net value subtracts the cost and burden of measurement, including privacy and opportunity costs (Raiffa, 1968).

Proposition 12.3 (Nonnegative expected value of optional information)**.** *Suppose the set of admissible actions is the same with and without new information Z, all relevant expectations exist, and observing Z is free and may be ignored. Then the EVSI in* (12.7) *is nonnegative.*

Proof. Let a_0 minimise expected loss given $\mathcal{I}$. After observing Z, the optimal conditional choice has loss no larger than choosing a_0 in each state of information: $V(\mathcal{I}, Z) \le \mathbb{E}[L(a_0, \theta) \mid \mathcal{I}, Z]$. Taking conditional expectations over Z and applying the tower property gives $\mathbb{E}_Z V(\mathcal{I}, Z) \le \mathbb{E}[L(a_0, \theta) \mid \mathcal{I}] = V(\mathcal{I})$. □

This proposition gives a mathematical justification for a modest but important innovation principle: learning the right fact can be more valuable than computing an answer more accurately from the wrong facts.

12.20 From the book to a functioning local course

An institution using this book need not teach all twelve chapters in one semester. A problem-definition and measurement course can use Part I. An applied modelling and computing course

can use Part II, selecting cases relevant to students' backgrounds. A final-year project or professional laboratory can use Part III to demand impact-evaluation thinking, maintenance and transfer. The complete capstone provides continuity across these modules.

For students with limited programming experience, begin with hand arithmetic and a spreadsheet before moving to Python. Make each variable and unit explicit. For more advanced students, require proofs or counterexamples, independent computational verification and sensitivity analysis. For a professional team, require a decision owner, realistic data provenance and a written institutional handover. The same example can support different levels of study without sacrificing intellectual honesty.

12.21 A final guard against impressive but empty claims

The final report should answer five questions in one page. What was the exact decision? What information supports it, and which observations are missing? Which mathematical and computational checks were passed? What real-world outcome was measured, and what alternative explanation remains? Who can operate, challenge or withdraw the solution after the original team leaves?

If the answer to the fourth question is "no field outcome was measured," write that. If a mathematical theorem is proved, present the assumptions and proof. If a software test passes, show the conditions and result. If an intervention reduces measured harm under a credible comparison, report the magnitude and uncertainty. These distinctions do not weaken a strong book; they make its examples safe to learn from and its reasoning more widely reusable.

12.22 A second look at the capstone numbers

The small water example is deliberately simple enough to check by hand. That simplicity makes it a useful testing ground for serious habits. A solver should not be trusted because it returns a decimal. It should be trusted to the extent that its constraints represent the decision, its solution can be checked, and its limitations are understood.

For S1+S3, the opening cost is $48 + 35 = 83$, total capacity is $45 + 55 = 100$, and total demand is 90. The pair is affordable and has sufficient aggregate capacity. For S2+S3, opening cost is $41 + 35 = 76$ and capacity $50 + 55 = 105$. This pair is also feasible. S1+S2 costs $48 + 41 = 89$, which exceeds the budget of 85 although it has enough capacity. Thus a simple resource calculation eliminates one option before any travel optimisation.

For S2+S3, the optimal continuous assignment follows savings relative to S3. A unit of B saves $10 - 3 = 7$ travel units at S2; a unit of A saves $12 - 6 = 6$; a unit of C saves $4 - 2 = 2$; a unit of D would cost $6 - 2 = 4$ more. Site S2 has capacity 50, so allocate all 25 units of B and 25 units of A to it. The remaining 5 units of A, 20 units of C and 15 units of D go to S3. The weighted travel burden is

$$25(3) + 25(6) + 5(12) + 20(4) + 15(2) = 395.$$

Compare this with 315 for S1+S3. The more expensive opening pair reduces assumed aggregate travel by 80 person-kilometres per service cycle. The capital difference is seven invented budget

units. Neither the arithmetic nor the book can determine whether seven budget units are worth saving 80 person-kilometres without a locally defined time, welfare and cost model. That is the policy choice the model makes visible.

12.22.1 The danger in a false notion of optimality

The continuous allocation splits A between two water points in the cheaper plan. This is mathematically legitimate when individual users may choose different locations or the settlement represents multiple households. It is not legitimate if regulations or operational arrangements require all of a settlement's residents to use one designated point. In the latter problem, assignments are binary at group level, and the optimum may change. A research report must not describe a fractional solution as an implementable household-level instruction without checking its meaning.

Likewise, the opening cost is not the total cost of water service. It excludes land acquisition, pipes, pumping energy, water treatment, staffing, testing and maintenance. It would be irresponsible to announce that the district can solve its water problem for 83 units. The exact claim supported here is that one *specified* and *invented* facility-location model has a feasible solution satisfying its declared cost and capacity constraints.

12.23 What happens when the policy demands more fairness?

Suppose the local authority decides that no household should be assigned a route longer than eight kilometres. In the S1+S3 configuration, settlement B's units beyond the 45-unit S1 capacity would have to use S3 at ten kilometres. The demand of A and B totals 55, but S1 can take only 45. Thus S1+S3 cannot satisfy the eight-kilometre ceiling while serving everyone. In the S2+S3 configuration, A can reach S2 at six kilometres but not S3 within the limit, while B can reach only S2 at three kilometres (S3 is ten). The total of A and B is 55, above S2's capacity of 50. This configuration also fails. The mathematical conclusion is infeasibility under the additional limit, not that the eight-kilometre principle is unreasonable.

The next step is to ask which change could restore feasibility. The authority might raise the budget, increase site capacity, consider a fourth site, improve a route, or negotiate a staged service target. Each action has different costs and equity implications. A good optimiser can identify the violated constraint, but only stakeholders and evidence can determine which change is acceptable.

Proposition 12.4 (Simple necessary capacity condition for a distance ceiling)**.** *For a set $H \subseteq I$ of settlement groups and a maximum permitted distance L, let $N_L(H)$ be the set of open sites reachable within L by at least one group in H. If a feasible allocation serves every group's full demand using only routes of length at most L, then*

$$\sum_{i \in H} d_i \leq \sum_{j \in N_L(H)} K_j.$$

Proof. Every unit of demand from a group in H must be assigned to a site in $N_L(H)$. Summing allocations from H across those sites gives $\sum_{i \in H} d_i$. The total allocation into these sites,

including any demand from outside H, cannot exceed their combined capacities. Therefore the displayed inequality is necessary. □

The condition is a capacity test related to familiar network-flow feasibility ideas (Hillier and Lieberman, 2021). It is only necessary as stated; it is not an implementation algorithm. For S1+S3 and $L = 8$, take $H = \{A, B\}$: both can use only S1 within the ceiling, so $55 > 45$. For S2+S3, the same H can use only S2, so $55 > 50$. The proof identifies exactly why each option fails and what additional capacity would be required.

12.24 Constructing the data audit in a real service

The field survey should not start from a spreadsheet with prefilled "demand" values. It should ask what is to be measured and why. If daily volume is the target, then household number alone is insufficient: the calculation needs consumption, use patterns, alternative sources and seasonal changes. If travel time is the target, map-based straight-line distance is not a substitute. Test a sample of mapped routes with actual journeys at relevant times, noting whether walking, cycling or vehicle access is assumed.

The observations should be recorded in an auditable data dictionary. A recommended entry includes variable name, precise meaning, unit, allowed range, collection method, observation date, geographic resolution, source organisation, licence and likely errors. Retain missing values as missing; do not replace a settlement's absent demand with zero because a solver requires a number. A missing high-demand settlement could make the whole plan look cheaply feasible while leaving its residents unserved.

12.24.1 A second model for physical feasibility

The abstract planning units of the toy optimisation must eventually connect to a hydraulic and operational model. For example, a storage tank obeys a balance like the one derived in Chapter 2, while a network may be constrained by pipe capacity, pressure and source yield. Demand served at a point may depend on operating hours and queueing as well as nominal installation capacity. If the facility-location stage chooses sites whose source cannot provide the required water, the planning solution is physically invalid even though its linear constraints are satisfied.

A staged model hierarchy is often better than an unnecessarily complex monolithic system. Use a simple location model to explore alternatives, then verify shortlisted sites with hydraulic and field calculations, updating the optimisation constraints as necessary. Document when the simpler model was sufficient and when more detailed information changed the decision.

12.25 Testing the code against the mathematics

The companion Python script is intentionally auditable rather than optimised for scale. It enumerates opening pairs, rejects those that exceed budget, checks combined capacity and solves the two-open-site continuous allocation by a savings ordering justified in Proposition 12.1.

Its output contains both the summary and each nonzero assignment. The test checks that each settlement's allocation sums to the declared demand, each open site respects capacity, and tightening the budget does not add feasible options.

A useful student extension is to implement a completely independent enumeration of settlement-level binary assignments. This will generally produce a different feasible set, allowing the student to discover the consequence of changing decision variables. Another extension is to use a standard linear or mixed-integer solver and compare its solution against the hand-checked small instance. The point is not to celebrate agreement as proof of every implementation detail, but to establish layered evidence: a mathematical formulation, an independent calculation, an executable test and a documented limitation.

12.25.1 A failed test should teach something

Delete one settlement from the demand file and run the optimiser. Without an input completeness check, it may happily recommend a cheaper plan. The program has not broken; the interpretation has. A production system must know which settlements are required, recognise an omitted record, and refuse to present the incomplete result as a district-wide solution. Similarly, changing a distance's unit from kilometres to metres without conversion can dominate the objective while leaving all the code syntactically valid.

This is why software tests must include semantic checks, not only a successful exit status. A project dossier should list expected records, valid units, acceptable missingness and circumstances that require human review.

12.26 How the twelve chapters join in a field investigation

The book's arc can be read as three repeated questions at increasing levels of responsibility. The first is: *what is the problem and what is known?* It demands careful stakeholder engagement, a model and data evidence. The second is: *what should be done, and will the proposed system function?* It demands comparison, optimisation, domain expertise, computing and reliable engineering. The third is: *did it improve anything, will the institution sustain it, and does it apply elsewhere?* It demands evaluation, governance and adaptation.

A team may discover at any point that the correct next step is not another algorithm. The data may be inadequate; the proposed site may have no water rights; the institution may lack funds for testing; or a comparison may reveal no worthwhile impact. Each is an intellectually valid result, provided it is documented. The desire to produce a successful project should never override evidence.

12.27 A final practical template for a project team

Start a new project folder with these files or equivalents: `problem.md` for the decision record; `model.tex` for equations and assumptions; `data_dictionary.csv` for field meanings and licences; `audit.py` for input checks; `baseline.py` and `candidate.py` for competing methods; `tests/` for verification; `results/` for reproducible outputs; `evaluation.md` for the causal

evidence plan; `operations.md` for support and maintenance; and `adaptation.md` for target-context differences. These filenames are examples, not a mandated platform or a newly invented scientific standard.

Record the software environment, random seeds, data access conditions and the exact command required to reproduce each table. Where private data cannot be redistributed, supply a synthetic test fixture and a clear procedure that authorised reviewers can execute on the protected source. A package that needs undocumented manual edits is not reproducible simply because it contains Python code.

The final dossier should end with a clear decision statement: what may proceed now, what requires measurement or approval, and what must not be claimed. This is a practical discipline applicable to a university project, a ministry analysis, a community innovation or a startup. It is how a model becomes part of a solution without being mistaken for the solution itself.

12.28 Field laboratory: write the actual solution dossier

This laboratory makes the book's central task tangible. A student group or working institution should choose a problem with a real decision owner, even if access to operational data is not yet possible. The project may produce a fully verified teaching prototype, an analysis using documented public data, a partner-tested tool, or a formally evaluated intervention. These are legitimate but different achievements. The final report must make the evidence level clear.

12.28.1 Part A: the first conversation with a decision maker

Before writing code, conduct a structured interview or simulate one with a documented role description. Ask: what outcome is currently unsatisfactory; who experiences it; how is it observed; what decisions can change it; which options are already under consideration; what money, staff, legal permissions and time are available; and what would count as a worthwhile improvement? Write down disagreements as well as agreements. If the problem is "poor clinic service", narrow it to something observable and actionable rather than building a complex system around a slogan.

The resulting problem record should be short enough for a service manager to read and precise enough for a mathematician to formalise. Include the named decision owner and an agreed date for revisiting assumptions. In a classroom exercise, fictional roles must be identified as such; an imaginary stakeholder interview is not field research.

12.28.2 Part B: make a model that can be checked without software

For the chosen problem, state variables, units, parameters, decisions and constraints. Derive at least one conservation, accounting or probabilistic identity with visible assumptions. Construct a small instance that can be solved or checked by hand. In the water capstone, the sum of allocated units equals 90, opening cost must be at most 85, and no site load may exceed its declared capacity. Those are independent checks on any code.

An important part of this exercise is to create a counterexample. Change one assumption so

that the proposed solution fails: a road closes, a supplier doubles its lead time, a school loses a teacher, or a tank level sensor develops bias. Explain whether the old model becomes invalid, the optimal decision changes, or only the uncertainty increases. The ability to identify failure conditions is evidence of understanding.

12.28.3 Part C: prepare a defensible data audit

Make a data dictionary with every field's definition, unit, source, sampling frame, collection time, missingness rule and permitted uses. If there is no operational dataset, say that explicitly and use labelled synthetic fixtures. If a public dataset is used, preserve the original download, its licence and checksum, and distinguish its estimates from the institution's own observations.

For survey data, inspect weighting and clustering; for monitoring data, inspect sensor calibration and time alignment; for routine service logs, inspect whether records are censored by service failure. Explain which planned conclusions are not identifiable from the available measurements. Inadequate data may justify a new sampling plan rather than a more complicated prediction model.

12.28.4 Part D: compare a simple method with the proposed method

Every substantial computational proposal should have a relevant baseline. Examples include a fixed reorder threshold, a moving-average forecast, a simple assignment rule, or a spreadsheet that applies a verified formula. Compare it with a more complex alternative on exactly defined tasks and held-out evidence. Measure not just prediction error but the resulting decision loss, computational time, memory use, device requirements and support workload when relevant.

If a sophisticated method offers no useful improvement over a simpler one, the project should be allowed to recommend the simpler method. The aim is not to demonstrate that AI can be used, but to find a sound and affordable solution to the stated problem.

12.28.5 Part E: produce one independently reproducible result

A final-year project should provide the command that reproduces its main table or figure from declared inputs. Fix random seeds where randomness is used; record versions; include an expected output check; and explain which observations are invented, publicly obtained or operationally measured. The Chapter 12 example can be rerun from a command line and produces two CSV files: one for feasible site combinations and one for each allocation. An instructor can compare those files with the hand derivation of 315 and 395 person-distance units.

Reproducibility does not mean every reader has permission to publish confidential source data. A protected project can release its code, synthetic fixtures and verification instructions while making actual observations available only to authorised reviewers. The final report should not imply that a synthetic fixture is a de-identified real dataset.

12.28.6 Part F: demonstrate the tool in a failure scenario

A practical application must survive the environment in which it is intended to operate. Choose a failure: loss of internet during an entry, duplicate submission, incorrect user permission, power interruption, invalid units or missing records. Before running the test, state the expected safe behaviour. Afterwards, record observed behaviour and any discrepancy. A system that rejects unsafe input with an intelligible explanation may be more valuable than one that continues producing a precise but false number.

The demonstration should be carried out on suitable low-cost hardware where possible. If only a development laptop is available, call the result a software test, not evidence of performance on a specific field device. Hardware performance claims require measurements on that hardware.

12.28.7 Part G: design the impact evaluation before claiming impact

Define what an improvement would look like to actual service users. For water access, possible outcomes include travel time, water quality, service uptime and household cost; for medicine inventory, availability, unfilled requests, expired stock and patient burden. Identify a feasible comparison and its assumptions, the planned measurement period, missingness rules, potentially affected groups and the institution's permission to collect data. Write a plan for what would count as negative or inconclusive evidence.

If the project has not reached deployment, the evaluation protocol is a deliverable but not an evaluation result. An impressive simulation does not justify changing the wording to "demonstrated improvement." That distinction must survive editing and publication.

12.28.8 Part H: institutional ownership and another context

Prepare a one-year operating plan showing who will run the service, maintain the hardware and software, correct records, answer incidents, fund recurring costs and safely withdraw it if necessary. Then choose one realistic alternative setting and ask which data, constraints, language, permissions and infrastructure would have to change. The adaptation record should distinguish reusable mathematics and code from empirical assumptions that require new evidence.

A final report should give a manager an answer and give another researcher a way to challenge it. It is not sufficient merely to present equations and screenshots; the work must show what it does not establish.

12.29 What a complete project submission looks like

A model solution should contain the following narrative sequence: the observed or proposed problem; the decision and competing objectives; the verified mathematical structure; the measurement and data limits; the baseline and selected method; the feasibility and uncertainty results; the tested software; the impact-evaluation evidence or protocol; the deployment obligations; and the transfer and limitations statement. Equations, code, figures and tables should be placed where they answer a question in that story.

A useful oral examination asks each student to explain an important choice without reading a slide: why a particular decision variable was used, why a piece of data could be trusted, why the baseline was appropriate, what test could falsify the conclusion, and what the institution would need to do next. Someone who can answer those questions has learned more than a sequence of Python commands.

12.30 An example of a defensible final recommendation

For the invented water case, a transparent recommendation would be: "Under the stated planning units, capital budget, candidate capacities and divided-demand assumption, S1+S3 minimises weighted travel among the two affordable pairs. S2+S3 is cheaper but has larger weighted travel. Neither configuration can meet an eight-kilometre maximum-assigned-distance requirement with all planned demand served. The inputs are invented, so no physical site should be selected from this calculation. A real decision requires verified demand, actual route accessibility, water quality and yield tests, site permissions, maintenance costing and consultation with residents."

That statement contains a meaningful mathematical result, its policy trade-off, its exact limitation and an actionable next step. It embodies the book's aim better than a dramatic conclusion unsupported by observations. When field evidence is eventually available, the same structure can support a stronger recommendation with measured outcomes.

12.31 Final exercises and fieldwork

1. Reproduce the two-site feasibility check using the invented demand, costs and capacities. List all three two-site opening pairs and explain which is financially infeasible.
2. Prove Proposition 12.1 by a pairwise exchange argument. Explain why the proof does not handle indivisible households.
3. Run the capstone script and compare feasible opening pairs. Explain why a lower capital cost need not mean lower travel burden.
4. Impose a distance ceiling of eight kilometres. Identify which routes cannot carry demand and analyse whether the remaining capacity is sufficient.
5. Increase settlement A's assumed demand by ten units and recalculate a feasible allocation or show why the old one must change.
6. Convert the aggregate allocation variables into binary whole-settlement assignments and compare the feasible set. Which model is more suitable when a village must use a single service point?
7. Prepare a data dictionary for a real water-service study. Identify units, sampling frame, provenance, errors and licence conditions.
8. Choose one possible prediction model and one simple baseline. Specify a geographically independent test of their use in the siting decision.

9. Write five acceptance tests for a small offline application collecting site capacity, access and cost estimates. Include failure and rollback behaviour.
10. Draft an impact-evaluation protocol that distinguishes shorter journeys from improved availability and safe water quality.
11. Develop a three-year maintenance budget and identify the institution responsible for every recurring cost.
12. Write a Country Adaptation Record for a different setting, naming five assumptions that must be rechecked.
13. **Final submission.** Produce the twelve-part dossier in Table 12.3. Mark each claim as mathematical, simulated, retrospective empirical, prototype-tested or impact-evaluated. Supply code, data permissions, tests, evidence limits and a decision about what may proceed next.

A final word

A working solution begins as a question that somebody can answer and ends as a responsibility somebody can carry. Between the two stand models, observations, mistakes, calculations, software, people and institutions. None should be hidden. An elegant result has value even if it tells us that a proposed service is not yet affordable, the data cannot identify demand, or the benefit cannot yet be established. Such answers can prevent waste and protect people.

The enduring skill is not knowing a particular algorithm. It is being able to explain, in language a colleague can check, what problem was addressed, why the method was chosen, what was observed, what remains uncertain, and what it would take for the result to improve someone's life. That discipline is not limited by national income. It is the foundation of useful innovation anywhere.

Appendix A

Printable Records for a Working Solution

This appendix provides forms that a student, lecturer or practitioner can adapt. They are not substitutes for local institutional policies or professional judgement. Their purpose is to keep decisions and evidence visible. The same forms can be maintained as plain text files or spreadsheets when printing is expensive.

A.1 Project cover and claim register

Every project should begin with its status. A mathematical construction, software demonstration and measured intervention effect must not be given the same label.

Project title	
Decision owner	
Prepared by and date	
Institution and intended users	
Population or service boundary	
Current decision and proposed alternatives	
Evidence level	Mathematical / synthetic simulation / retrospective observed / prototype-tested / impact-evaluated
Approval and access restrictions	
Next decision and required evidence	

For every central claim, add a row to a claim register:

Claim	Supporting evidence	What would disprove or weaken it?

A.2 Problem Definition Record

Question	Response to complete before choosing a method
What is going wrong?	Observable outcome, not just a proposed solution
Who is affected?	Groups, locations and possible unequal consequences
Who can act?	Named authority or decision maker
What can change?	Set of feasible actions, including doing nothing
What matters?	Outcomes, constraints, safety standards and equity
What is already known?	Data, measurements, expert and local knowledge
What is missing?	Most decision-relevant unknowns
What would count as success?	Concrete measure and meaningful threshold
What could fail?	Model, data, software, institution and unintended harms
What is the next step?	Measurement, modelling, trial, deployment or withdrawal

A.3 Mathematical Model Record

The model record must give its purpose, state variables, units, parameters, inputs, decisions and constraints. Include a small example that can be checked independently.

Item	Required entry
Target decision	
Variables and units	
Parameters and provenance	
Governing equations	
Objective and constraints	
Assumptions	
Identifiability limits	
Verified special case	
Counterexample or failure condition	
Numerical method and tolerance	

A.4 Data Dictionary and Quality Record

For each field, include an unambiguous definition. A short worked pattern is:

Field	Definition	Unit / type	Quality and permission
`stock`	Physical units counted at agreed time	count	Count date, identifier, verifier
`issue`	Units dispensed in recorded interval	count	Censored by shortages; not demand
`lead_time`	Delay from approved order to receipt	days	Record definition and supplier

The data quality record includes source and retrieval date; permissions and licensing; intended

population; missing-value mechanism; sensor or survey uncertainty; temporal and geographical resolution; units; cleaning actions; and what claims remain unsupported. Store the original data separately from cleaned files. Do not overwrite the source after transformations.

A.5 Prediction and Decision Audit

Question	Evidence or answer
Which decision uses the prediction?	
What is the simplest relevant baseline?	
Which information exists at decision time?	
How are training and test separated?	
Which subgroups or locations are held out?	
What calibration and uncertainty are reported?	
What decision costs are compared?	
Which resource costs were measured?	
When must the model abstain or be withdrawn?	

A.6 Evaluation Protocol

Required item	Pre-specified description
Intervention and implementation	
Comparison and assignment mechanism	
Population, unit and follow-up period	
Estimand and primary outcome	
Minimum worthwhile effect	
Missingness and measurement plan	
Statistical method and clustering	
Confounding and spillover risks	
Cost and equity consequences	
Ethics, consent and permissions	
Criteria for proceeding or stopping	

A.7 Deployment, Maintenance and Transfer Records

A handover record identifies the service owner, technical custodian, data steward, domain reviewer, ordinary support process, security permissions, recovery objectives, tested backup, annual budget and withdrawal procedure. A transfer record identifies the source and new setting, unchanged mathematical structure, changed measurements, target population, new constraints,

external validation, adaptation costs and the final adoption decision. These documents can be short, but they should be owned and revised by named people rather than living only in a researcher's notebook.

One rule for every form. A blank unknown is preferable to an invented fact. Write *unknown* and state how it could be measured. Reliable innovation often begins when an institution can see exactly what it does not yet know.

Appendix B

Reproducing the Computational Examples

This appendix describes the supplied source package. The book's numerical files are teaching demonstrations unless a specific chapter explicitly documents observed data. No programme downloads confidential records or silently replaces invented examples with real observations.

B.1 Directory structure and basic environment

The editable package contains `main.tex`, chapter files in `chapters/`, figures in `figures/`, Python programmes in `code/`, generated tabular outputs in `generated/`, the reference list and a requirements file. The source package includes its compiled PDF so that readers can inspect the edition without compiling it first.

A standard Python 3 installation with `numpy`, `pandas`, `matplotlib` and `scikit-learn`, in versions compatible with `requirements.txt`, supports the earlier examples. The four new chapter programmes use Python's standard library. The LaTeX document uses a conventional pdfLaTeX toolchain with common mathematical and graphical packages. Package installation depends on the local operating system; institutions should test a reproducible environment before teaching.

Listing B.1: Basic reproduction commands from the source package directory.

```
python -m pip install -r requirements.txt
python code/reproduce_chapters.py
python code/reproduce_extensions.py
python code/ch09_impact_evaluation.py
python code/ch10_service_reliability.py
python code/ch11_transport_audit.py
python code/ch12_integrated_capstone.py
python code/test_chapters_09_12.py
latexmk -pdf -interaction=nonstopmode main.tex
```

For an offline installation, download approved installation files in advance and verify their provenance; install locally with the Python packaging tool's offline options. A facility is not

required to use a paid cloud platform or GPU for the supplied teaching scripts. Some live-data extensions require data access and separate permission; they are not prerequisites for the synthetic demonstrations.

B.2 Expected values for independent checks

The new scripts print the word `SYNTHETIC` or `INVENTED` where appropriate. Their basic expected results include:

Script	Core check
Chapter 9	Known simulated effect is 8; estimated DiD varies with seeded noise, and changes if untreated group trends differ
Chapter 10	$A(240, 12) \approx 0.952381$, $A(240, 3) \approx 0.987654$; duplicate transaction replay does not change balance
Chapter 11	Source risk 0.14, target covariate-shift risk 0.24; conditional shift raises illustrated target risk to 0.345
Chapter 12	S1+S3: opening cost 83, travel 315; S2+S3: opening cost 76, travel 395; S1+S2 exceeds budget

The exact seeded output for Chapter 9 is recorded in generated CSV files; its sample estimate is not expected to equal the data-generating effect exactly. Chapter 10's simulated availability values approximate, but need not exactly equal, the analytic values. The checks are designed to catch accidental changes and distinguish deterministic mathematical results from Monte Carlo estimates.

B.3 Evidence classification in filenames and captions

Do not name a synthetic results file `real_impact.csv` or describe it as field-tested in a figure caption. Suggested prefixes are `synthetic_`, `public_observed_`, `partner_restricted_` and `pilot_evaluated_`, used only when their meaning is actually supported. A filename is not evidence, but clear naming reduces confusion when tables are copied between drafts.

When a real public dataset replaces a teaching fixture, record its source, licence, retrieval date, original checksum and transformation script. When protected institutional data are analysed, keep raw records in an authorised environment and distribute only permitted aggregates, synthetic fixtures and reproducible methods. A synthetic fixture generated to imitate a real system is still synthetic.

B.4 A compact reproducibility report

For each computational result report:

1. the question, outcome and mathematical formulation;
2. the provenance of all inputs and the reason each is admissible;

3. software and dependency versions, machine characteristics where performance is claimed, and random seeds;
4. an exact executable command and expected output file;
5. comparisons with a hand calculation or independent implementation;
6. sensitivity, missingness and failure-case checks;
7. what is proved, measured, simulated or merely hypothesised;
8. the limits on use in real service decisions.

A result is reproducible when an independent reader can follow the steps and obtain the claimed output from authorised inputs. It is not necessarily accurate about the physical world. Verification of computation and validation of the underlying model remain distinct (Sandve et al., 2013; Oberkampf and Roy, 2010).

B.5 Working with low-bandwidth teaching laboratories

A lecturer can distribute the source package by local storage, extract it on student computers and use the synthetic fixtures without internet. Begin with a printed table and a short hand calculation. Students should then run the script, compare outputs and explain a deliberate failure condition. This sequence remains educational even if a machine cannot install a large scientific library. More advanced work with live environmental or survey data can be an extension for institutions with appropriate access and permissions.

Appendix C

A Catalogue of Common Failure Modes

A useful handbook should tell readers how apparently successful projects go wrong. The following cases recur across disciplines. Each case gives a misleading conclusion and a better test.

C.1 The outcome is not the quantity of interest

Recorded medicine issues may represent available stock rather than actual demand. A traffic counter may miss vehicles when the sensor fails. A school attendance sheet may be filled only on days when the teacher is present. Ask what process produces the measured number and whether it changes under the proposed intervention.

C.2 The units are wrong

A water-volume balance may mix litres with litres per second, or a cost model may add annual and daily expenditure. List the units of every term and check dimensional consistency before numerical optimisation. Mathematically precise arithmetic with inconsistent units is not a model of the intended system.

C.3 A missing record becomes a zero

A blank stock count is treated as zero stock; an unvisited settlement is treated as zero demand; an absent survey answer is treated as no service use. Preserve missingness explicitly and investigate whether it depends on the outcome. Report bounds or scenarios when the quantity is not identifiable.

C.4 The test set leaks future information

A forecast for tomorrow uses a feature calculated from next week's deliveries. A spatial prediction is tested by randomly withholding rows collected at the same monitoring station. Use a validation design that matches the proposed deployment decision, including time and site separation.

C.5 An optimiser solves the wrong problem

A facility-location solution minimises average distance but violates a maximum access requirement. A schedule satisfies a solver's time budget but ignores travel or cleaning time. Revisit the problem record and write explicit feasibility checks with the affected service users.

C.6 A high score hides costly mistakes

A classifier is accurate overall but misses the rare cases that trigger serious harm. Evaluate errors in relation to operational consequences, calibration, uncertainty and affected groups. Compare the model with a simple baseline and permit abstention when essential information is missing.

C.7 Code tests pass but the operation fails

An offline application saves correctly in one session but duplicates a stock deduction when the same event is uploaded twice. A backup exists but cannot be restored by local staff. Test failure and recovery on intended devices and require independently verified reconciliation.

C.8 A pilot improves but the cause is unclear

Service improves after installation while new budgets, staff or supplier contracts change at the same time. Record the assignment mechanism and choose a defensible comparison. If a causal design is not available, report the change as an observation with explicit alternative explanations.

C.9 A tool has no owner after the project ends

A student or external developer leaves, maintenance stops and the only copy of the data becomes inaccessible. Specify ownership, documentation, export, training, spare parts, funding and a withdrawal procedure before scaling.

C.10 Success is copied into a different context

An algorithm that worked in one district is deployed elsewhere without checking data definitions, travel conditions, regulatory duties or service costs. Distinguish transferable mathematics from locally estimated inputs and intervention effects. Revalidate in the target environment.

C.11 A manuscript describes invented figures as field impact

Synthetic data are useful for explaining a method and checking a computation. They do not demonstrate real-world benefits. Label them clearly, use measured data only where authorised and available, and distinguish mathematical proofs, retrospective observations, operational tests and credible causal evidence.

The recurring remedy. Start with the actual decision, ask what information and resources it requires, construct a checkable method, and make its limits visible. This is the common discipline joining all twelve chapters.

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